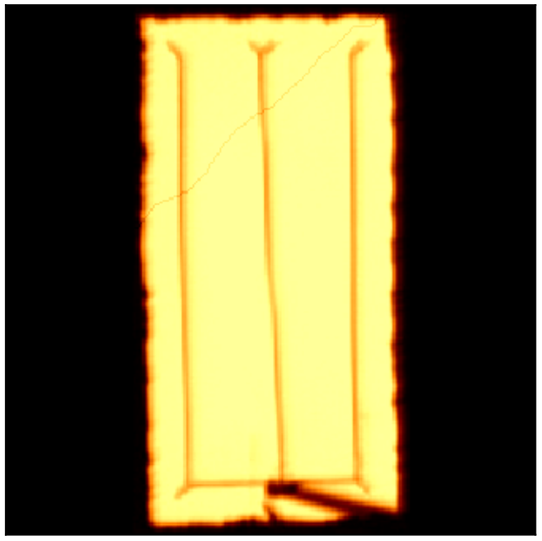
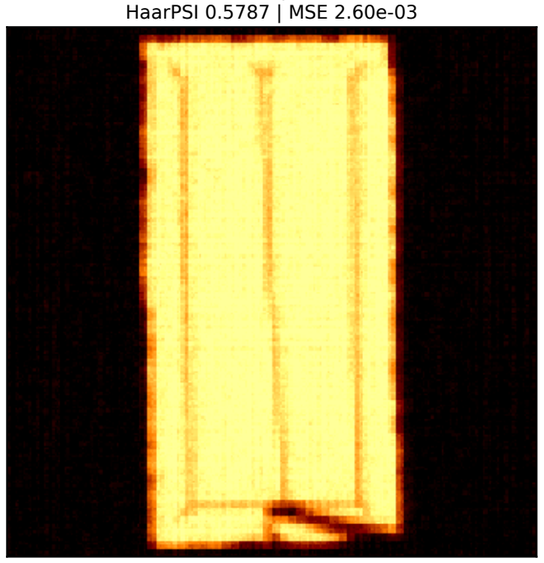
CIGS | test image 0068 | 20% Hadamard + NPL-default noise

Target
Ground truth



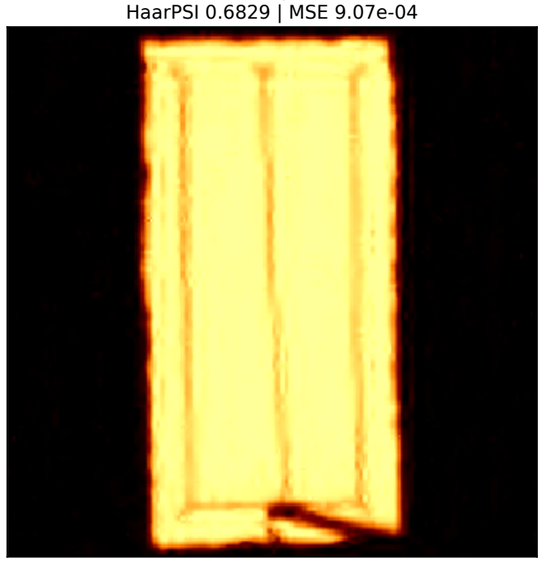
Zero-filled
PSNR 25.84 dB | SSIM 0.4814
HaarPSI 0.5787 | MSE 2.60e-03



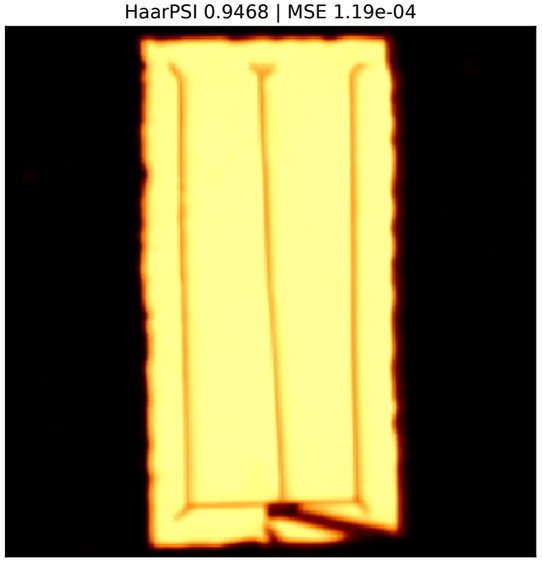
SPGL1-BP
PSNR 26.44 dB | SSIM 0.4456
HaarPSI 0.6652 | MSE 2.27e-03



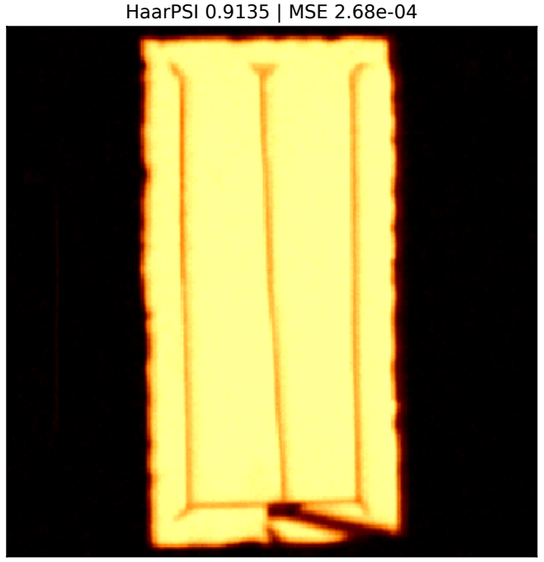
SPGL1-BPDN
PSNR 30.42 dB | SSIM 0.7745
HaarPSI 0.6829 | MSE 9.07e-04



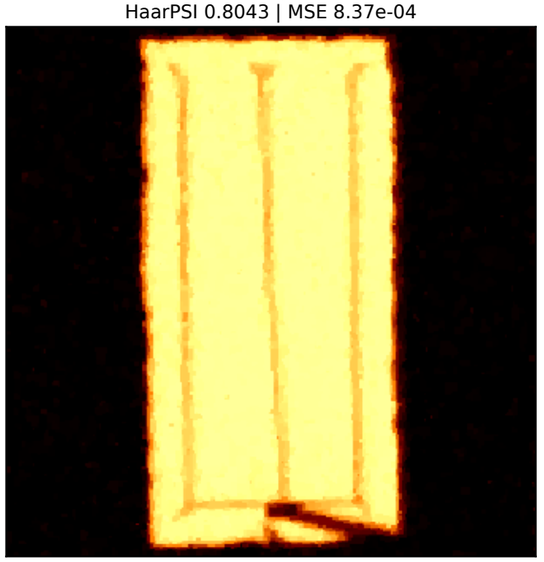
PnP-ADMM tuned-5
PSNR 39.23 dB | SSIM 0.9757
HaarPSI 0.9468 | MSE 1.19e-04



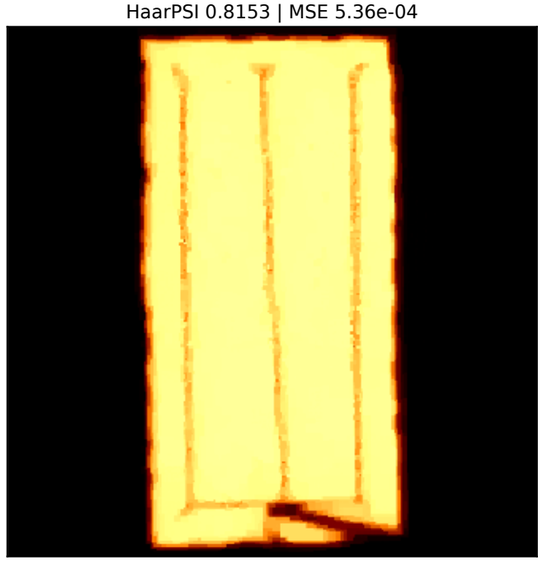
GS-PnP oracle
PSNR 35.72 dB | SSIM 0.8351
HaarPSI 0.9135 | MSE 2.68e-04



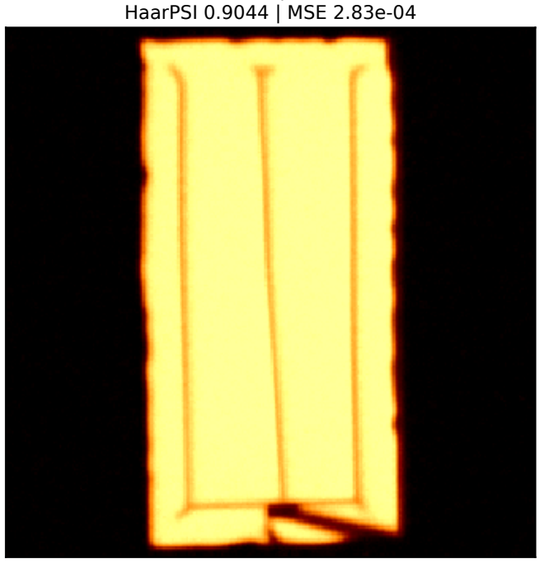
Scalar TV
PSNR 30.77 dB | SSIM 0.7427
HaarPSI 0.8043 | MSE 8.37e-04



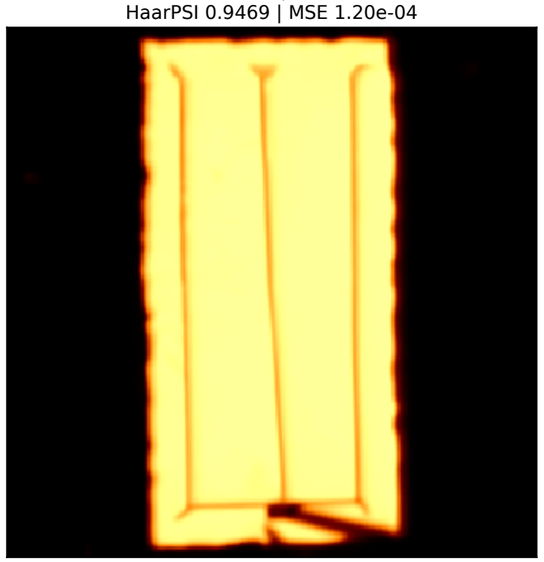
UTV-C60
PSNR 32.71 dB | SSIM 0.9347
HaarPSI 0.8153 | MSE 5.36e-04



GS-PnP tuned-5
PSNR 35.48 dB | SSIM 0.8618
HaarPSI 0.9044 | MSE 2.83e-04



PnP-ADMM Tune-1 (0768)
PSNR 39.22 dB | SSIM 0.9773
HaarPSI 0.9469 | MSE 1.20e-04

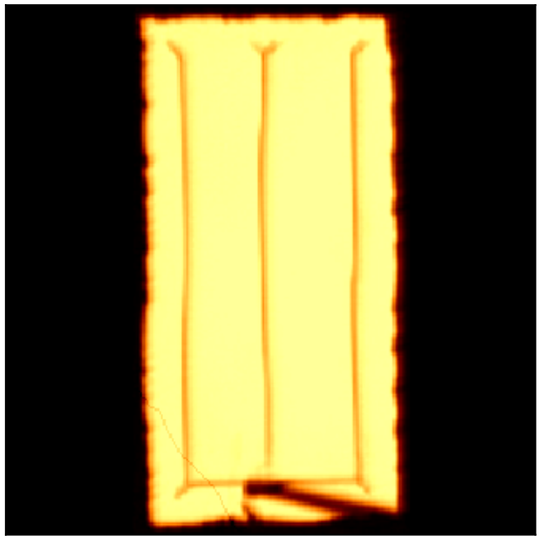


GS-PnP Tune-1 (0530)
PSNR 35.33 dB | SSIM 0.8225
HaarPSI 0.9122 | MSE 2.93e-04

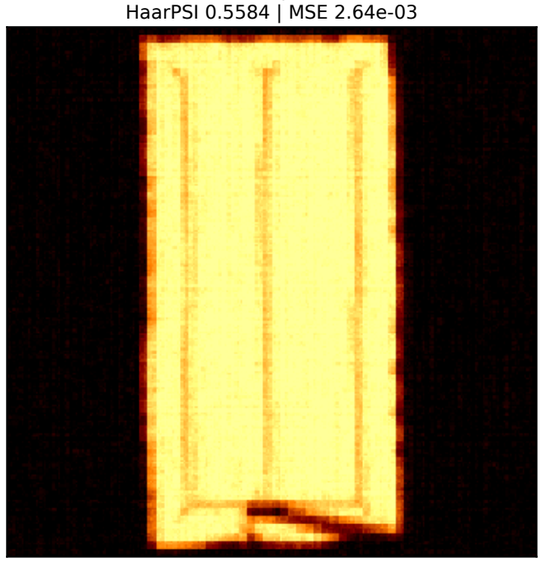


CIGS | test image 0095 | 20% Hadamard + NPL-default noise

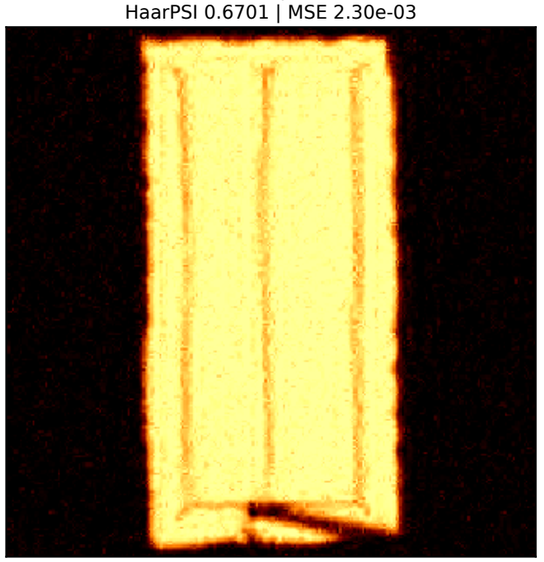
Target
Ground truth



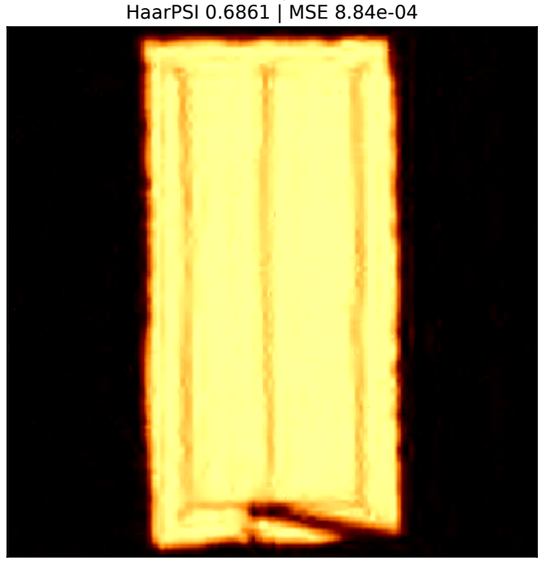
Zero-filled
PSNR 25.79 dB | SSIM 0.4868
HaarPSI 0.5584 | MSE 2.64e-03



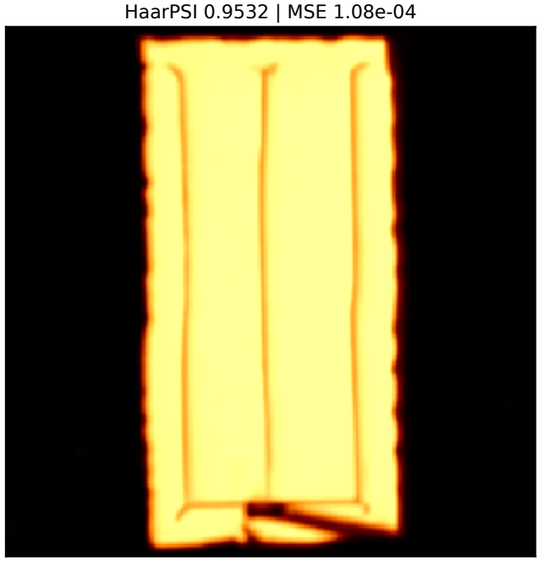
SPGL1-BP
PSNR 26.38 dB | SSIM 0.4419
HaarPSI 0.6701 | MSE 2.30e-03



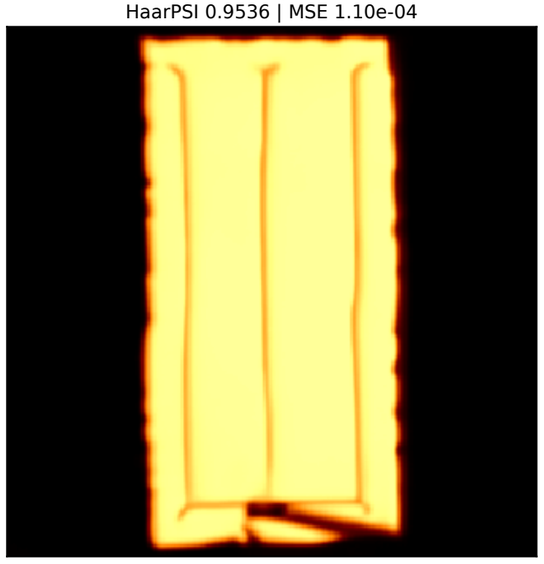
SPGL1-BPDN
PSNR 30.53 dB | SSIM 0.7770
HaarPSI 0.6861 | MSE 8.84e-04



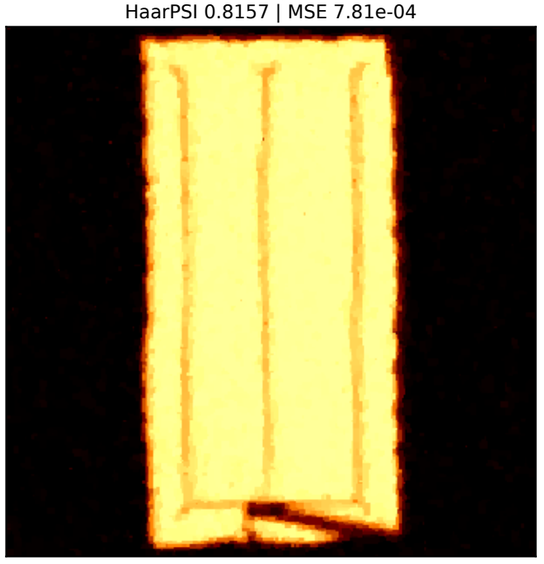
PnP-ADMM tuned-5
PSNR 39.68 dB | SSIM 0.9808
HaarPSI 0.9532 | MSE 1.08e-04



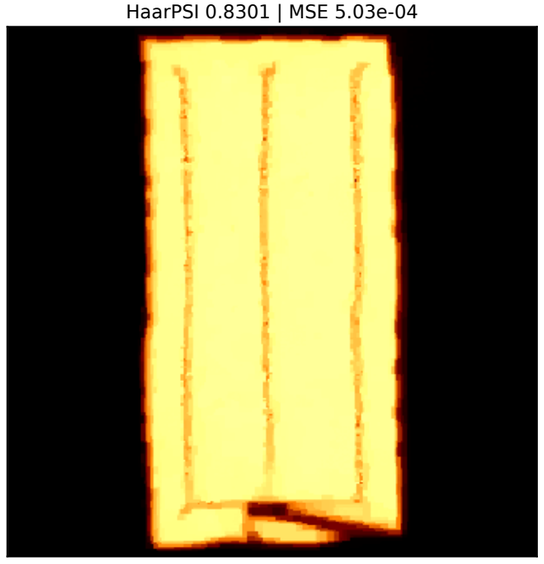
GS-PnP oracle
PSNR 39.61 dB | SSIM 0.9791
HaarPSI 0.9536 | MSE 1.10e-04



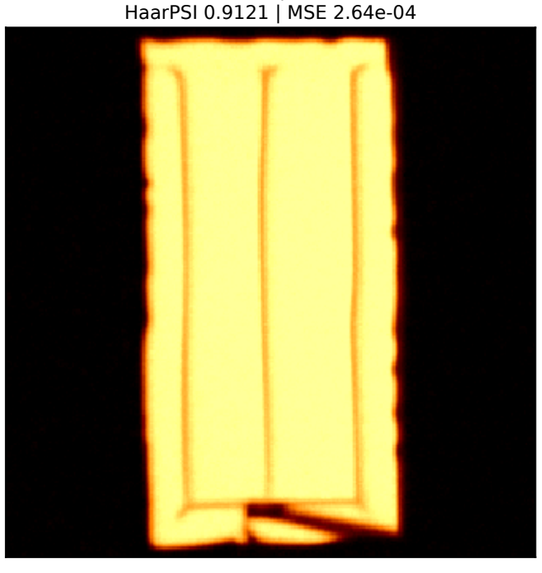
Scalar TV
PSNR 31.07 dB | SSIM 0.7550
HaarPSI 0.8157 | MSE 7.81e-04



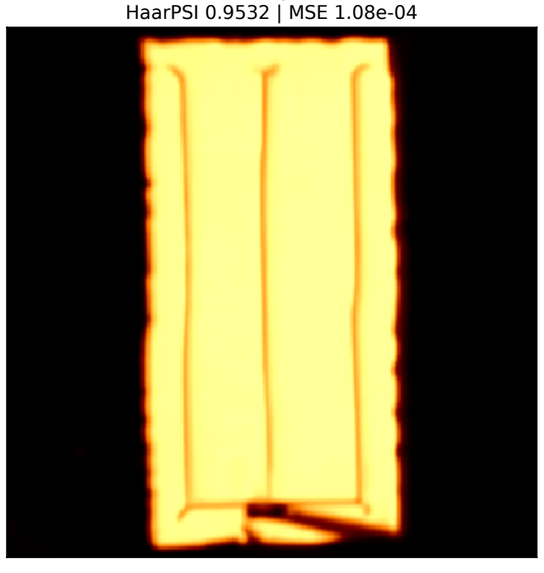
UTV-C60
PSNR 32.98 dB | SSIM 0.9383
HaarPSI 0.8301 | MSE 5.03e-04



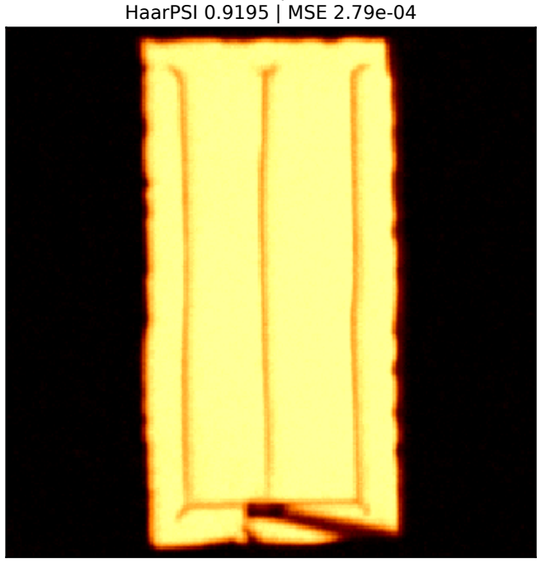
GS-PnP tuned-5
PSNR 35.79 dB | SSIM 0.8697
HaarPSI 0.9121 | MSE 2.64e-04



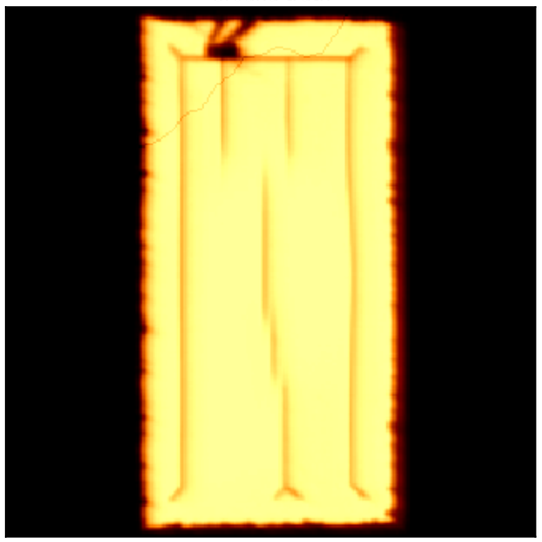
PnP-ADMM Tune-1 (0768)
PSNR 39.69 dB | SSIM 0.9821
HaarPSI 0.9532 | MSE 1.08e-04

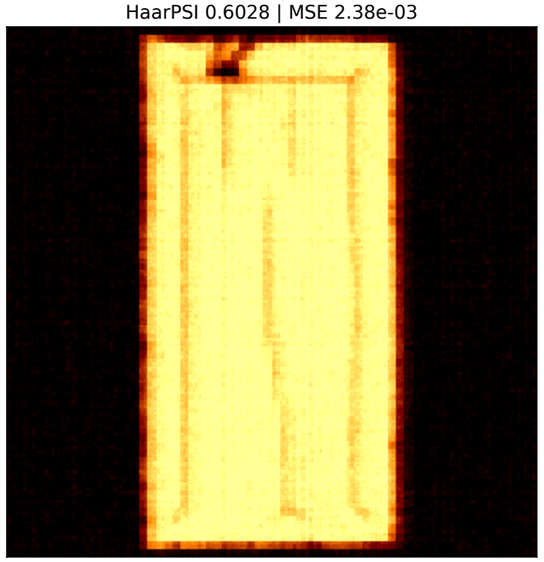


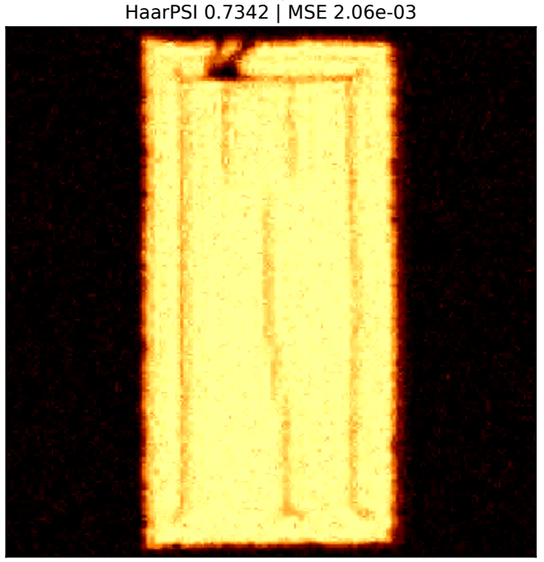
GS-PnP Tune-1 (0530)
PSNR 35.54 dB | SSIM 0.8310
HaarPSI 0.9195 | MSE 2.79e-04

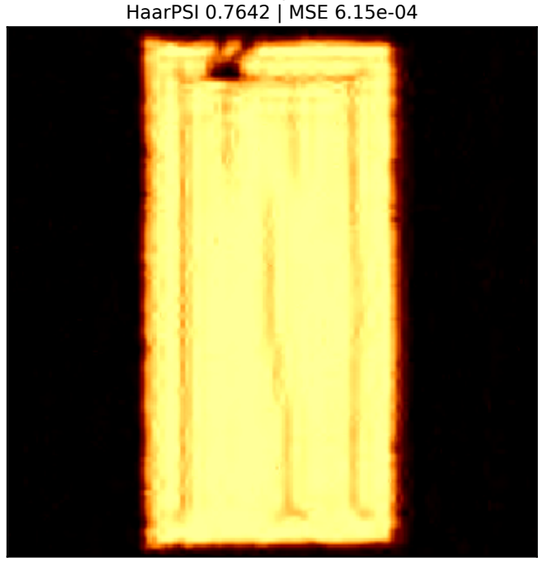


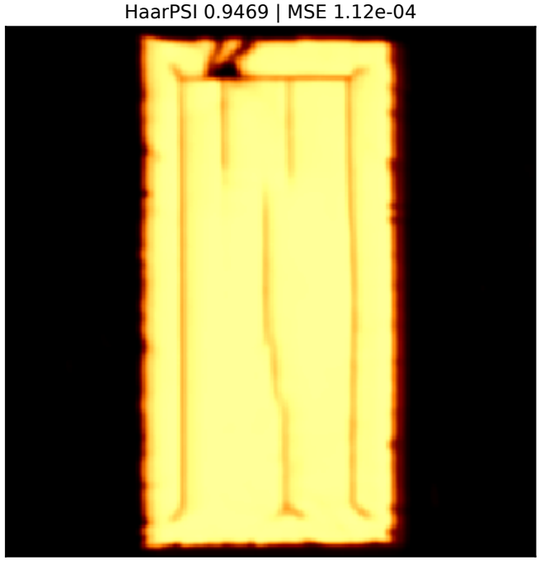
CIGS | test image 0254 | 20% Hadamard + NPL-default noise

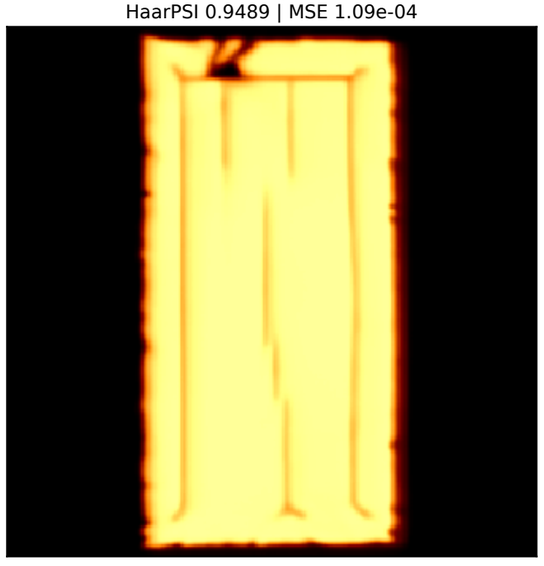

Target
Ground truth

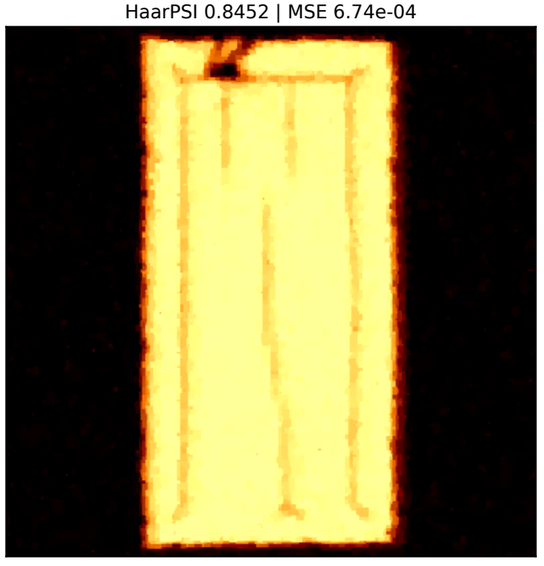

Zero-filled
PSNR 26.23 dB | SSIM 0.5066
HaarPSI 0.6028 | MSE 2.38e-03

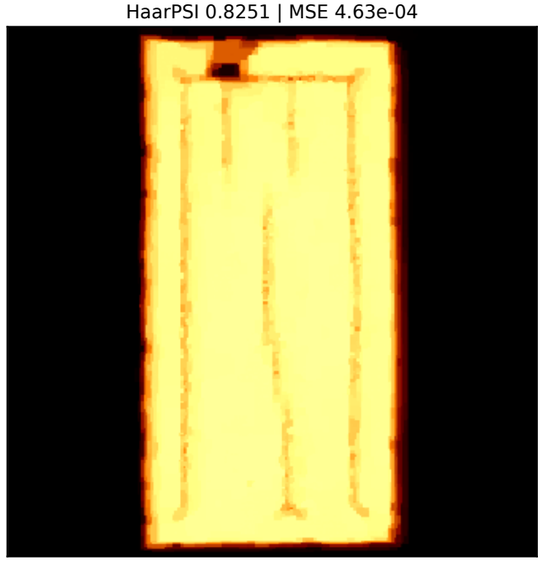

SPGL1-BP
PSNR 26.87 dB | SSIM 0.4628
HaarPSI 0.7342 | MSE 2.06e-03

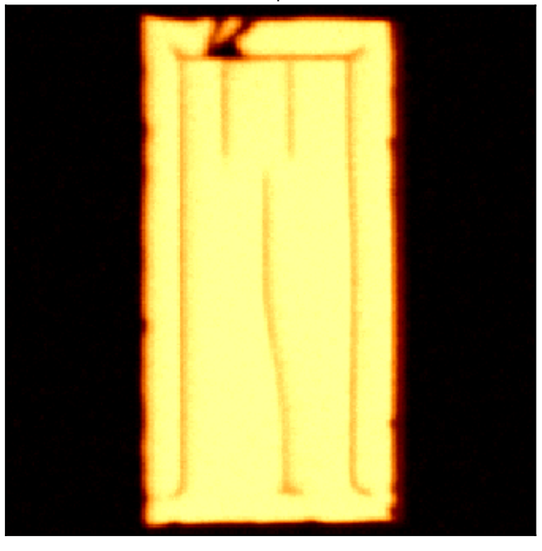

SPGL1-BPDN
PSNR 32.11 dB | SSIM 0.8242
HaarPSI 0.7642 | MSE 6.15e-04

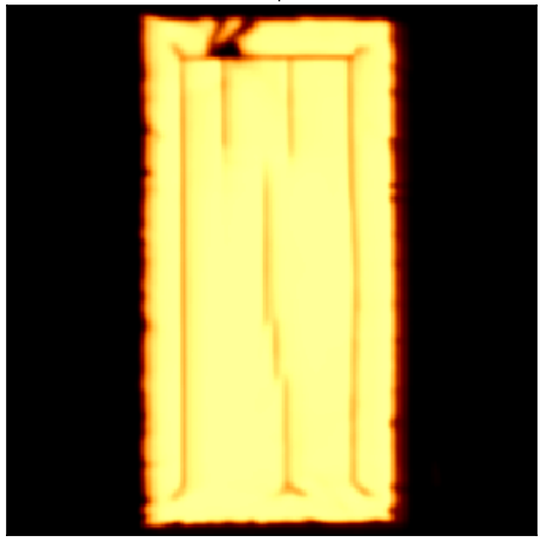

PnP-ADMM tuned-5
PSNR 39.50 dB | SSIM 0.9835
HaarPSI 0.9469 | MSE 1.12e-04

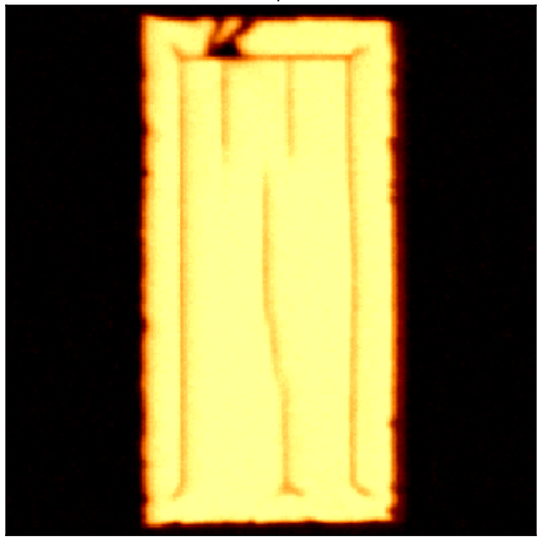

GS-PnP oracle
PSNR 39.62 dB | SSIM 0.9844
HaarPSI 0.9489 | MSE 1.09e-04


Scalar TV
PSNR 31.72 dB | SSIM 0.7655
HaarPSI 0.8452 | MSE 6.74e-04


UTV-C60
PSNR 33.34 dB | SSIM 0.9383
HaarPSI 0.8251 | MSE 4.63e-04

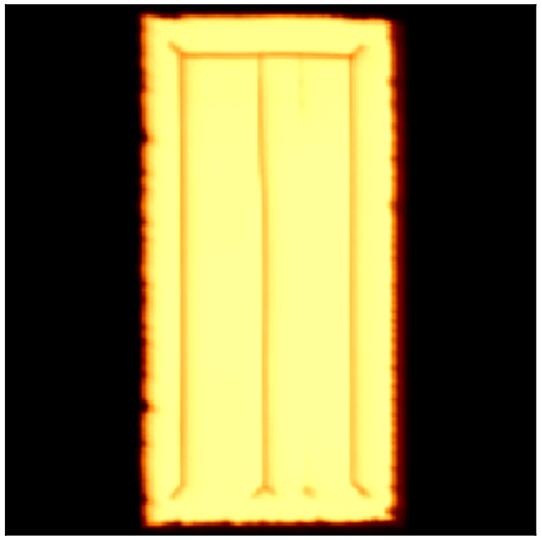

GS-PnP tuned-5
PSNR 34.12 dB | SSIM 0.8061
HaarPSI 0.8865 | MSE 3.88e-04


PnP-ADMM Tune-1 (0768)
PSNR 39.63 dB | SSIM 0.9854
HaarPSI 0.9491 | MSE 1.09e-04

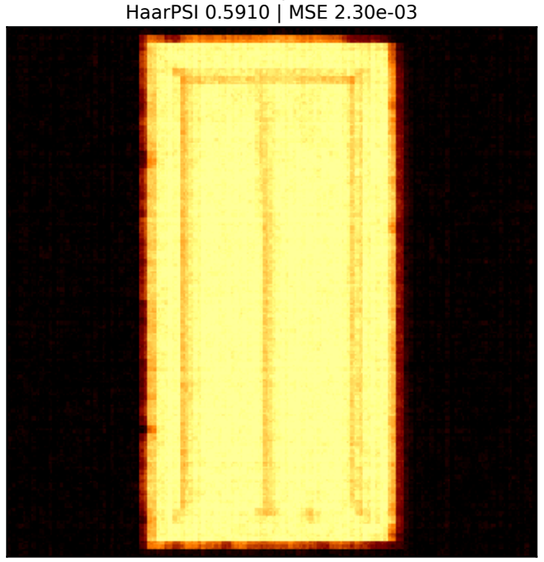

GS-PnP Tune-1 (0530)
PSNR 35.52 dB | SSIM 0.8347
HaarPSI 0.9148 | MSE 2.81e-04

CIGS | test image 0292 | 20% Hadamard + NPL-default noise

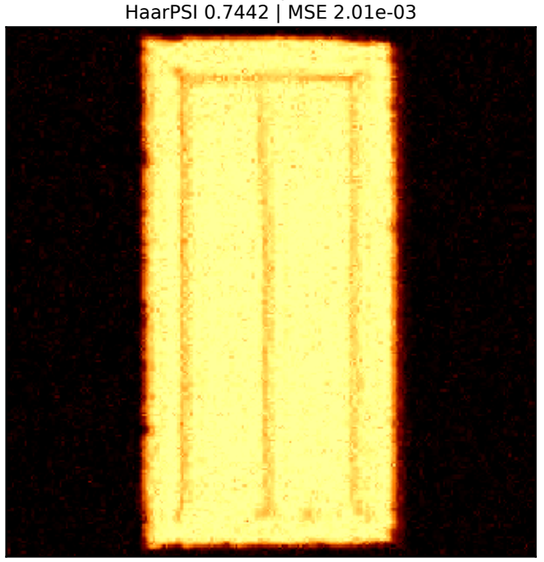
Target
Ground truth



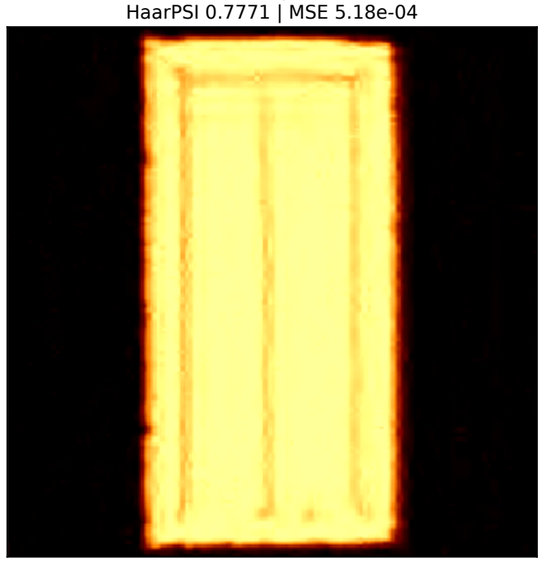
Zero-filled
PSNR 26.38 dB | SSIM 0.5003
HaarPSI 0.5910 | MSE 2.30e-03



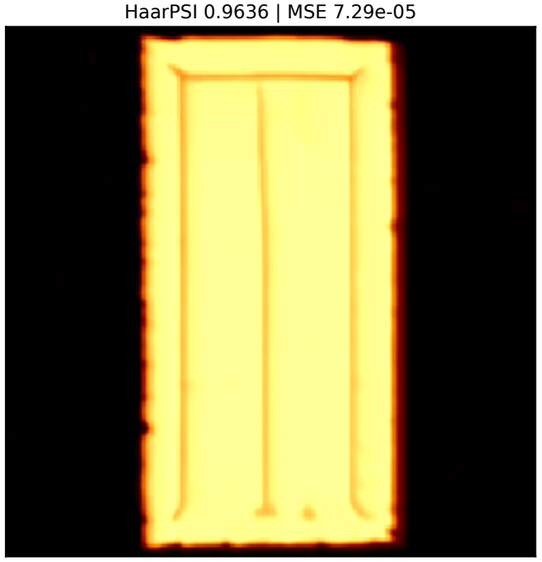
SPGL1-BP
PSNR 26.97 dB | SSIM 0.4527
HaarPSI 0.7442 | MSE 2.01e-03



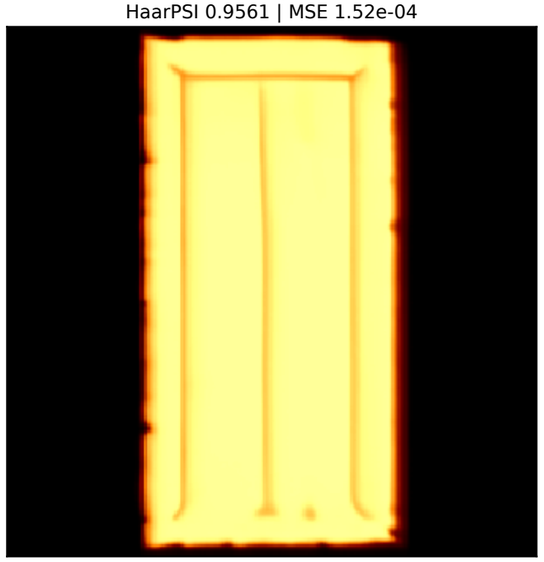
SPGL1-BPDN
PSNR 32.86 dB | SSIM 0.8220
HaarPSI 0.7771 | MSE 5.18e-04



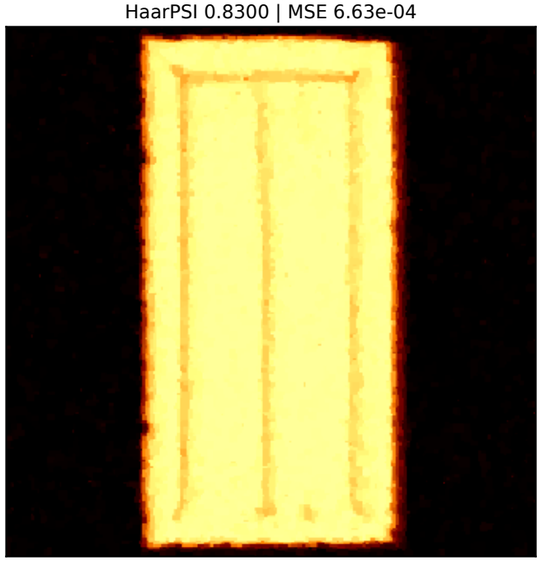
PnP-ADMM tuned-5
PSNR 41.37 dB | SSIM 0.9828
HaarPSI 0.9636 | MSE 7.29e-05



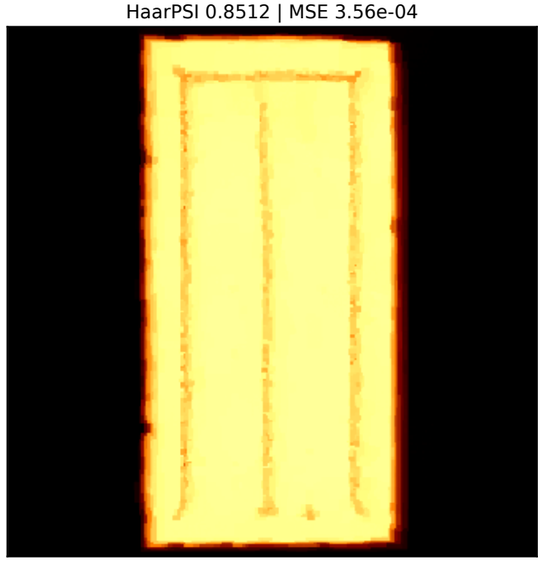
GS-PnP oracle
PSNR 38.17 dB | SSIM 0.9843
HaarPSI 0.9561 | MSE 1.52e-04



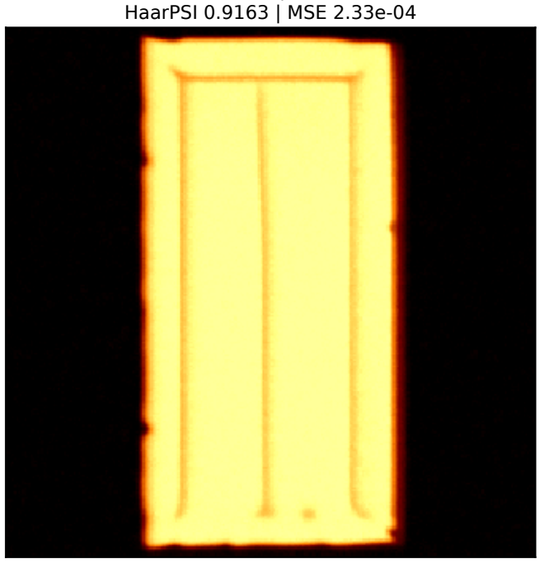
Scalar TV
PSNR 31.78 dB | SSIM 0.7519
HaarPSI 0.8300 | MSE 6.63e-04



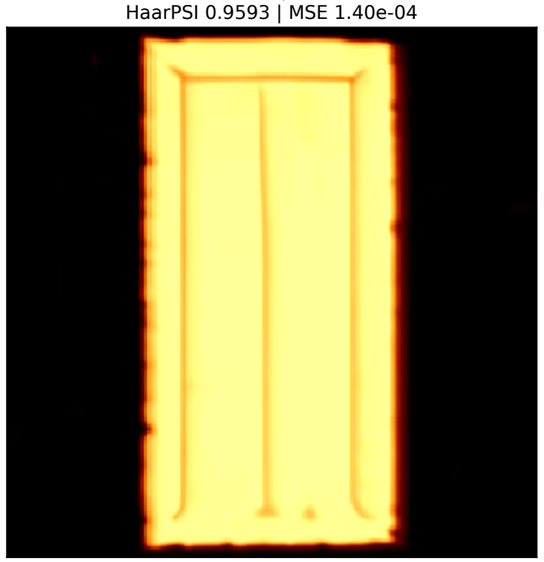
UTV-C60
PSNR 34.48 dB | SSIM 0.9366
HaarPSI 0.8512 | MSE 3.56e-04



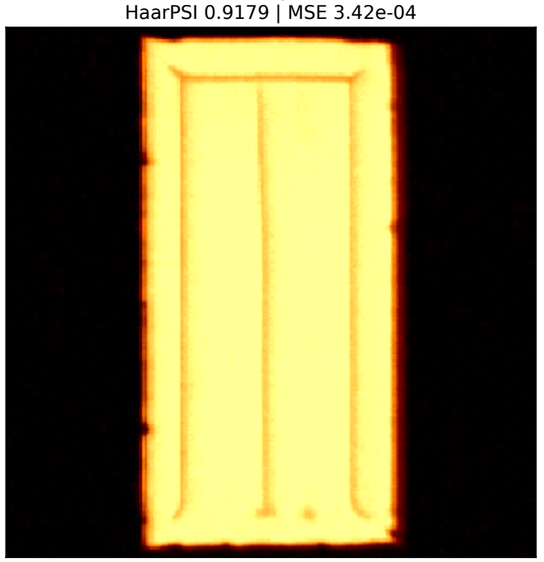
GS-PnP tuned-5
PSNR 36.33 dB | SSIM 0.8749
HaarPSI 0.9163 | MSE 2.33e-04



PnP-ADMM Tune-1 (0768)
PSNR 38.55 dB | SSIM 0.9837
HaarPSI 0.9593 | MSE 1.40e-04

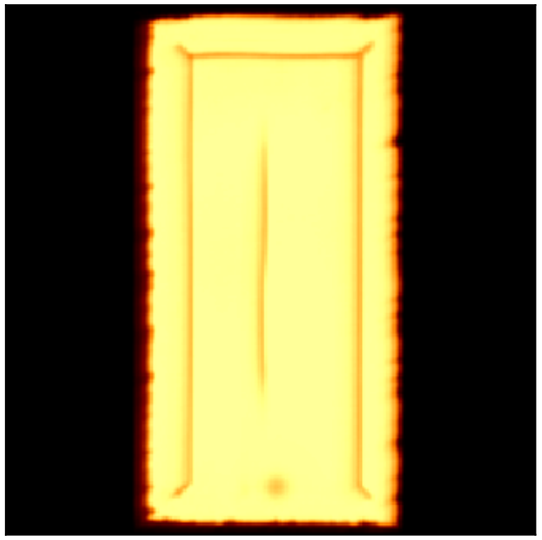


GS-PnP Tune-1 (0530)
PSNR 34.66 dB | SSIM 0.8293
HaarPSI 0.9179 | MSE 3.42e-04

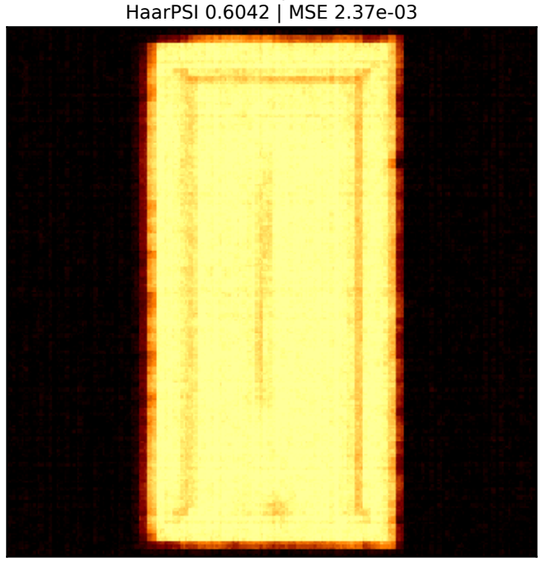


CIGS | test image 0304 | 20% Hadamard + NPL-default noise

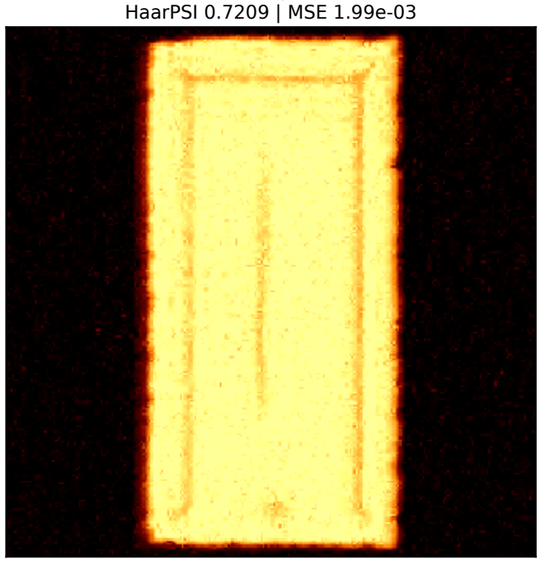
Target
Ground truth



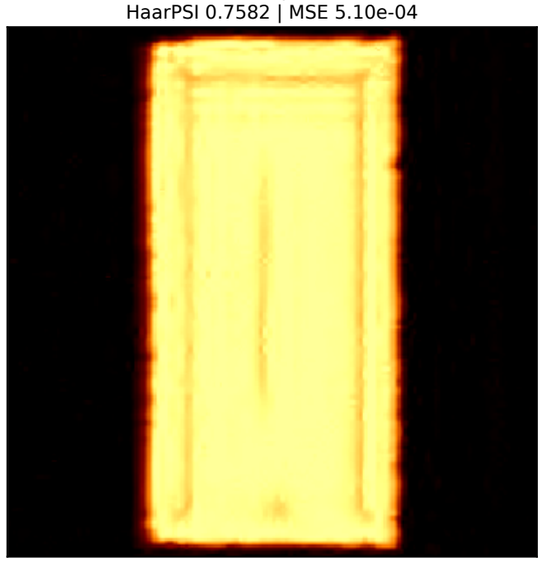
Zero-filled
PSNR 26.26 dB | SSIM 0.4933
HaarPSI 0.6042 | MSE 2.37e-03



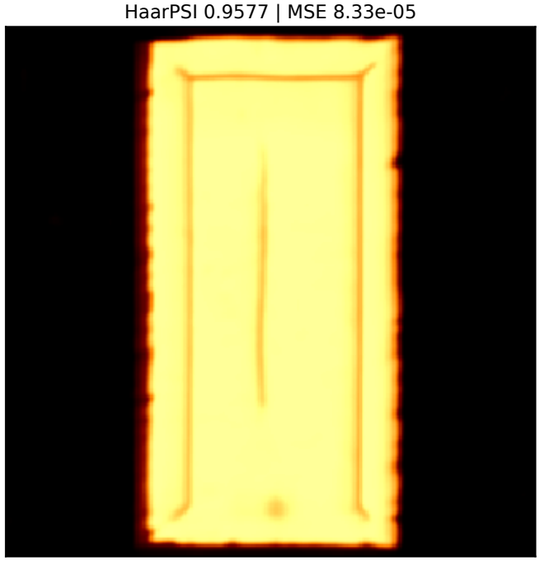
SPGL1-BP
PSNR 27.02 dB | SSIM 0.4510
HaarPSI 0.7209 | MSE 1.99e-03



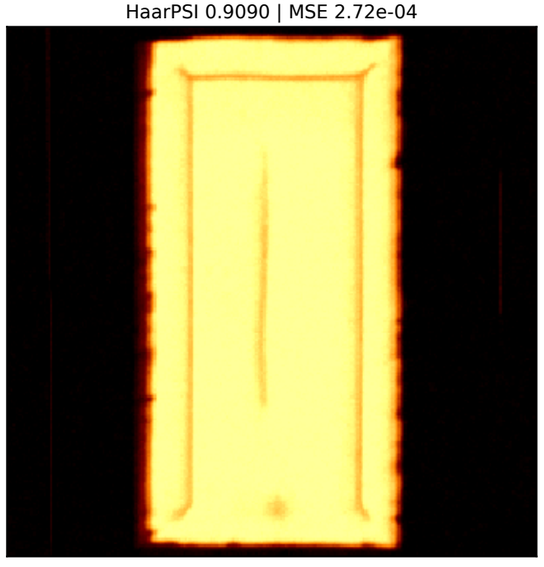
SPGL1-BPDN
PSNR 32.93 dB | SSIM 0.8257
HaarPSI 0.7582 | MSE 5.10e-04



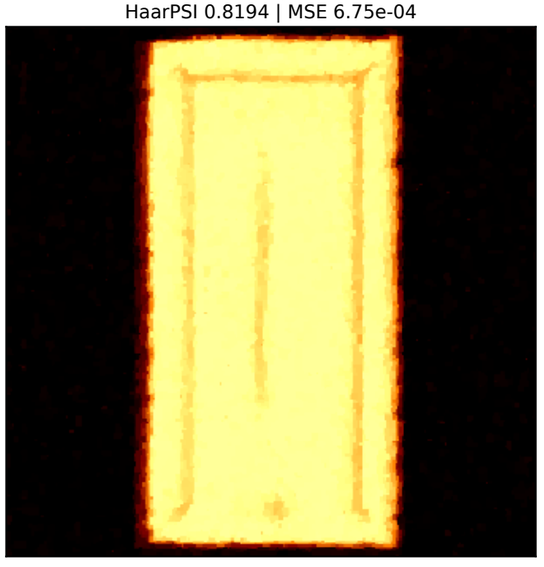
PnP-ADMM tuned-5
PSNR 40.80 dB | SSIM 0.9877
HaarPSI 0.9577 | MSE 8.33e-05



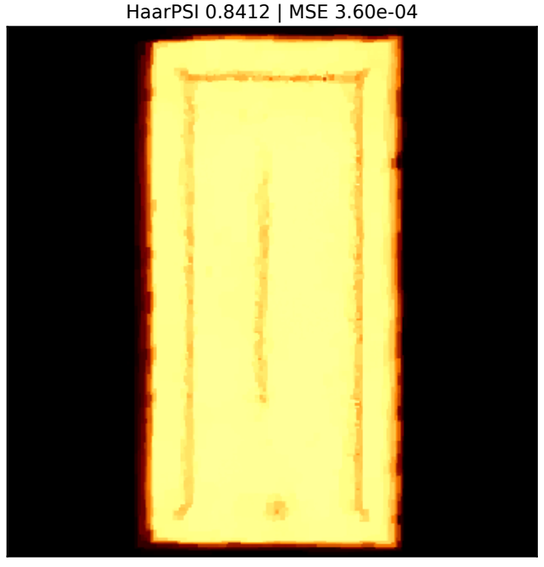
GS-PnP oracle
PSNR 35.65 dB | SSIM 0.8331
HaarPSI 0.9090 | MSE 2.72e-04



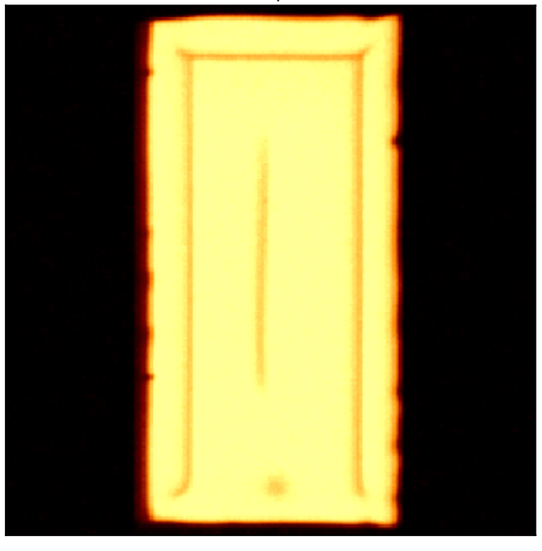
Scalar TV
PSNR 31.71 dB | SSIM 0.7592
HaarPSI 0.8194 | MSE 6.75e-04



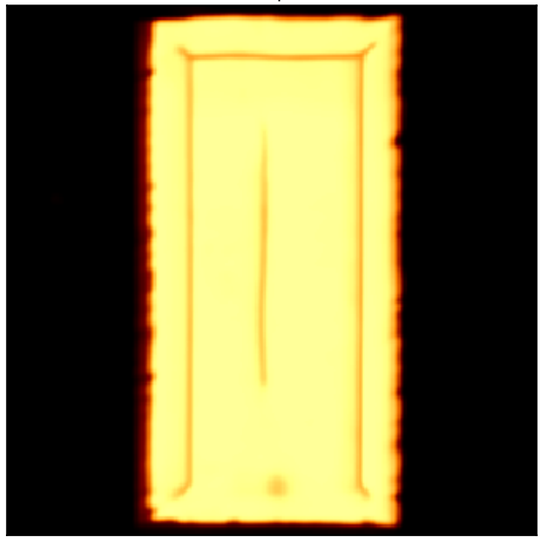
UTV-C60
PSNR 34.44 dB | SSIM 0.9453
HaarPSI 0.8412 | MSE 3.60e-04



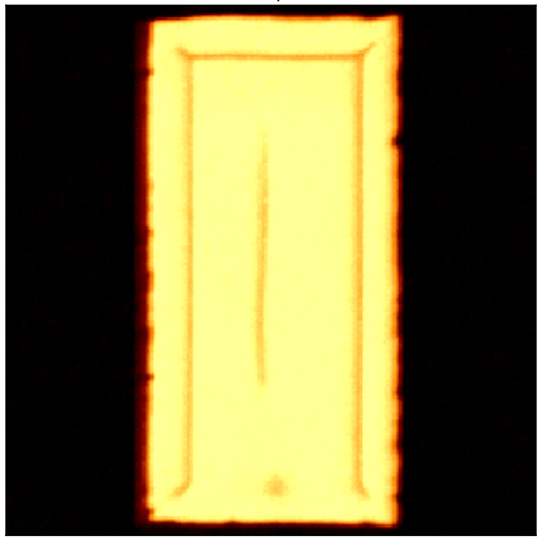
GS-PnP tuned-5
PSNR 36.24 dB | SSIM 0.8776
HaarPSI 0.9158 | MSE 2.38e-04



PnP-ADMM Tune-1 (0768)
PSNR 40.99 dB | SSIM 0.9898
HaarPSI 0.9603 | MSE 7.97e-05

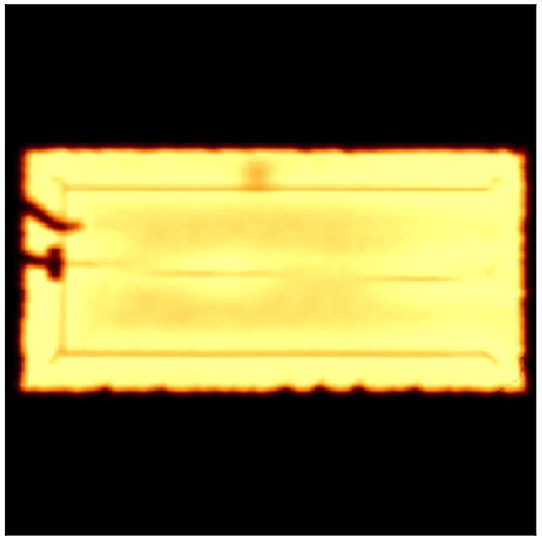


GS-PnP Tune-1 (0530)
PSNR 35.94 dB | SSIM 0.8328
HaarPSI 0.9207 | MSE 2.55e-04

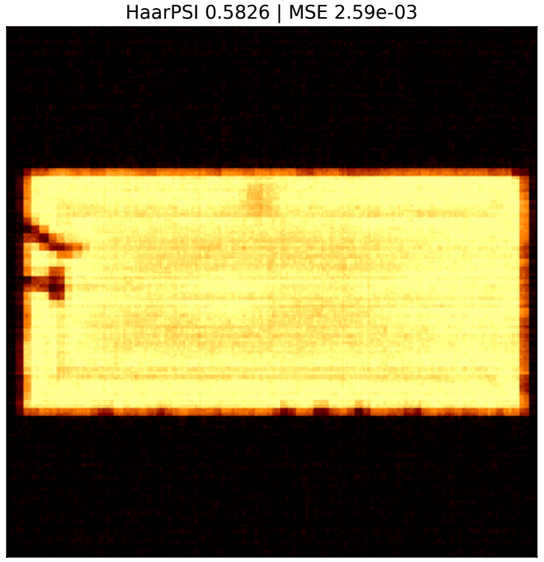


CIGS | test image 0507 | 20% Hadamard + NPL-default noise

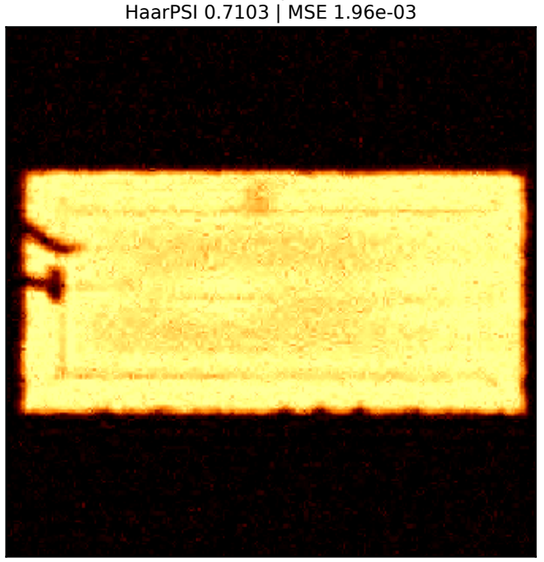
Target
Ground truth



Zero-filled
PSNR 25.86 dB | SSIM 0.4718
HaarPSI 0.5826 | MSE 2.59e-03



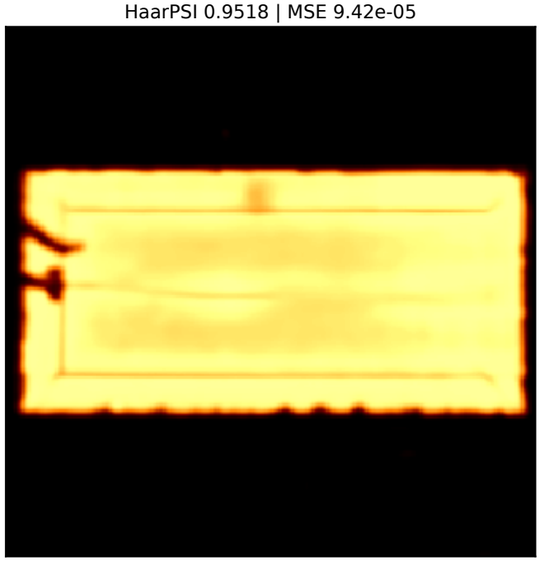
SPGL1-BP
PSNR 27.07 dB | SSIM 0.4447
HaarPSI 0.7103 | MSE 1.96e-03



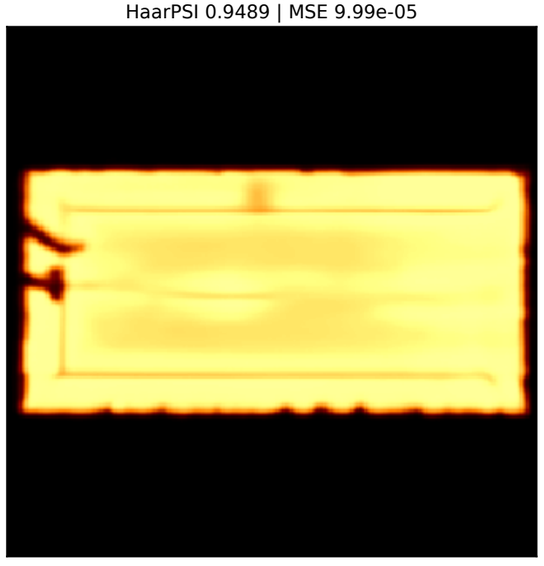
SPGL1-BPDN
PSNR 32.57 dB | SSIM 0.8041
HaarPSI 0.7202 | MSE 5.53e-04



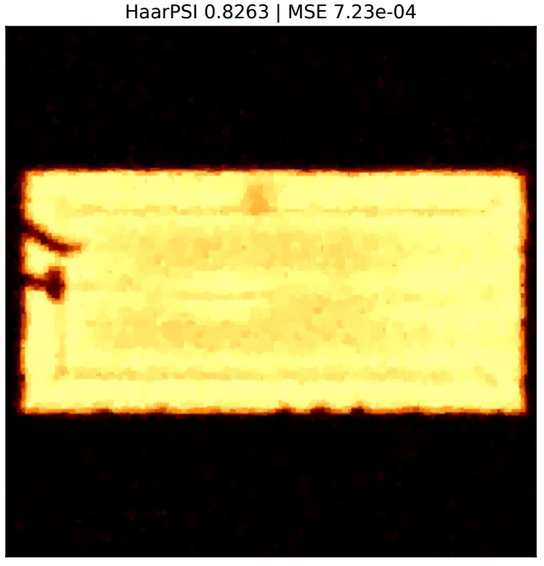
PnP-ADMM tuned-5
PSNR 40.26 dB | SSIM 0.9827
HaarPSI 0.9518 | MSE 9.42e-05



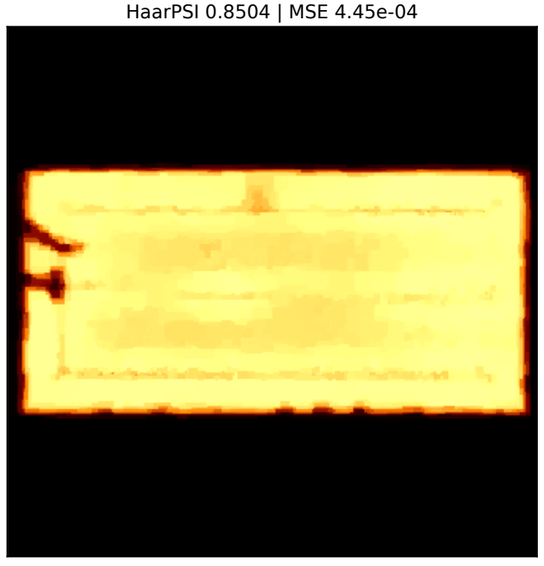
GS-PnP oracle
PSNR 40.00 dB | SSIM 0.9768
HaarPSI 0.9489 | MSE 9.99e-05



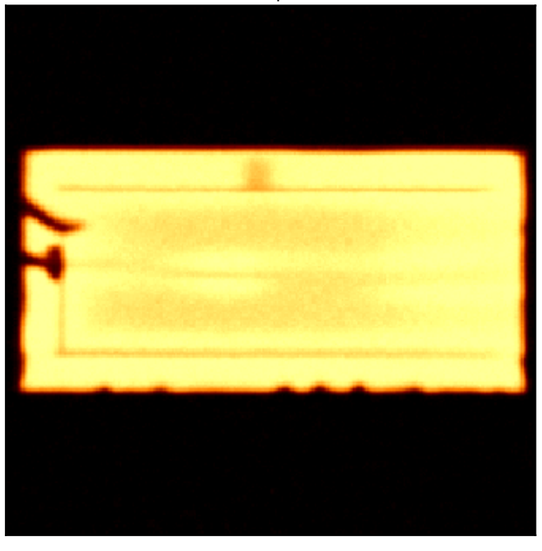
Scalar TV
PSNR 31.41 dB | SSIM 0.7440
HaarPSI 0.8263 | MSE 7.23e-04



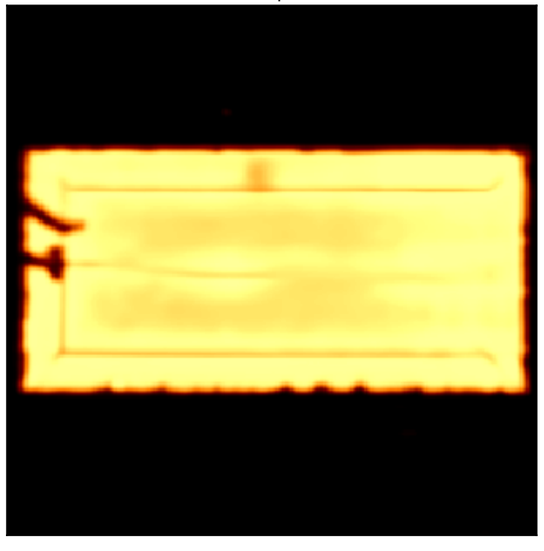
UTV-C60
PSNR 33.52 dB | SSIM 0.9374
HaarPSI 0.8504 | MSE 4.45e-04



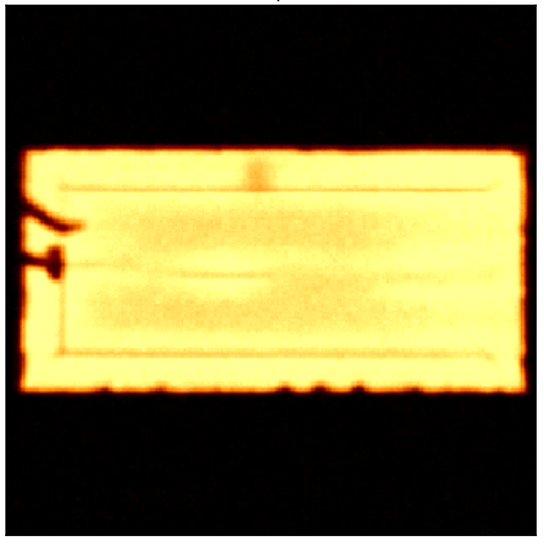
GS-PnP tuned-5
PSNR 35.60 dB | SSIM 0.8637
HaarPSI 0.8927 | MSE 2.75e-04



PnP-ADMM Tune-1 (0768)
PSNR 40.31 dB | SSIM 0.9839
HaarPSI 0.9530 | MSE 9.30e-05

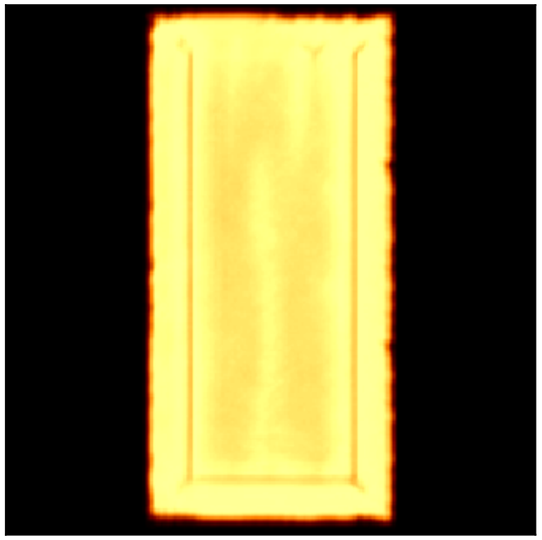


GS-PnP Tune-1 (0530)
PSNR 35.64 dB | SSIM 0.8259
HaarPSI 0.9172 | MSE 2.73e-04

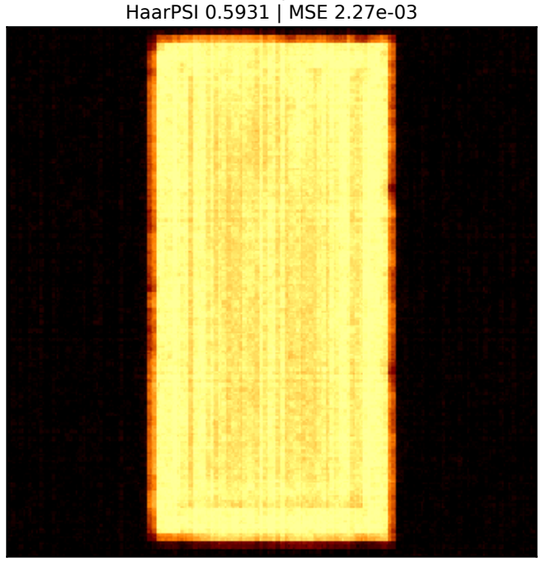


CIGS | test image 0527 | 20% Hadamard + NPL-default noise

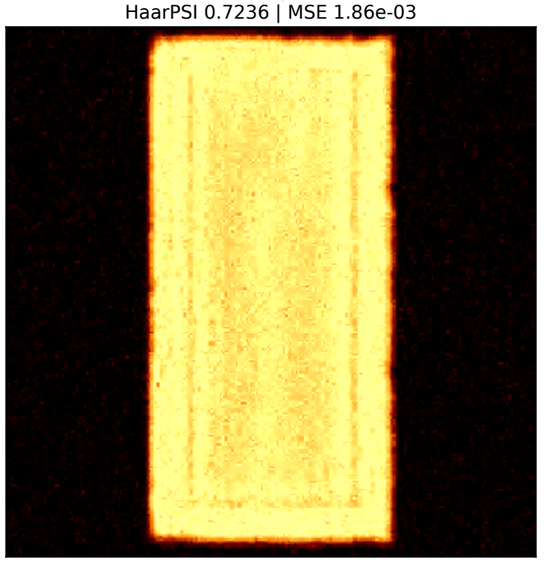
Target
Ground truth



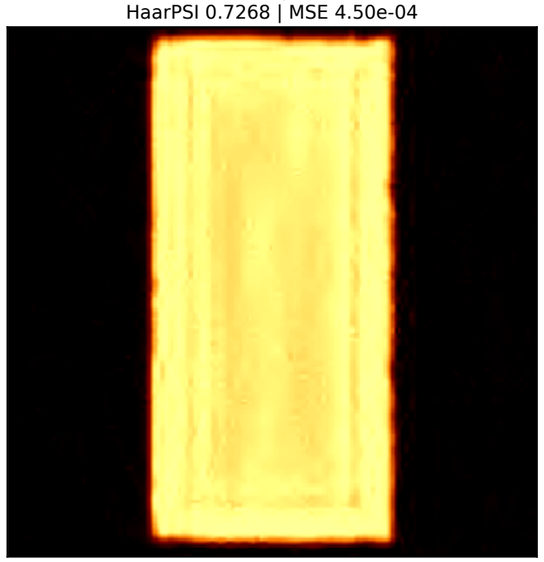
Zero-filled
PSNR 26.45 dB | SSIM 0.4706
HaarPSI 0.5931 | MSE 2.27e-03



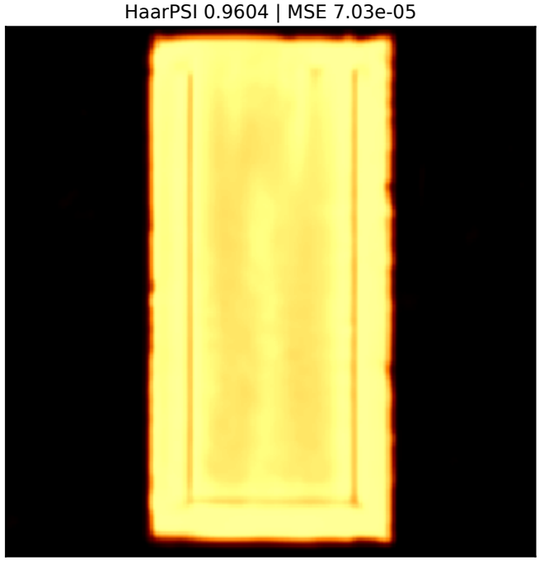
SPGL1-BP
PSNR 27.30 dB | SSIM 0.4341
HaarPSI 0.7236 | MSE 1.86e-03



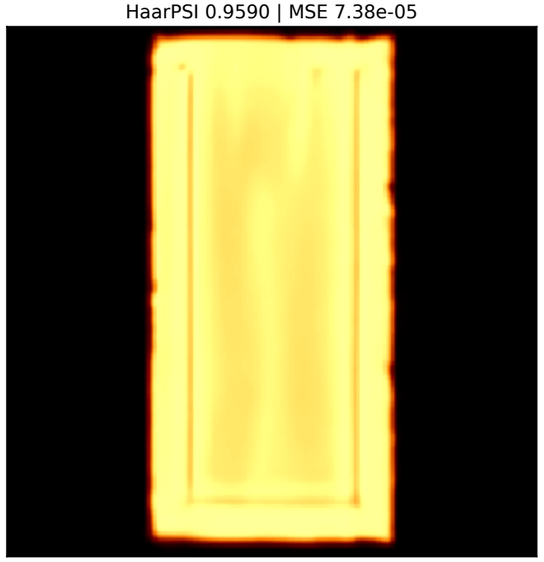
SPGL1-BPDN
PSNR 33.47 dB | SSIM 0.8122
HaarPSI 0.7268 | MSE 4.50e-04



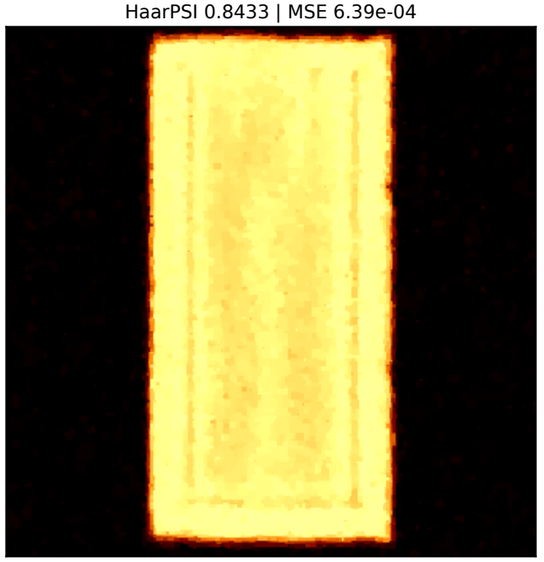
PnP-ADMM tuned-5
PSNR 41.53 dB | SSIM 0.9807
HaarPSI 0.9604 | MSE 7.03e-05



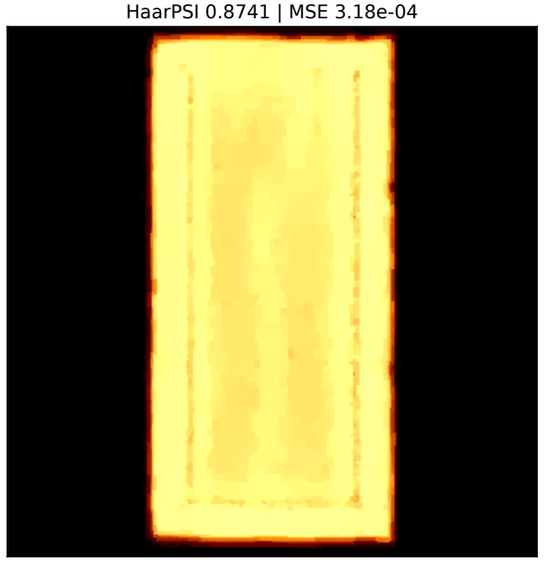
GS-PnP oracle
PSNR 41.32 dB | SSIM 0.9789
HaarPSI 0.9590 | MSE 7.38e-05



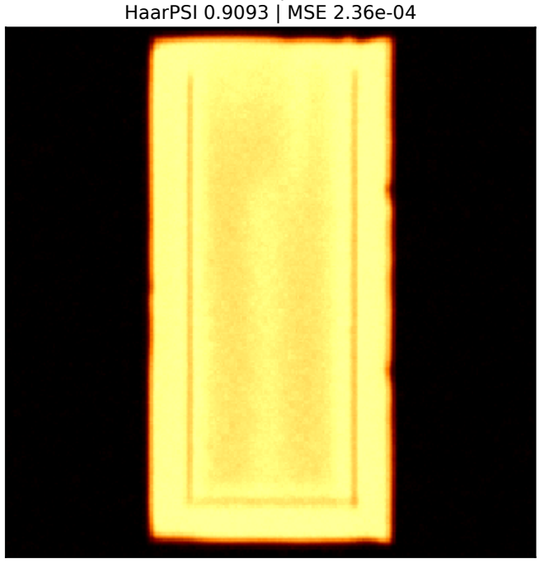
Scalar TV
PSNR 31.95 dB | SSIM 0.7401
HaarPSI 0.8433 | MSE 6.39e-04



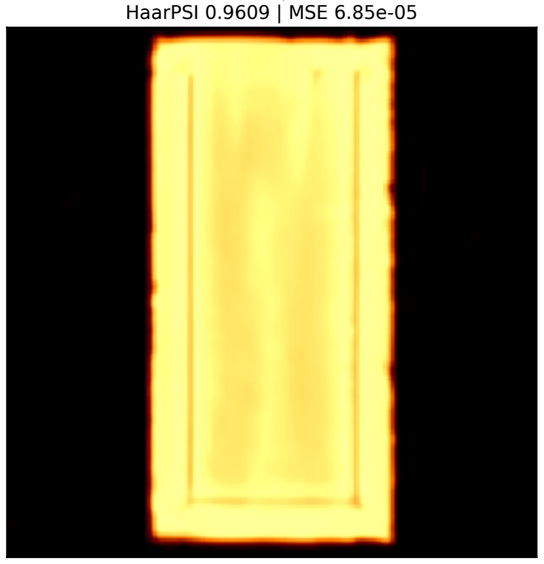
UTV-C60
PSNR 34.97 dB | SSIM 0.9443
HaarPSI 0.8741 | MSE 3.18e-04



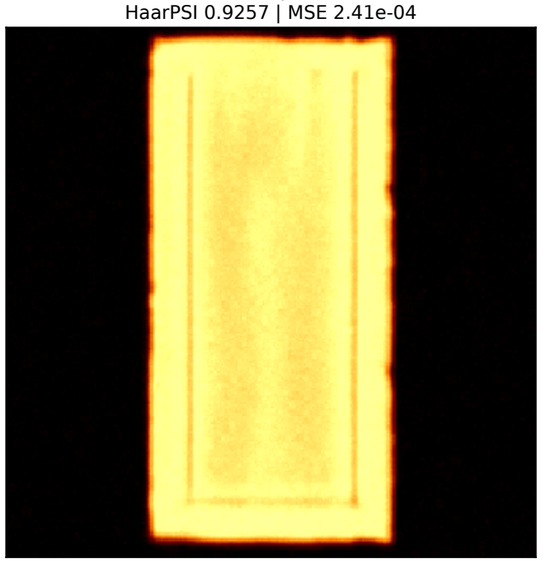
GS-PnP tuned-5
PSNR 36.27 dB | SSIM 0.8634
HaarPSI 0.9093 | MSE 2.36e-04



PnP-ADMM Tune-1 (0768)
PSNR 41.64 dB | SSIM 0.9831
HaarPSI 0.9609 | MSE 6.85e-05

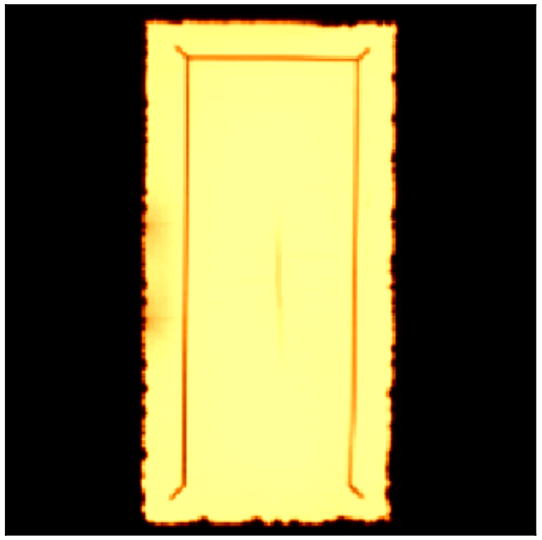


GS-PnP Tune-1 (0530)
PSNR 36.18 dB | SSIM 0.8233
HaarPSI 0.9257 | MSE 2.41e-04

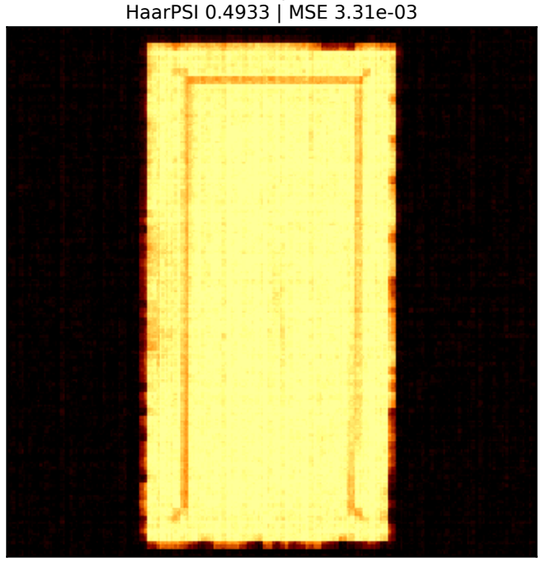


CIGS | test image 0710 | 20% Hadamard + NPL-default noise

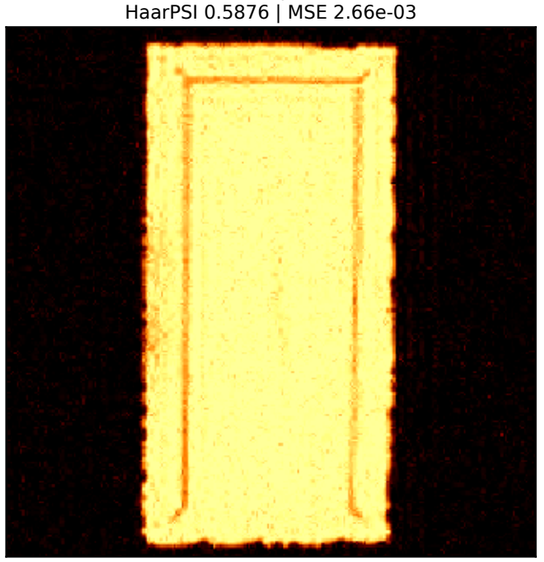
Target
Ground truth



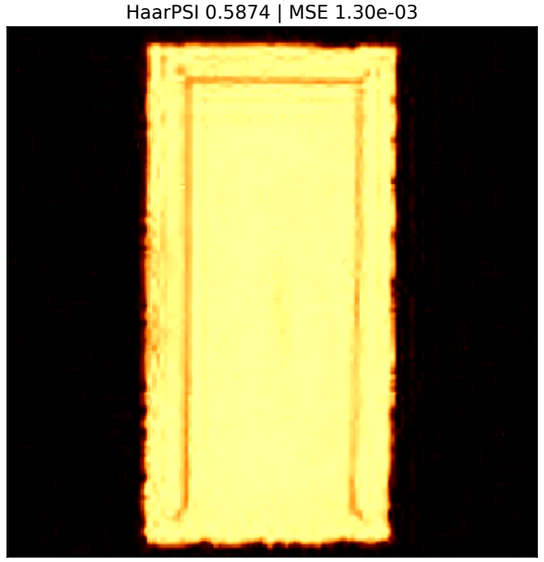
Zero-filled
PSNR 24.80 dB | SSIM 0.4112
HaarPSI 0.4933 | MSE 3.31e-03



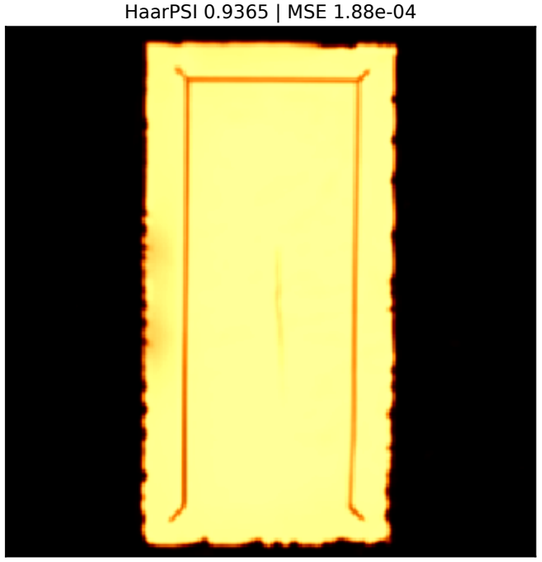
SPGL1-BP
PSNR 25.75 dB | SSIM 0.4019
HaarPSI 0.5876 | MSE 2.66e-03



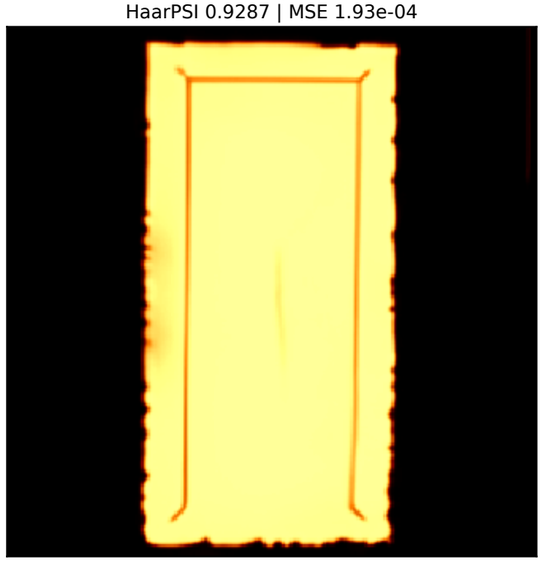
SPGL1-BPDN
PSNR 28.87 dB | SSIM 0.7561
HaarPSI 0.5874 | MSE 1.30e-03



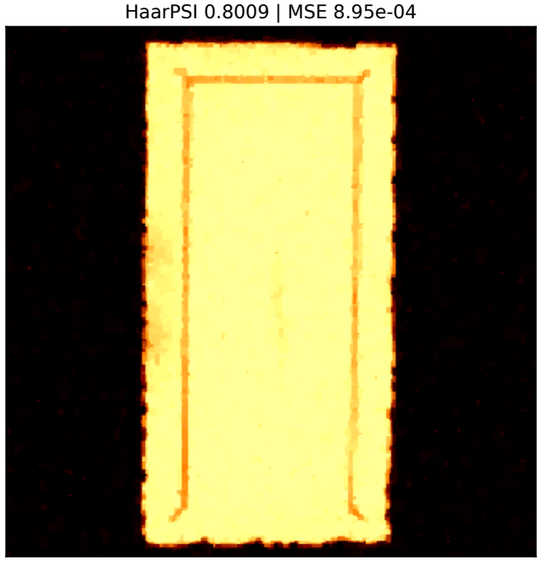
PnP-ADMM tuned-5
PSNR 37.25 dB | SSIM 0.9851
HaarPSI 0.9365 | MSE 1.88e-04



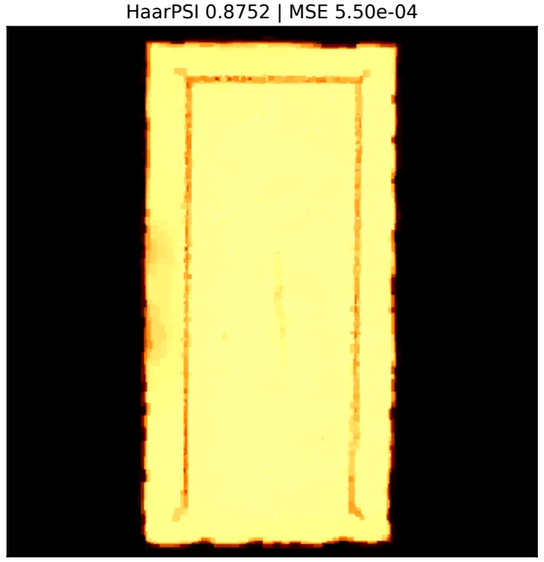
GS-PnP oracle
PSNR 37.15 dB | SSIM 0.9737
HaarPSI 0.9287 | MSE 1.93e-04



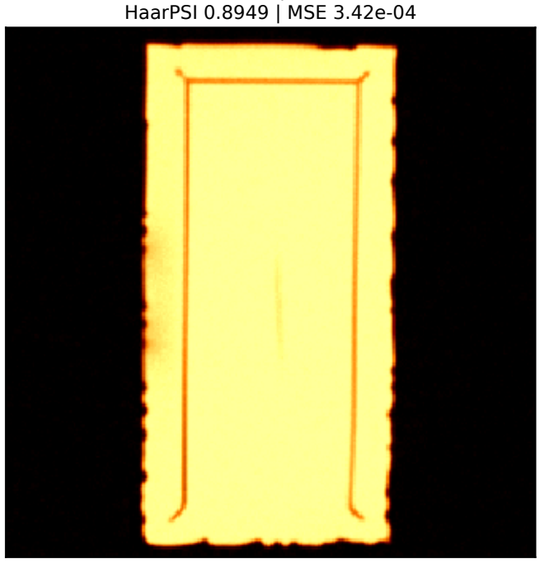
Scalar TV
PSNR 30.48 dB | SSIM 0.7384
HaarPSI 0.8009 | MSE 8.95e-04



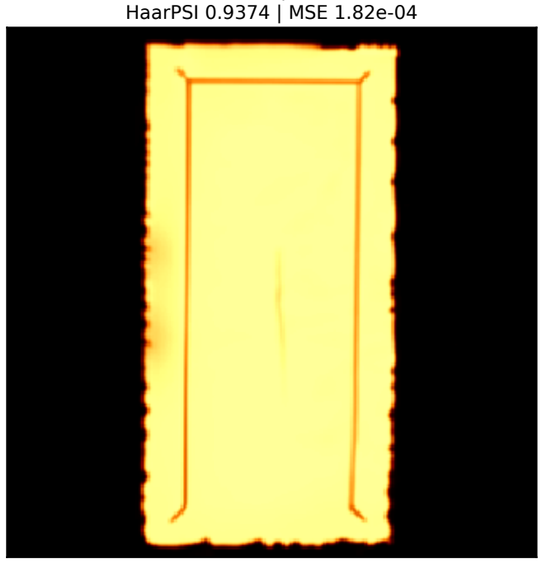
UTV-C60
PSNR 32.60 dB | SSIM 0.9572
HaarPSI 0.8752 | MSE 5.50e-04



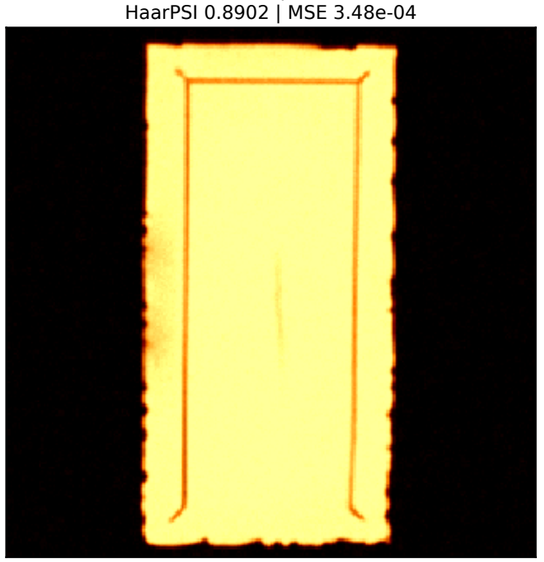
GS-PnP tuned-5
PSNR 34.66 dB | SSIM 0.8635
HaarPSI 0.8949 | MSE 3.42e-04



PnP-ADMM Tune-1 (0768)
PSNR 37.40 dB | SSIM 0.9868
HaarPSI 0.9374 | MSE 1.82e-04

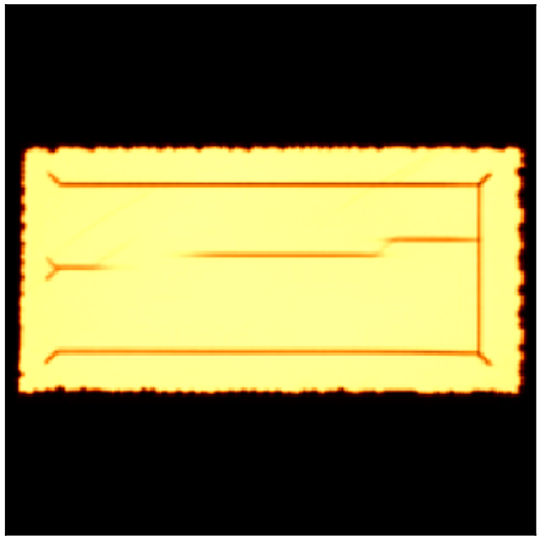


GS-PnP Tune-1 (0530)
PSNR 34.59 dB | SSIM 0.8183
HaarPSI 0.8902 | MSE 3.48e-04

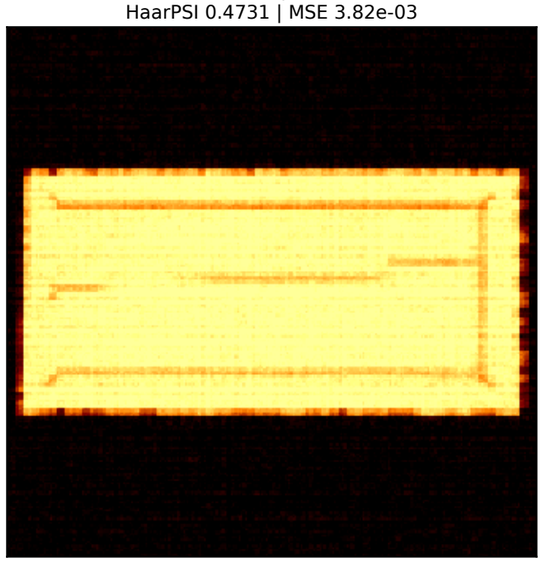


CIGS | test image 0935 | 20% Hadamard + NPL-default noise

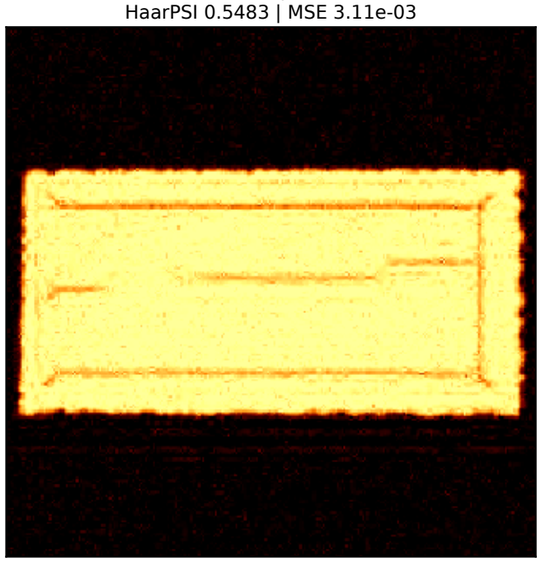
Target
Ground truth



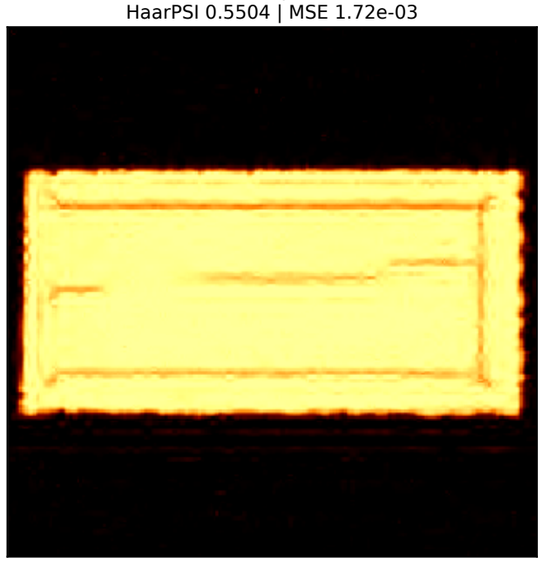
Zero-filled
PSNR 24.18 dB | SSIM 0.4132
HaarPSI 0.4731 | MSE 3.82e-03



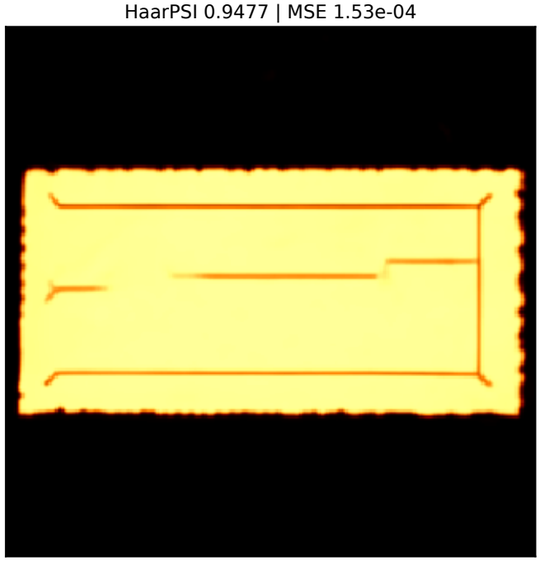
SPGL1-BP
PSNR 25.07 dB | SSIM 0.3940
HaarPSI 0.5483 | MSE 3.11e-03



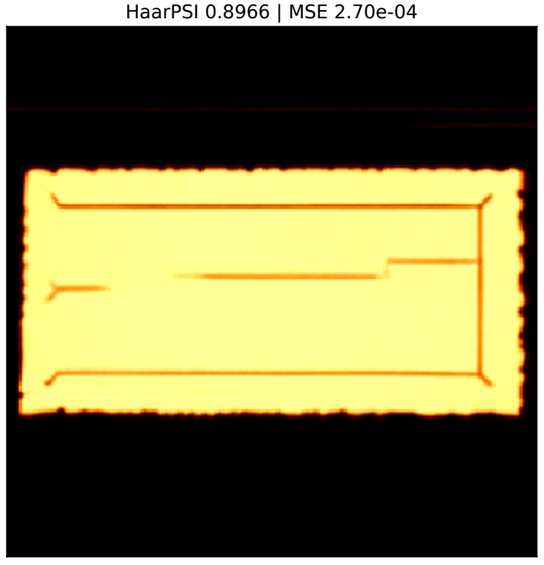
SPGL1-BPDN
PSNR 27.64 dB | SSIM 0.7168
HaarPSI 0.5504 | MSE 1.72e-03



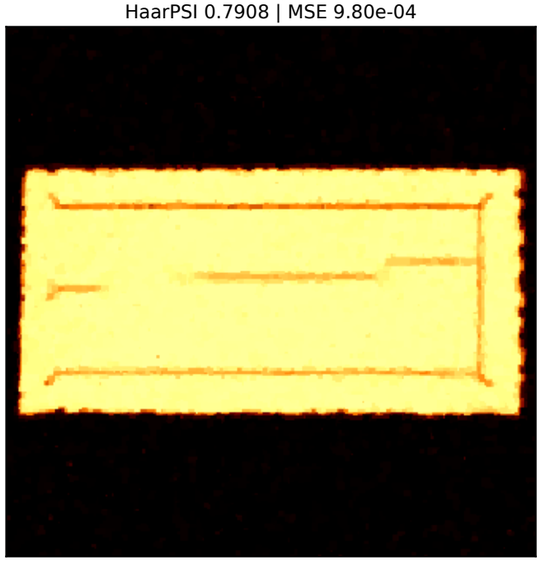
PnP-ADMM tuned-5
PSNR 38.15 dB | SSIM 0.9866
HaarPSI 0.9477 | MSE 1.53e-04



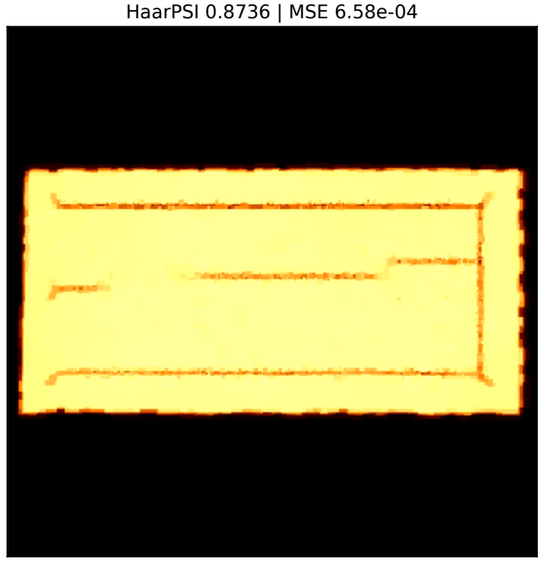
GS-PnP oracle
PSNR 35.69 dB | SSIM 0.8922
HaarPSI 0.8966 | MSE 2.70e-04



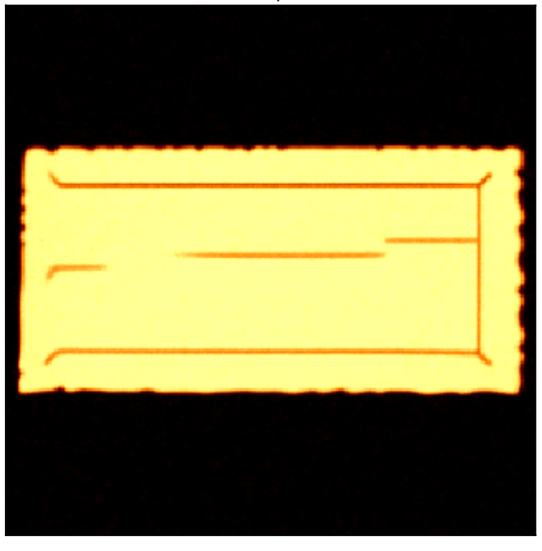
Scalar TV
PSNR 30.09 dB | SSIM 0.7387
HaarPSI 0.7908 | MSE 9.80e-04



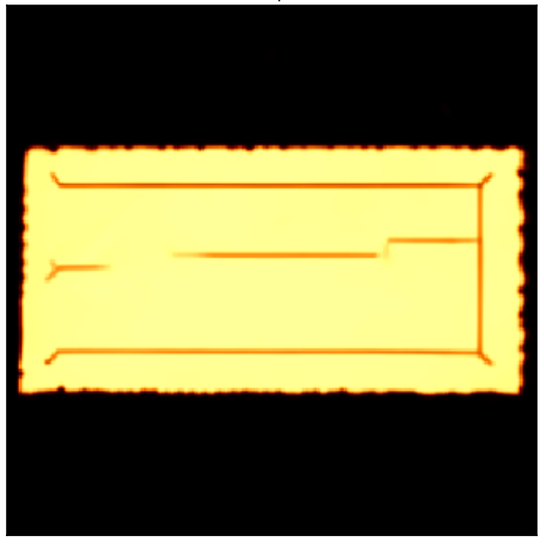
UTV-C60
PSNR 31.82 dB | SSIM 0.9519
HaarPSI 0.8736 | MSE 6.58e-04



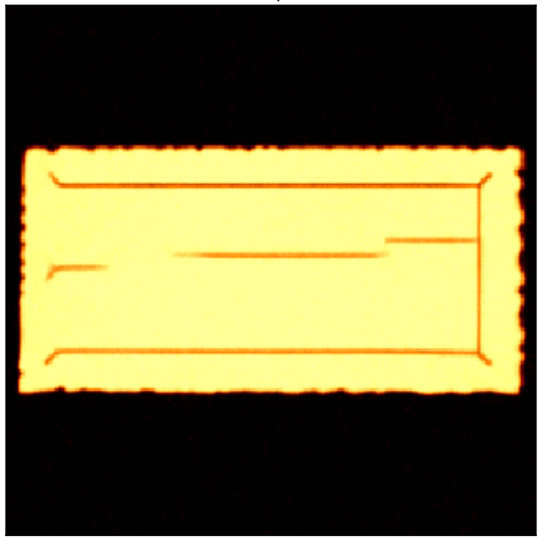
GS-PnP tuned-5
PSNR 33.60 dB | SSIM 0.7982
HaarPSI 0.8786 | MSE 4.36e-04



PnP-ADMM Tune-1 (0768)
PSNR 38.32 dB | SSIM 0.9875
HaarPSI 0.9467 | MSE 1.47e-04

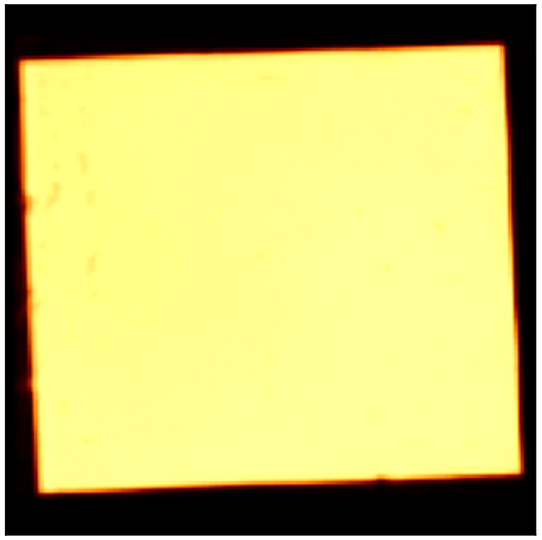


GS-PnP Tune-1 (0530)
PSNR 34.72 dB | SSIM 0.8233
HaarPSI 0.9053 | MSE 3.37e-04

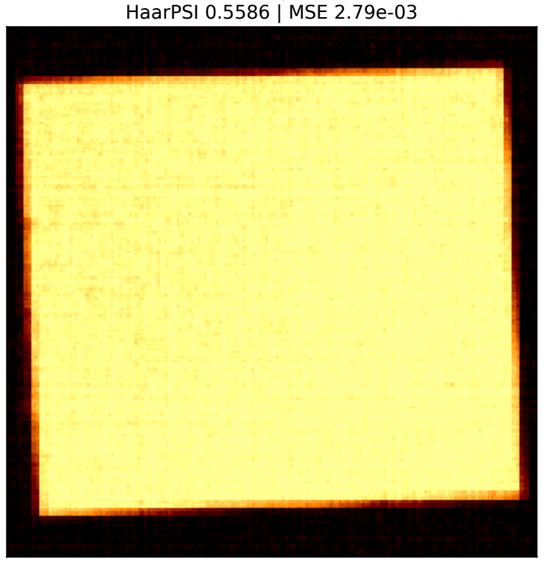


Organic | test image 0014 | 20% Hadamard + NPL-default noise

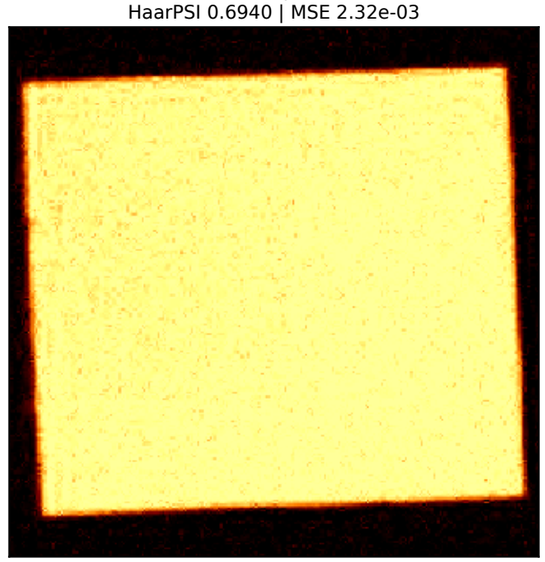
Target
Ground truth



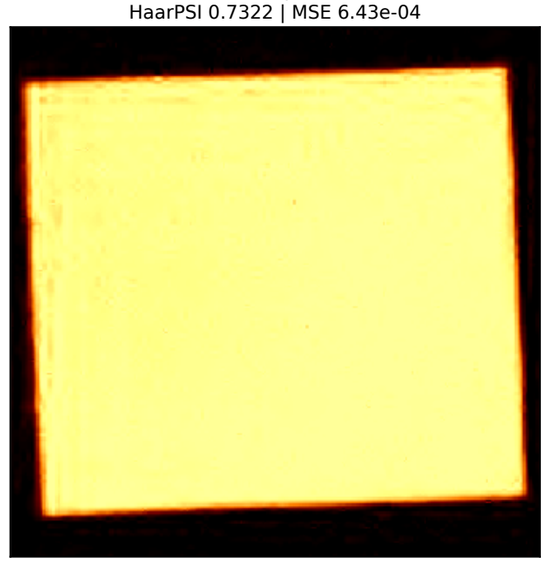
Zero-filled
PSNR 25.54 dB | SSIM 0.4741
HaarPSI 0.5586 | MSE 2.79e-03



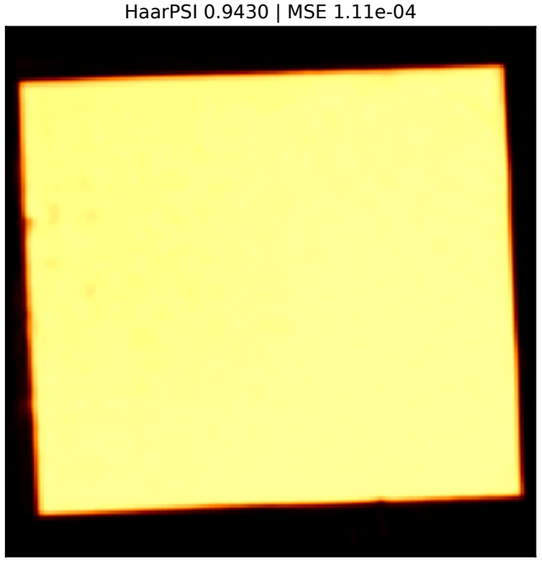
SPGL1-BP
PSNR 26.35 dB | SSIM 0.4204
HaarPSI 0.6940 | MSE 2.32e-03



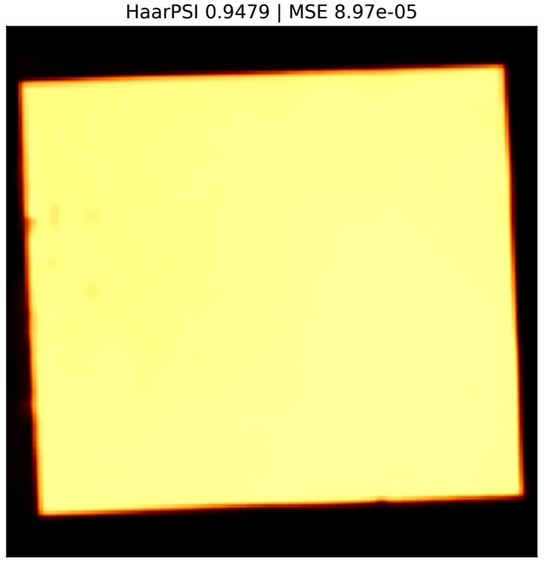
SPGL1-BPDN
PSNR 31.92 dB | SSIM 0.8249
HaarPSI 0.7322 | MSE 6.43e-04



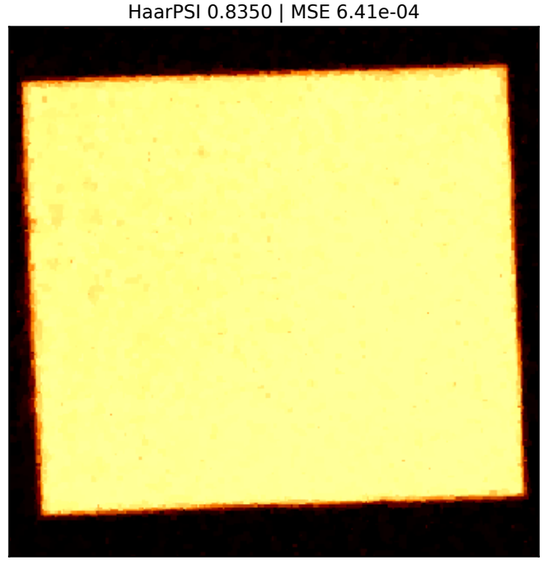
PnP-ADMM tuned-5
PSNR 39.53 dB | SSIM 0.9658
HaarPSI 0.9430 | MSE 1.11e-04



GS-PnP oracle
PSNR 40.47 dB | SSIM 0.9731
HaarPSI 0.9479 | MSE 8.97e-05



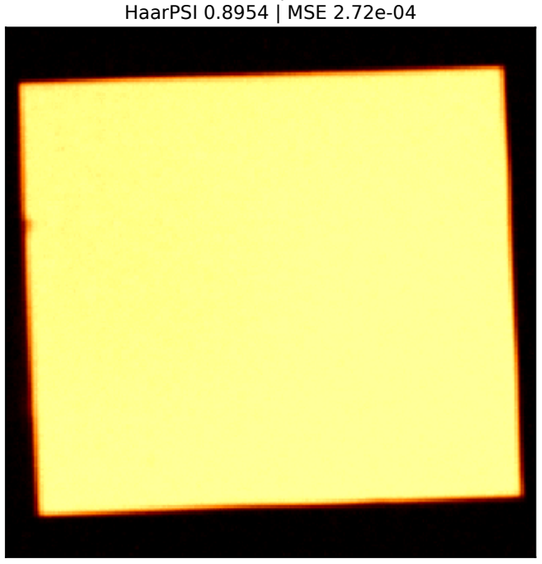
Scalar TV
PSNR 31.93 dB | SSIM 0.7828
HaarPSI 0.8350 | MSE 6.41e-04



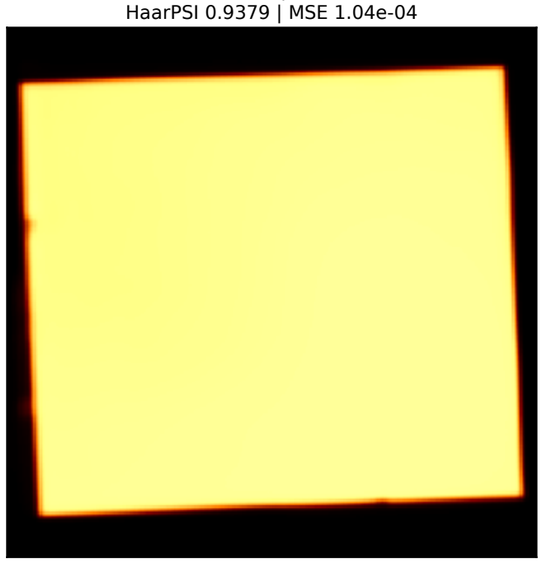
UTV-C60
PSNR 36.65 dB | SSIM 0.9440
HaarPSI 0.8700 | MSE 2.16e-04



GS-PnP tuned-5
PSNR 35.66 dB | SSIM 0.8558
HaarPSI 0.8954 | MSE 2.72e-04



PnP-ADMM Tune-1 (0768)
PSNR 39.83 dB | SSIM 0.9723
HaarPSI 0.9379 | MSE 1.04e-04

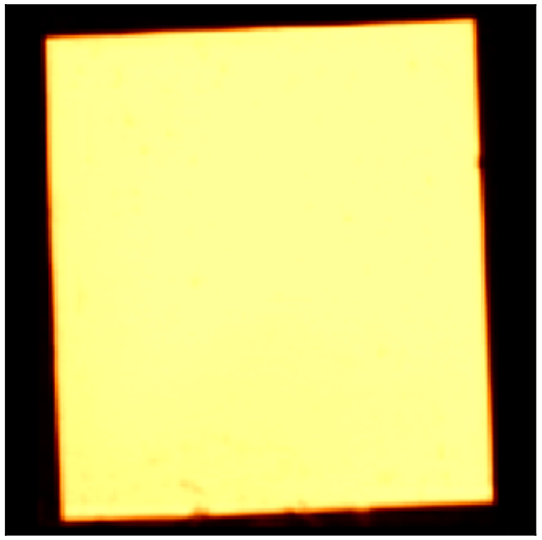


GS-PnP Tune-1 (0158)
PSNR 34.21 dB | SSIM 0.7703
HaarPSI 0.8910 | MSE 3.79e-04

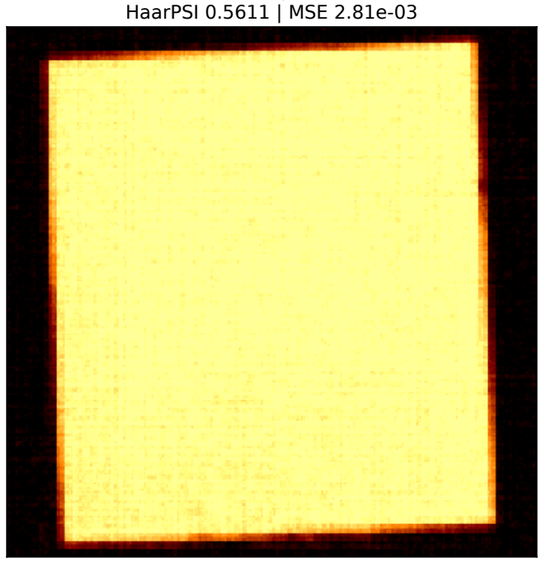


Organic | test image 0055 | 20% Hadamard + NPL-default noise

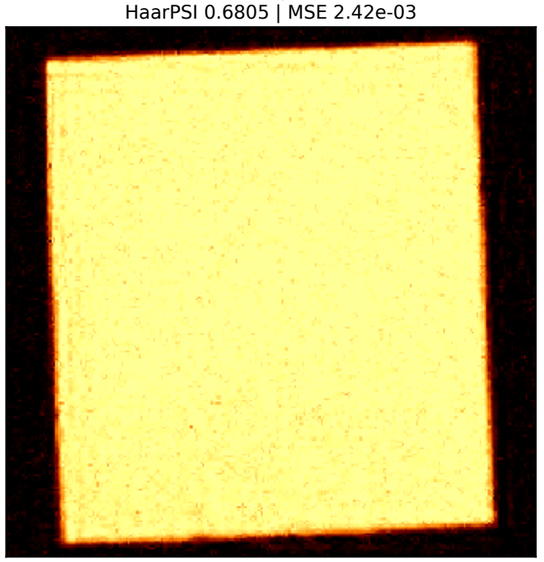
Target
Ground truth



Zero-filled
PSNR 25.51 dB | SSIM 0.4715
HaarPSI 0.5611 | MSE 2.81e-03



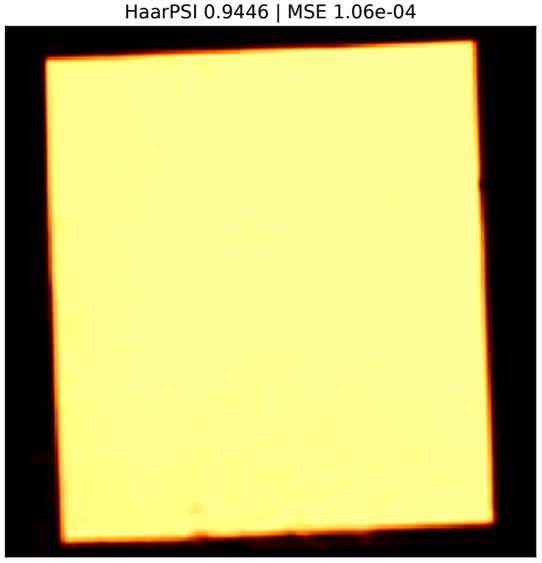
SPGL1-BP
PSNR 26.17 dB | SSIM 0.4140
HaarPSI 0.6805 | MSE 2.42e-03



SPGL1-BPDN
PSNR 31.60 dB | SSIM 0.8212
HaarPSI 0.7288 | MSE 6.92e-04



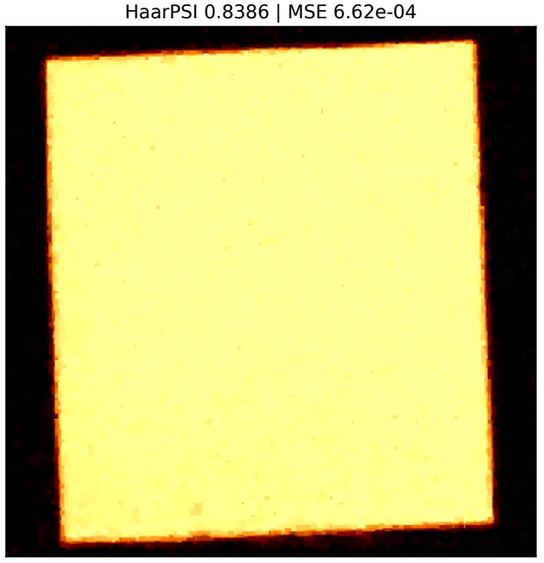
PnP-ADMM tuned-5
PSNR 39.76 dB | SSIM 0.9691
HaarPSI 0.9446 | MSE 1.06e-04



GS-PnP oracle
PSNR 40.63 dB | SSIM 0.9781
HaarPSI 0.9521 | MSE 8.64e-05



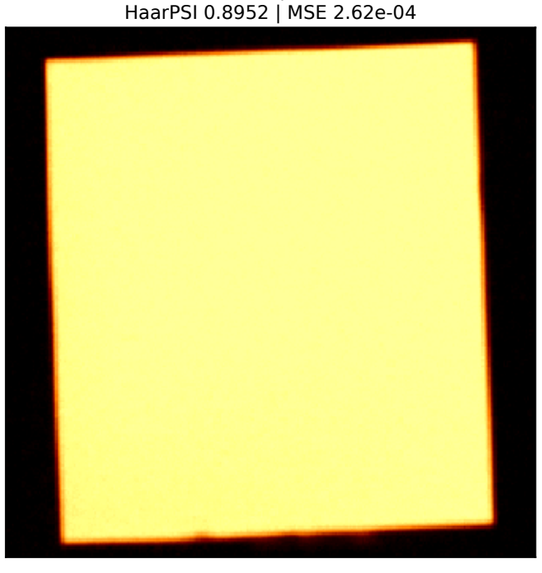
Scalar TV
PSNR 31.79 dB | SSIM 0.7848
HaarPSI 0.8386 | MSE 6.62e-04



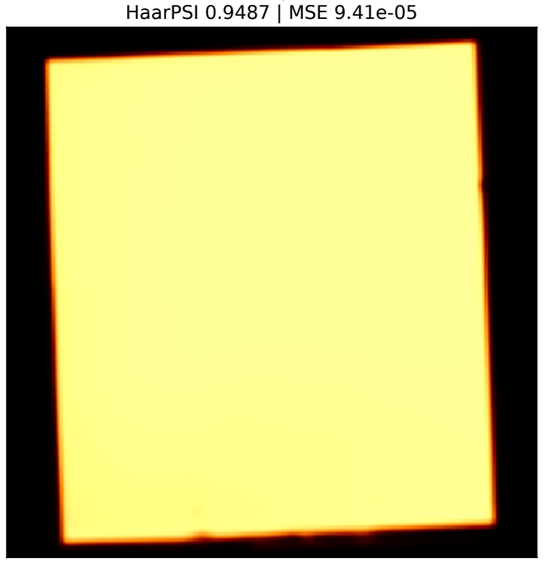
UTV-C60
PSNR 36.34 dB | SSIM 0.9489
HaarPSI 0.8961 | MSE 2.32e-04



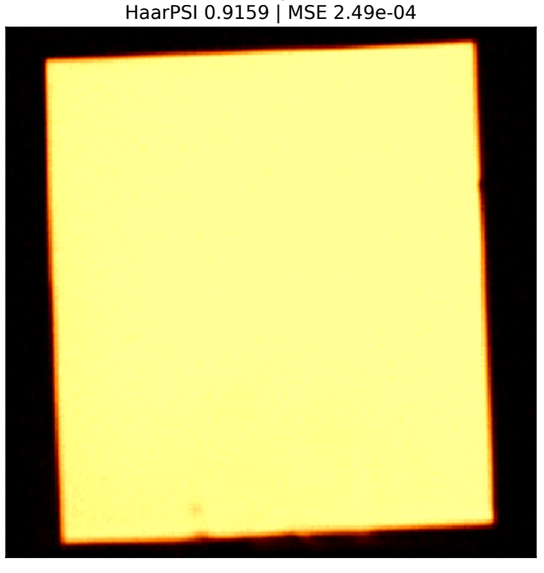
GS-PnP tuned-5
PSNR 35.82 dB | SSIM 0.8597
HaarPSI 0.8952 | MSE 2.62e-04



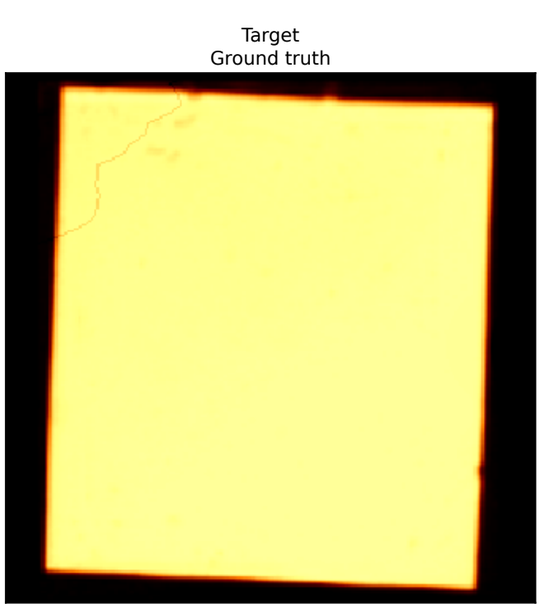
PnP-ADMM Tune-1 (0768)
PSNR 40.26 dB | SSIM 0.9806
HaarPSI 0.9487 | MSE 9.41e-05

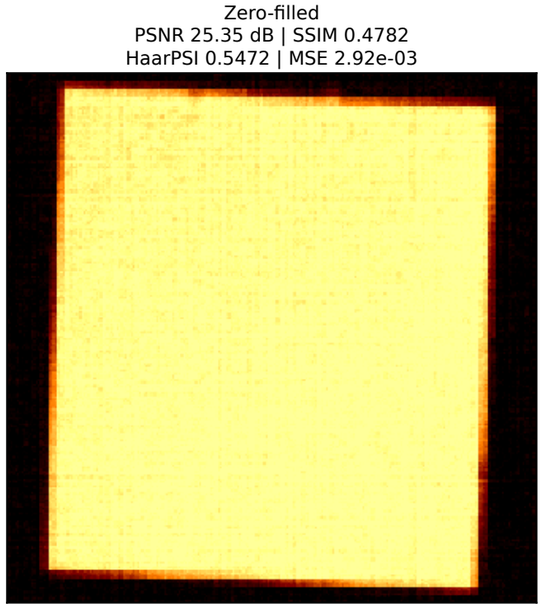


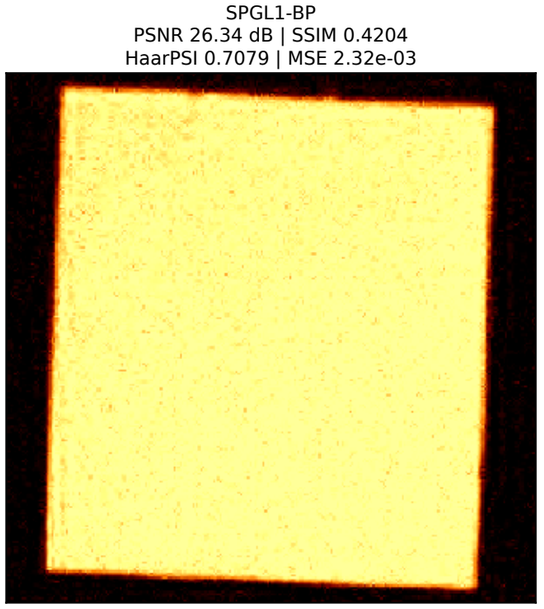
GS-PnP Tune-1 (0158)
PSNR 36.03 dB | SSIM 0.8498
HaarPSI 0.9159 | MSE 2.49e-04

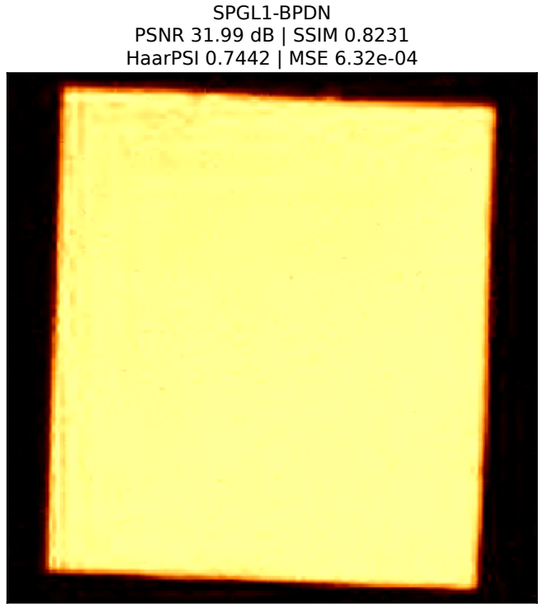


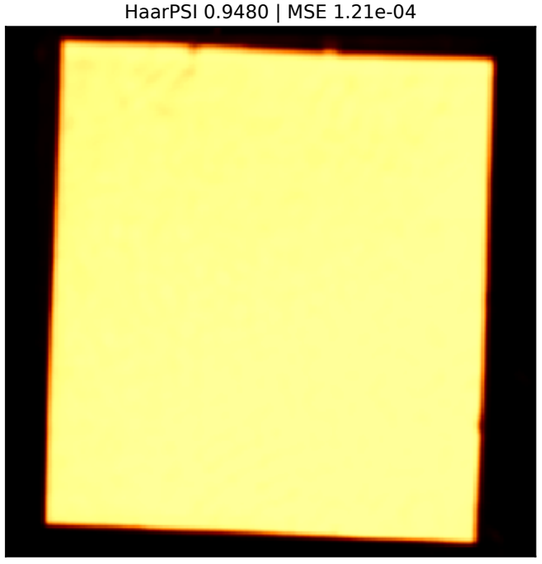
Organic | test image 0116 | 20% Hadamard + NPL-default noise

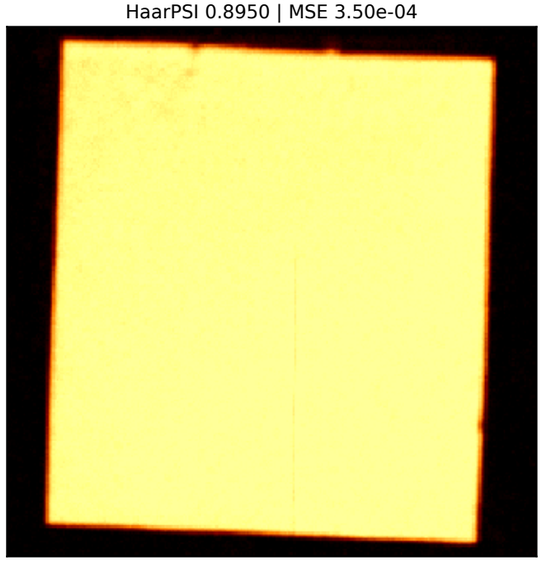

Target
Ground truth

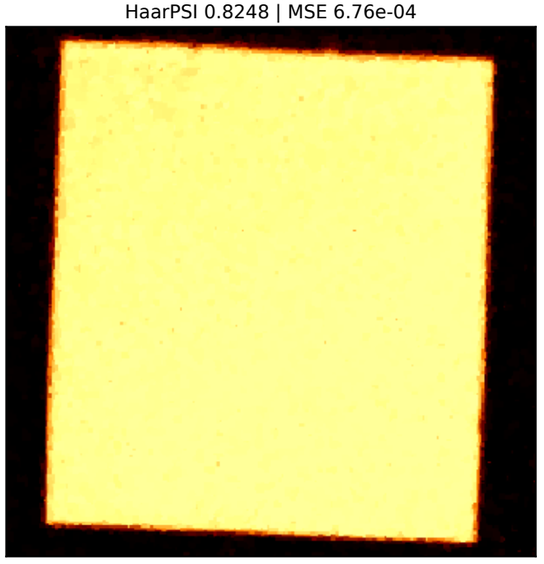

Zero-filled
PSNR 25.35 dB | SSIM 0.4782
HaarPSI 0.5472 | MSE 2.92e-03

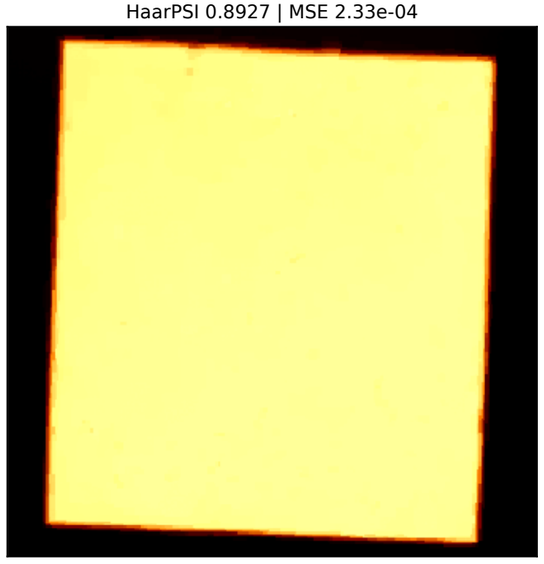

SPGL1-BP
PSNR 26.34 dB | SSIM 0.4204
HaarPSI 0.7079 | MSE 2.32e-03

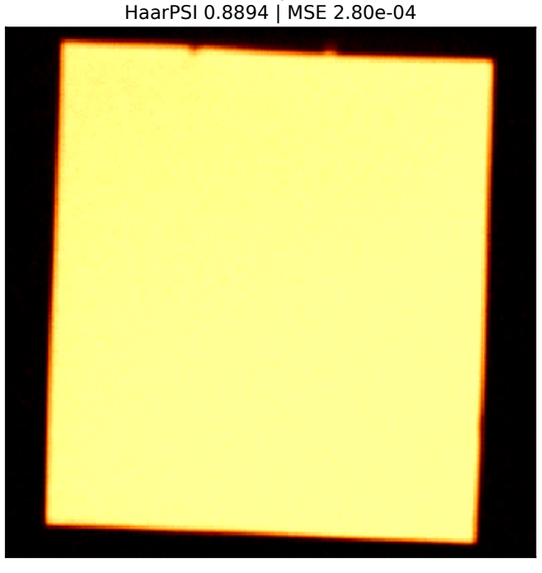

SPGL1-BPDN
PSNR 31.99 dB | SSIM 0.8231
HaarPSI 0.7442 | MSE 6.32e-04

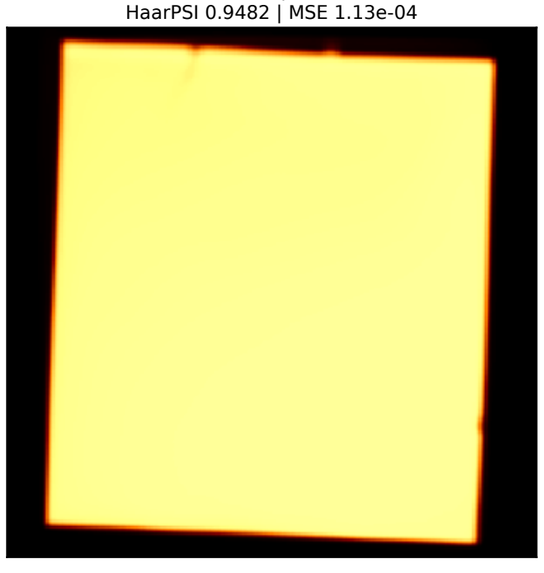

PnP-ADMM tuned-5
PSNR 39.17 dB | SSIM 0.9613
HaarPSI 0.9480 | MSE 1.21e-04

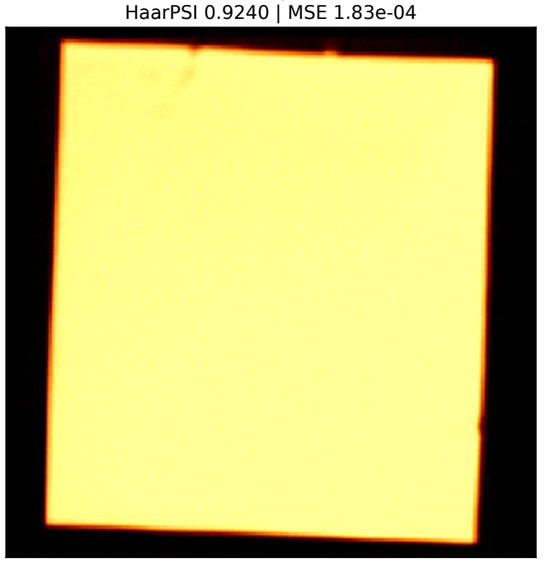

GS-PnP oracle
PSNR 34.55 dB | SSIM 0.7881
HaarPSI 0.8950 | MSE 3.50e-04


Scalar TV
PSNR 31.70 dB | SSIM 0.7790
HaarPSI 0.8248 | MSE 6.76e-04


UTV-C60
PSNR 36.34 dB | SSIM 0.9412
HaarPSI 0.8927 | MSE 2.33e-04

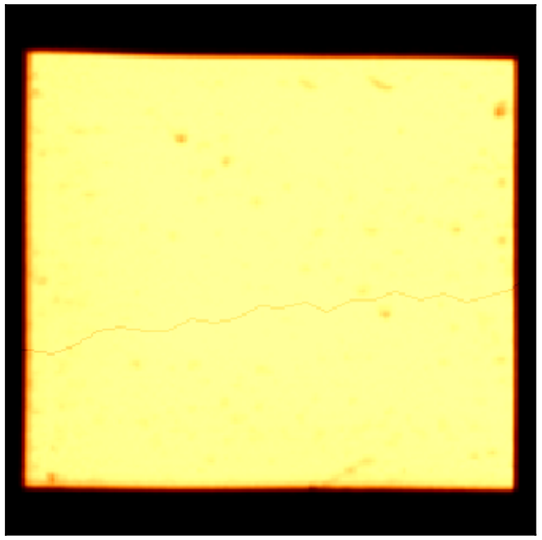

GS-PnP tuned-5
PSNR 35.53 dB | SSIM 0.8517
HaarPSI 0.8894 | MSE 2.80e-04


PnP-ADMM Tune-1 (0768)
PSNR 39.49 dB | SSIM 0.9731
HaarPSI 0.9482 | MSE 1.13e-04

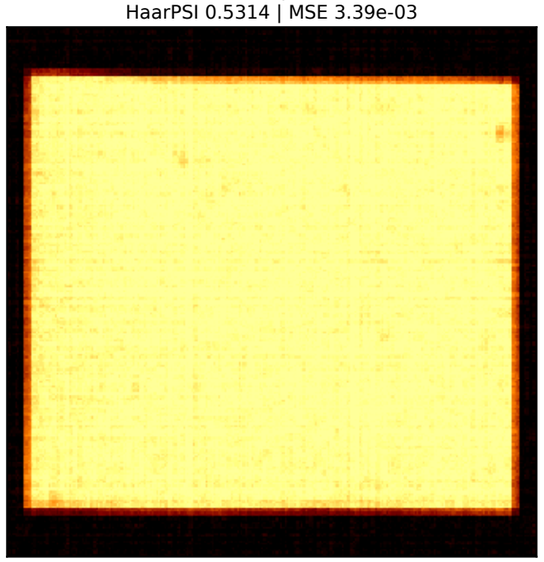

GS-PnP Tune-1 (0158)
PSNR 37.38 dB | SSIM 0.9046
HaarPSI 0.9240 | MSE 1.83e-04

Organic | test image 0279 | 20% Hadamard + NPL-default noise

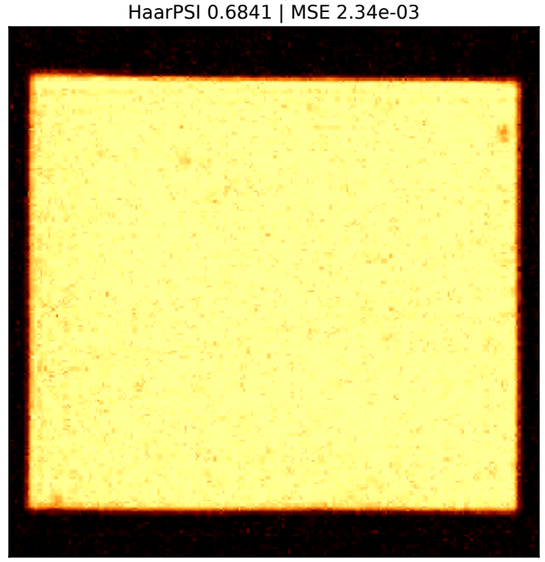
Target
Ground truth



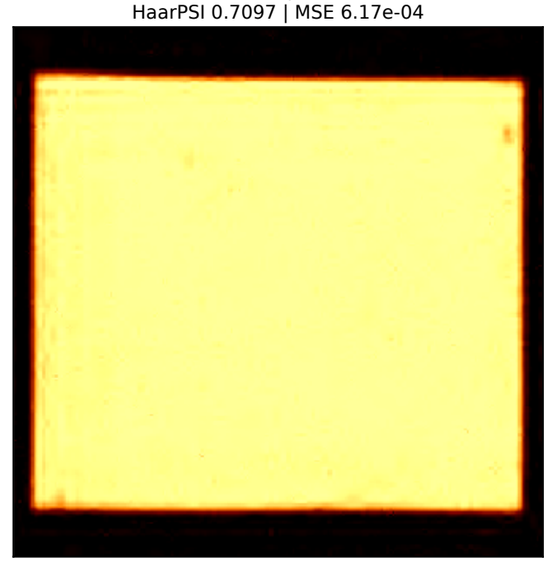
Zero-filled
PSNR 24.69 dB | SSIM 0.4615
HaarPSI 0.5314 | MSE 3.39e-03



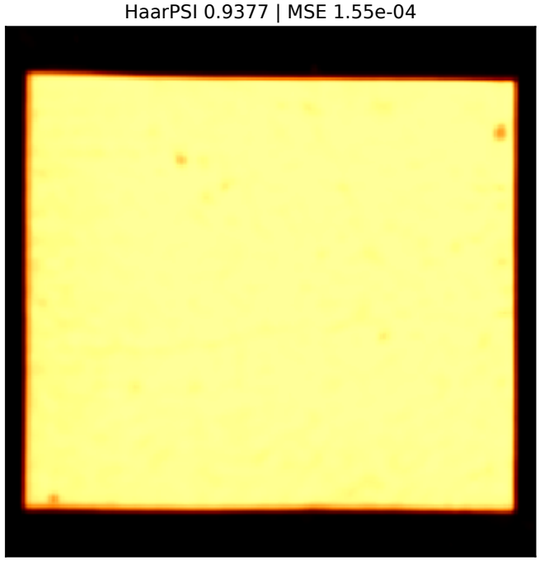
SPGL1-BP
PSNR 26.31 dB | SSIM 0.4242
HaarPSI 0.6841 | MSE 2.34e-03



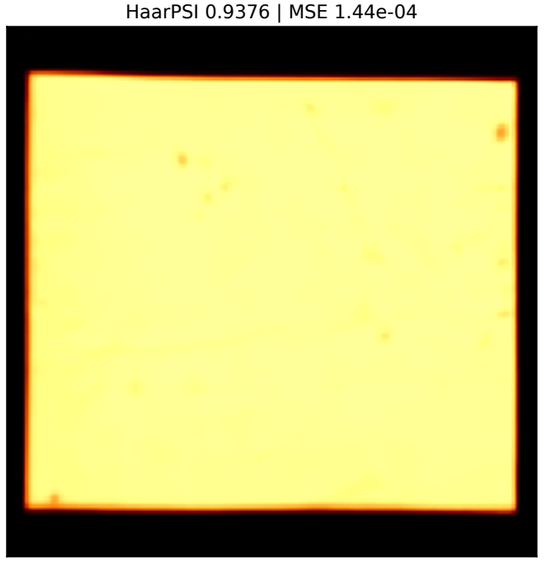
SPGL1-BPDN
PSNR 32.10 dB | SSIM 0.7995
HaarPSI 0.7097 | MSE 6.17e-04



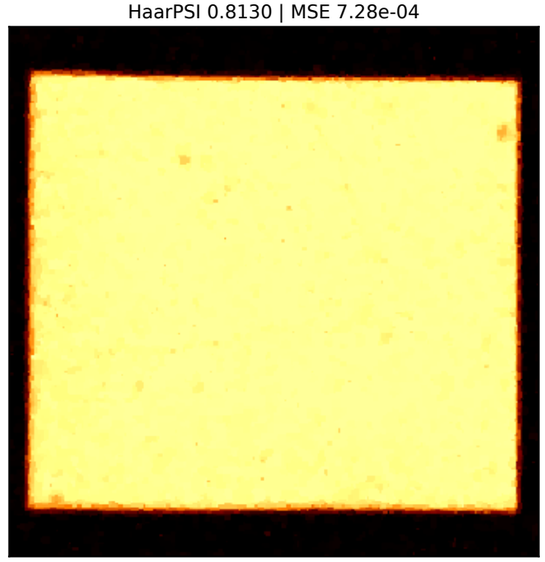
PnP-ADMM tuned-5
PSNR 38.11 dB | SSIM 0.9414
HaarPSI 0.9377 | MSE 1.55e-04



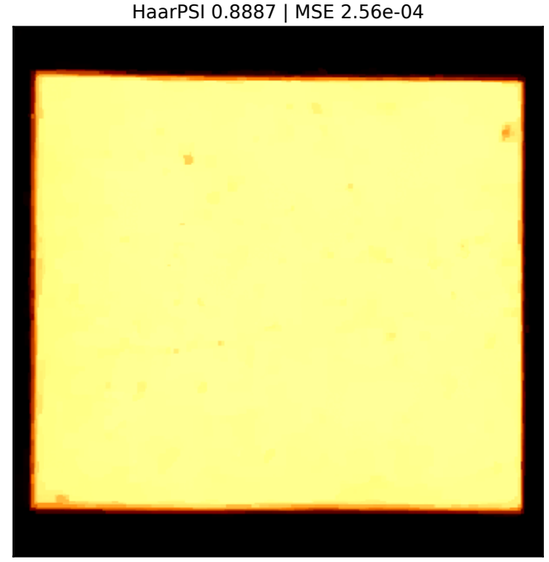
GS-PnP oracle
PSNR 38.43 dB | SSIM 0.9429
HaarPSI 0.9376 | MSE 1.44e-04



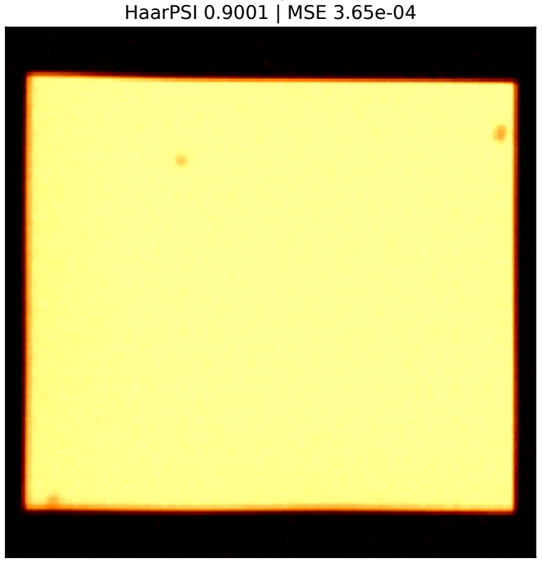
Scalar TV
PSNR 31.38 dB | SSIM 0.7673
HaarPSI 0.8130 | MSE 7.28e-04



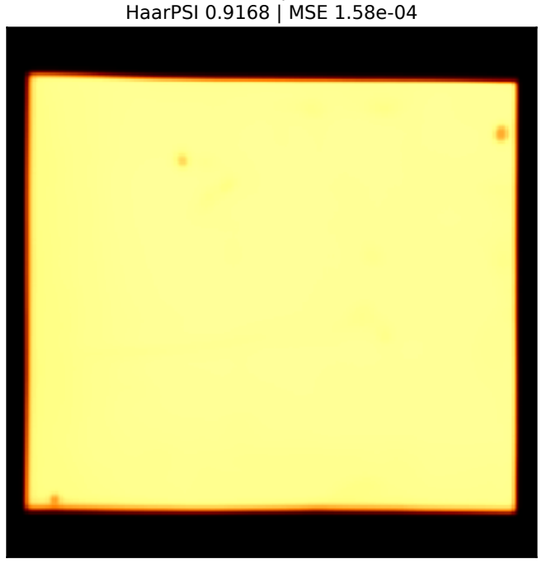
UTV-C60
PSNR 35.92 dB | SSIM 0.9246
HaarPSI 0.8887 | MSE 2.56e-04



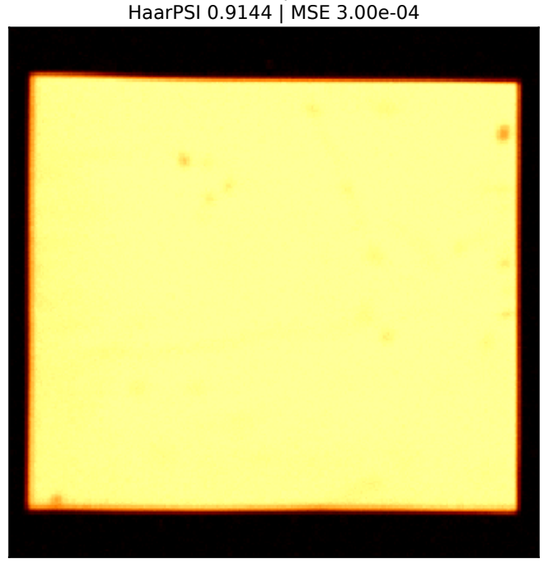
GS-PnP tuned-5
PSNR 34.37 dB | SSIM 0.8114
HaarPSI 0.9001 | MSE 3.65e-04



PnP-ADMM Tune-1 (0768)
PSNR 38.02 dB | SSIM 0.9416
HaarPSI 0.9168 | MSE 1.58e-04

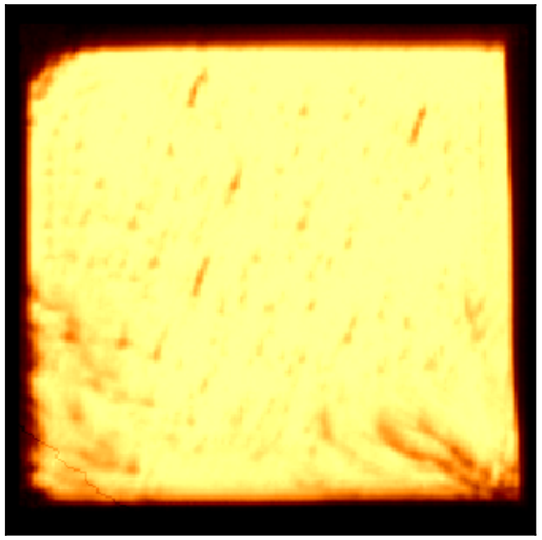


GS-PnP Tune-1 (0158)
PSNR 35.23 dB | SSIM 0.8248
HaarPSI 0.9144 | MSE 3.00e-04

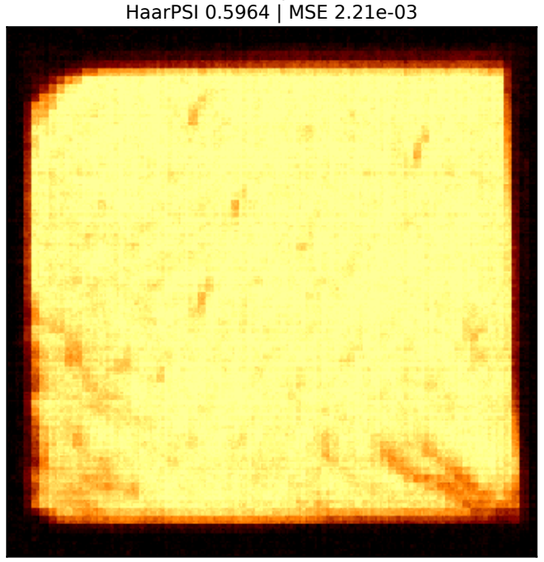


Organic | test image 0470 | 20% Hadamard + NPL-default noise

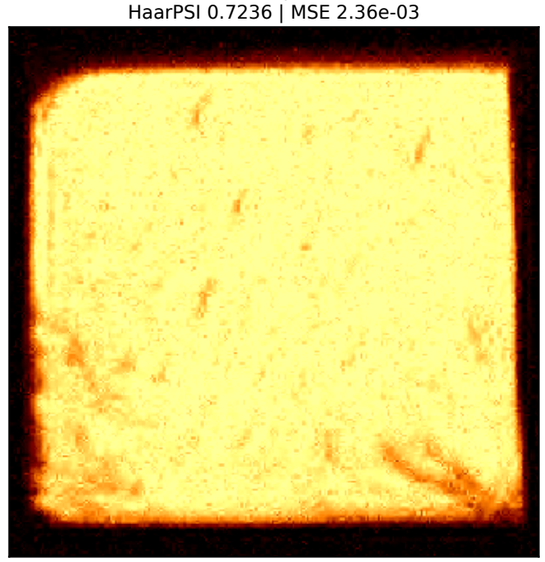
Target
Ground truth



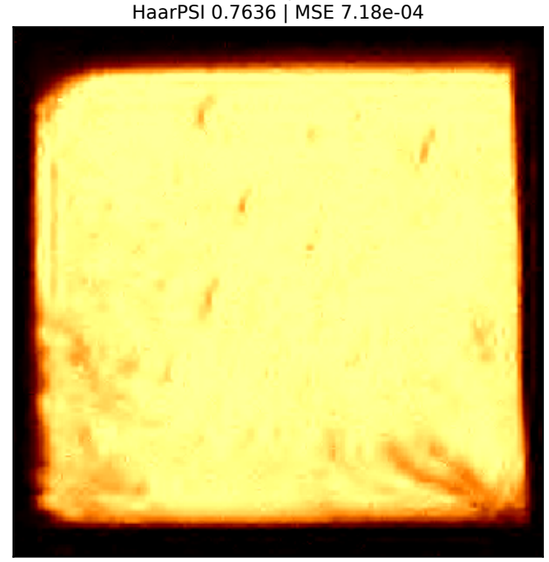
Zero-filled
PSNR 26.55 dB | SSIM 0.5712
HaarPSI 0.5964 | MSE 2.21e-03



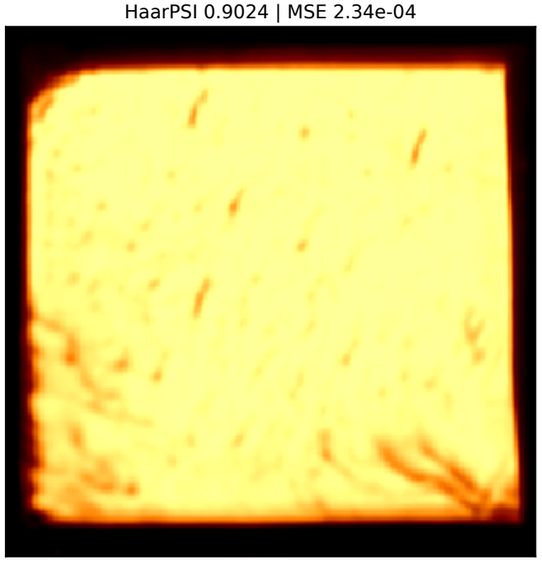
SPGL1-BP
PSNR 26.27 dB | SSIM 0.5014
HaarPSI 0.7236 | MSE 2.36e-03



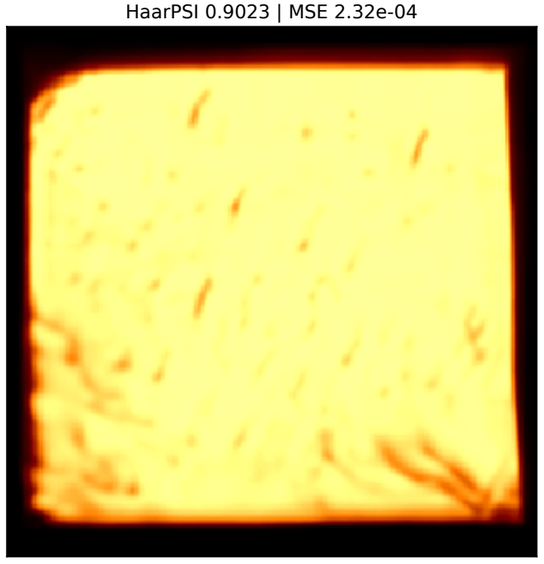
SPGL1-BPDN
PSNR 31.44 dB | SSIM 0.7887
HaarPSI 0.7636 | MSE 7.18e-04



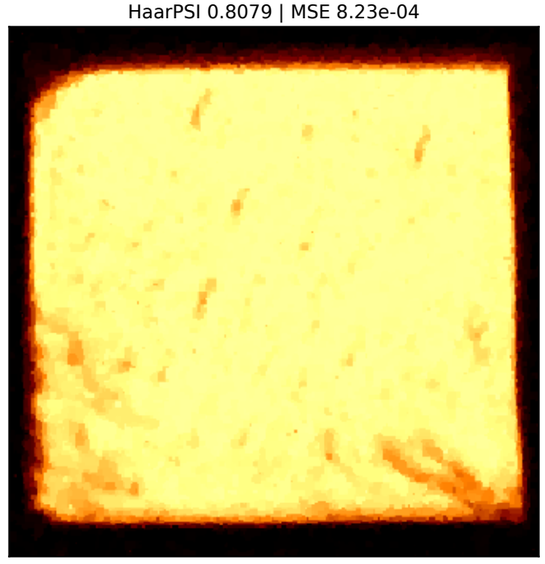
PnP-ADMM tuned-5
PSNR 36.30 dB | SSIM 0.9238
HaarPSI 0.9024 | MSE 2.34e-04



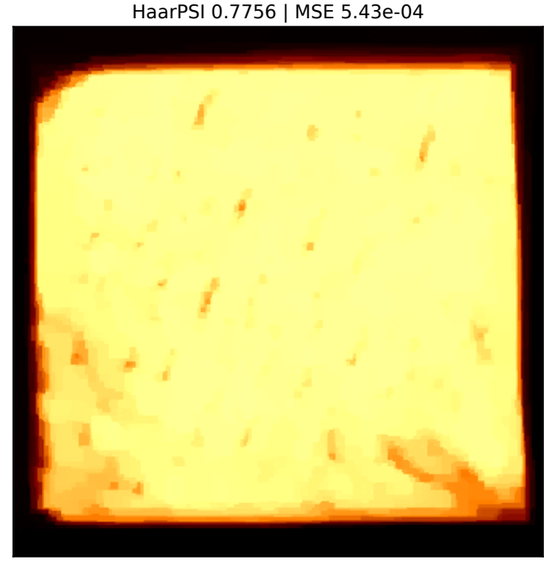
GS-PnP oracle
PSNR 36.34 dB | SSIM 0.9217
HaarPSI 0.9023 | MSE 2.32e-04



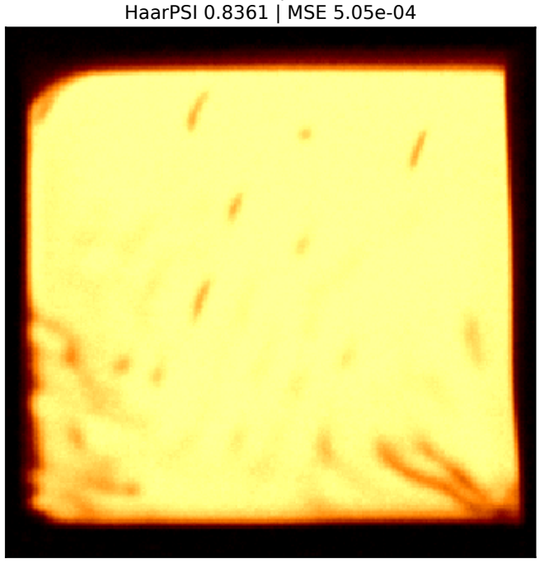
Scalar TV
PSNR 30.84 dB | SSIM 0.7963
HaarPSI 0.8079 | MSE 8.23e-04



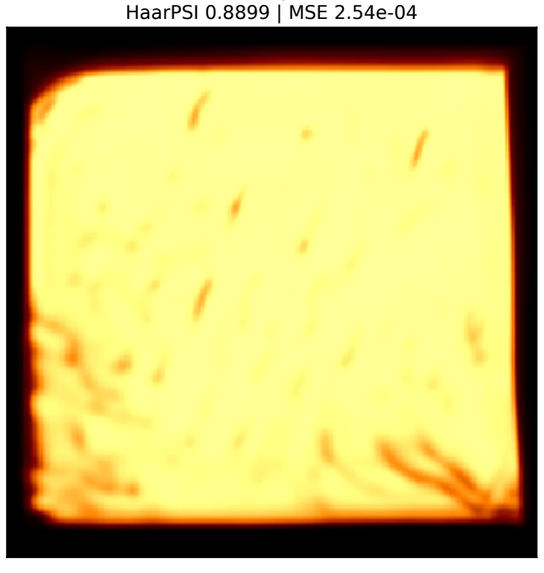
UTV-C60
PSNR 32.65 dB | SSIM 0.8569
HaarPSI 0.7756 | MSE 5.43e-04



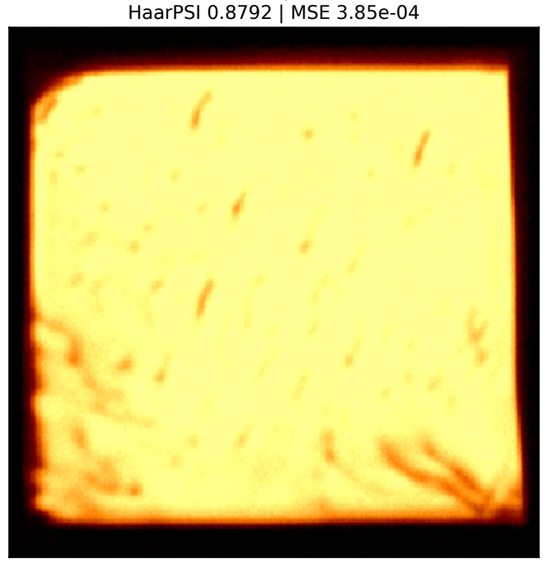
GS-PnP tuned-5
PSNR 32.97 dB | SSIM 0.7990
HaarPSI 0.8361 | MSE 5.05e-04



PnP-ADMM Tune-1 (0768)
PSNR 35.95 dB | SSIM 0.9169
HaarPSI 0.8899 | MSE 2.54e-04

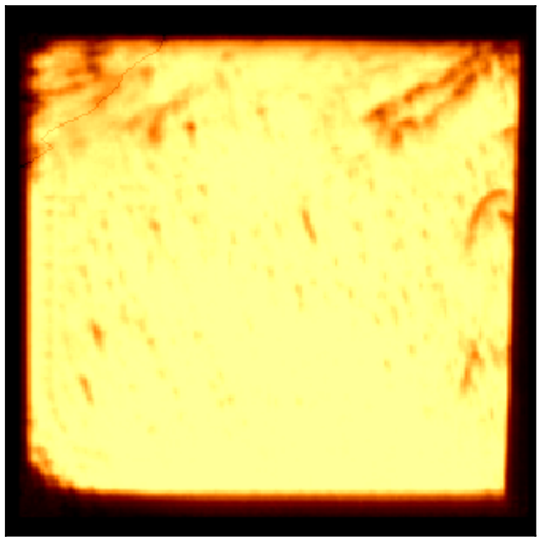


GS-PnP Tune-1 (0158)
PSNR 34.14 dB | SSIM 0.8378
HaarPSI 0.8792 | MSE 3.85e-04

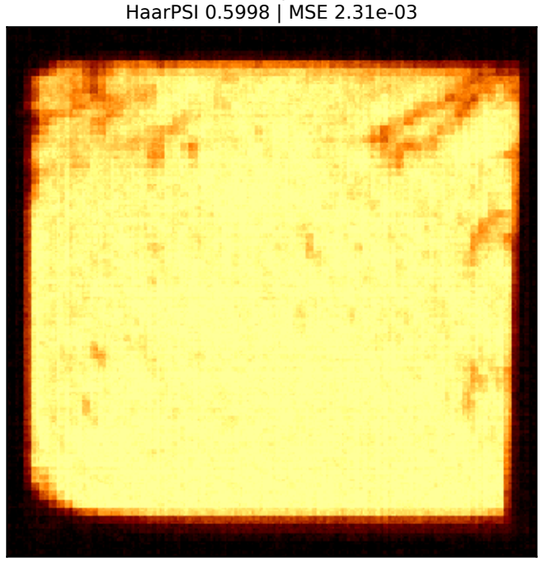


Organic | test image 0518 | 20% Hadamard + NPL-default noise

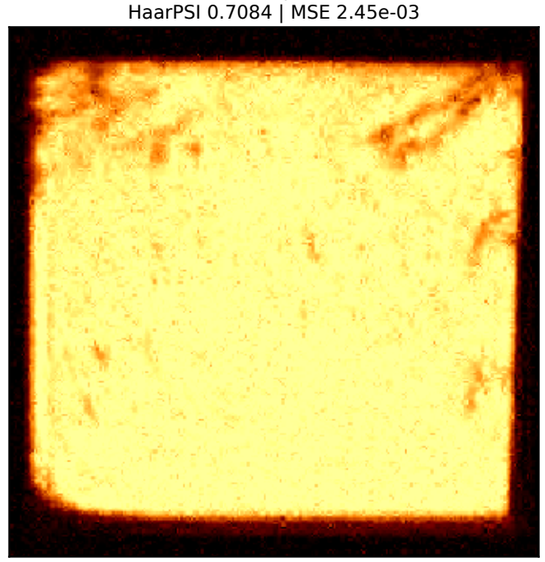
Target
Ground truth



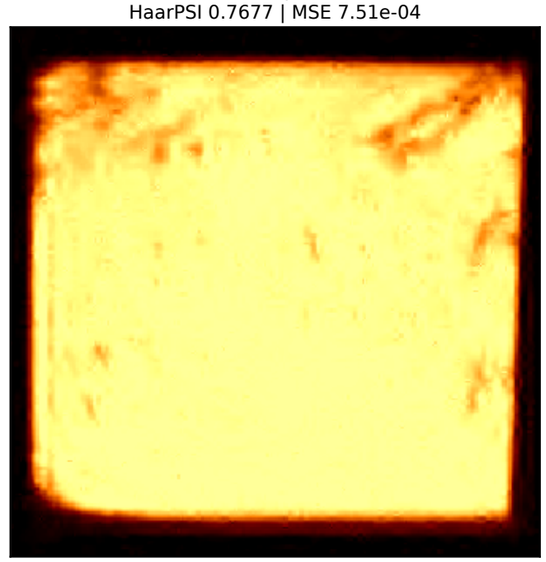
Zero-filled
PSNR 26.37 dB | SSIM 0.5707
HaarPSI 0.5998 | MSE 2.31e-03



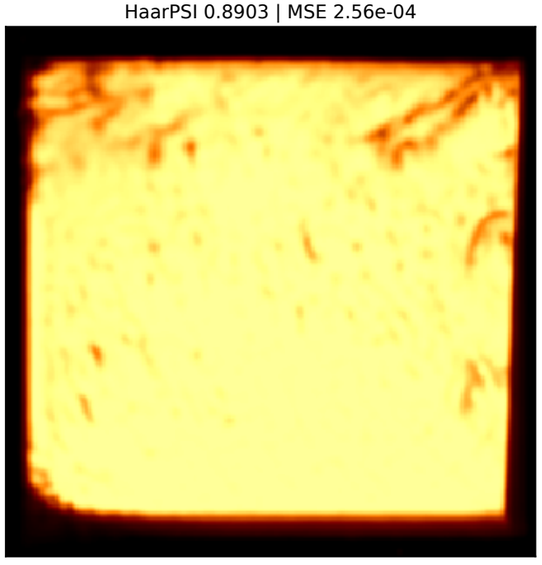
SPGL1-BP
PSNR 26.11 dB | SSIM 0.5061
HaarPSI 0.7084 | MSE 2.45e-03



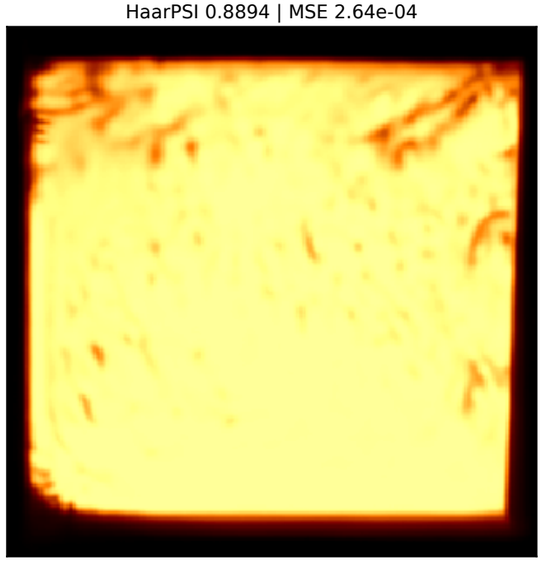
SPGL1-BPDN
PSNR 31.24 dB | SSIM 0.8024
HaarPSI 0.7677 | MSE 7.51e-04



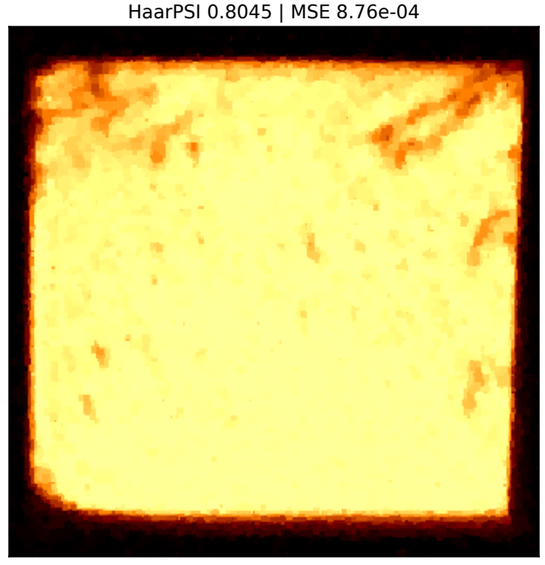
PnP-ADMM tuned-5
PSNR 35.92 dB | SSIM 0.9252
HaarPSI 0.8903 | MSE 2.56e-04



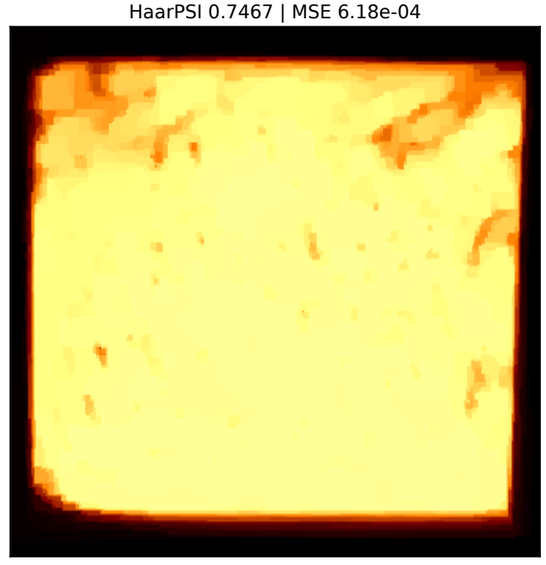
GS-PnP oracle
PSNR 35.79 dB | SSIM 0.9242
HaarPSI 0.8894 | MSE 2.64e-04



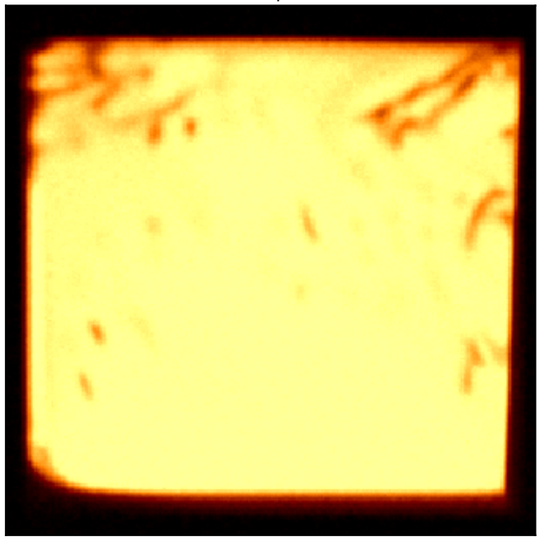
Scalar TV
PSNR 30.57 dB | SSIM 0.7954
HaarPSI 0.8045 | MSE 8.76e-04



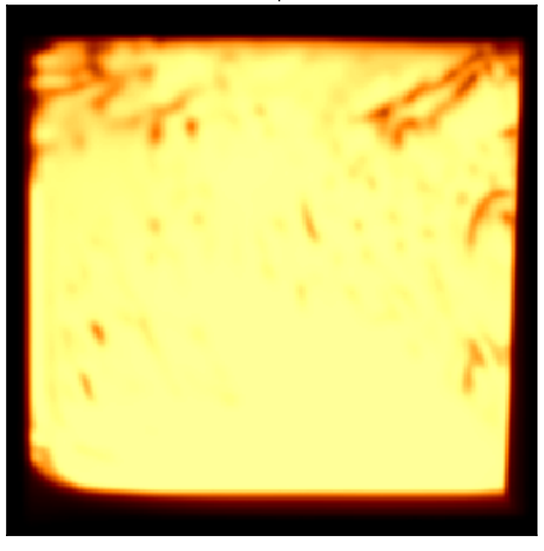
UTV-C60
PSNR 32.09 dB | SSIM 0.8644
HaarPSI 0.7467 | MSE 6.18e-04



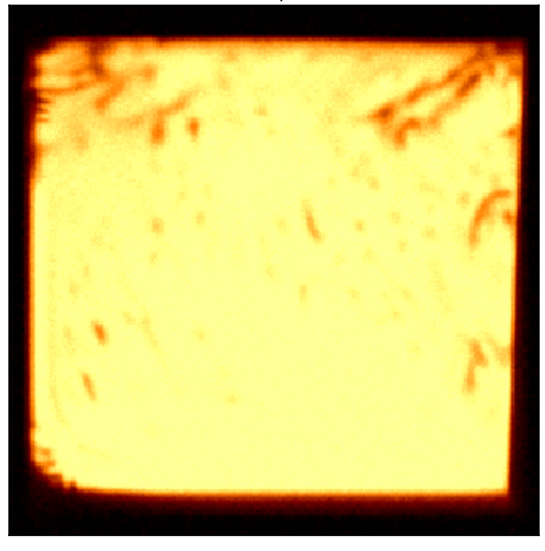
GS-PnP tuned-5
PSNR 32.68 dB | SSIM 0.7981
HaarPSI 0.8195 | MSE 5.39e-04



PnP-ADMM Tune-1 (0768)
PSNR 35.54 dB | SSIM 0.9179
HaarPSI 0.8726 | MSE 2.79e-04

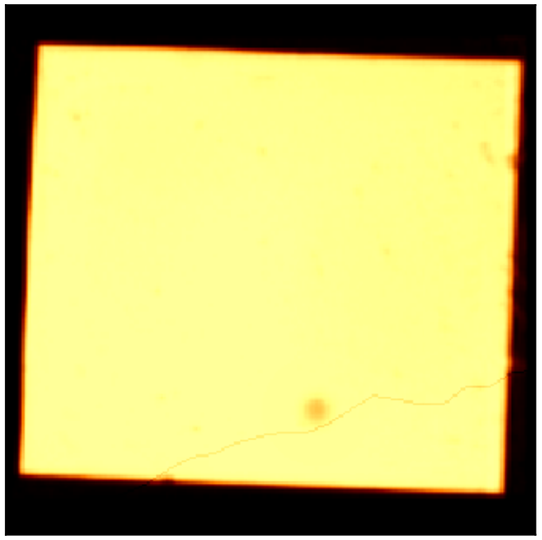


GS-PnP Tune-1 (0158)
PSNR 32.67 dB | SSIM 0.7842
HaarPSI 0.8553 | MSE 5.41e-04

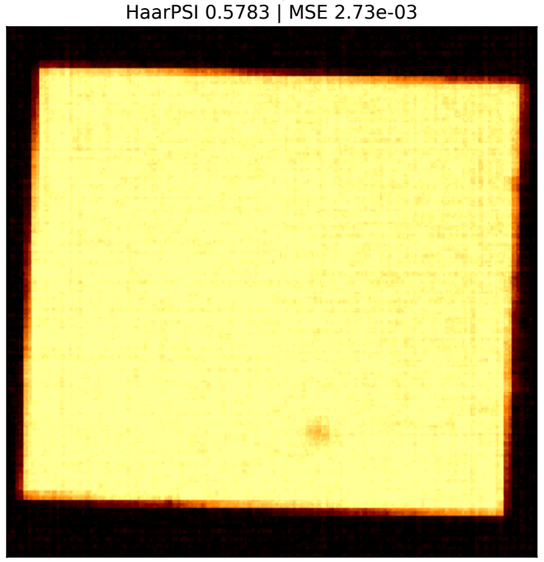


Organic | test image 0862 | 20% Hadamard + NPL-default noise

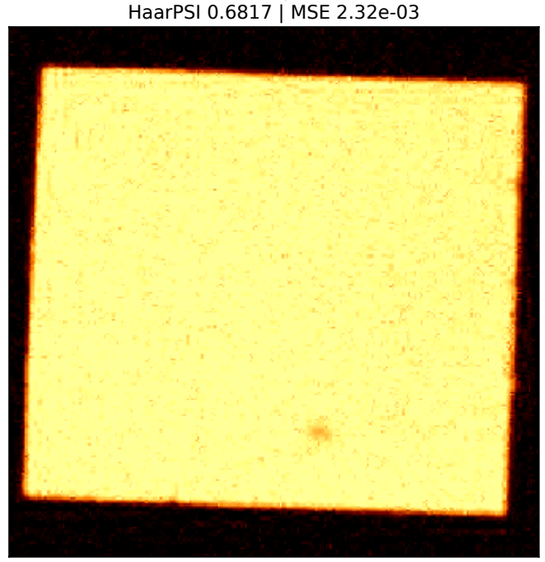
Target
Ground truth



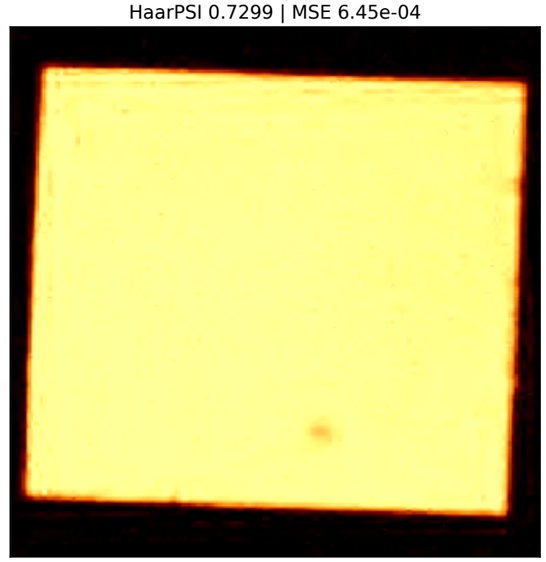
Zero-filled
PSNR 25.64 dB | SSIM 0.4645
HaarPSI 0.5783 | MSE 2.73e-03



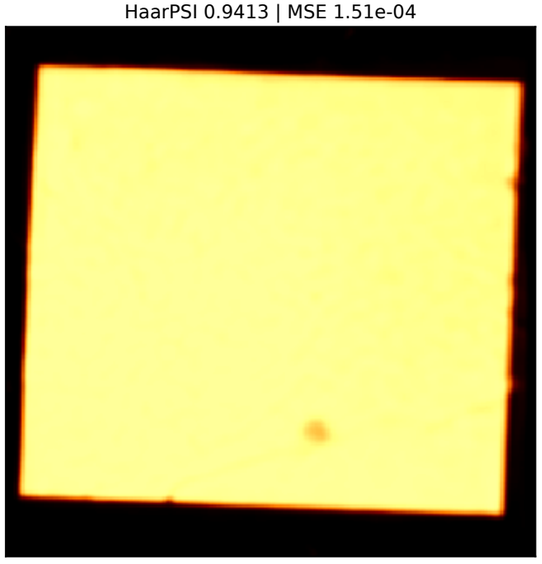
SPGL1-BP
PSNR 26.35 dB | SSIM 0.4196
HaarPSI 0.6817 | MSE 2.32e-03



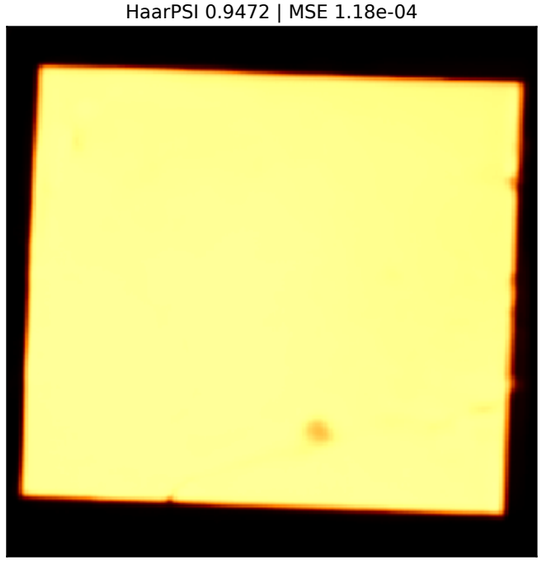
SPGL1-BPDN
PSNR 31.91 dB | SSIM 0.8072
HaarPSI 0.7299 | MSE 6.45e-04



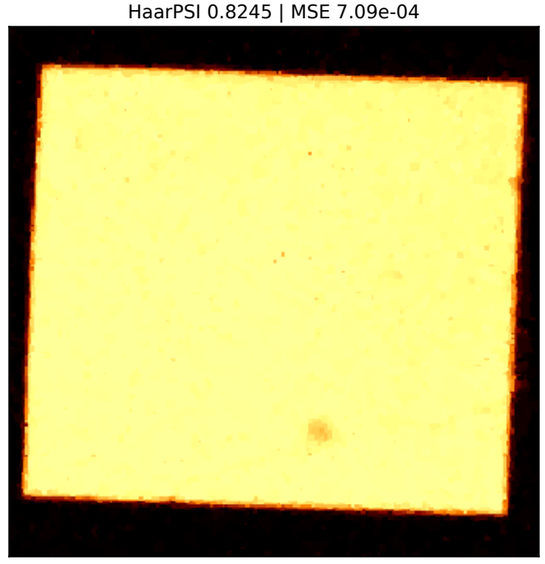
PnP-ADMM tuned-5
PSNR 38.22 dB | SSIM 0.9516
HaarPSI 0.9413 | MSE 1.51e-04



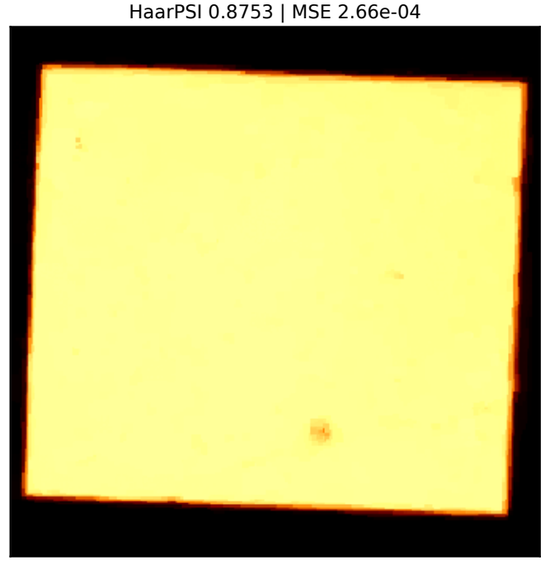
GS-PnP oracle
PSNR 39.27 dB | SSIM 0.9647
HaarPSI 0.9472 | MSE 1.18e-04



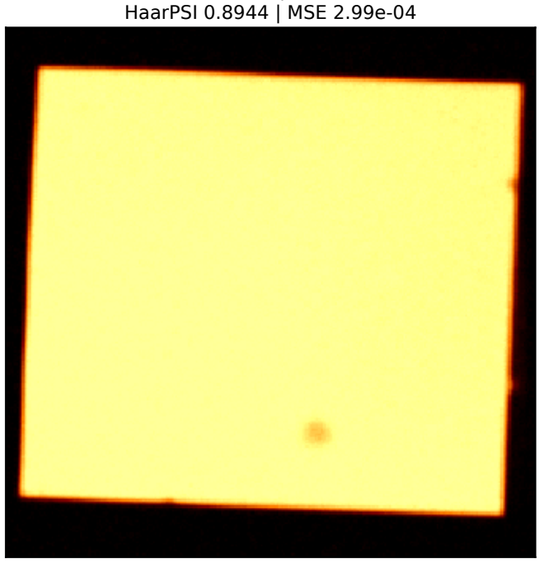
Scalar TV
PSNR 31.49 dB | SSIM 0.7727
HaarPSI 0.8245 | MSE 7.09e-04



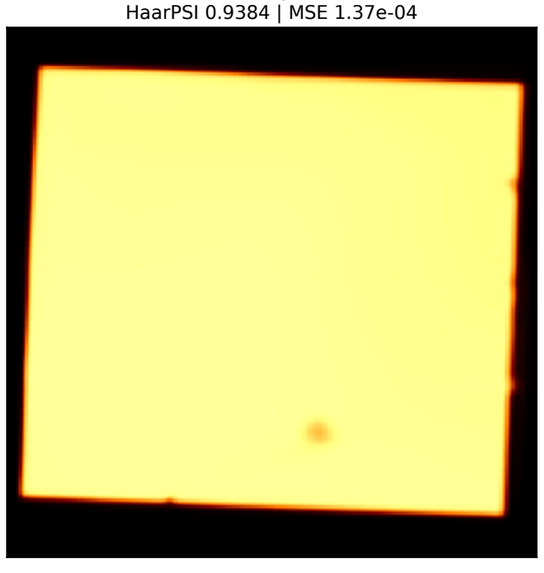
UTV-C60
PSNR 35.75 dB | SSIM 0.9351
HaarPSI 0.8753 | MSE 2.66e-04



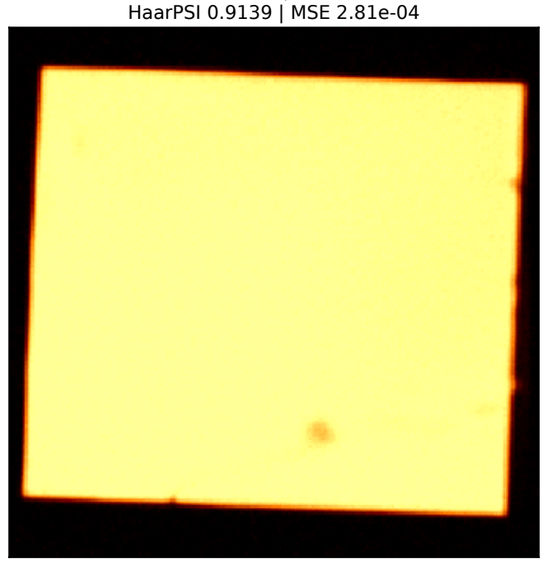
GS-PnP tuned-5
PSNR 35.25 dB | SSIM 0.8443
HaarPSI 0.8944 | MSE 2.99e-04



PnP-ADMM Tune-1 (0768)
PSNR 38.65 dB | SSIM 0.9645
HaarPSI 0.9384 | MSE 1.37e-04

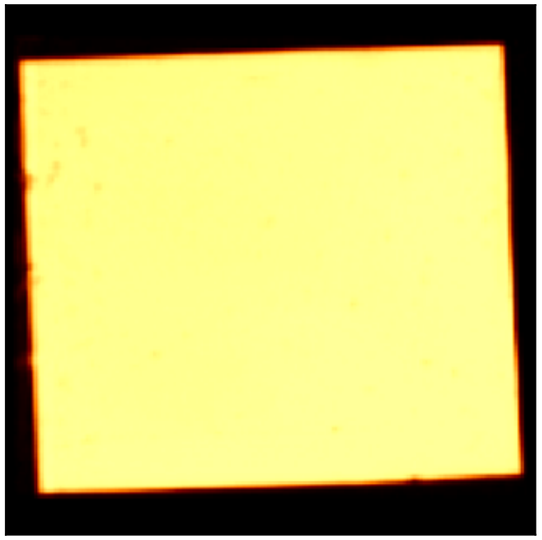


GS-PnP Tune-1 (0158)
PSNR 35.51 dB | SSIM 0.8384
HaarPSI 0.9139 | MSE 2.81e-04

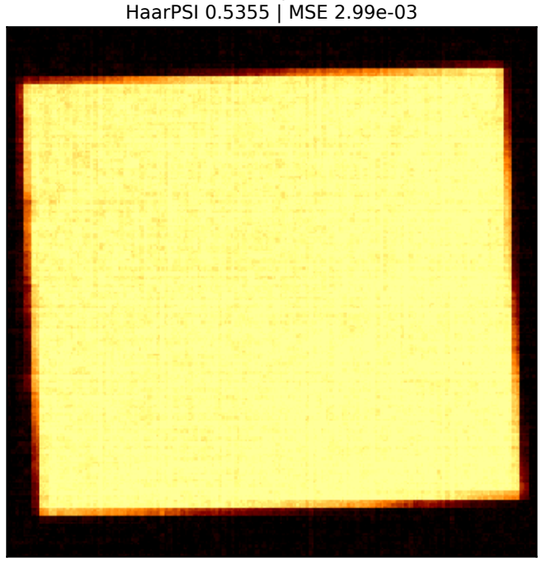


Organic | test image 0935 | 20% Hadamard + NPL-default noise

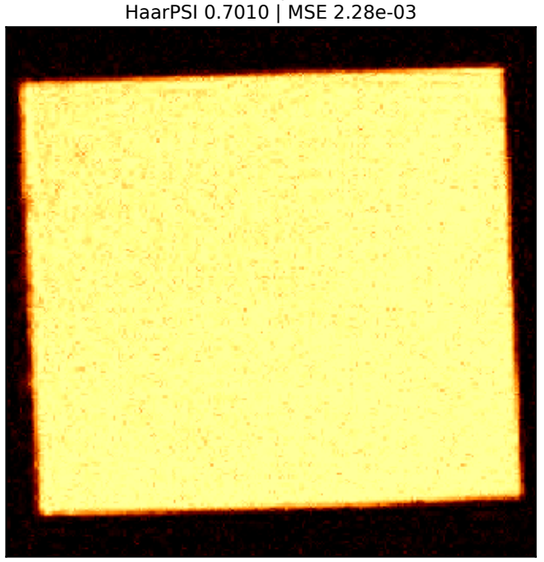
Target
Ground truth



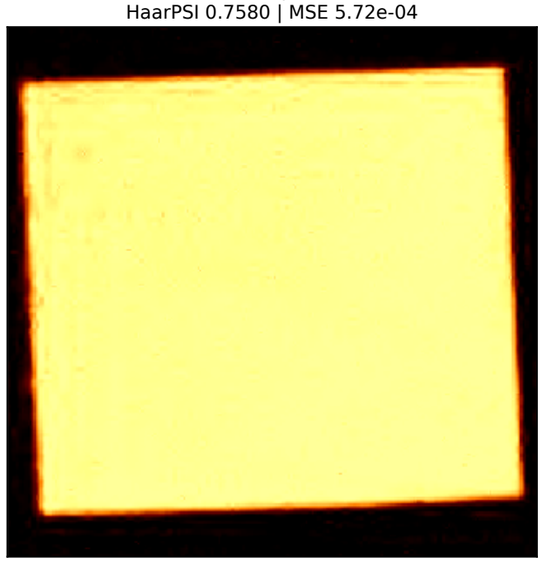
Zero-filled
PSNR 25.24 dB | SSIM 0.4741
HaarPSI 0.5355 | MSE 2.99e-03



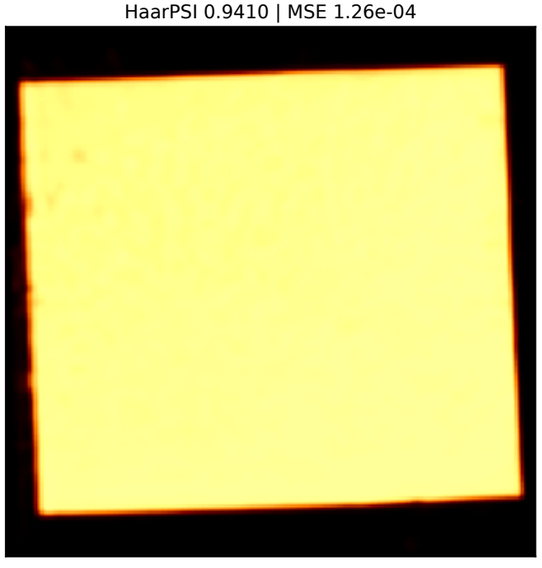
SPGL1-BP
PSNR 26.42 dB | SSIM 0.4178
HaarPSI 0.7010 | MSE 2.28e-03



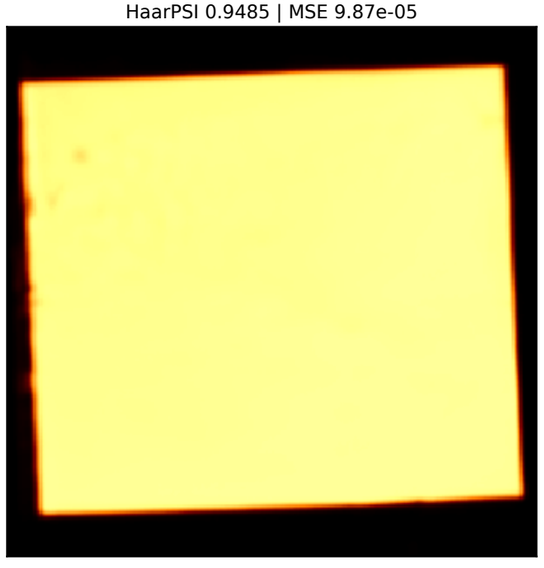
SPGL1-BPDN
PSNR 32.42 dB | SSIM 0.8242
HaarPSI 0.7580 | MSE 5.72e-04



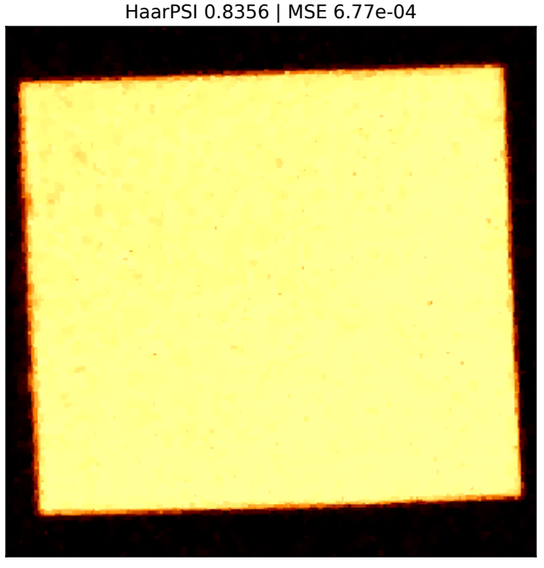
PnP-ADMM tuned-5
PSNR 38.99 dB | SSIM 0.9605
HaarPSI 0.9410 | MSE 1.26e-04



GS-PnP oracle
PSNR 40.06 dB | SSIM 0.9719
HaarPSI 0.9485 | MSE 9.87e-05



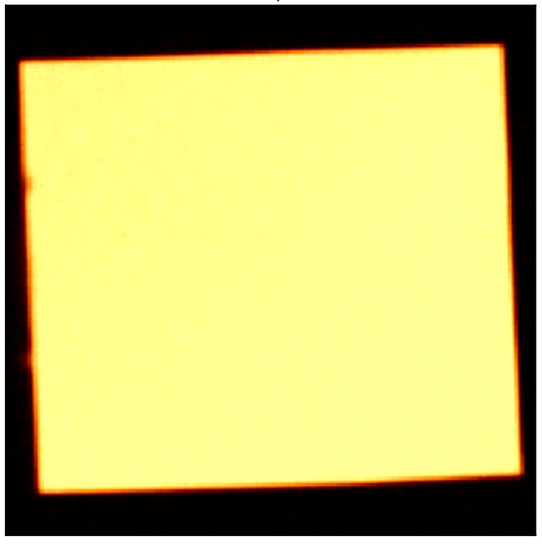
Scalar TV
PSNR 31.69 dB | SSIM 0.7757
HaarPSI 0.8356 | MSE 6.77e-04



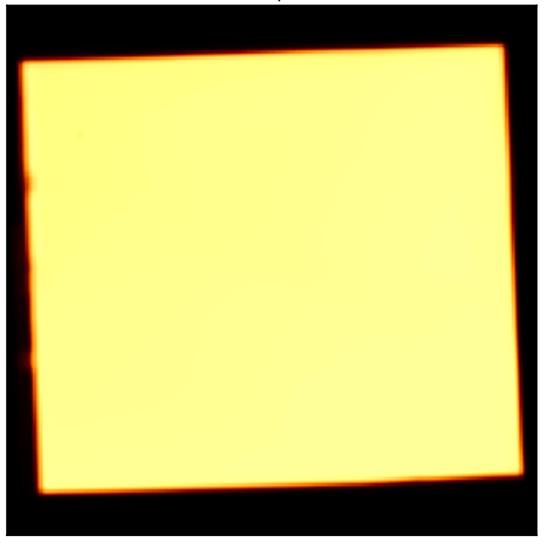
UTV-C60
PSNR 36.20 dB | SSIM 0.9460
HaarPSI 0.8695 | MSE 2.40e-04



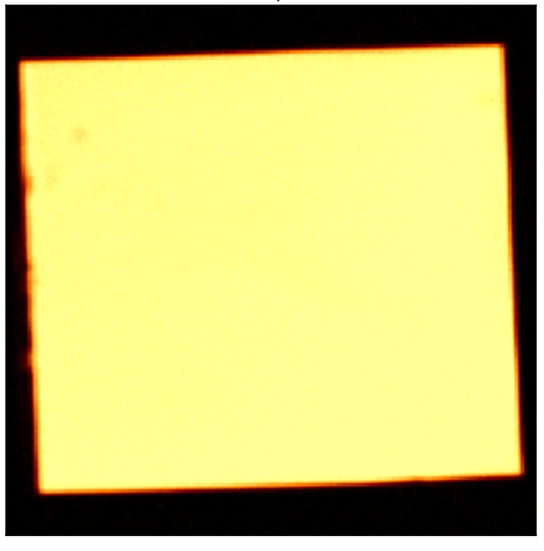
GS-PnP tuned-5
PSNR 35.63 dB | SSIM 0.8520
HaarPSI 0.9001 | MSE 2.74e-04



PnP-ADMM Tune-1 (0768)
PSNR 39.41 dB | SSIM 0.9723
HaarPSI 0.9444 | MSE 1.15e-04

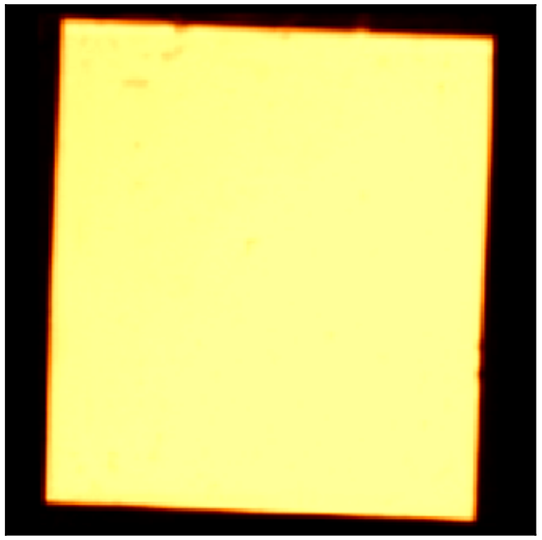


GS-PnP Tune-1 (0158)
PSNR 35.90 dB | SSIM 0.8454
HaarPSI 0.9159 | MSE 2.57e-04

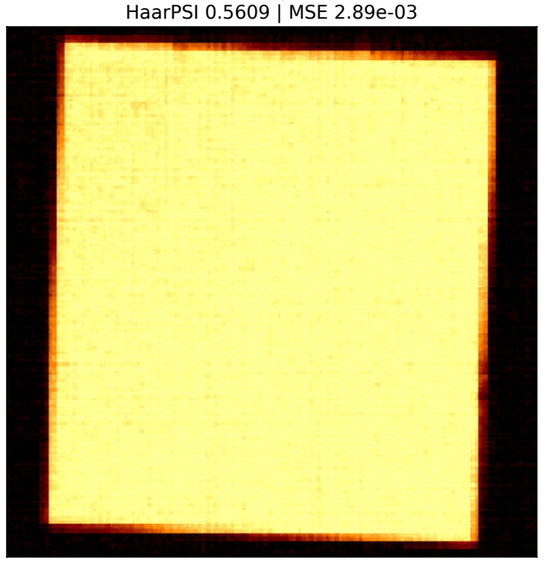


Organic | test image 0946 | 20% Hadamard + NPL-default noise

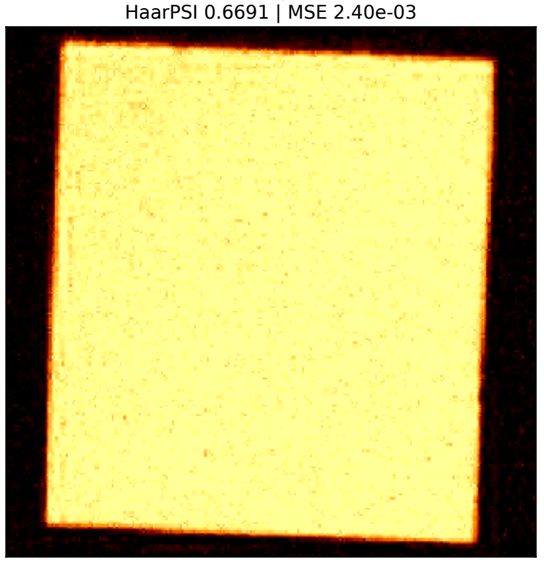
Target
Ground truth



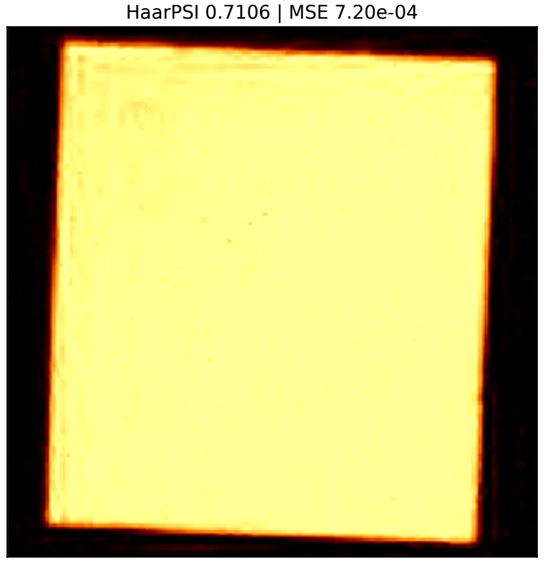
Zero-filled
PSNR 25.39 dB | SSIM 0.4747
HaarPSI 0.5609 | MSE 2.89e-03



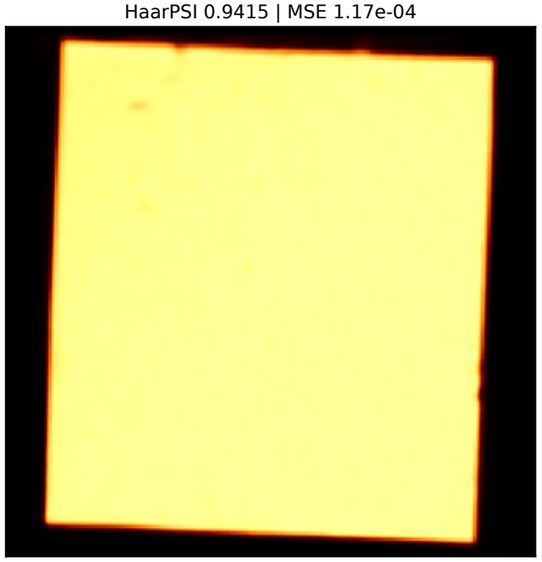
SPGL1-BP
PSNR 26.20 dB | SSIM 0.4155
HaarPSI 0.6691 | MSE 2.40e-03



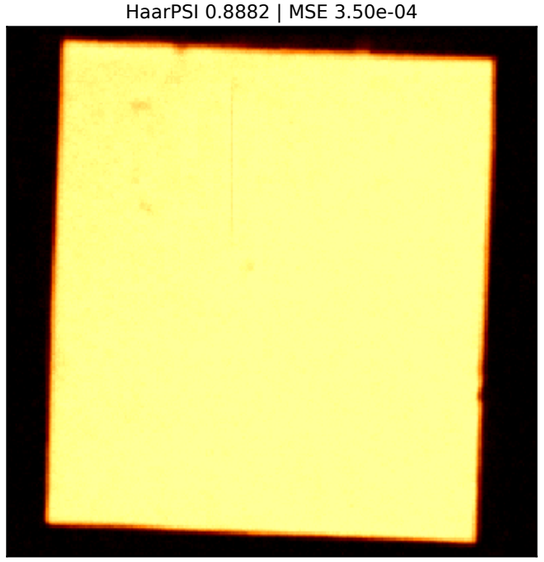
SPGL1-BPDN
PSNR 31.43 dB | SSIM 0.8143
HaarPSI 0.7106 | MSE 7.20e-04



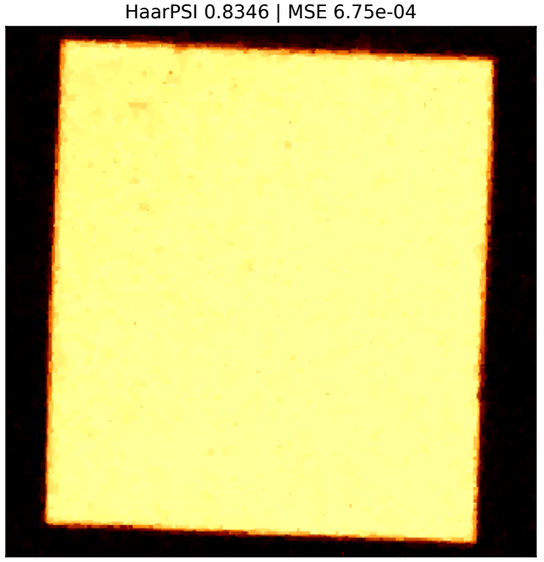
PnP-ADMM tuned-5
PSNR 39.30 dB | SSIM 0.9665
HaarPSI 0.9415 | MSE 1.17e-04



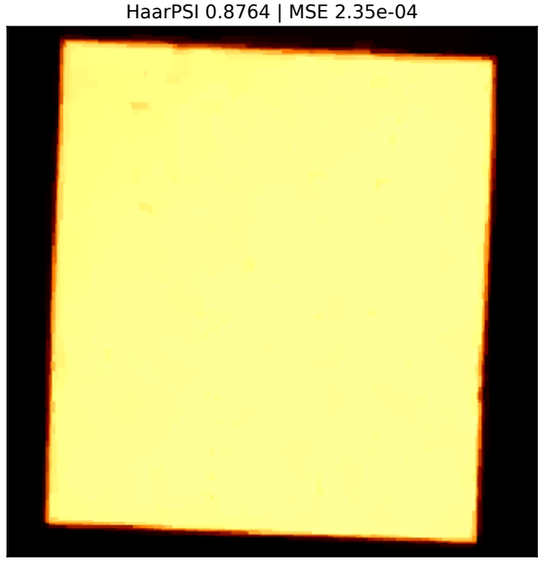
GS-PnP oracle
PSNR 34.56 dB | SSIM 0.7939
HaarPSI 0.8882 | MSE 3.50e-04



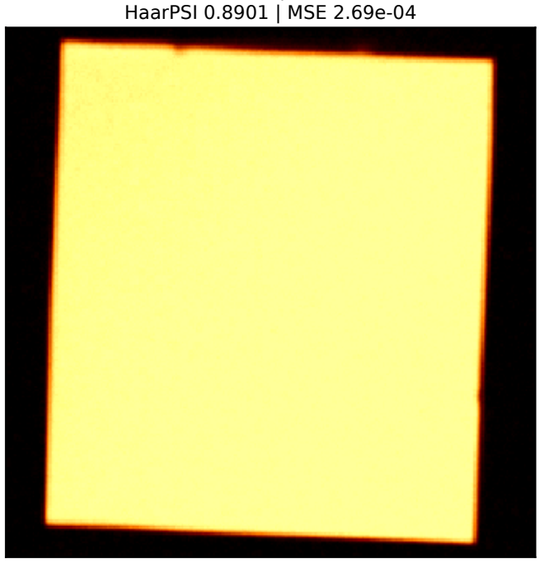
Scalar TV
PSNR 31.70 dB | SSIM 0.7818
HaarPSI 0.8346 | MSE 6.75e-04



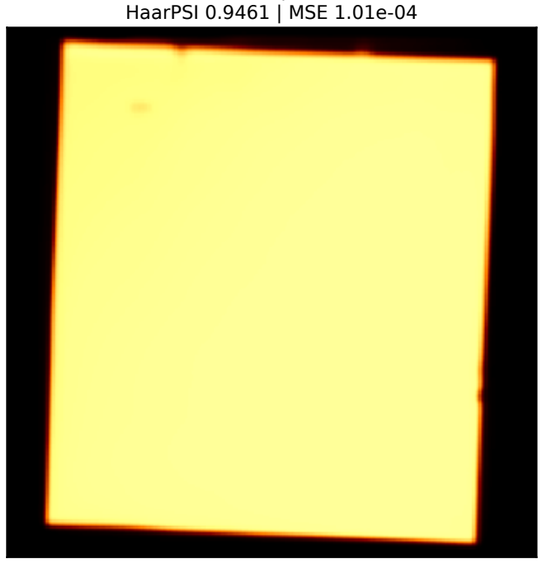
UTV-C60
PSNR 36.29 dB | SSIM 0.9440
HaarPSI 0.8764 | MSE 2.35e-04



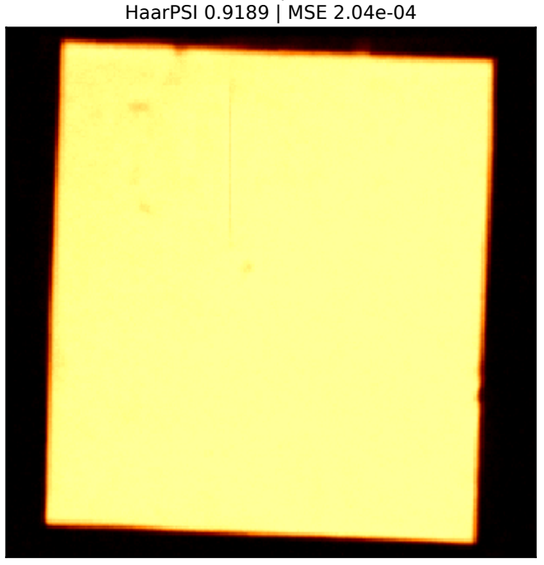
GS-PnP tuned-5
PSNR 35.70 dB | SSIM 0.8582
HaarPSI 0.8901 | MSE 2.69e-04



PnP-ADMM Tune-1 (0768)
PSNR 39.94 dB | SSIM 0.9801
HaarPSI 0.9461 | MSE 1.01e-04

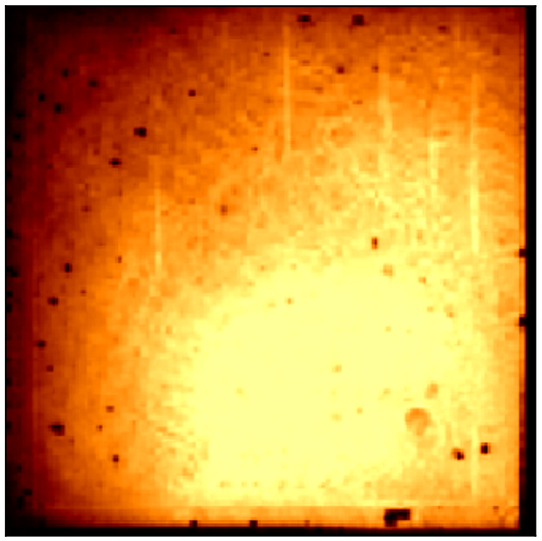


GS-PnP Tune-1 (0158)
PSNR 36.91 dB | SSIM 0.8897
HaarPSI 0.9189 | MSE 2.04e-04

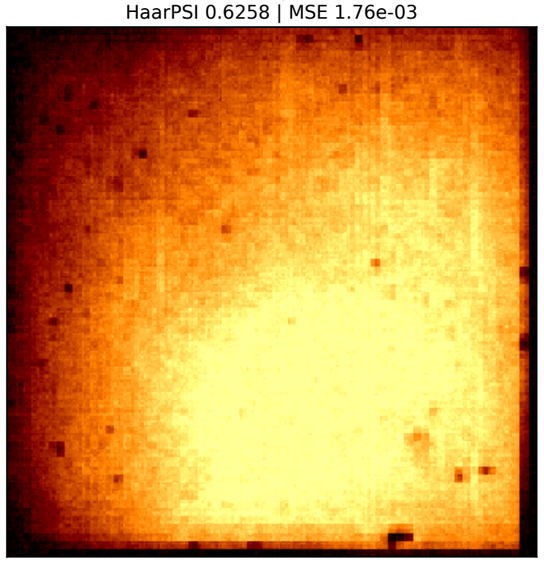


Perovskite | test image 0021 | 20% Hadamard + NPL-default noise

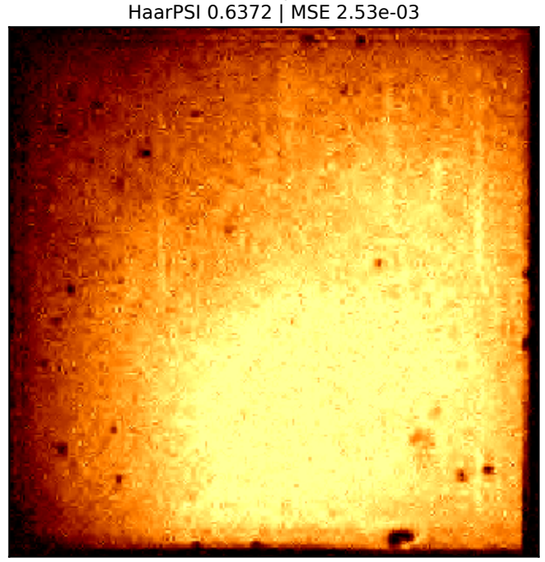
Target
Ground truth



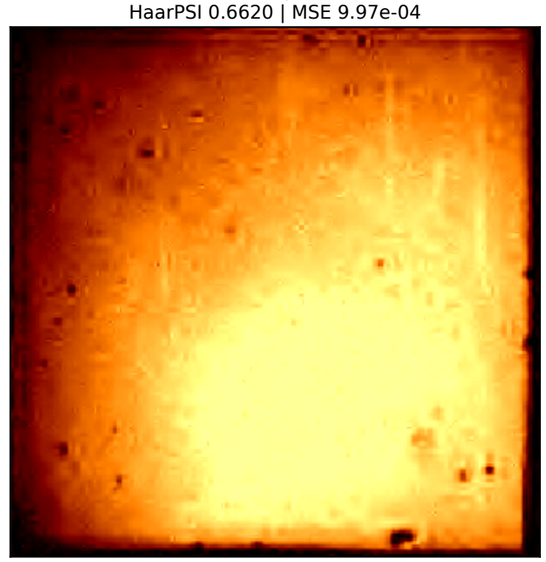
Zero-filled
PSNR 27.54 dB | SSIM 0.6188
HaarPSI 0.6258 | MSE 1.76e-03



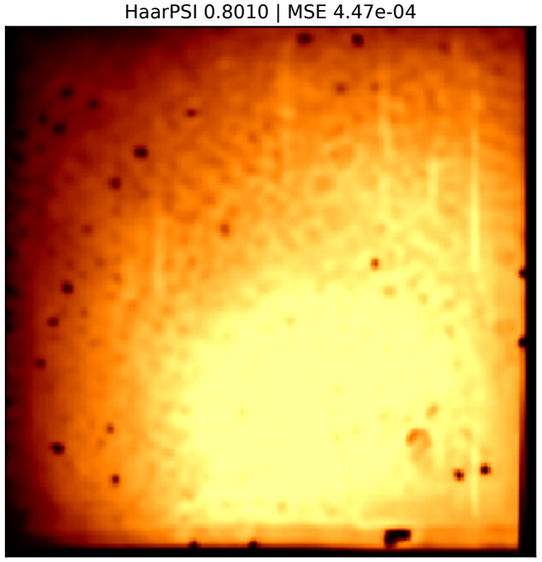
SPGL1-BP
PSNR 25.96 dB | SSIM 0.5073
HaarPSI 0.6372 | MSE 2.53e-03



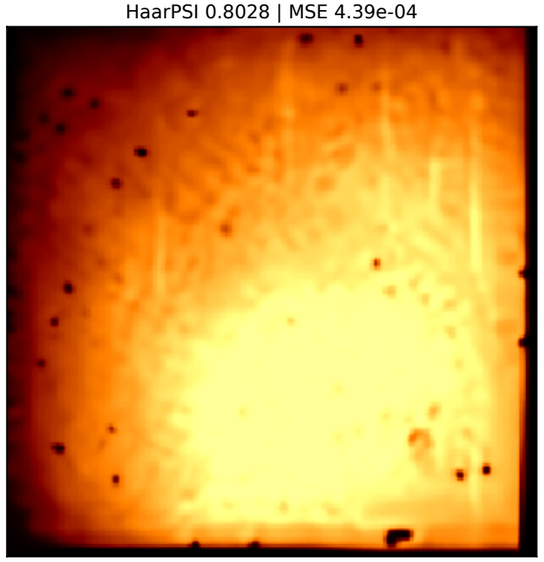
SPGL1-BPDN
PSNR 30.01 dB | SSIM 0.7672
HaarPSI 0.6620 | MSE 9.97e-04



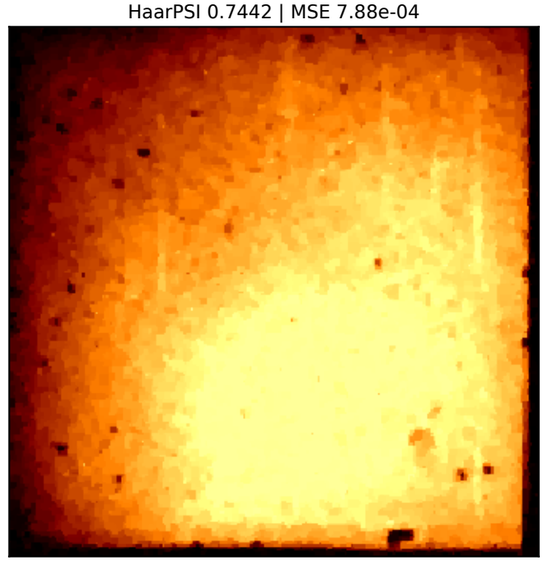
PnP-ADMM tuned-5
PSNR 33.50 dB | SSIM 0.8662
HaarPSI 0.8010 | MSE 4.47e-04



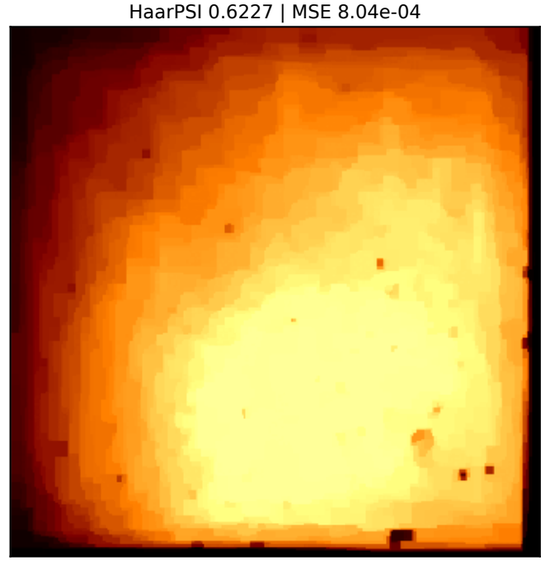
GS-PnP oracle
PSNR 33.58 dB | SSIM 0.8652
HaarPSI 0.8028 | MSE 4.39e-04



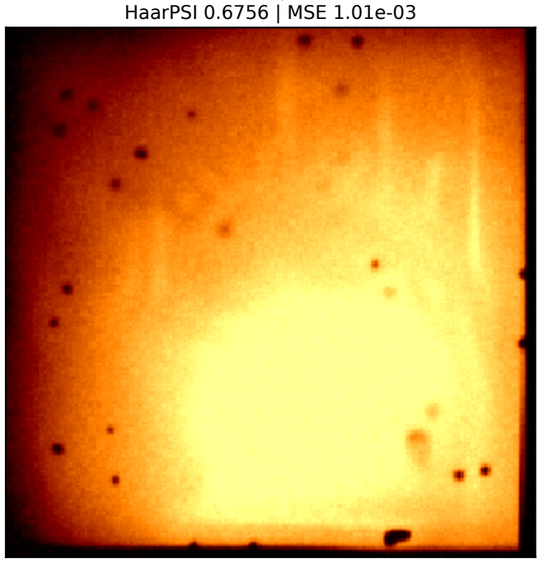
Scalar TV
PSNR 31.04 dB | SSIM 0.7853
HaarPSI 0.7442 | MSE 7.88e-04



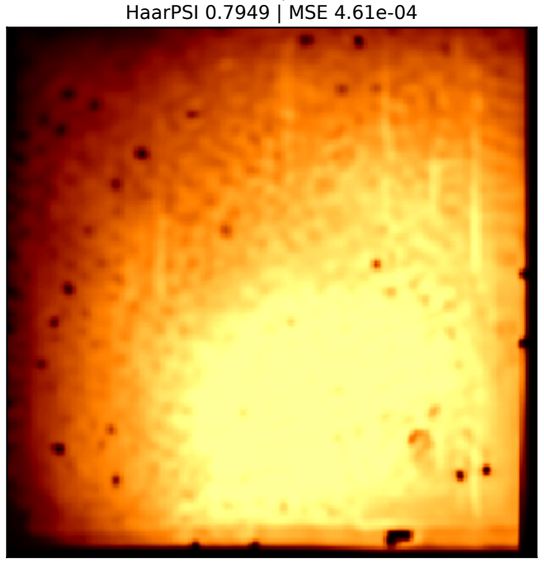
UTV-C60
PSNR 30.95 dB | SSIM 0.7955
HaarPSI 0.6227 | MSE 8.04e-04



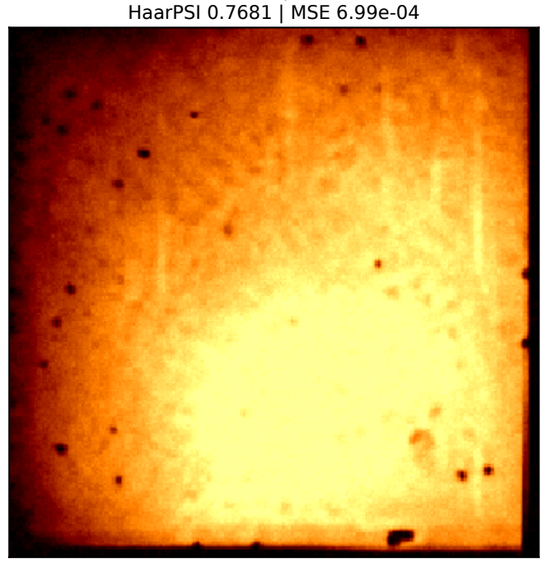
GS-PnP tuned-5
PSNR 29.97 dB | SSIM 0.6677
HaarPSI 0.6756 | MSE 1.01e-03



PnP-ADMM Tune-1 (0008)
PSNR 33.36 dB | SSIM 0.8633
HaarPSI 0.7949 | MSE 4.61e-04

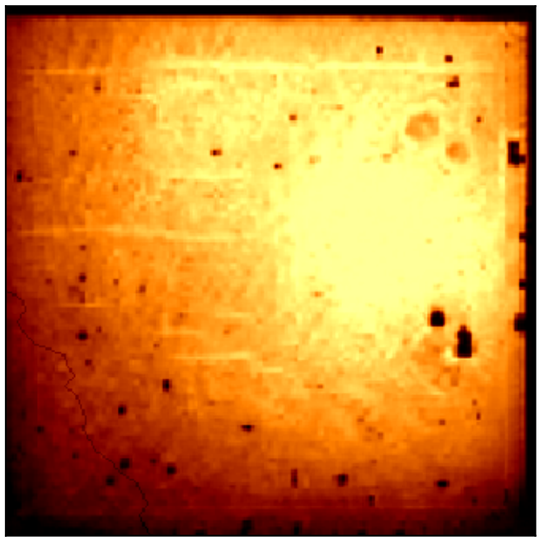


GS-PnP Tune-1 (0768)
PSNR 31.55 dB | SSIM 0.7667
HaarPSI 0.7681 | MSE 6.99e-04

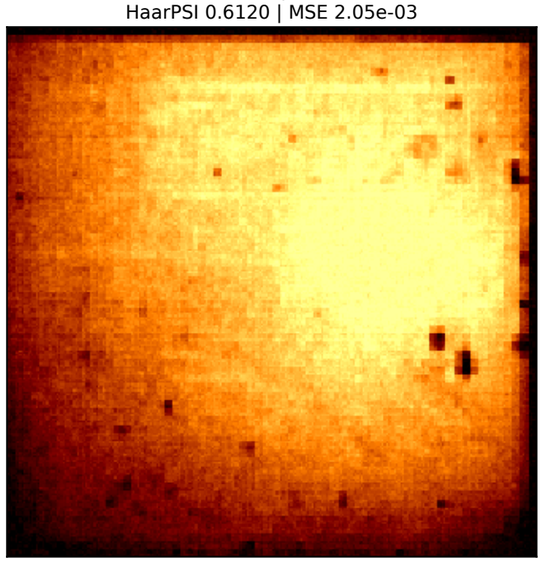


Perovskite | test image 0187 | 20% Hadamard + NPL-default noise

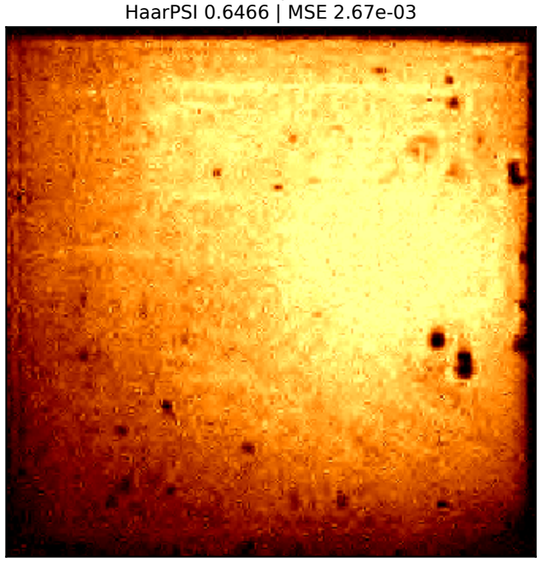
Target
Ground truth



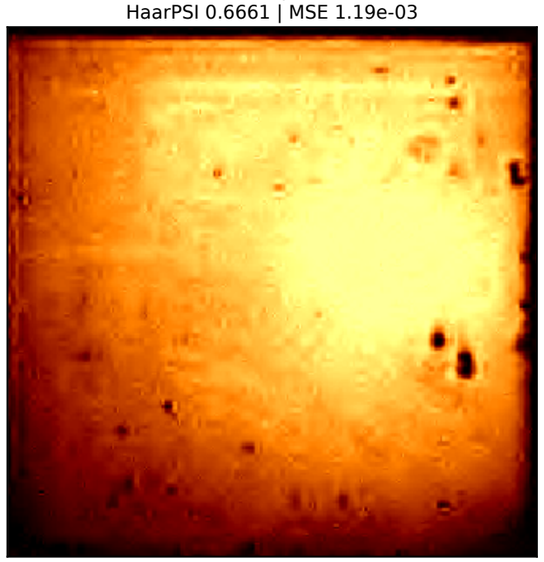
Zero-filled
PSNR 26.88 dB | SSIM 0.6148
HaarPSI 0.6120 | MSE 2.05e-03



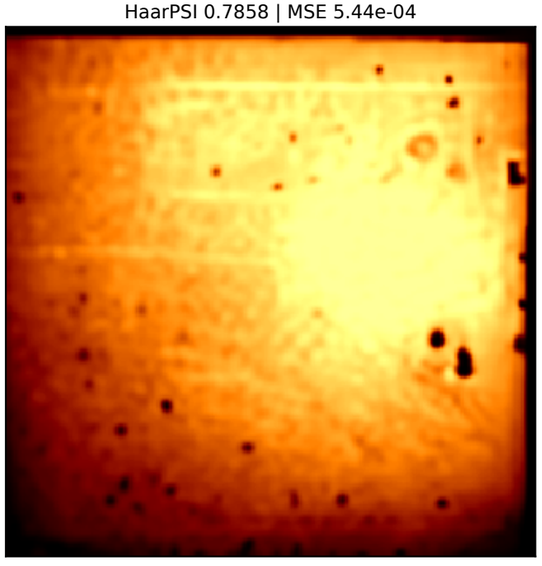
SPGL1-BP
PSNR 25.73 dB | SSIM 0.5154
HaarPSI 0.6466 | MSE 2.67e-03



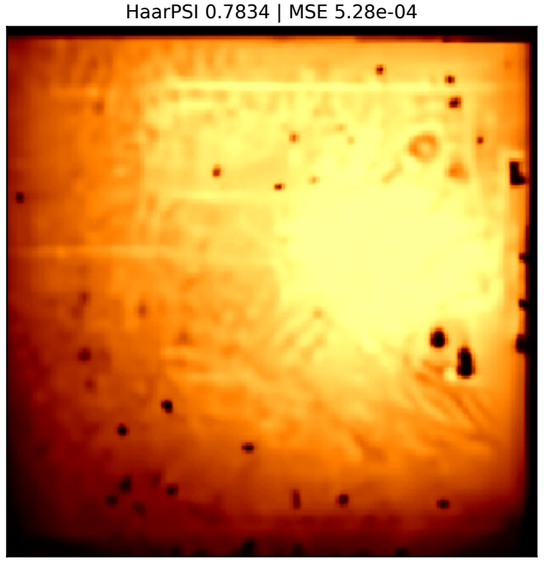
SPGL1-BPDN
PSNR 29.25 dB | SSIM 0.7450
HaarPSI 0.6661 | MSE 1.19e-03



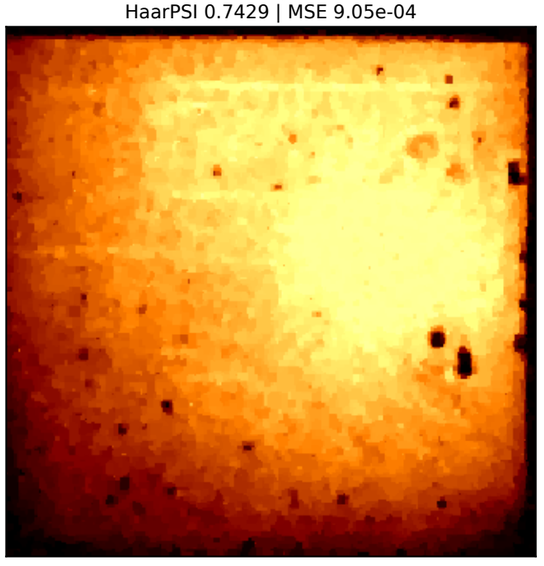
PnP-ADMM tuned-5
PSNR 32.65 dB | SSIM 0.8504
HaarPSI 0.7858 | MSE 5.44e-04



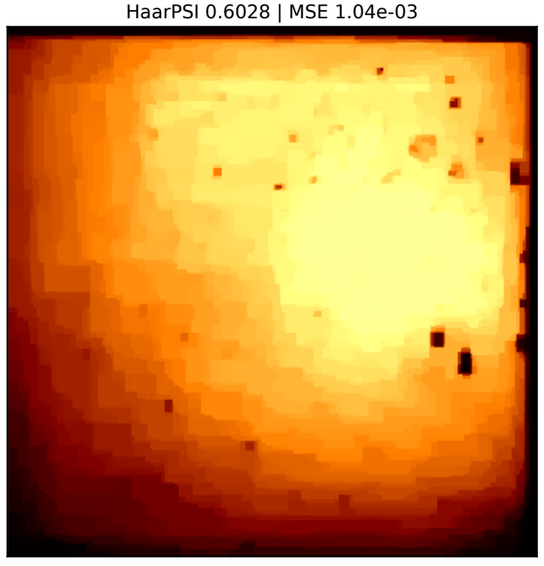
GS-PnP oracle
PSNR 32.78 dB | SSIM 0.8481
HaarPSI 0.7834 | MSE 5.28e-04



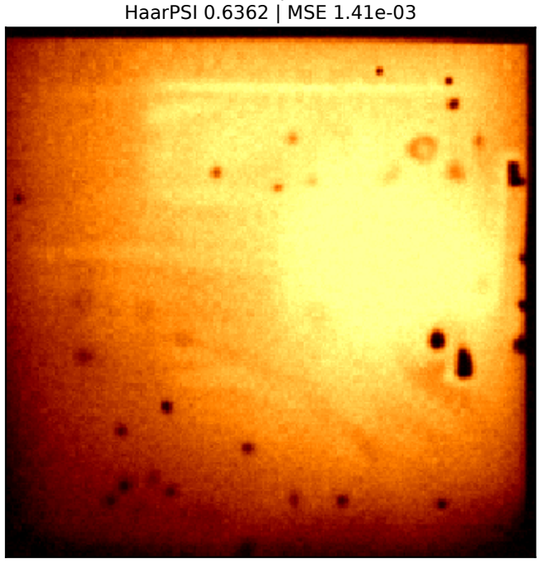
Scalar TV
PSNR 30.43 dB | SSIM 0.7758
HaarPSI 0.7429 | MSE 9.05e-04



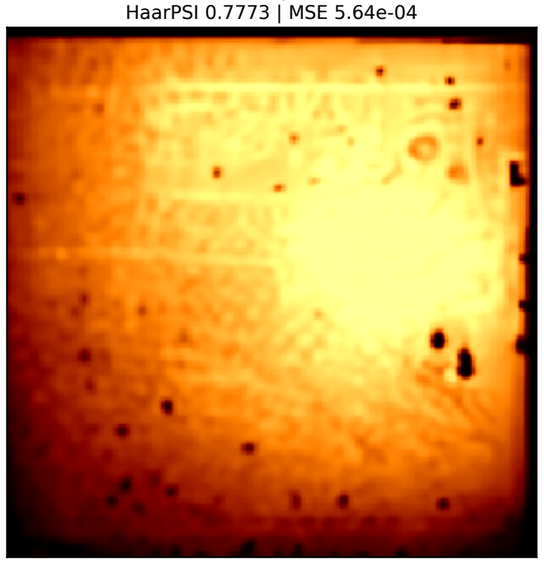
UTV-C60
PSNR 29.82 dB | SSIM 0.7657
HaarPSI 0.6028 | MSE 1.04e-03



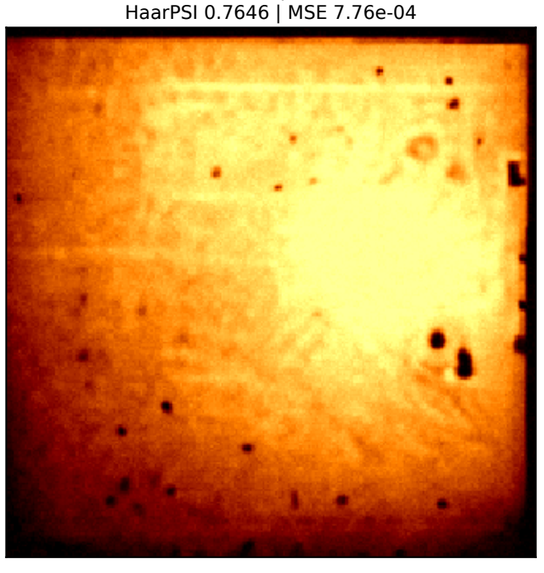
GS-PnP tuned-5
PSNR 28.50 dB | SSIM 0.5838
HaarPSI 0.6362 | MSE 1.41e-03



PnP-ADMM Tune-1 (0008)
PSNR 32.48 dB | SSIM 0.8464
HaarPSI 0.7773 | MSE 5.64e-04

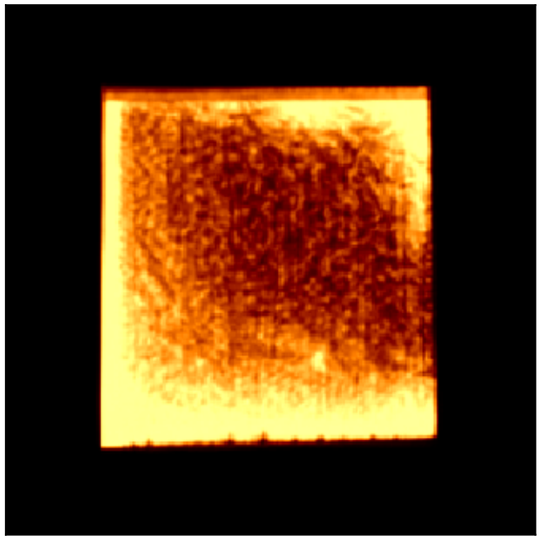


GS-PnP Tune-1 (0768)
PSNR 31.10 dB | SSIM 0.7635
HaarPSI 0.7646 | MSE 7.76e-04

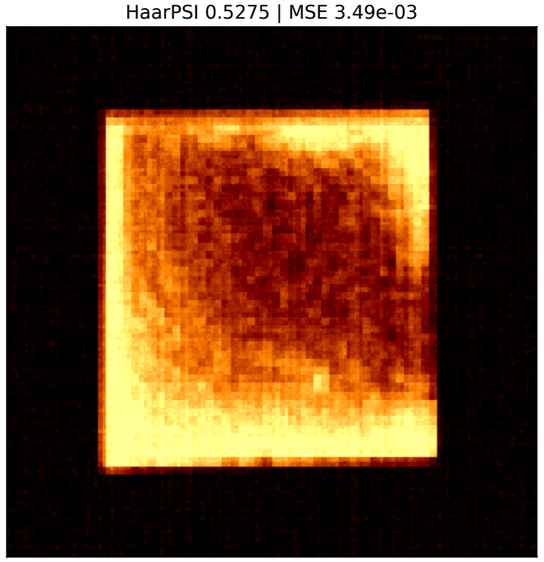


Perovskite | test image 0292 | 20% Hadamard + NPL-default noise

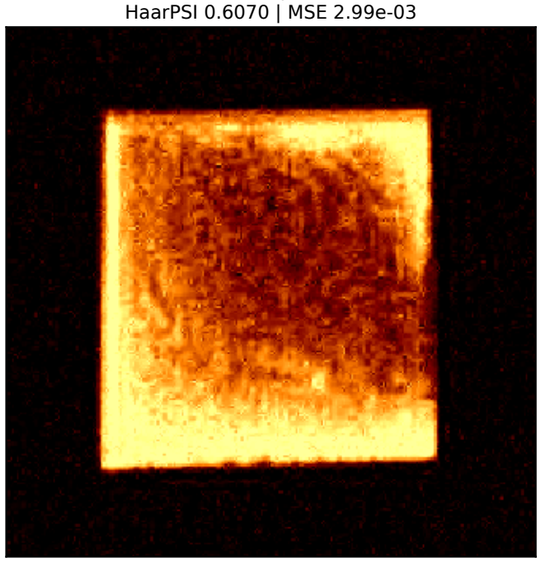
Target
Ground truth



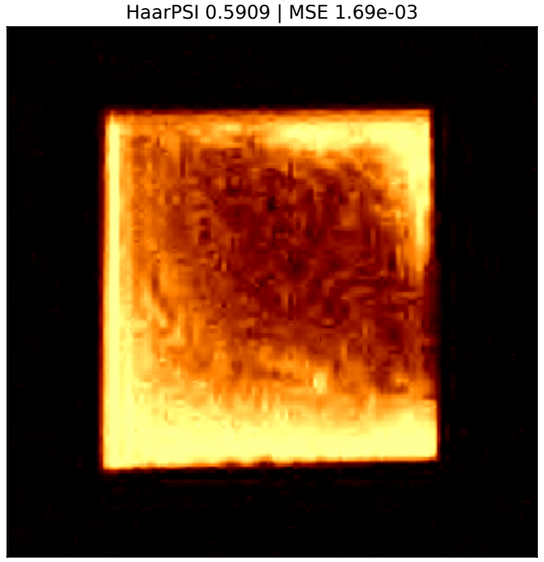
Zero-filled
PSNR 24.57 dB | SSIM 0.4757
HaarPSI 0.5275 | MSE 3.49e-03



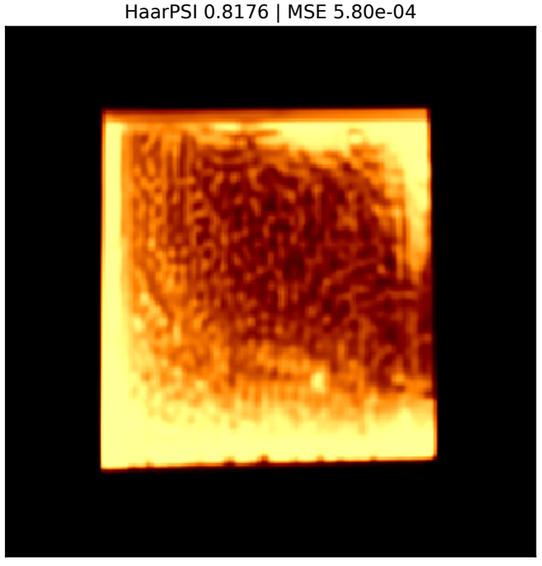
SPGL1-BP
PSNR 25.24 dB | SSIM 0.4540
HaarPSI 0.6070 | MSE 2.99e-03



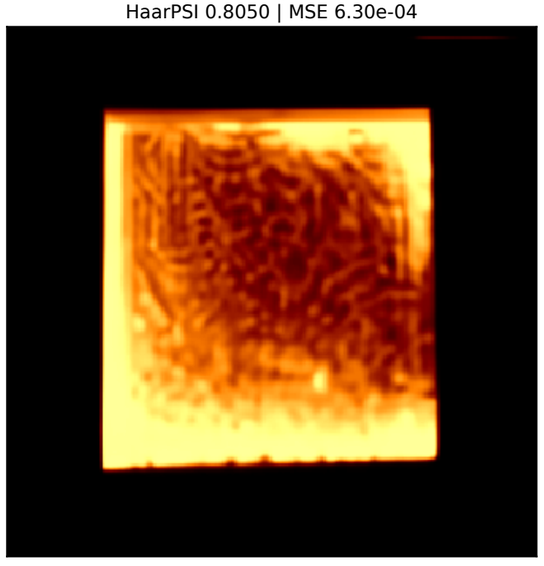
SPGL1-BPDN
PSNR 27.71 dB | SSIM 0.7021
HaarPSI 0.5909 | MSE 1.69e-03



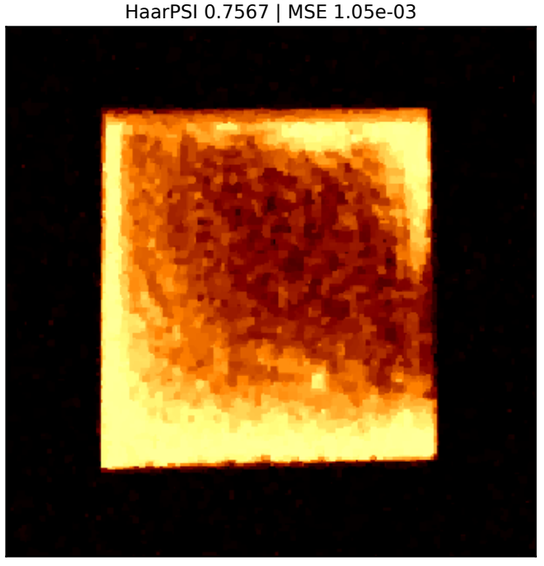
PnP-ADMM tuned-5
PSNR 32.37 dB | SSIM 0.9269
HaarPSI 0.8176 | MSE 5.80e-04



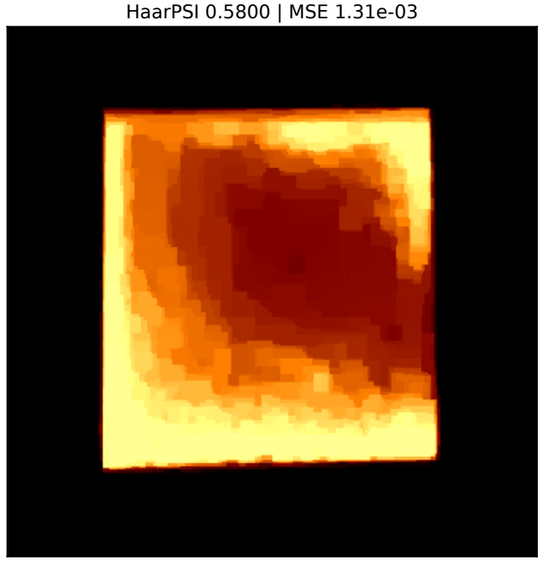
GS-PnP oracle
PSNR 32.01 dB | SSIM 0.9037
HaarPSI 0.8050 | MSE 6.30e-04



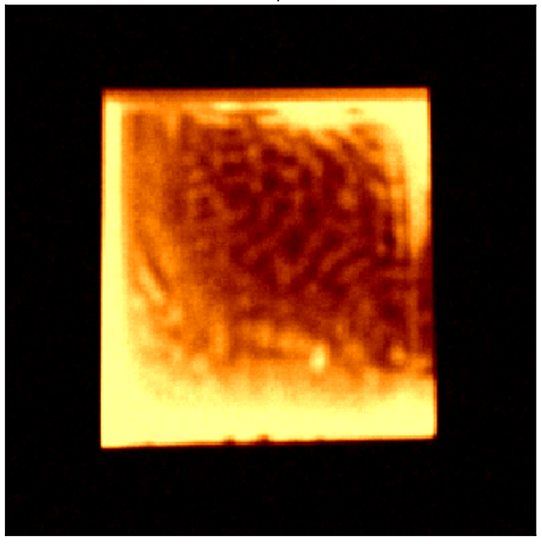
Scalar TV
PSNR 29.79 dB | SSIM 0.7172
HaarPSI 0.7567 | MSE 1.05e-03



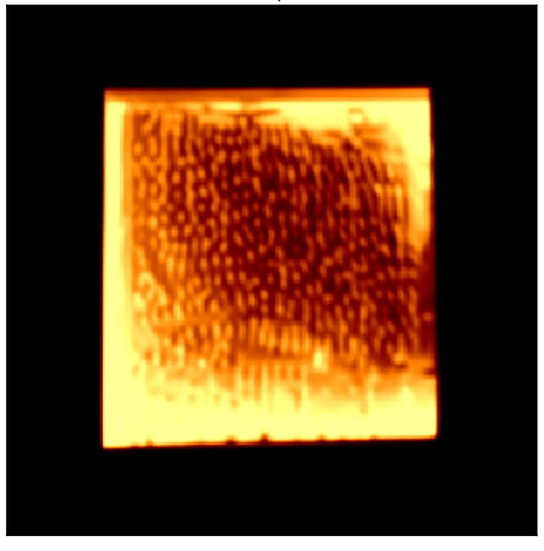
UTV-C60
PSNR 28.81 dB | SSIM 0.8093
HaarPSI 0.5800 | MSE 1.31e-03



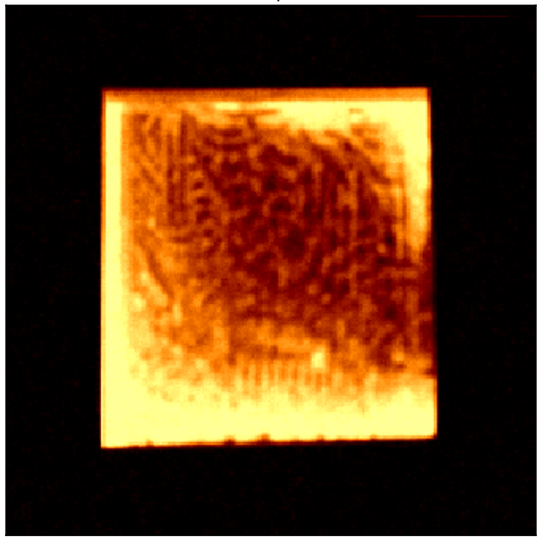
GS-PnP tuned-5
PSNR 29.20 dB | SSIM 0.6882
HaarPSI 0.7251 | MSE 1.20e-03



PnP-ADMM Tune-1 (0008)
PSNR 30.52 dB | SSIM 0.9071
HaarPSI 0.7860 | MSE 8.87e-04

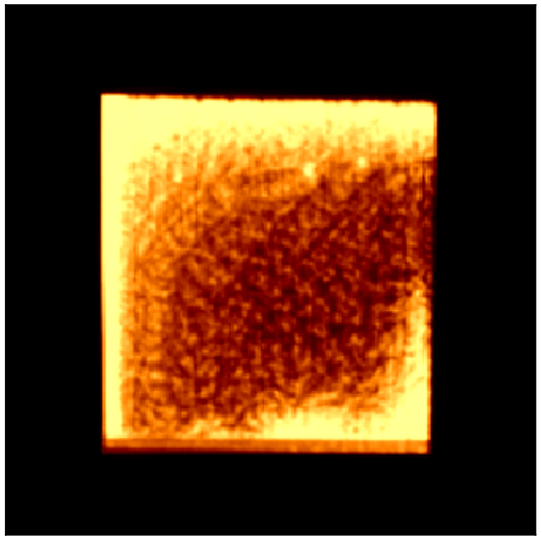


GS-PnP Tune-1 (0768)
PSNR 30.55 dB | SSIM 0.7126
HaarPSI 0.7958 | MSE 8.81e-04

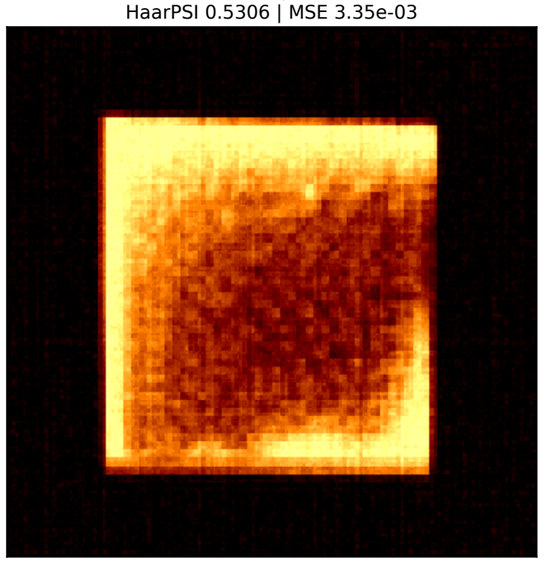


Perovskite | test image 0301 | 20% Hadamard + NPL-default noise

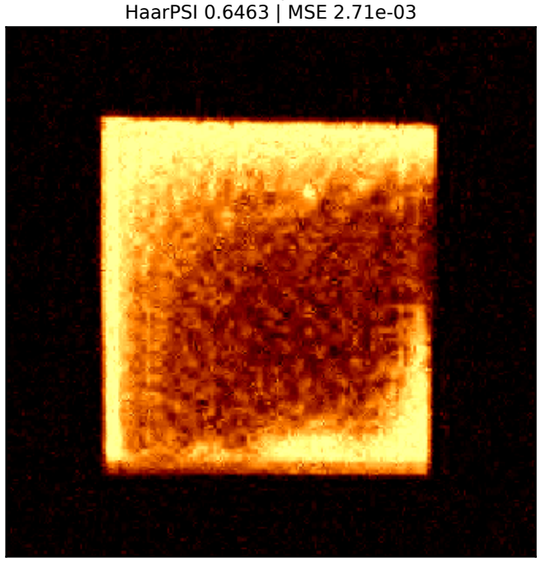
Target
Ground truth



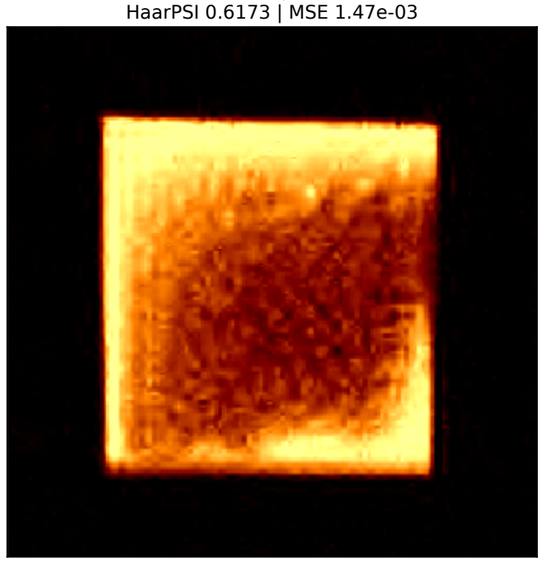
Zero-filled
PSNR 24.75 dB | SSIM 0.4699
HaarPSI 0.5306 | MSE 3.35e-03



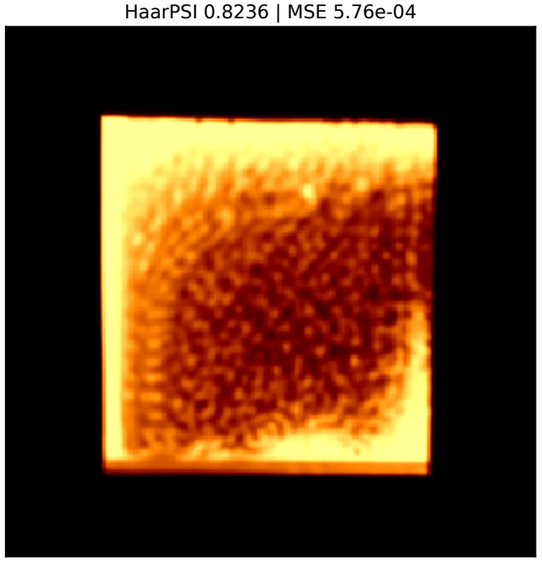
SPGL1-BP
PSNR 25.67 dB | SSIM 0.4620
HaarPSI 0.6463 | MSE 2.71e-03



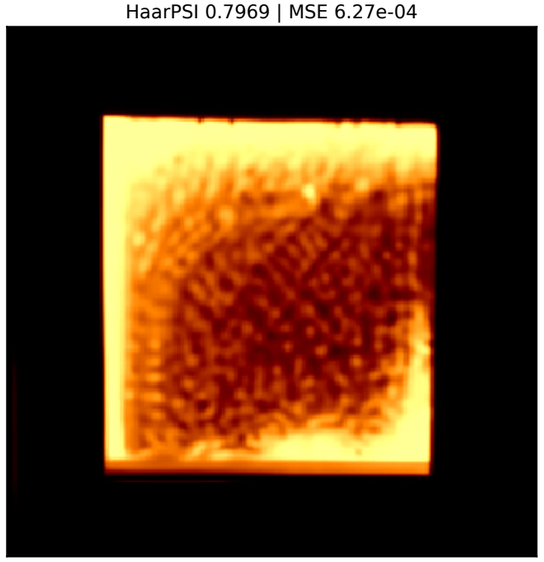
SPGL1-BPDN
PSNR 28.34 dB | SSIM 0.7152
HaarPSI 0.6173 | MSE 1.47e-03



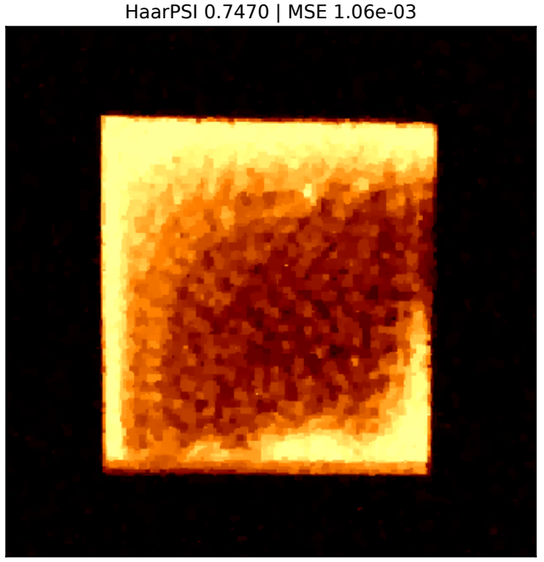
PnP-ADMM tuned-5
PSNR 32.39 dB | SSIM 0.9269
HaarPSI 0.8236 | MSE 5.76e-04



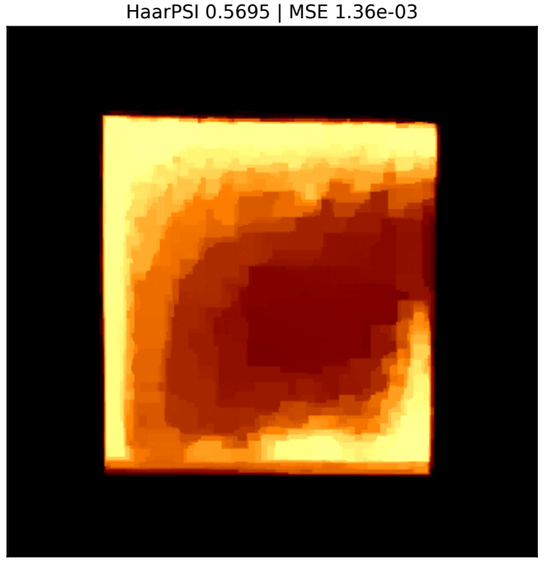
GS-PnP oracle
PSNR 32.03 dB | SSIM 0.9052
HaarPSI 0.7969 | MSE 6.27e-04



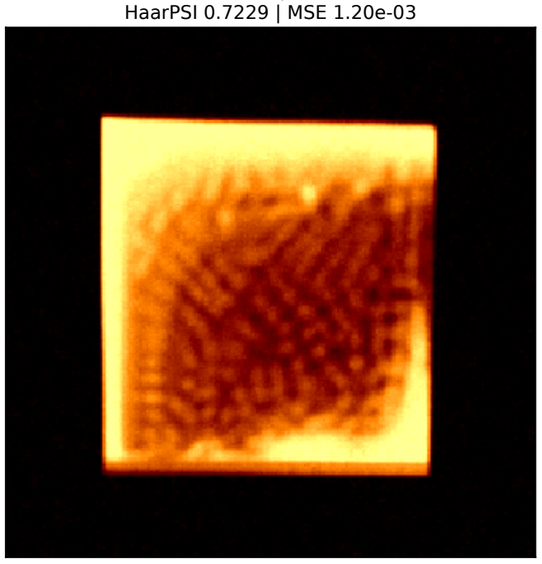
Scalar TV
PSNR 29.76 dB | SSIM 0.7178
HaarPSI 0.7470 | MSE 1.06e-03



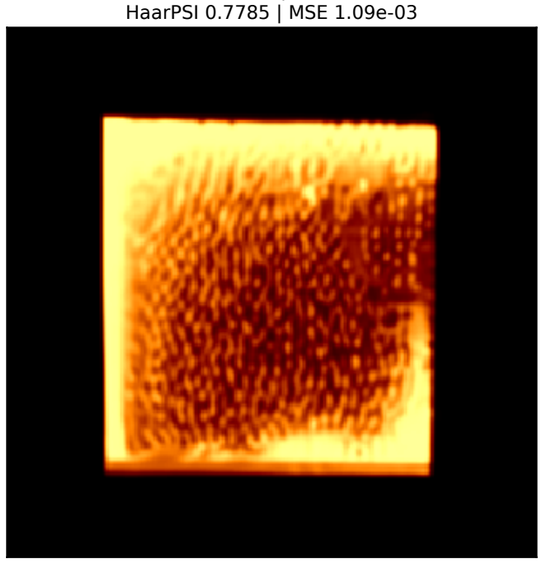
UTV-C60
PSNR 28.65 dB | SSIM 0.8029
HaarPSI 0.5695 | MSE 1.36e-03



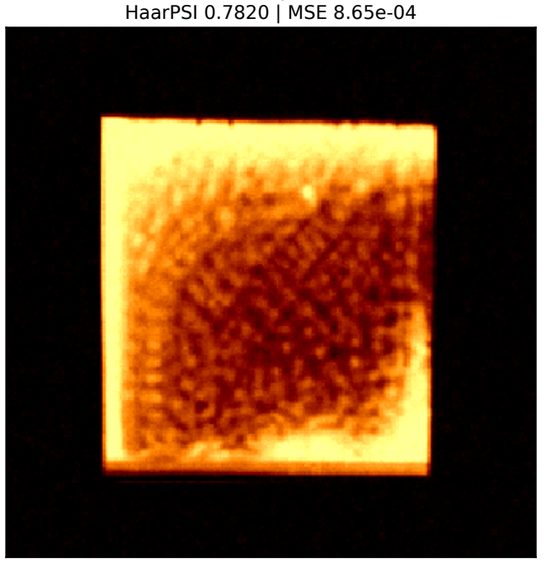
GS-PnP tuned-5
PSNR 29.21 dB | SSIM 0.7010
HaarPSI 0.7229 | MSE 1.20e-03



PnP-ADMM Tune-1 (0008)
PSNR 29.63 dB | SSIM 0.8956
HaarPSI 0.7785 | MSE 1.09e-03

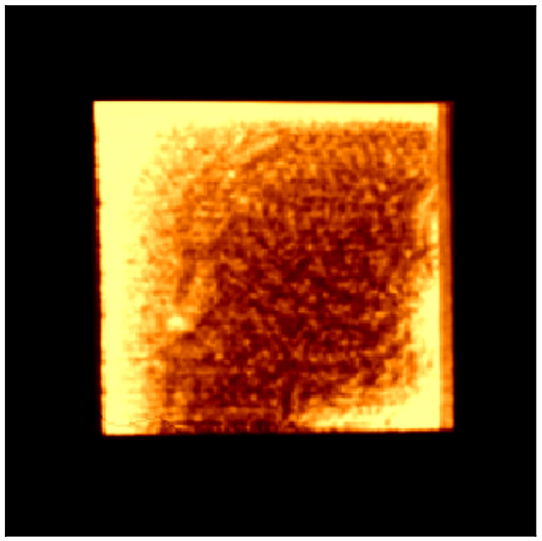


GS-PnP Tune-1 (0768)
PSNR 30.63 dB | SSIM 0.7159
HaarPSI 0.7820 | MSE 8.65e-04

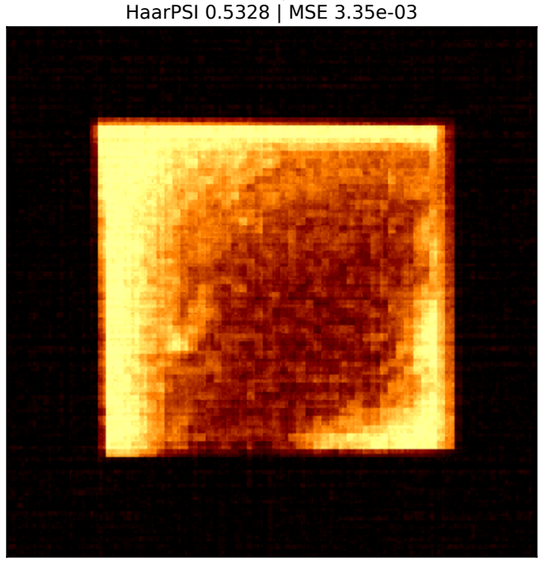


Perovskite | test image 0376 | 20% Hadamard + NPL-default noise

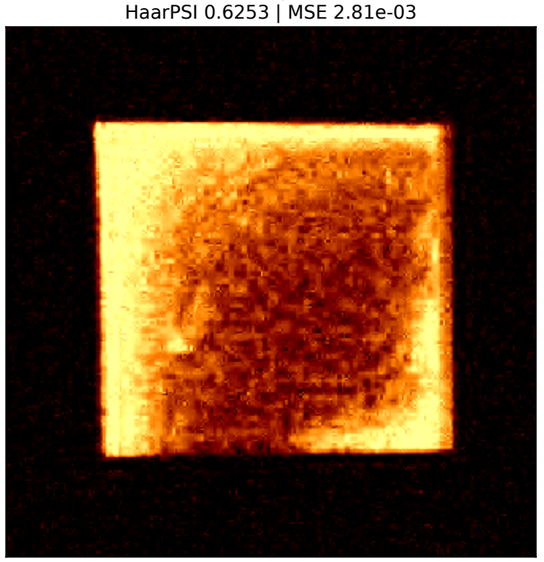
Target
Ground truth



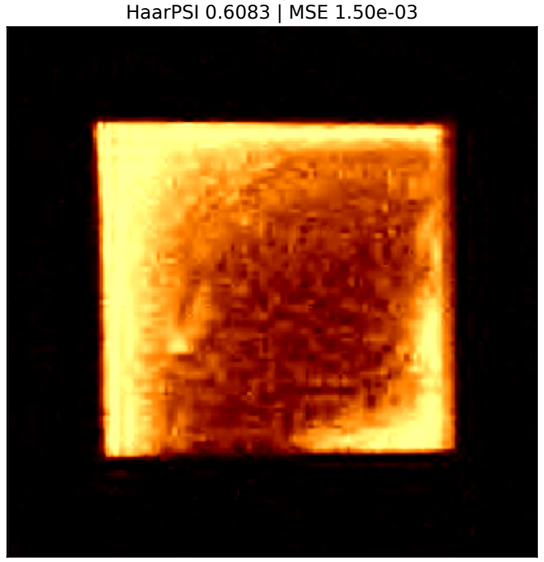
Zero-filled
PSNR 24.75 dB | SSIM 0.4601
HaarPSI 0.5328 | MSE 3.35e-03



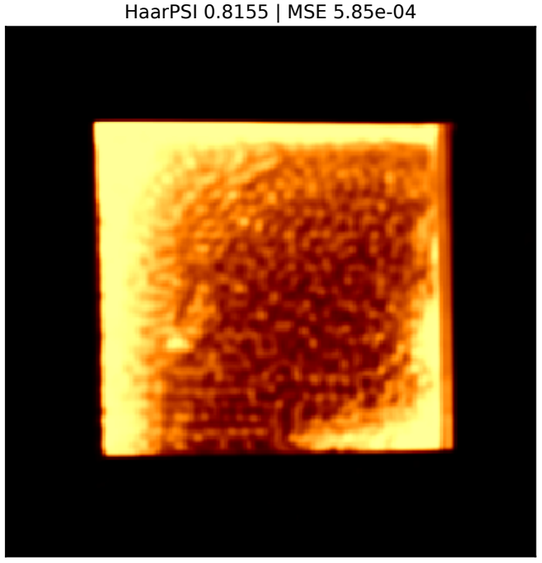
SPGL1-BP
PSNR 25.51 dB | SSIM 0.4590
HaarPSI 0.6253 | MSE 2.81e-03



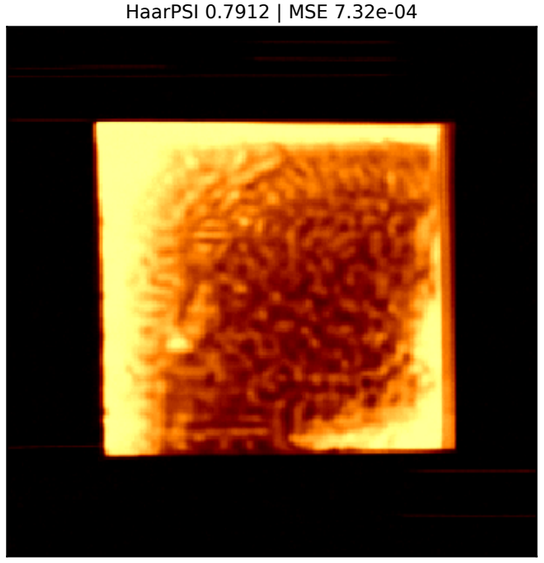
SPGL1-BPDN
PSNR 28.23 dB | SSIM 0.7043
HaarPSI 0.6083 | MSE 1.50e-03



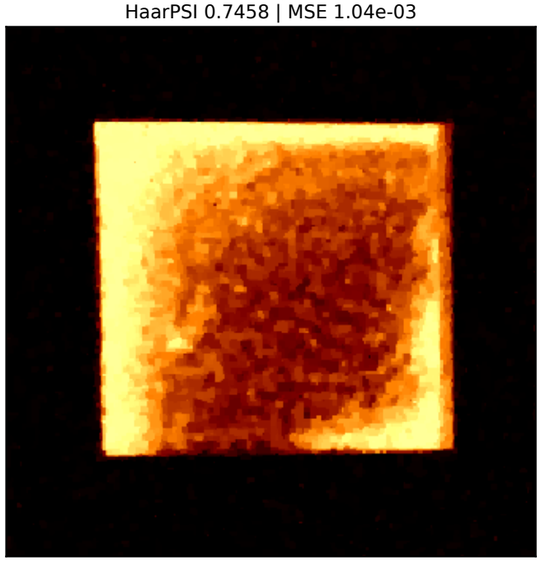
PnP-ADMM tuned-5
PSNR 32.33 dB | SSIM 0.9249
HaarPSI 0.8155 | MSE 5.85e-04



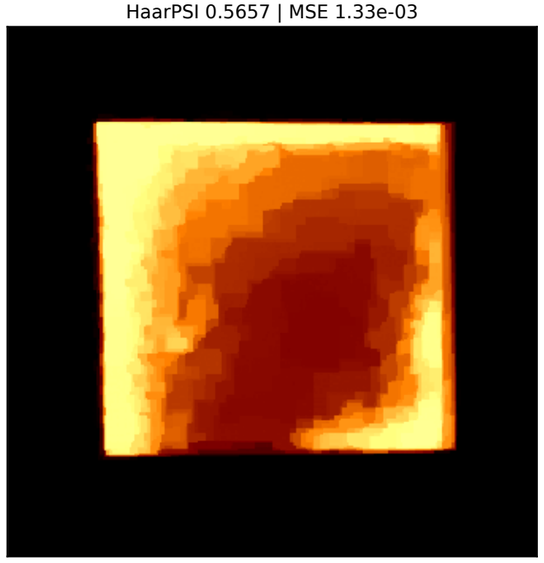
GS-PnP oracle
PSNR 31.36 dB | SSIM 0.8250
HaarPSI 0.7912 | MSE 7.32e-04



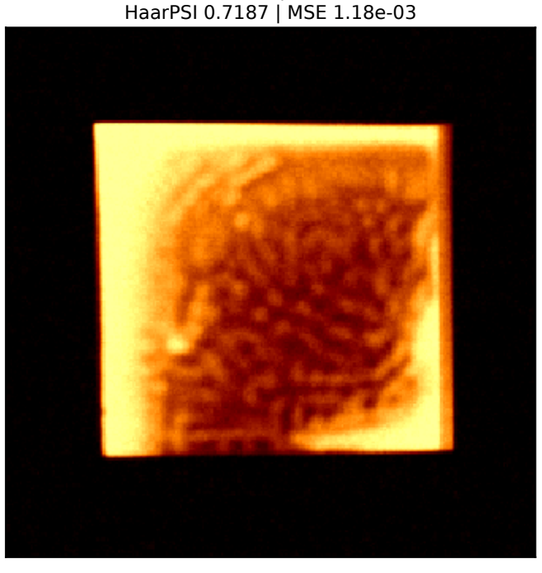
Scalar TV
PSNR 29.84 dB | SSIM 0.7245
HaarPSI 0.7458 | MSE 1.04e-03



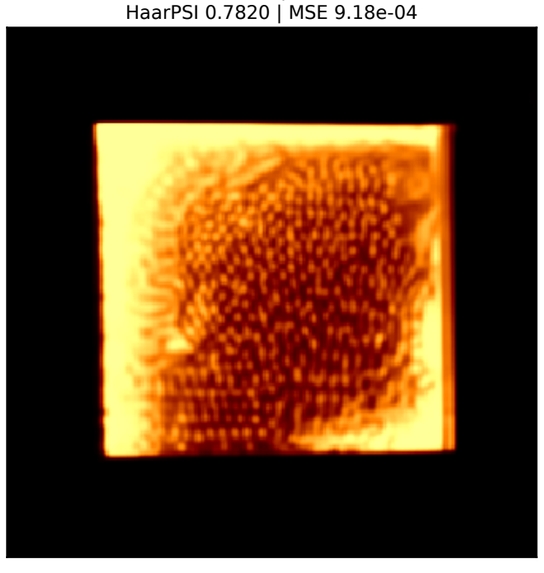
UTV-C60
PSNR 28.76 dB | SSIM 0.8037
HaarPSI 0.5657 | MSE 1.33e-03



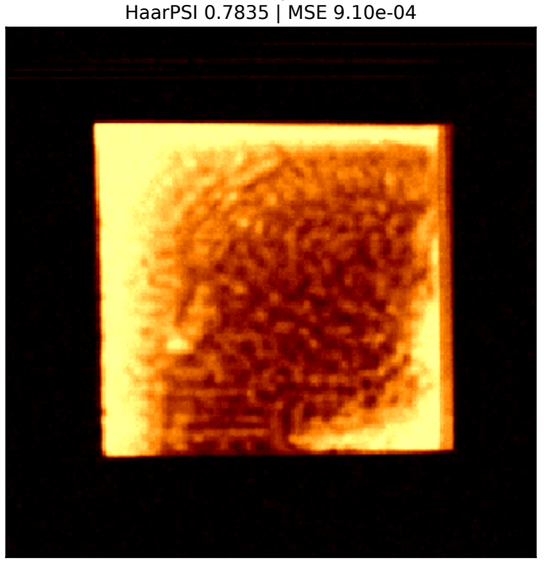
GS-PnP tuned-5
PSNR 29.27 dB | SSIM 0.6998
HaarPSI 0.7187 | MSE 1.18e-03



PnP-ADMM Tune-1 (0008)
PSNR 30.37 dB | SSIM 0.9057
HaarPSI 0.7820 | MSE 9.18e-04

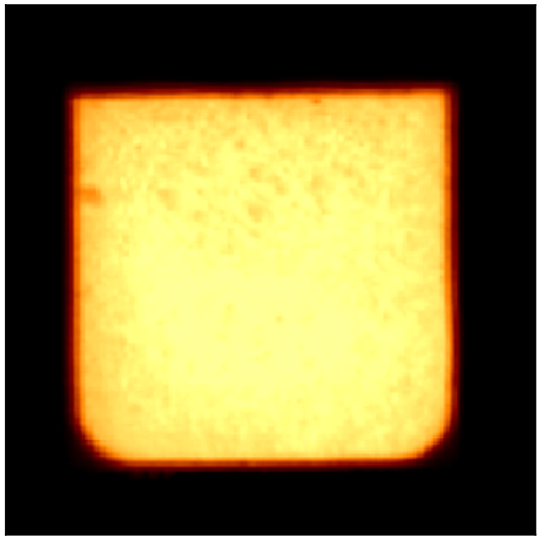


GS-PnP Tune-1 (0768)
PSNR 30.41 dB | SSIM 0.6968
HaarPSI 0.7835 | MSE 9.10e-04

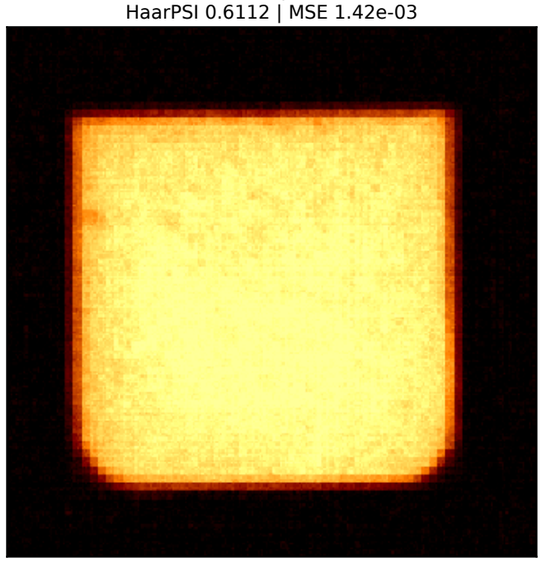


Perovskite | test image 0610 | 20% Hadamard + NPL-default noise

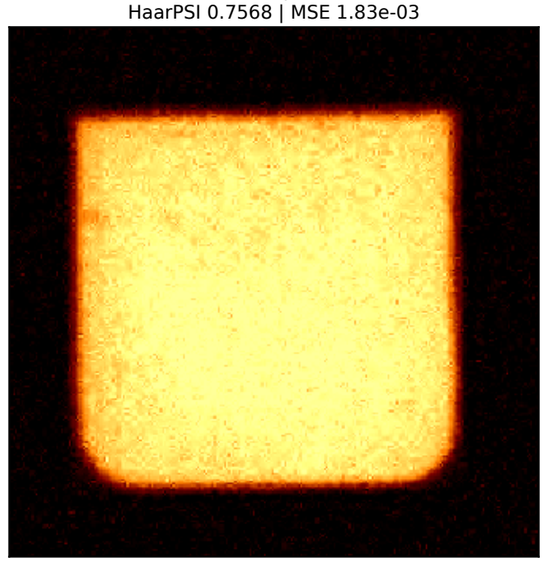
Target
Ground truth



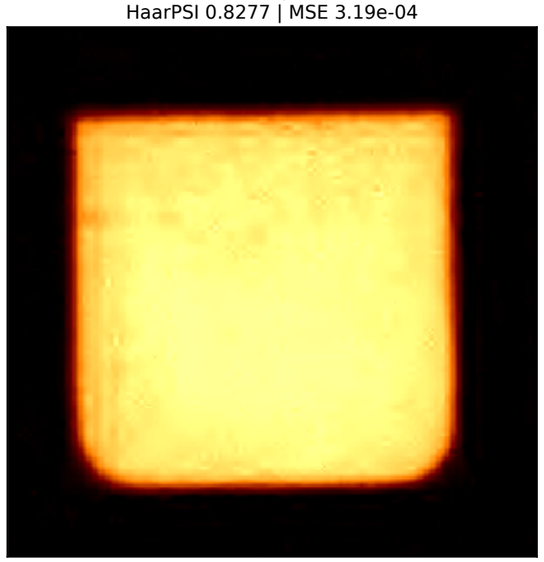
Zero-filled
PSNR 28.46 dB | SSIM 0.5458
HaarPSI 0.6112 | MSE 1.42e-03



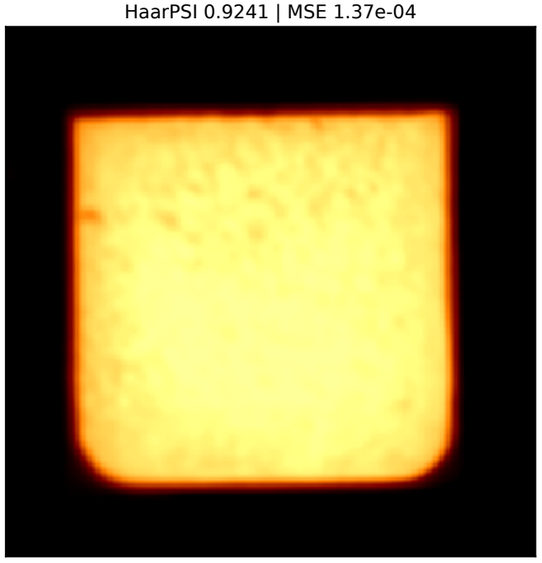
SPGL1-BP
PSNR 27.37 dB | SSIM 0.4479
HaarPSI 0.7568 | MSE 1.83e-03



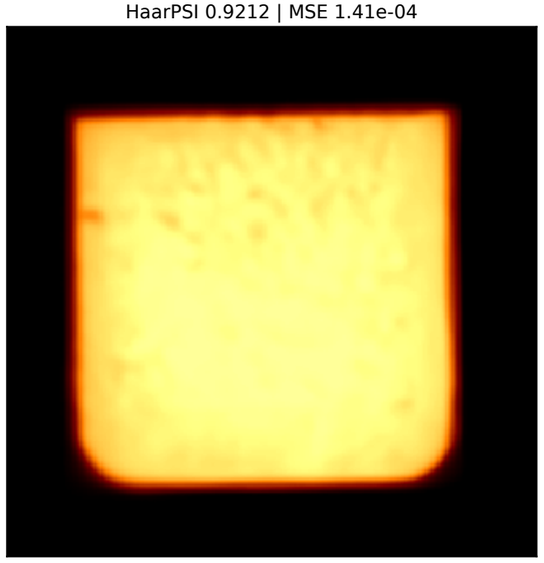
SPGL1-BPDN
PSNR 34.97 dB | SSIM 0.8246
HaarPSI 0.8277 | MSE 3.19e-04



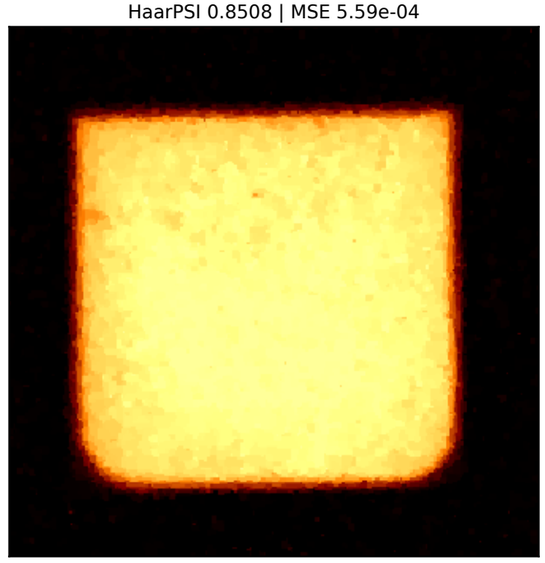
PnP-ADMM tuned-5
PSNR 38.63 dB | SSIM 0.9507
HaarPSI 0.9241 | MSE 1.37e-04



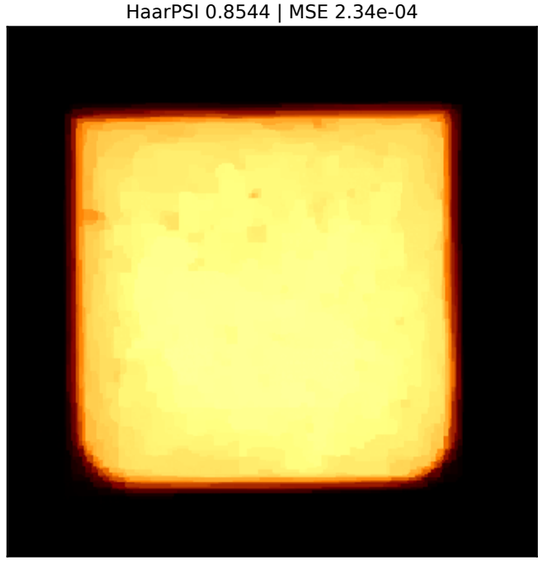
GS-PnP oracle
PSNR 38.52 dB | SSIM 0.9432
HaarPSI 0.9212 | MSE 1.41e-04



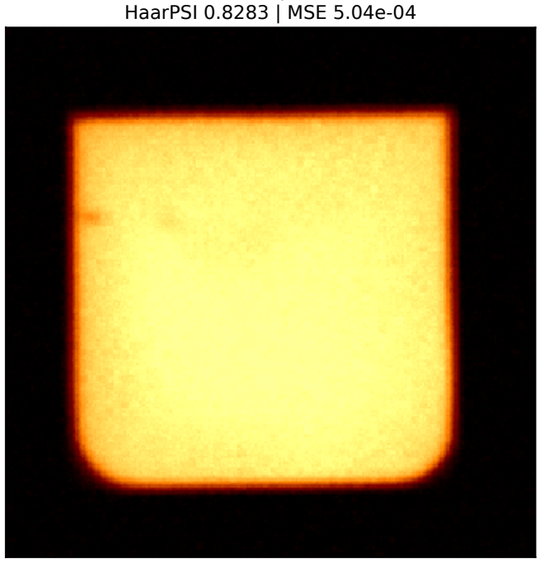
Scalar TV
PSNR 32.53 dB | SSIM 0.7593
HaarPSI 0.8508 | MSE 5.59e-04



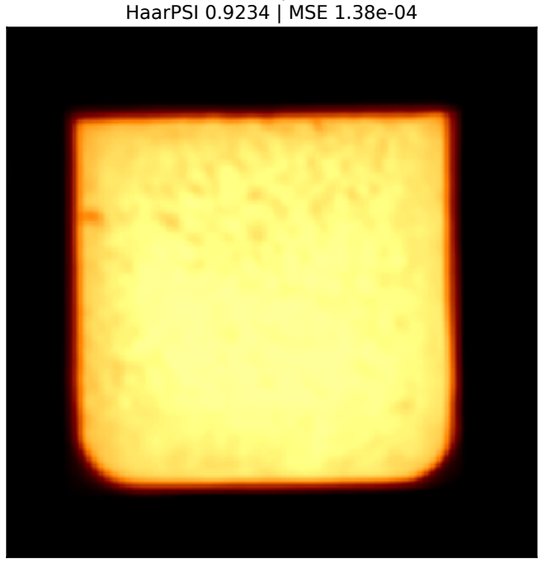
UTV-C60
PSNR 36.32 dB | SSIM 0.8986
HaarPSI 0.8544 | MSE 2.34e-04



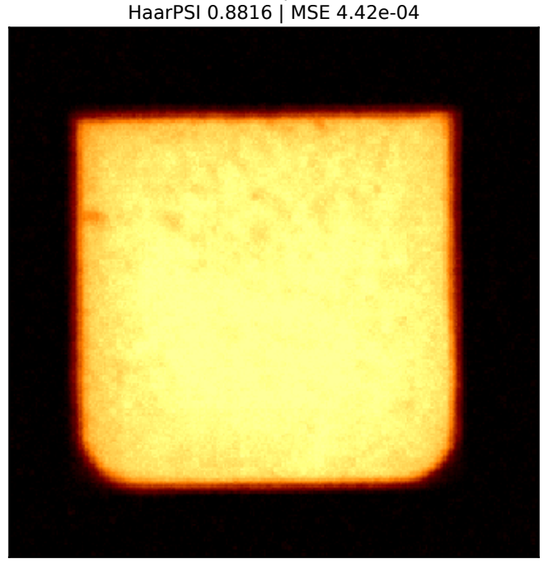
GS-PnP tuned-5
PSNR 32.98 dB | SSIM 0.7303
HaarPSI 0.8283 | MSE 5.04e-04



PnP-ADMM Tune-1 (0008)
PSNR 38.60 dB | SSIM 0.9508
HaarPSI 0.9234 | MSE 1.38e-04

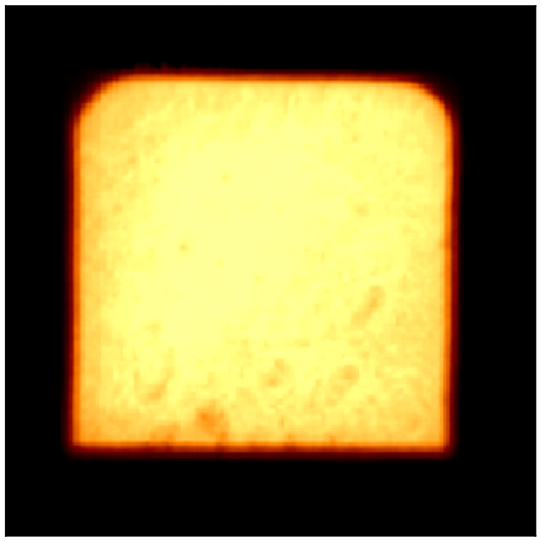


GS-PnP Tune-1 (0768)
PSNR 33.55 dB | SSIM 0.7281
HaarPSI 0.8816 | MSE 4.42e-04

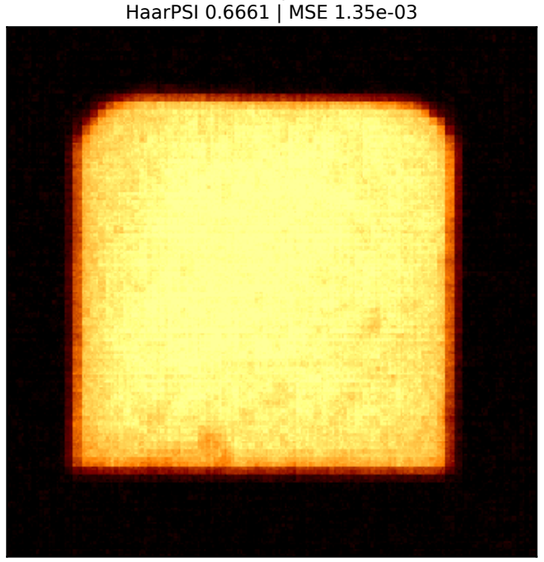


Perovskite | test image 0727 | 20% Hadamard + NPL-default noise

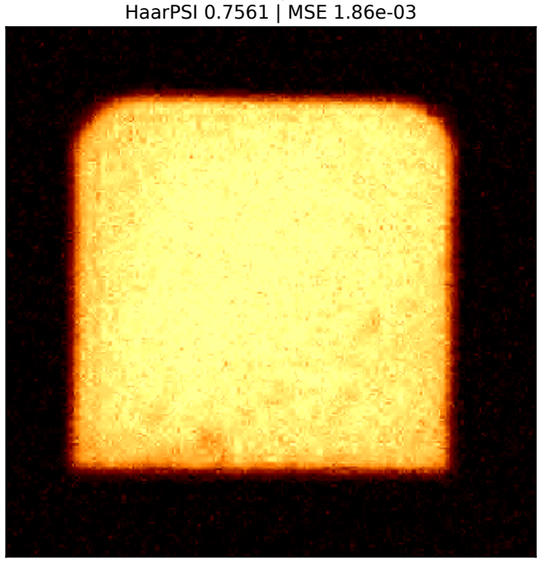
Target
Ground truth



Zero-filled
PSNR 28.71 dB | SSIM 0.5378
HaarPSI 0.6661 | MSE 1.35e-03



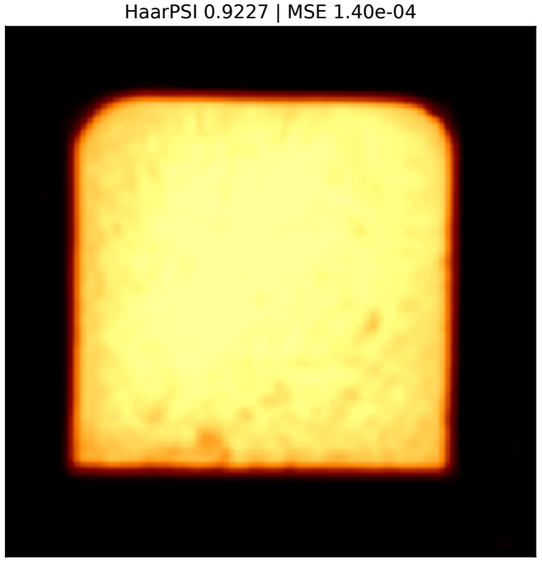
SPGL1-BP
PSNR 27.31 dB | SSIM 0.4448
HaarPSI 0.7561 | MSE 1.86e-03



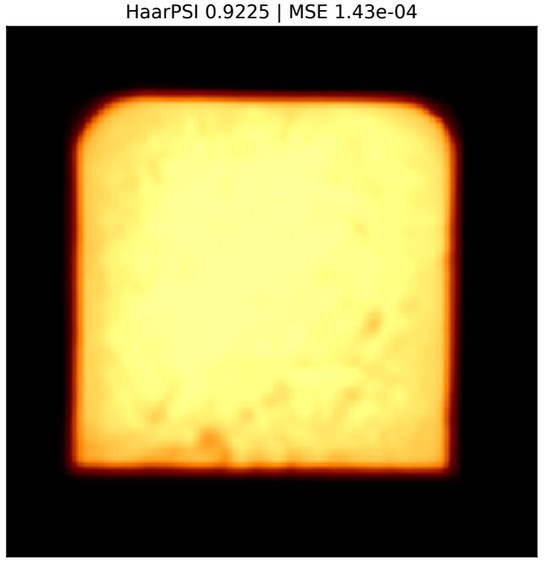
SPGL1-BPDN
PSNR 34.90 dB | SSIM 0.8233
HaarPSI 0.8305 | MSE 3.24e-04



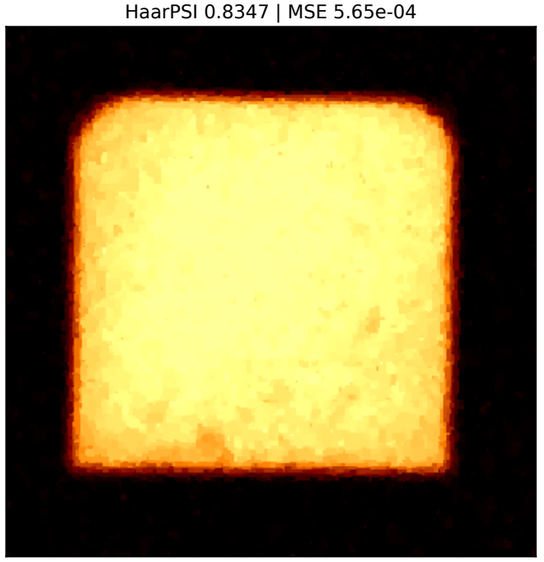
PnP-ADMM tuned-5
PSNR 38.53 dB | SSIM 0.9532
HaarPSI 0.9227 | MSE 1.40e-04



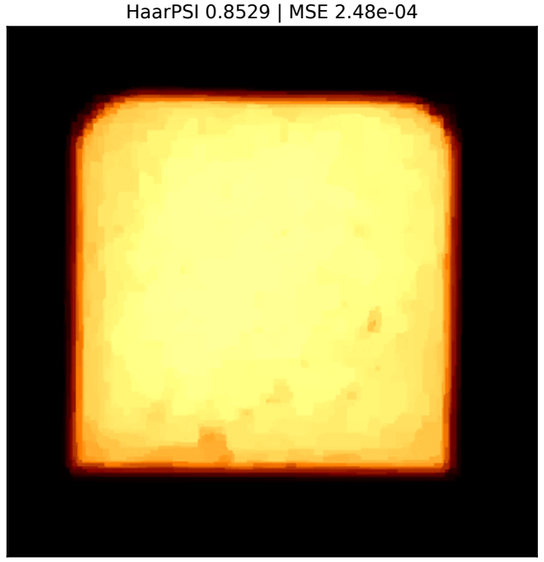
GS-PnP oracle
PSNR 38.46 dB | SSIM 0.9464
HaarPSI 0.9225 | MSE 1.43e-04



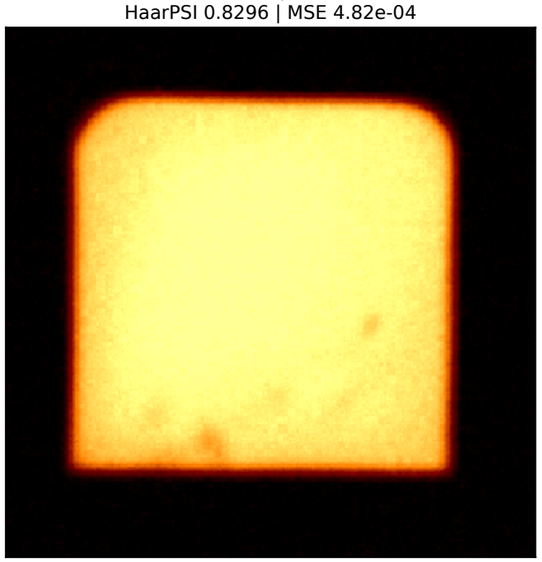
Scalar TV
PSNR 32.48 dB | SSIM 0.7543
HaarPSI 0.8347 | MSE 5.65e-04



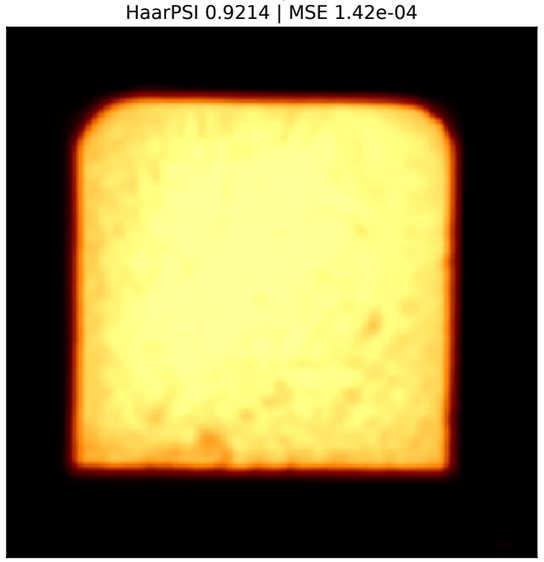
UTV-C60
PSNR 36.06 dB | SSIM 0.9024
HaarPSI 0.8529 | MSE 2.48e-04



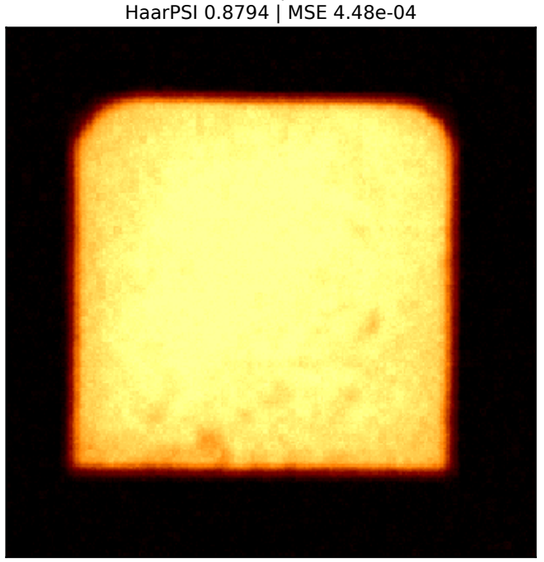
GS-PnP tuned-5
PSNR 33.17 dB | SSIM 0.7355
HaarPSI 0.8296 | MSE 4.82e-04



PnP-ADMM Tune-1 (0008)
PSNR 38.49 dB | SSIM 0.9537
HaarPSI 0.9214 | MSE 1.42e-04

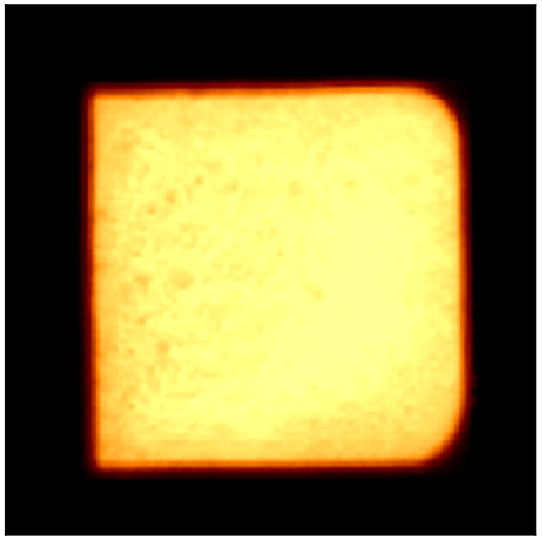


GS-PnP Tune-1 (0768)
PSNR 33.48 dB | SSIM 0.7250
HaarPSI 0.8794 | MSE 4.48e-04

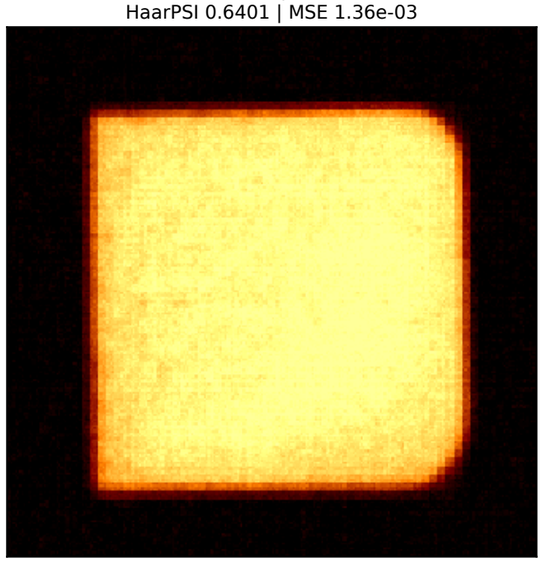


Perovskite | test image 0767 | 20% Hadamard + NPL-default noise

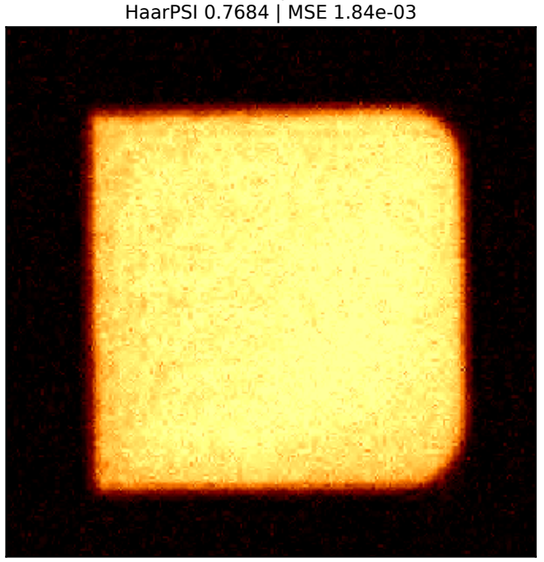
Target
Ground truth



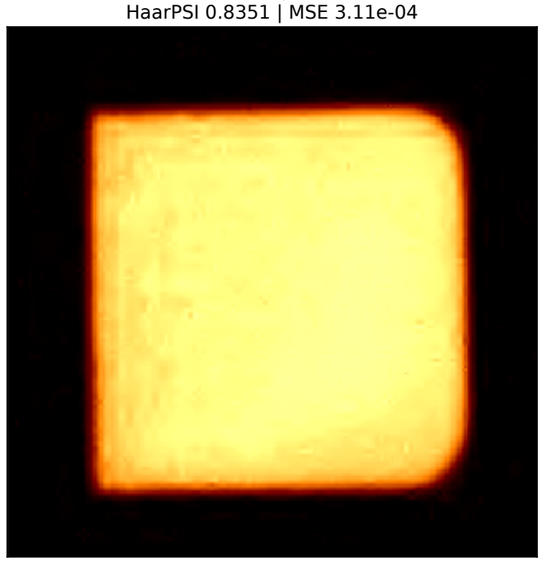
Zero-filled
PSNR 28.66 dB | SSIM 0.5443
HaarPSI 0.6401 | MSE 1.36e-03



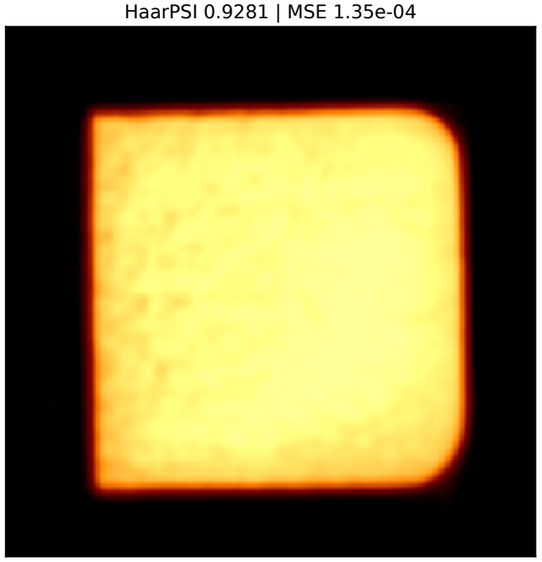
SPGL1-BP
PSNR 27.34 dB | SSIM 0.4448
HaarPSI 0.7684 | MSE 1.84e-03



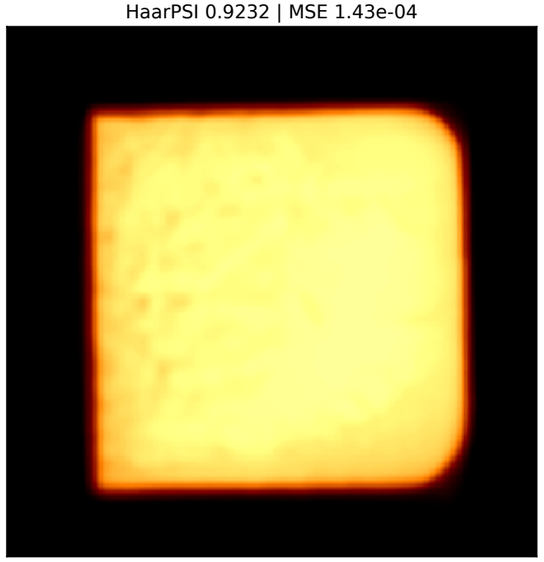
SPGL1-BPDN
PSNR 35.07 dB | SSIM 0.8204
HaarPSI 0.8351 | MSE 3.11e-04



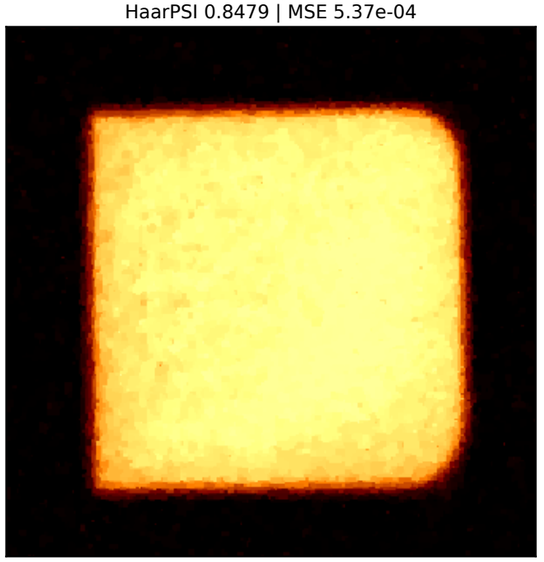
PnP-ADMM tuned-5
PSNR 38.69 dB | SSIM 0.9525
HaarPSI 0.9281 | MSE 1.35e-04



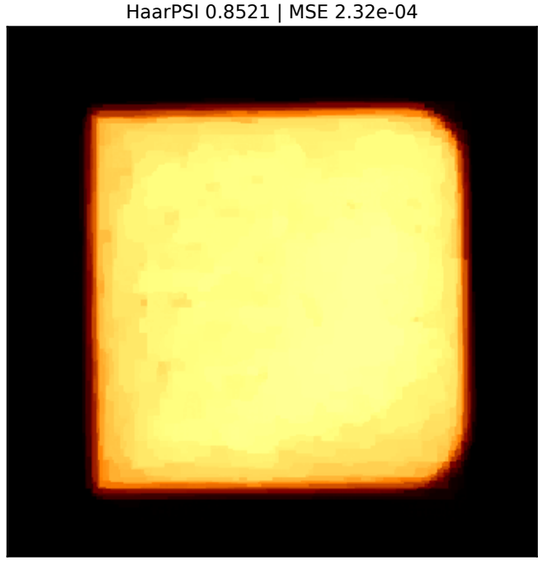
GS-PnP oracle
PSNR 38.46 dB | SSIM 0.9435
HaarPSI 0.9232 | MSE 1.43e-04



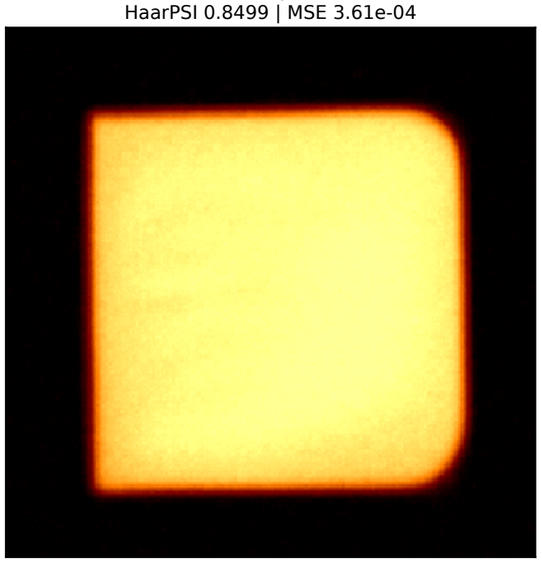
Scalar TV
PSNR 32.70 dB | SSIM 0.7563
HaarPSI 0.8479 | MSE 5.37e-04



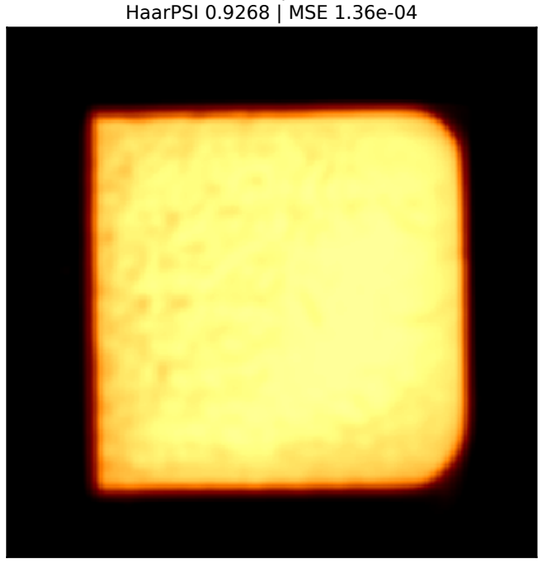
UTV-C60
PSNR 36.34 dB | SSIM 0.9049
HaarPSI 0.8521 | MSE 2.32e-04



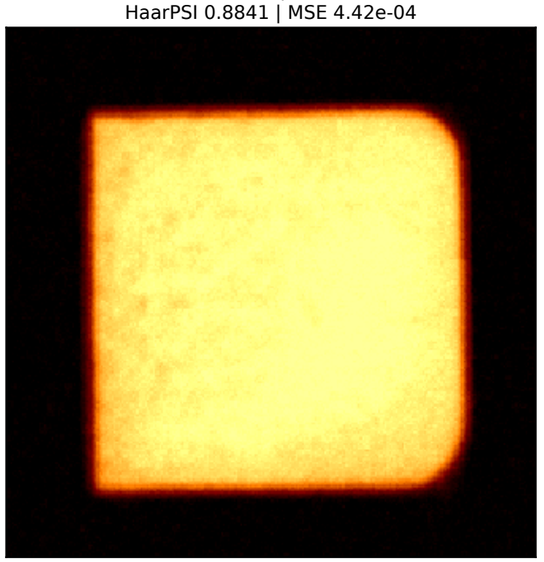
GS-PnP tuned-5
PSNR 34.43 dB | SSIM 0.8066
HaarPSI 0.8499 | MSE 3.61e-04



PnP-ADMM Tune-1 (0008)
PSNR 38.67 dB | SSIM 0.9530
HaarPSI 0.9268 | MSE 1.36e-04

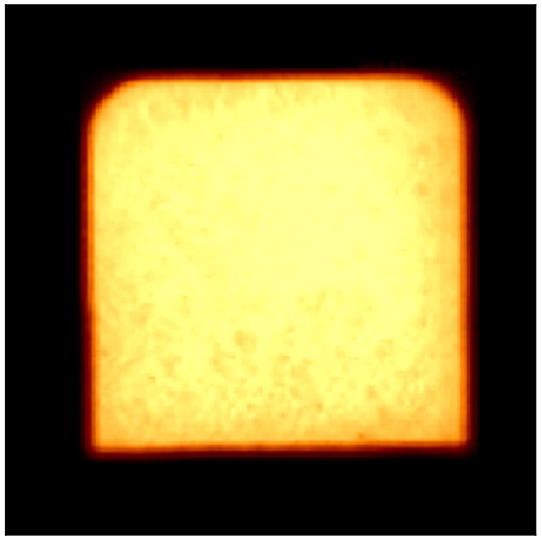


GS-PnP Tune-1 (0768)
PSNR 33.55 dB | SSIM 0.7260
HaarPSI 0.8841 | MSE 4.42e-04

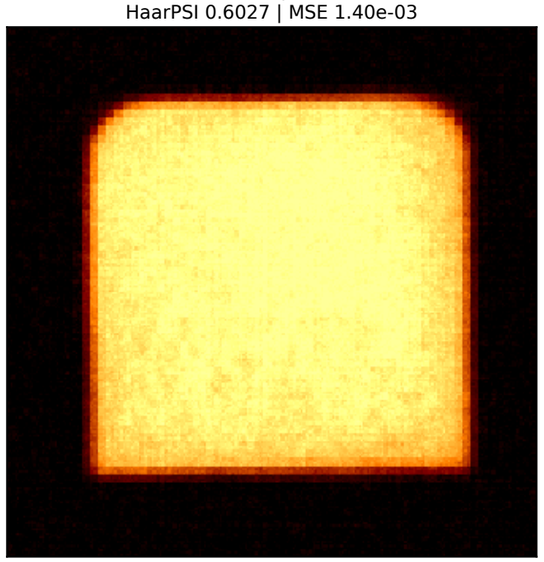


Perovskite | test image 0786 | 20% Hadamard + NPL-default noise

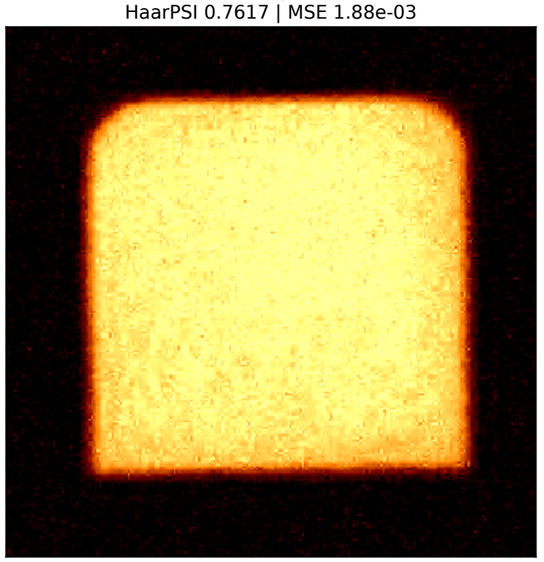
Target
Ground truth



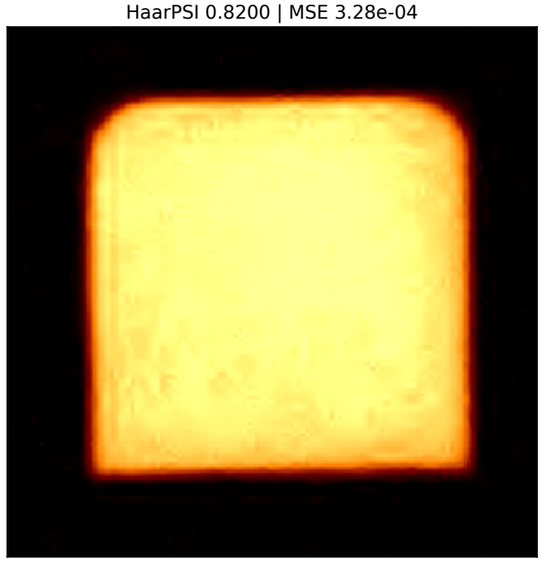
Zero-filled
PSNR 28.55 dB | SSIM 0.5481
HaarPSI 0.6027 | MSE 1.40e-03



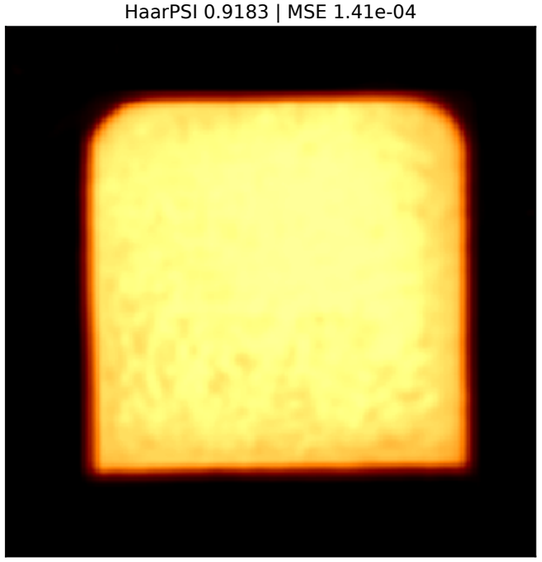
SPGL1-BP
PSNR 27.25 dB | SSIM 0.4442
HaarPSI 0.7617 | MSE 1.88e-03



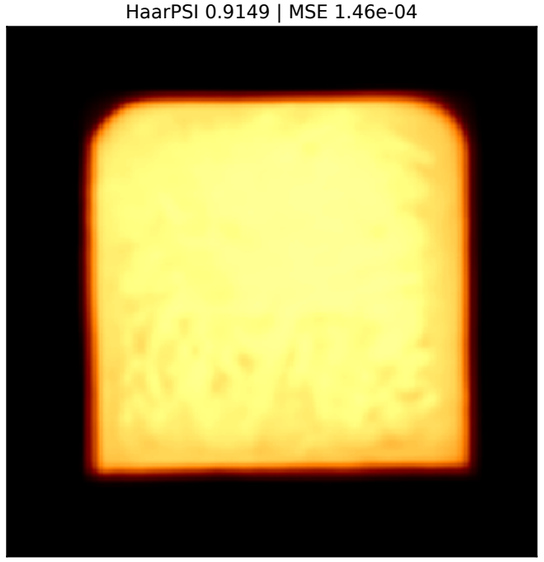
SPGL1-BPDN
PSNR 34.84 dB | SSIM 0.8168
HaarPSI 0.8200 | MSE 3.28e-04



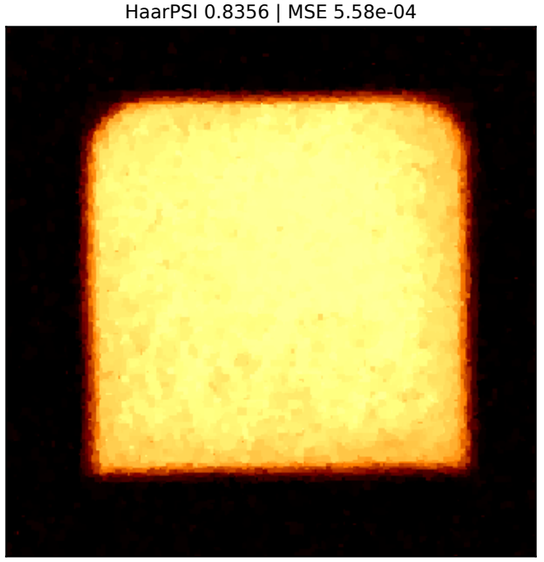
PnP-ADMM tuned-5
PSNR 38.50 dB | SSIM 0.9495
HaarPSI 0.9183 | MSE 1.41e-04



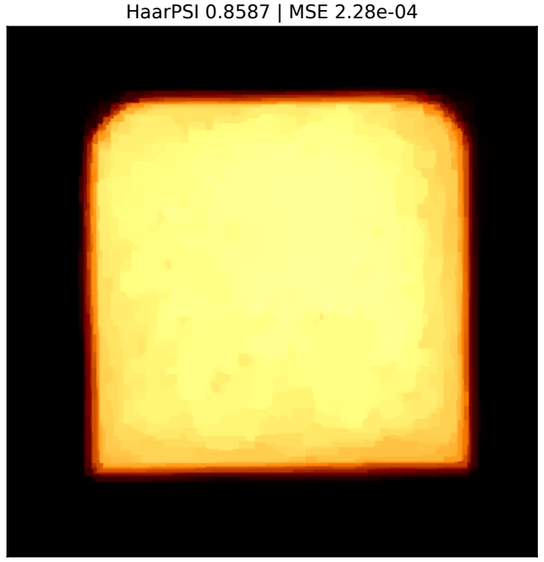
GS-PnP oracle
PSNR 38.37 dB | SSIM 0.9409
HaarPSI 0.9149 | MSE 1.46e-04



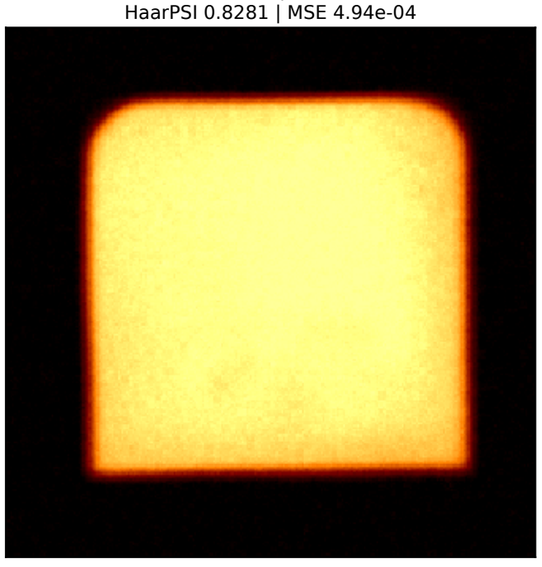
Scalar TV
PSNR 32.53 dB | SSIM 0.7544
HaarPSI 0.8356 | MSE 5.58e-04



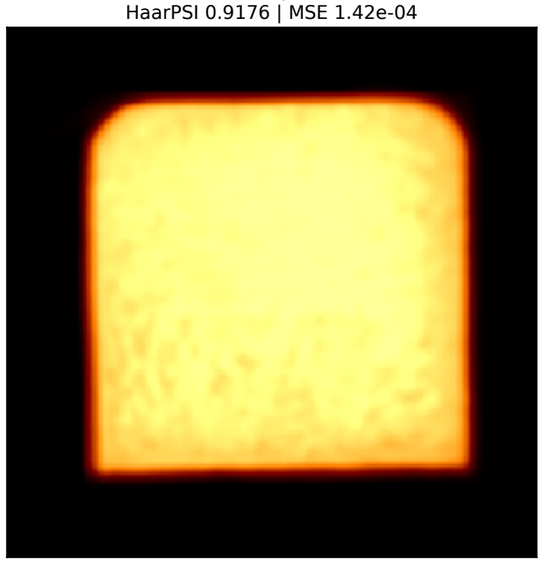
UTV-C60
PSNR 36.41 dB | SSIM 0.8960
HaarPSI 0.8587 | MSE 2.28e-04



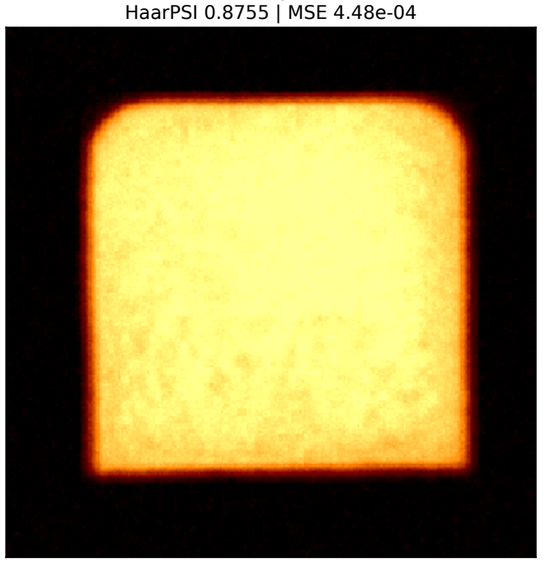
GS-PnP tuned-5
PSNR 33.06 dB | SSIM 0.7270
HaarPSI 0.8281 | MSE 4.94e-04



PnP-ADMM Tune-1 (0008)
PSNR 38.47 dB | SSIM 0.9498
HaarPSI 0.9176 | MSE 1.42e-04

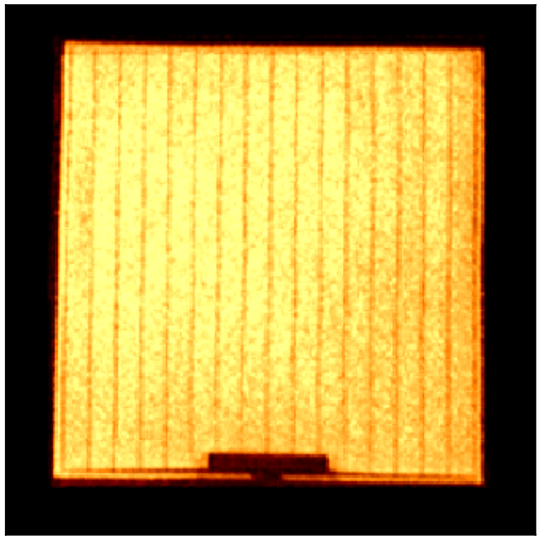


GS-PnP Tune-1 (0768)
PSNR 33.49 dB | SSIM 0.7251
HaarPSI 0.8755 | MSE 4.48e-04

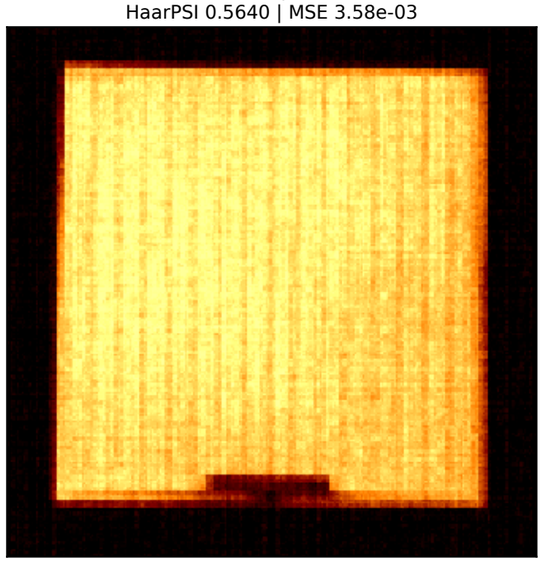


Reference | test image 0021 | 20% Hadamard + NPL-default noise

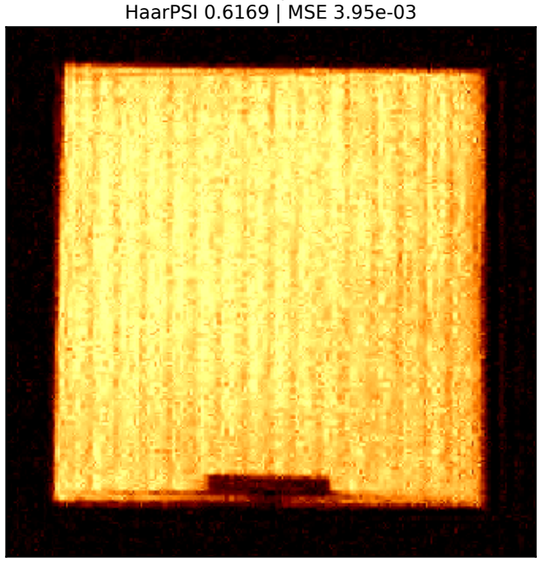
Target
Ground truth



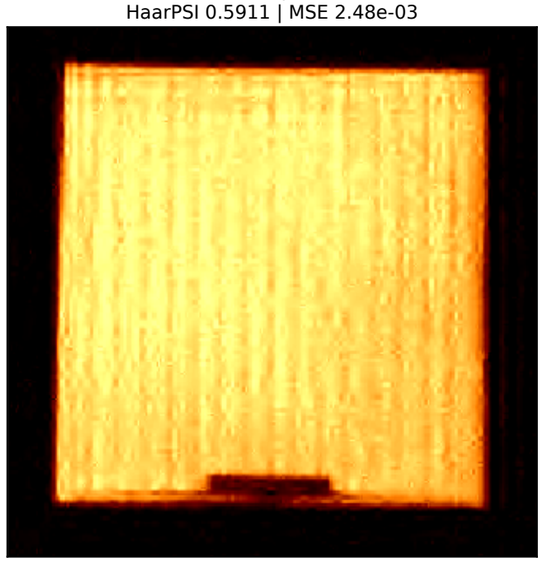
Zero-filled
PSNR 24.47 dB | SSIM 0.5090
HaarPSI 0.5640 | MSE 3.58e-03



SPGL1-BP
PSNR 24.04 dB | SSIM 0.4535
HaarPSI 0.6169 | MSE 3.95e-03



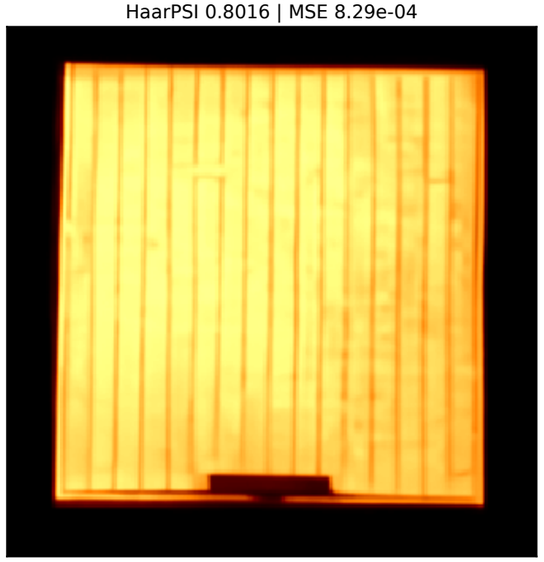
SPGL1-BPDN
PSNR 26.06 dB | SSIM 0.5797
HaarPSI 0.5911 | MSE 2.48e-03



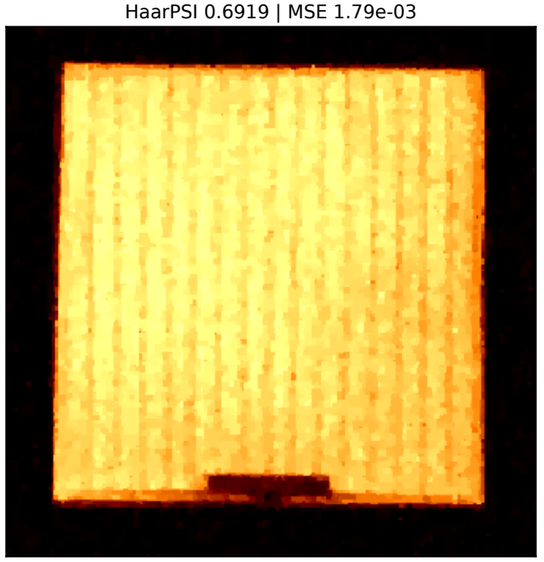
PnP-ADMM tuned-5
PSNR 31.07 dB | SSIM 0.8473
HaarPSI 0.8215 | MSE 7.82e-04



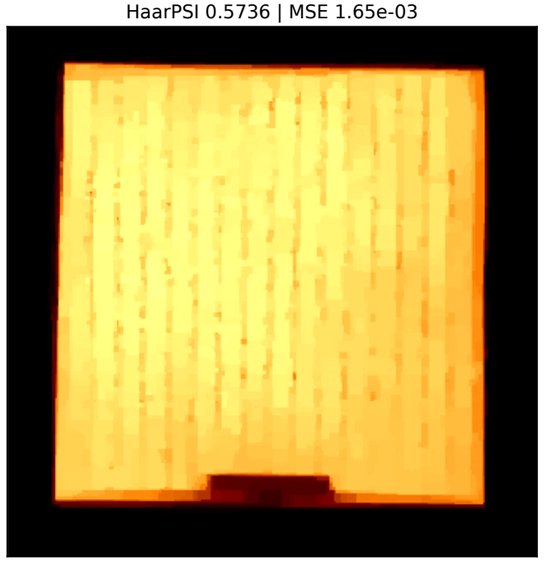
GS-PnP oracle
PSNR 30.81 dB | SSIM 0.8279
HaarPSI 0.8016 | MSE 8.29e-04



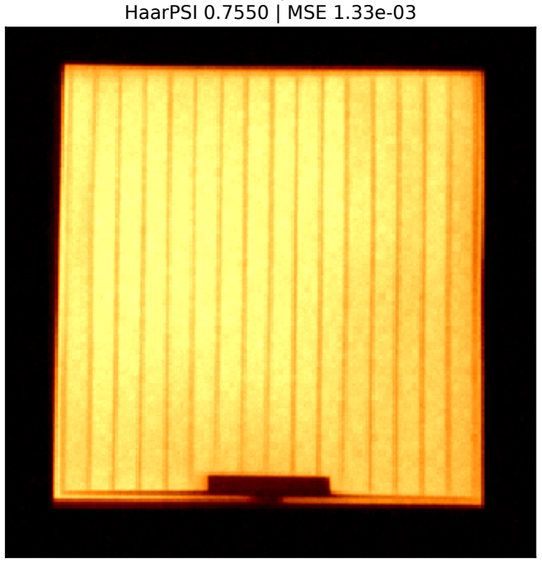
Scalar TV
PSNR 27.48 dB | SSIM 0.6419
HaarPSI 0.6919 | MSE 1.79e-03



UTV-C60
PSNR 27.83 dB | SSIM 0.6945
HaarPSI 0.5736 | MSE 1.65e-03



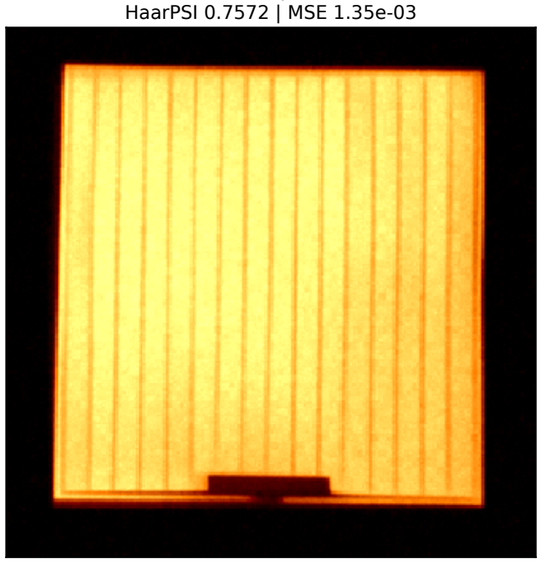
GS-PnP tuned-5
PSNR 28.76 dB | SSIM 0.7052
HaarPSI 0.7550 | MSE 1.33e-03



PnP-ADMM Tune-1 (0008)
PSNR 30.84 dB | SSIM 0.8401
HaarPSI 0.8074 | MSE 8.24e-04

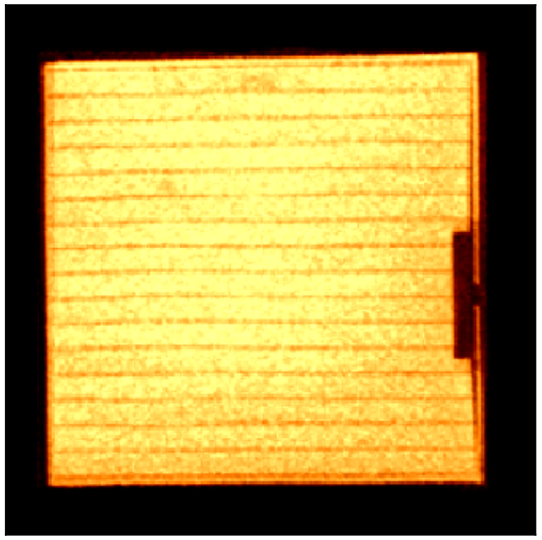


GS-PnP Tune-1 (0008)
PSNR 28.68 dB | SSIM 0.6942
HaarPSI 0.7572 | MSE 1.35e-03

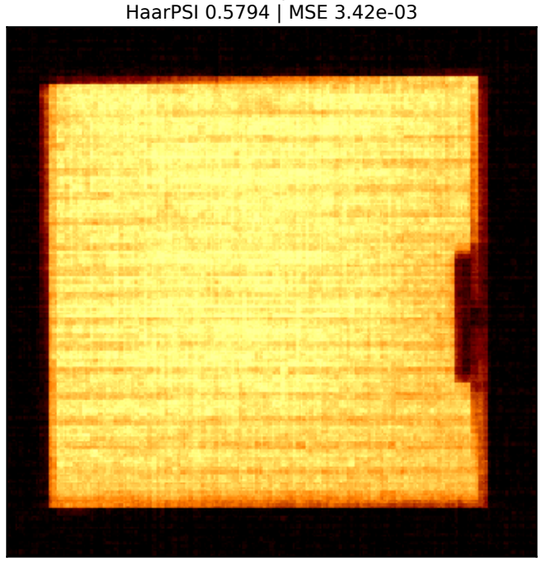


Reference | test image 0055 | 20% Hadamard + NPL-default noise

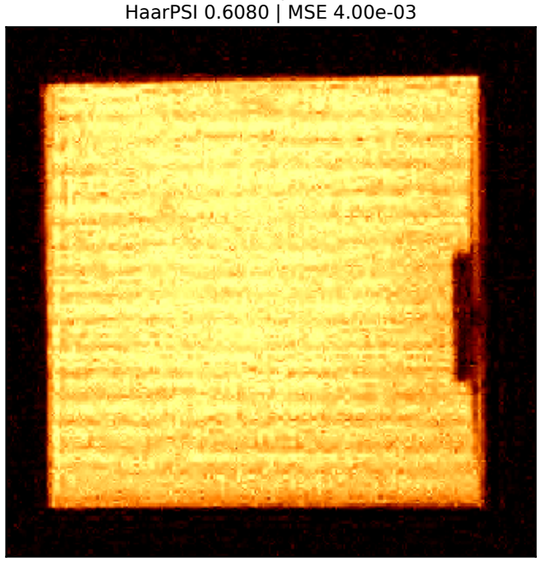
Target
Ground truth



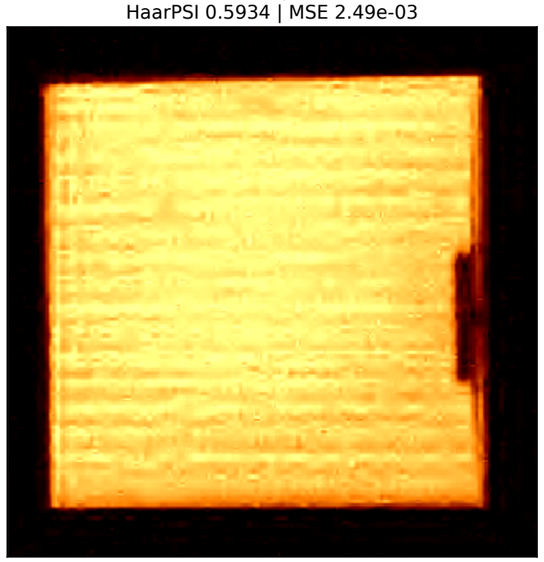
Zero-filled
PSNR 24.66 dB | SSIM 0.5125
HaarPSI 0.5794 | MSE 3.42e-03



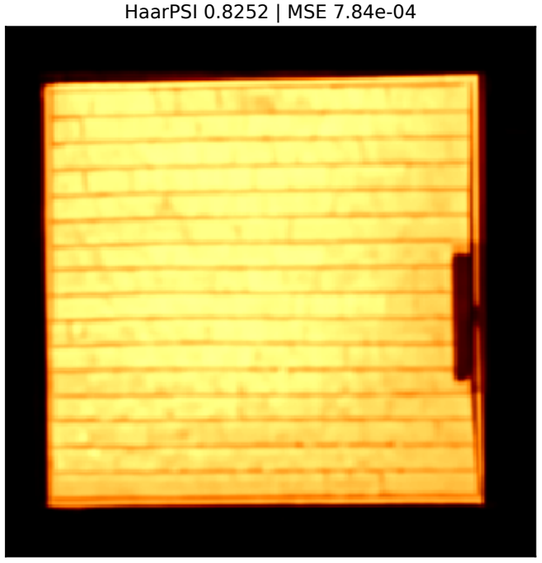
SPGL1-BP
PSNR 23.98 dB | SSIM 0.4484
HaarPSI 0.6080 | MSE 4.00e-03



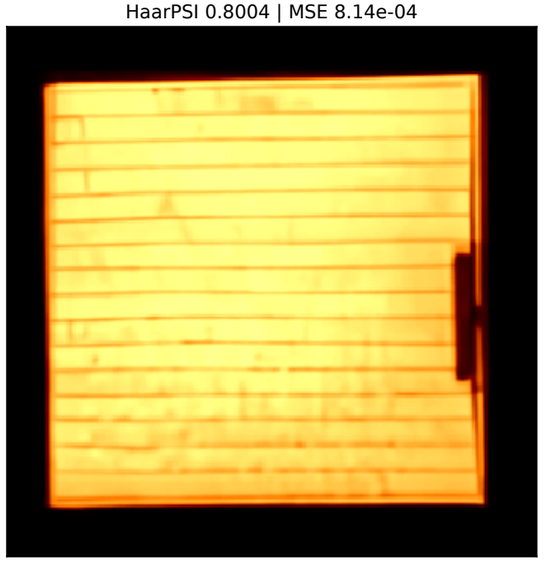
SPGL1-BPDN
PSNR 26.05 dB | SSIM 0.5724
HaarPSI 0.5934 | MSE 2.49e-03



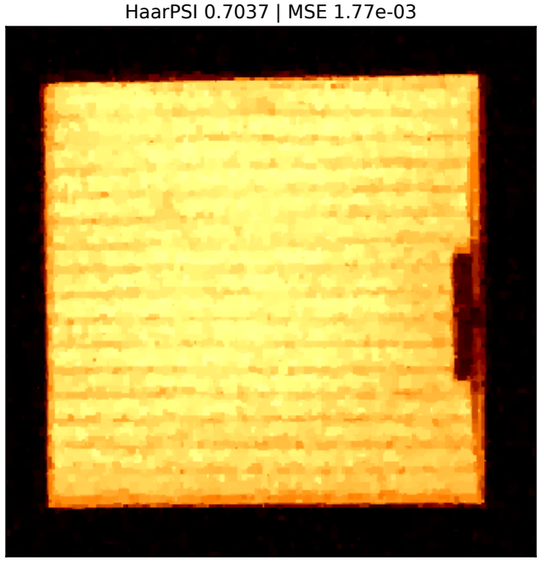
PnP-ADMM tuned-5
PSNR 31.06 dB | SSIM 0.8438
HaarPSI 0.8252 | MSE 7.84e-04



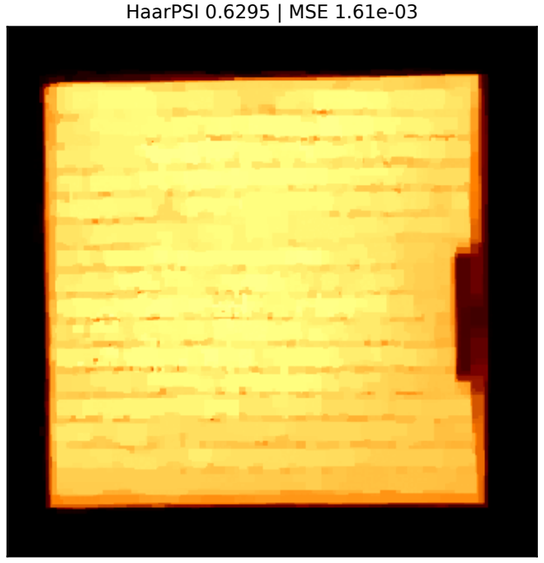
GS-PnP oracle
PSNR 30.89 dB | SSIM 0.8334
HaarPSI 0.8004 | MSE 8.14e-04



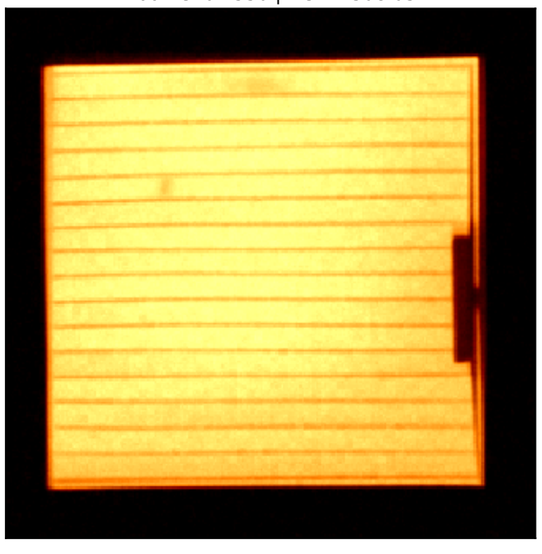
Scalar TV
PSNR 27.52 dB | SSIM 0.6508
HaarPSI 0.7037 | MSE 1.77e-03



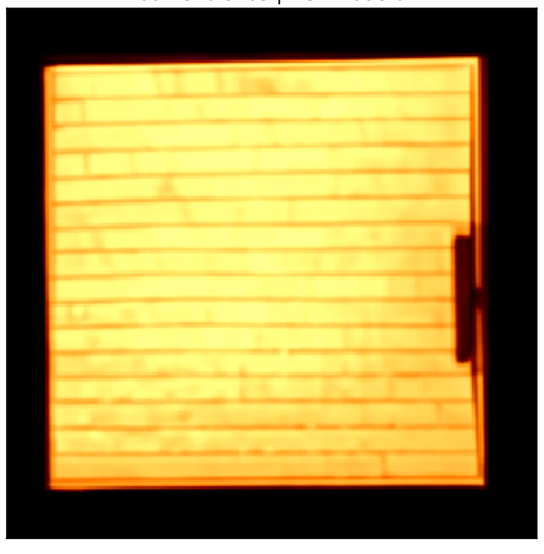
UTV-C60
PSNR 27.93 dB | SSIM 0.7048
HaarPSI 0.6295 | MSE 1.61e-03



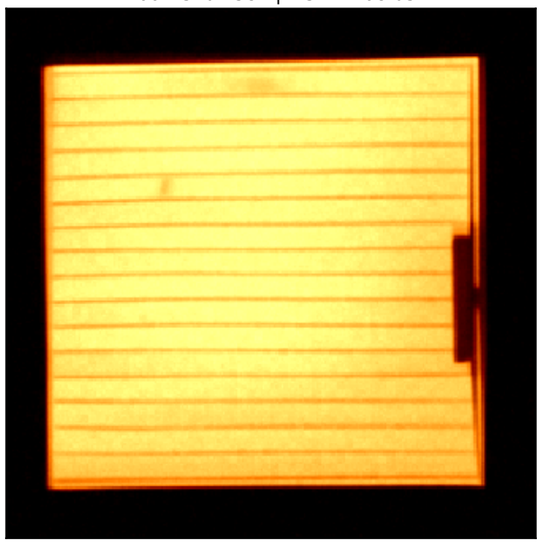
GS-PnP tuned-5
PSNR 28.87 dB | SSIM 0.7118
HaarPSI 0.7550 | MSE 1.30e-03



PnP-ADMM Tune-1 (0008)
PSNR 30.97 dB | SSIM 0.8387
HaarPSI 0.8163 | MSE 7.99e-04

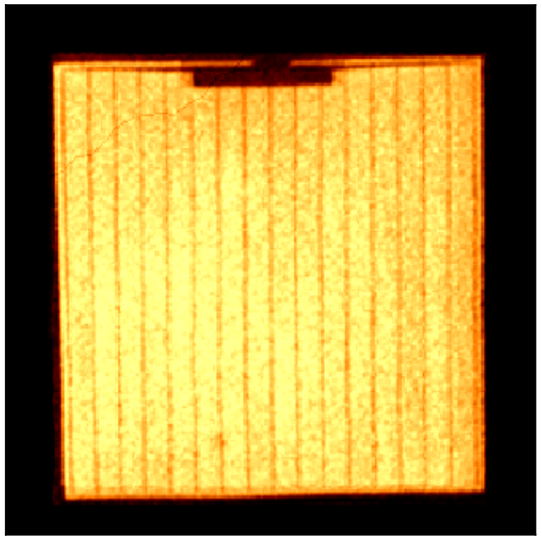


GS-PnP Tune-1 (0008)
PSNR 29.34 dB | SSIM 0.7446
HaarPSI 0.7582 | MSE 1.16e-03

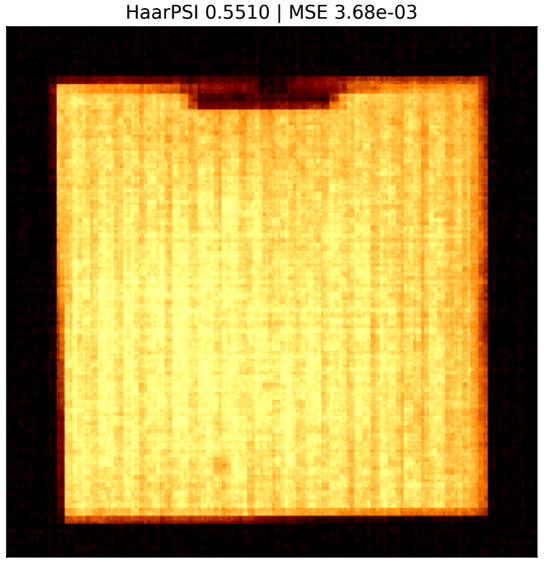


Reference | test image 0129 | 20% Hadamard + NPL-default noise

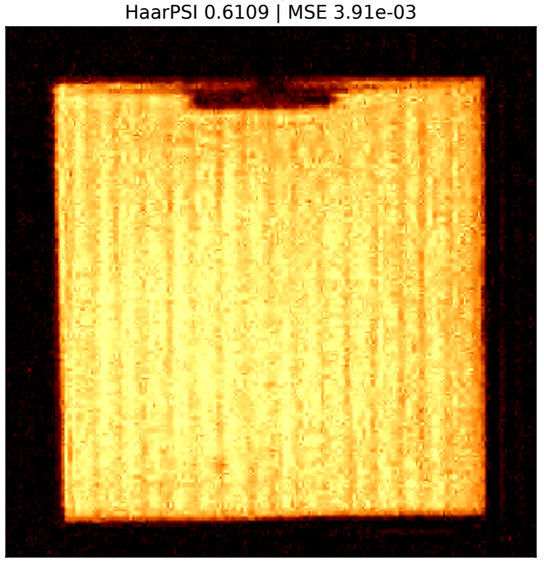
Target
Ground truth



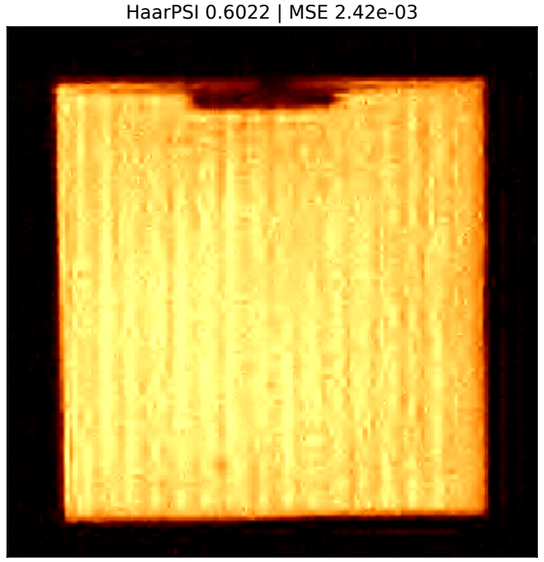
Zero-filled
PSNR 24.34 dB | SSIM 0.5054
HaarPSI 0.5510 | MSE 3.68e-03



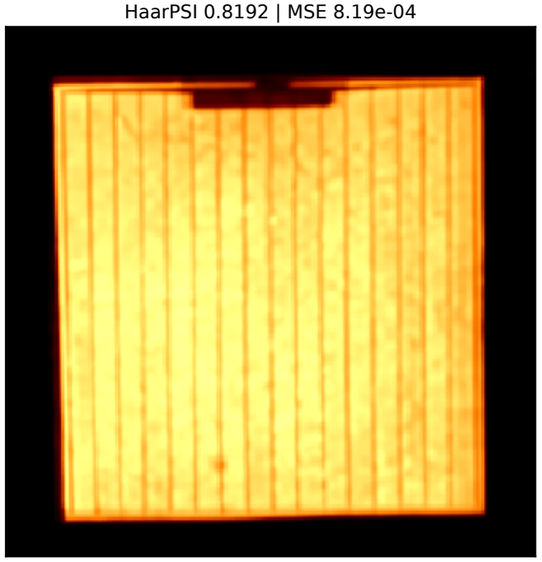
SPGL1-BP
PSNR 24.08 dB | SSIM 0.4570
HaarPSI 0.6109 | MSE 3.91e-03



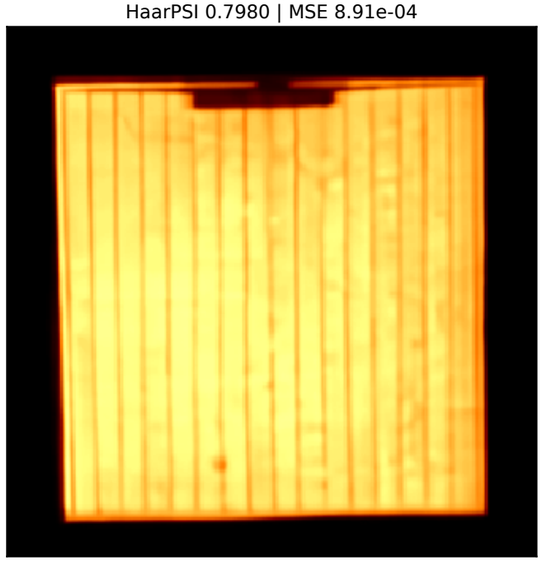
SPGL1-BPDN
PSNR 26.16 dB | SSIM 0.5782
HaarPSI 0.6022 | MSE 2.42e-03



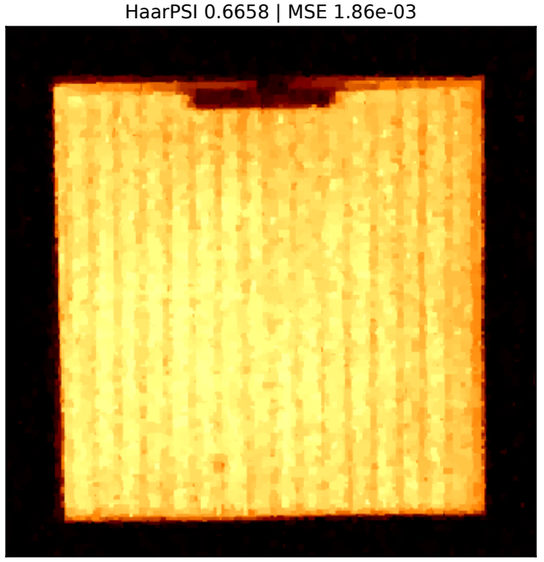
PnP-ADMM tuned-5
PSNR 30.87 dB | SSIM 0.8462
HaarPSI 0.8192 | MSE 8.19e-04



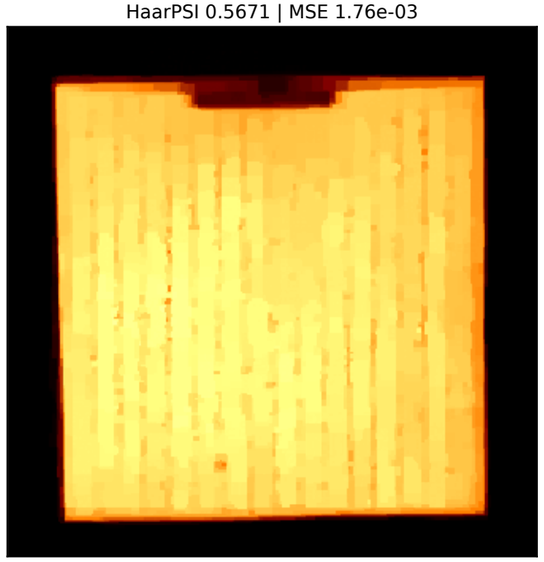
GS-PnP oracle
PSNR 30.50 dB | SSIM 0.8229
HaarPSI 0.7980 | MSE 8.91e-04



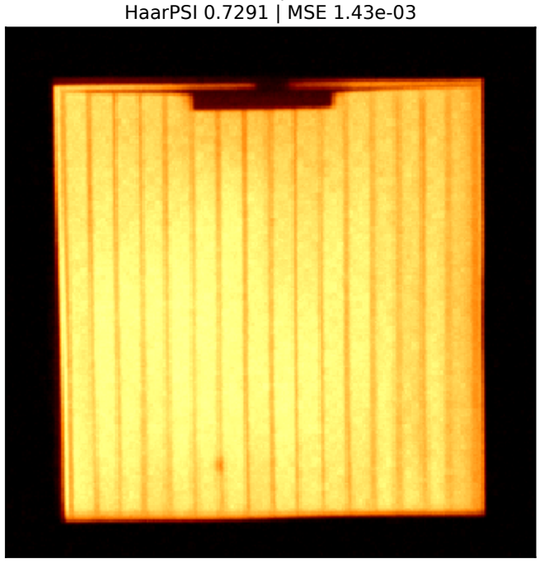
Scalar TV
PSNR 27.30 dB | SSIM 0.6419
HaarPSI 0.6658 | MSE 1.86e-03



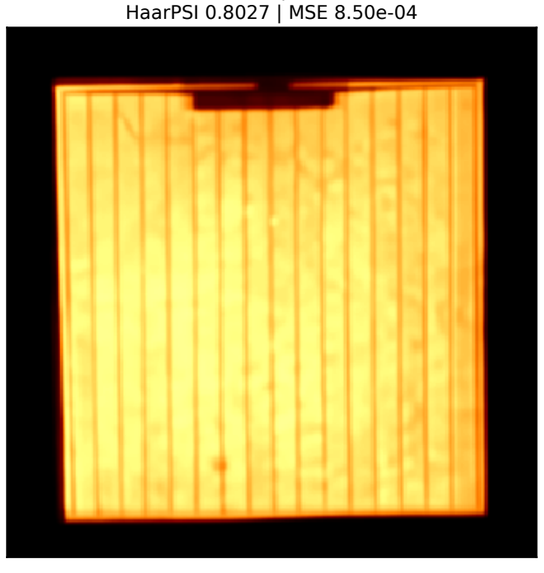
UTV-C60
PSNR 27.55 dB | SSIM 0.6792
HaarPSI 0.5671 | MSE 1.76e-03



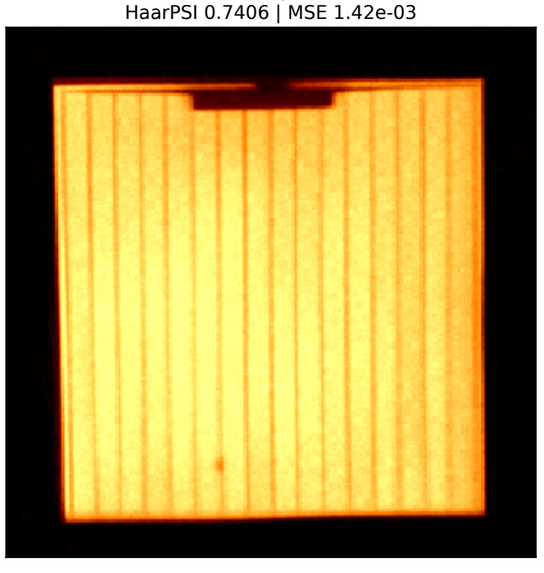
GS-PnP tuned-5
PSNR 28.46 dB | SSIM 0.6928
HaarPSI 0.7291 | MSE 1.43e-03



PnP-ADMM Tune-1 (0008)
PSNR 30.71 dB | SSIM 0.8384
HaarPSI 0.8027 | MSE 8.50e-04

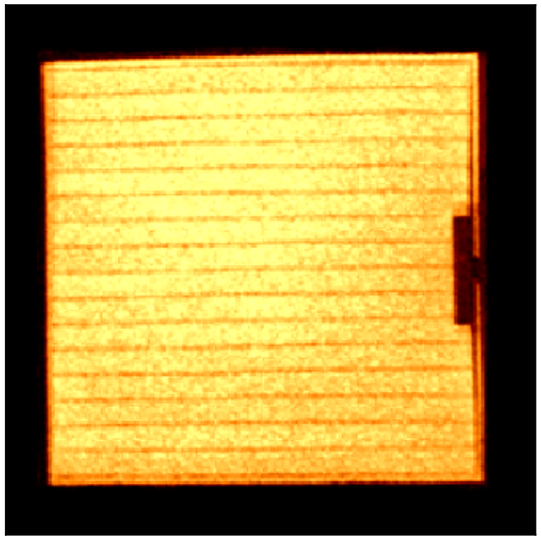


GS-PnP Tune-1 (0008)
PSNR 28.47 dB | SSIM 0.6882
HaarPSI 0.7406 | MSE 1.42e-03

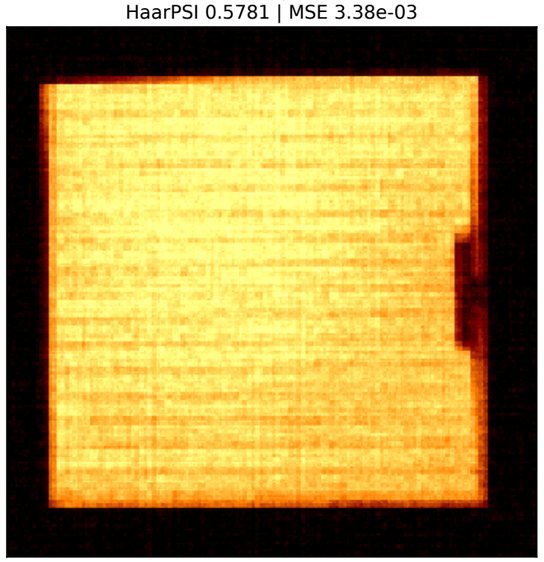


Reference | test image 0199 | 20% Hadamard + NPL-default noise

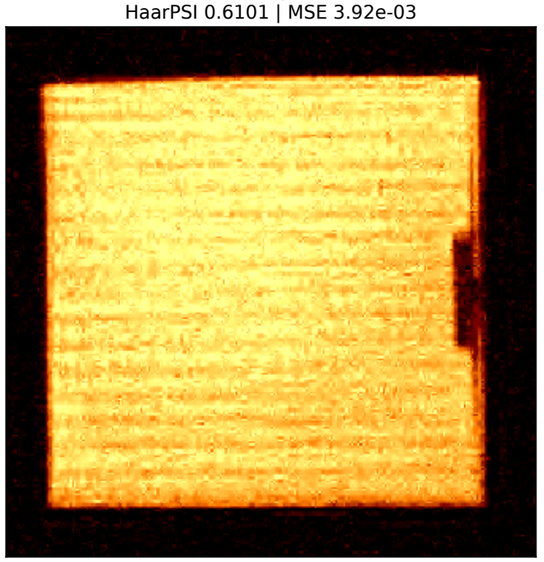
Target
Ground truth



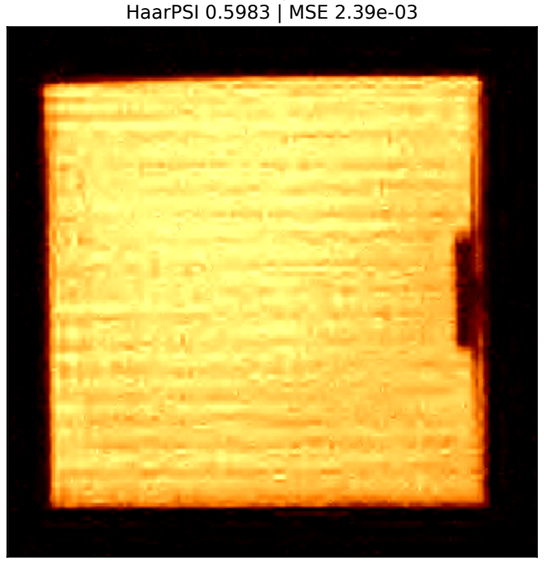
Zero-filled
PSNR 24.70 dB | SSIM 0.5059
HaarPSI 0.5781 | MSE 3.38e-03



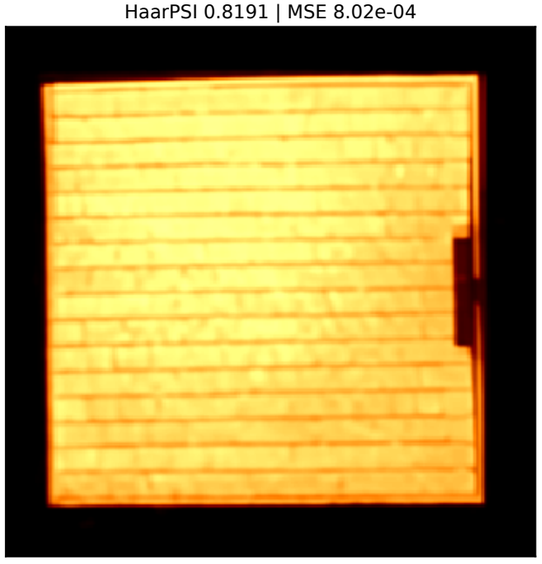
SPGL1-BP
PSNR 24.07 dB | SSIM 0.4492
HaarPSI 0.6101 | MSE 3.92e-03



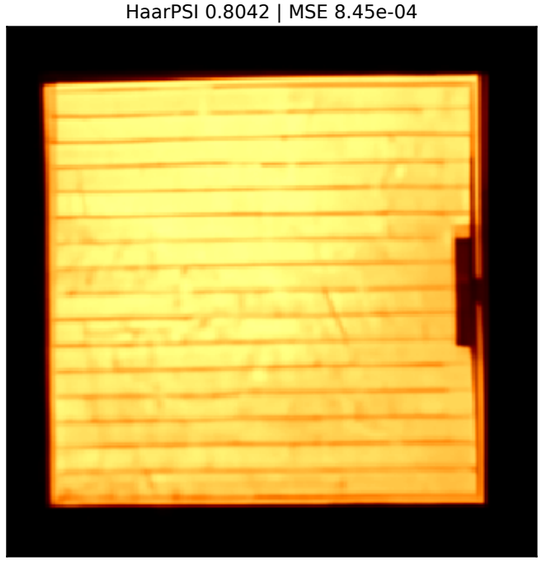
SPGL1-BPDN
PSNR 26.21 dB | SSIM 0.5786
HaarPSI 0.5983 | MSE 2.39e-03



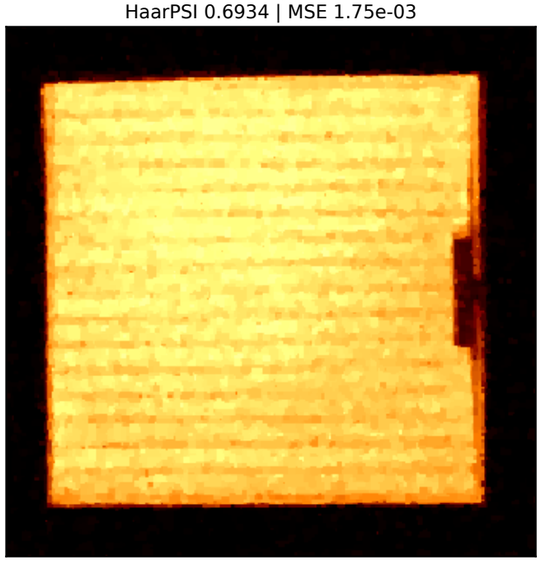
PnP-ADMM tuned-5
PSNR 30.96 dB | SSIM 0.8393
HaarPSI 0.8191 | MSE 8.02e-04



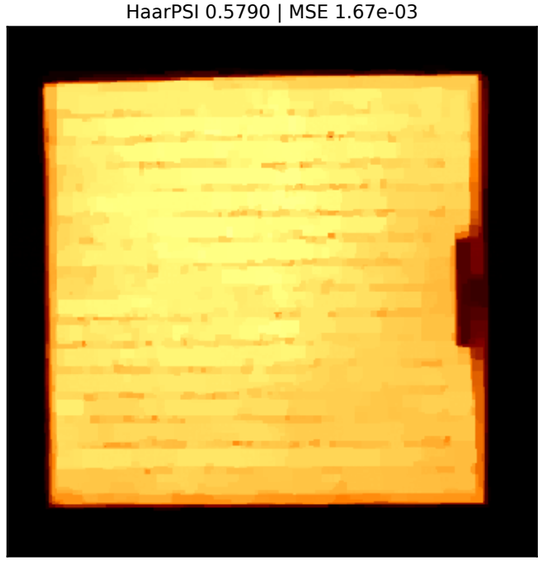
GS-PnP oracle
PSNR 30.73 dB | SSIM 0.8225
HaarPSI 0.8042 | MSE 8.45e-04



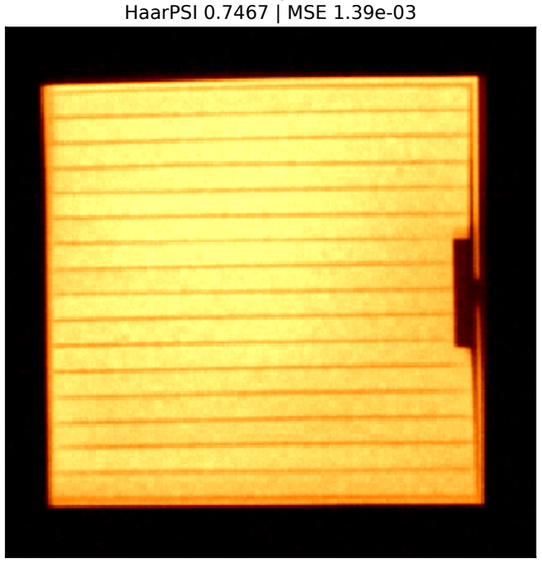
Scalar TV
PSNR 27.56 dB | SSIM 0.6395
HaarPSI 0.6934 | MSE 1.75e-03



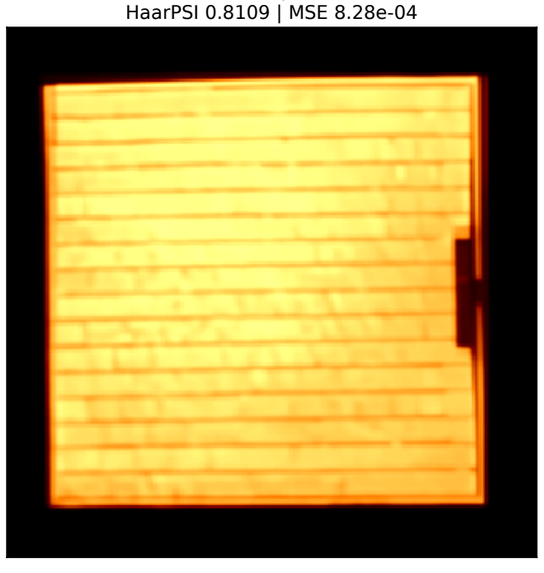
UTV-C60
PSNR 27.77 dB | SSIM 0.6877
HaarPSI 0.5790 | MSE 1.67e-03



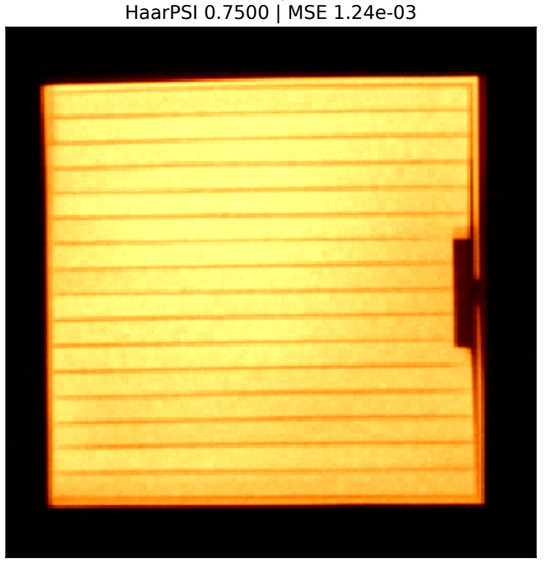
GS-PnP tuned-5
PSNR 28.58 dB | SSIM 0.6954
HaarPSI 0.7467 | MSE 1.39e-03



PnP-ADMM Tune-1 (0008)
PSNR 30.82 dB | SSIM 0.8320
HaarPSI 0.8109 | MSE 8.28e-04

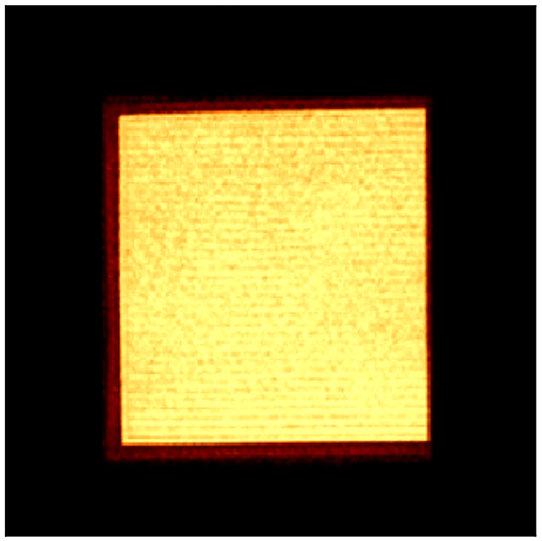


GS-PnP Tune-1 (0008)
PSNR 29.08 dB | SSIM 0.7294
HaarPSI 0.7500 | MSE 1.24e-03

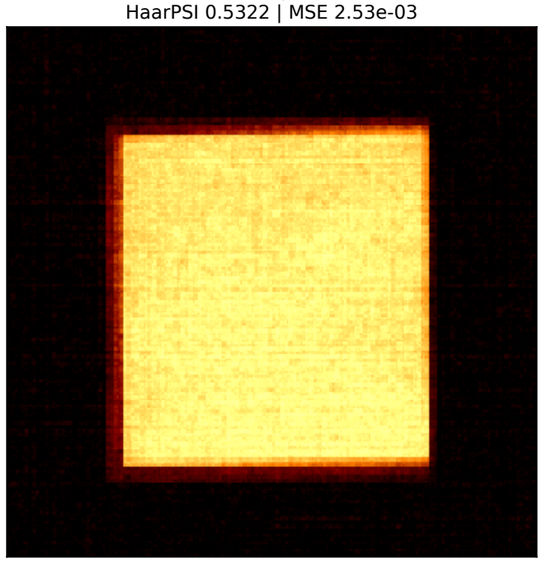


Reference | test image 0260 | 20% Hadamard + NPL-default noise

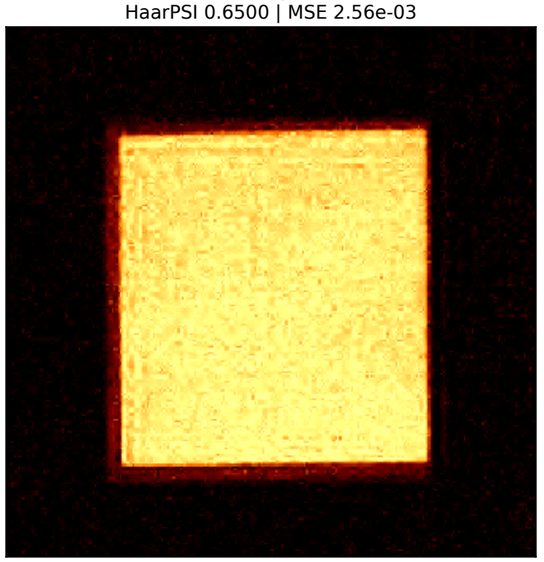
Target
Ground truth



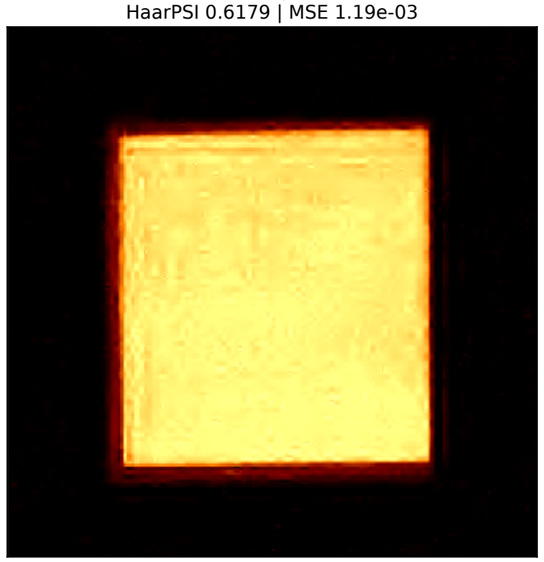
Zero-filled
PSNR 25.97 dB | SSIM 0.4425
HaarPSI 0.5322 | MSE 2.53e-03



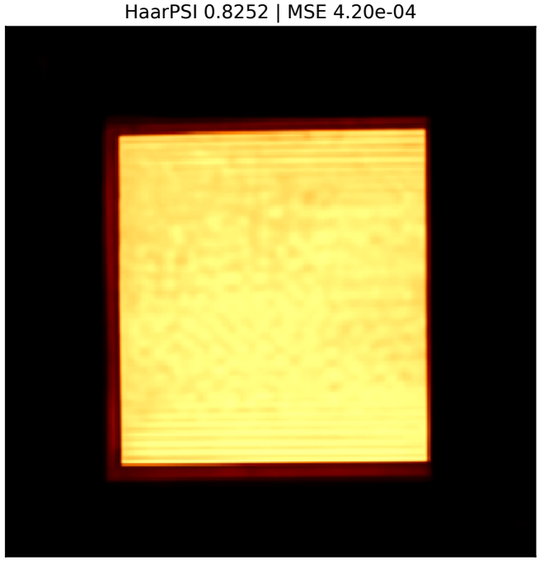
SPGL1-BP
PSNR 25.93 dB | SSIM 0.3871
HaarPSI 0.6500 | MSE 2.56e-03



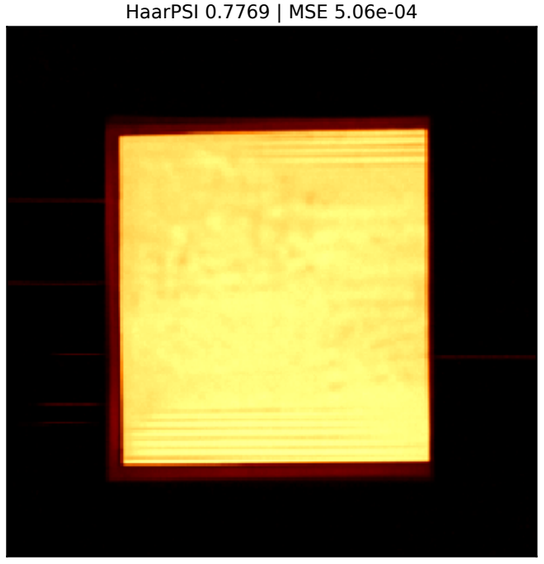
SPGL1-BPDN
PSNR 29.25 dB | SSIM 0.6791
HaarPSI 0.6179 | MSE 1.19e-03



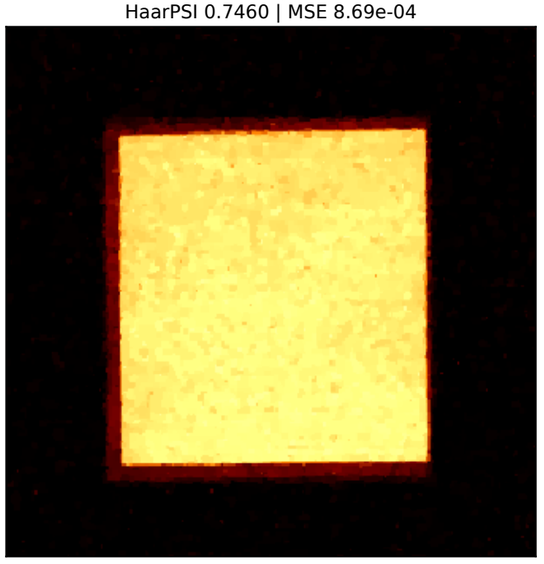
PnP-ADMM tuned-5
PSNR 33.76 dB | SSIM 0.8605
HaarPSI 0.8252 | MSE 4.20e-04



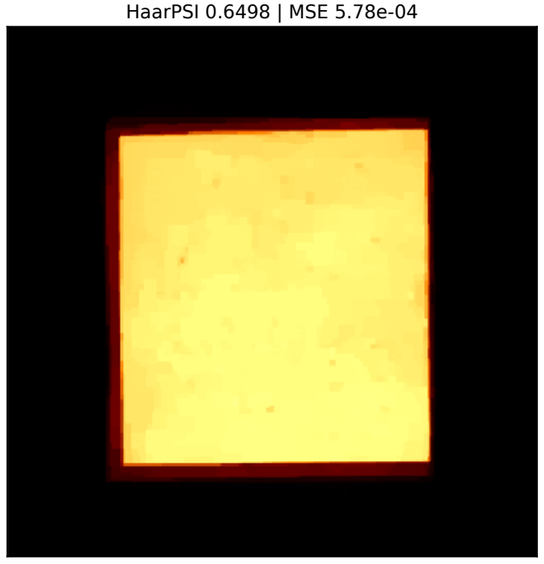
GS-PnP oracle
PSNR 32.96 dB | SSIM 0.8030
HaarPSI 0.7769 | MSE 5.06e-04



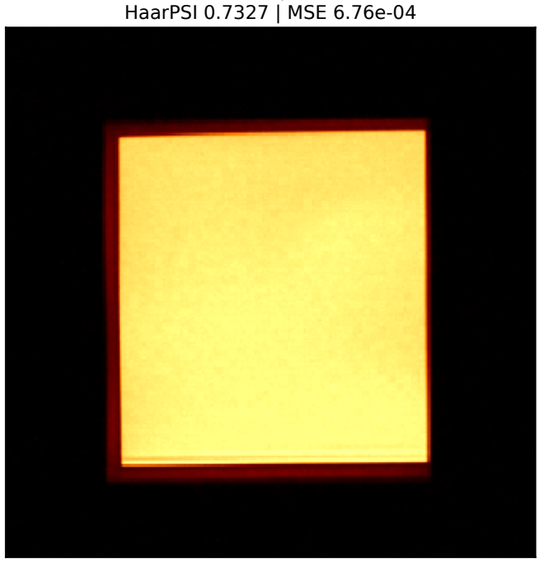
Scalar TV
PSNR 30.61 dB | SSIM 0.6376
HaarPSI 0.7460 | MSE 8.69e-04



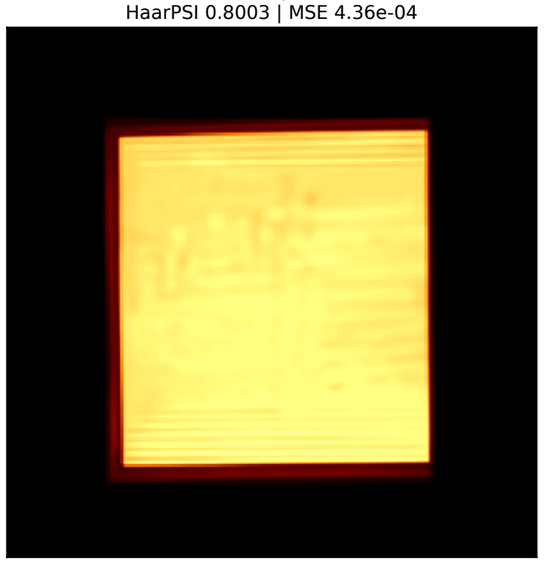
UTV-C60
PSNR 32.38 dB | SSIM 0.7909
HaarPSI 0.6498 | MSE 5.78e-04



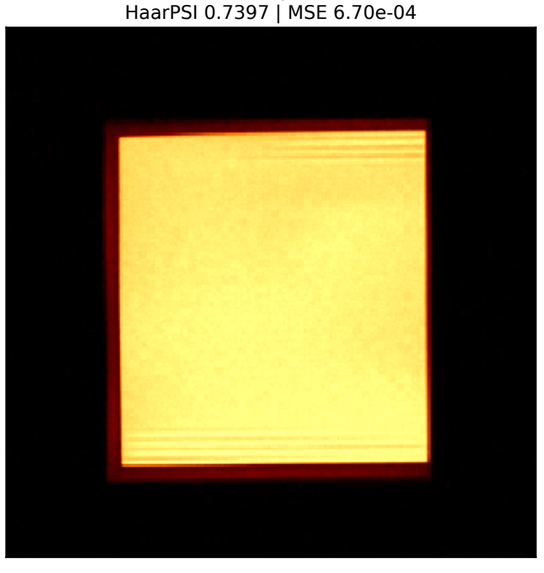
GS-PnP tuned-5
PSNR 31.70 dB | SSIM 0.7361
HaarPSI 0.7327 | MSE 6.76e-04



PnP-ADMM Tune-1 (0008)
PSNR 33.61 dB | SSIM 0.8539
HaarPSI 0.8003 | MSE 4.36e-04

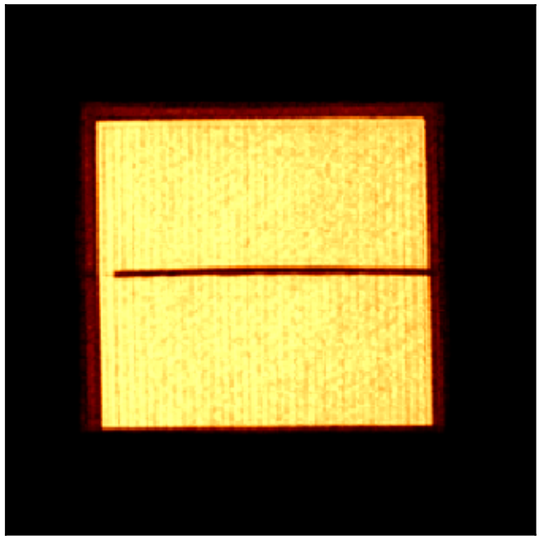


GS-PnP Tune-1 (0008)
PSNR 31.74 dB | SSIM 0.7313
HaarPSI 0.7397 | MSE 6.70e-04

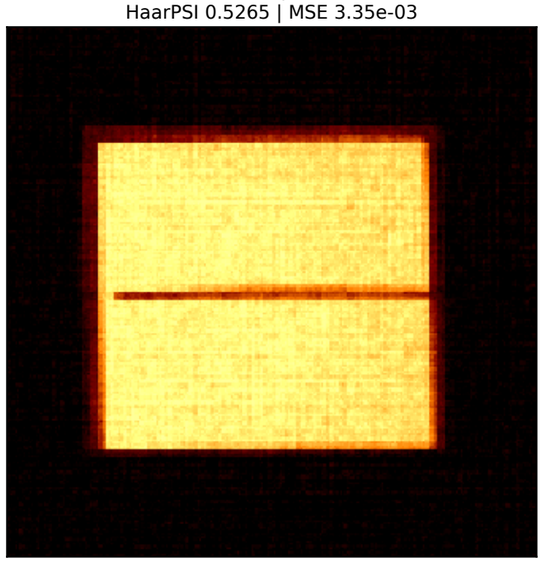


Reference | test image 0357 | 20% Hadamard + NPL-default noise

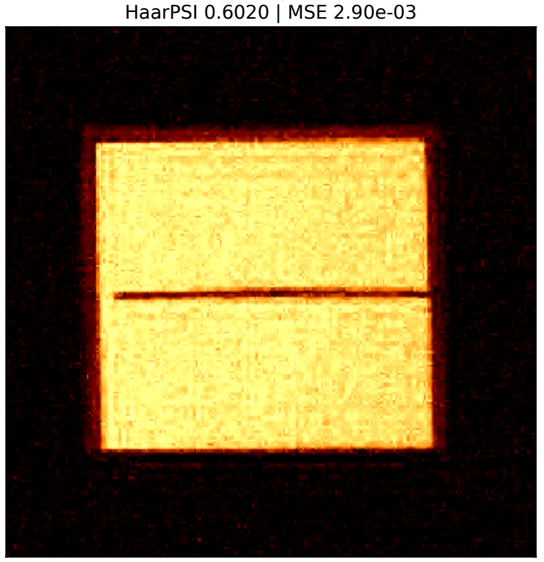
Target
Ground truth



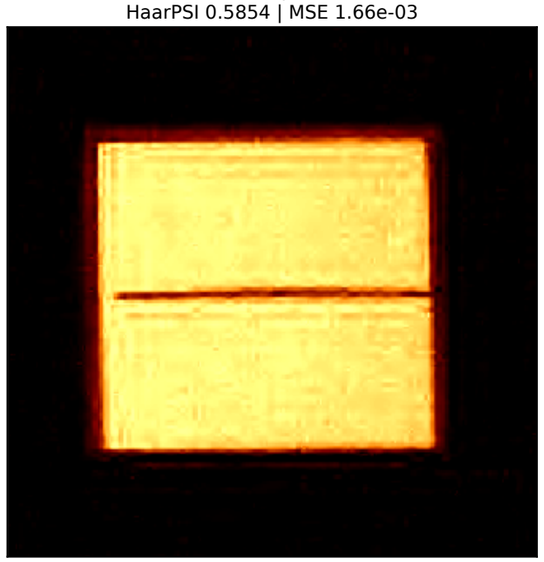
Zero-filled
PSNR 24.75 dB | SSIM 0.4316
HaarPSI 0.5265 | MSE 3.35e-03



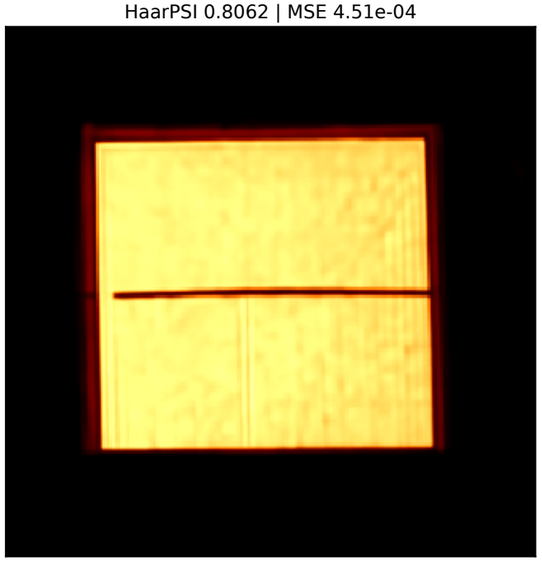
SPGL1-BP
PSNR 25.37 dB | SSIM 0.3870
HaarPSI 0.6020 | MSE 2.90e-03



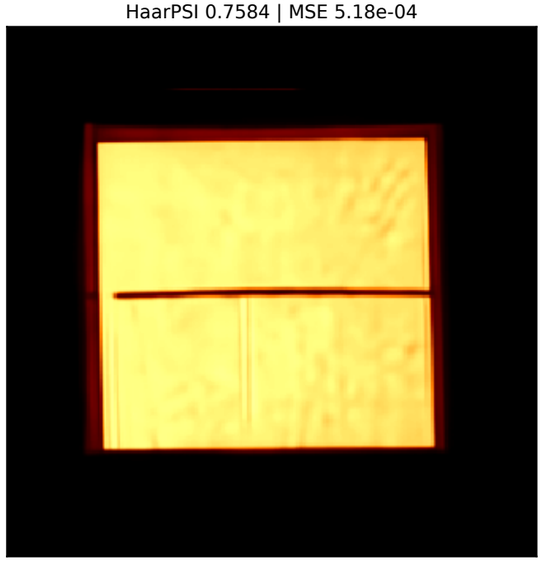
SPGL1-BPDN
PSNR 27.79 dB | SSIM 0.6636
HaarPSI 0.5854 | MSE 1.66e-03



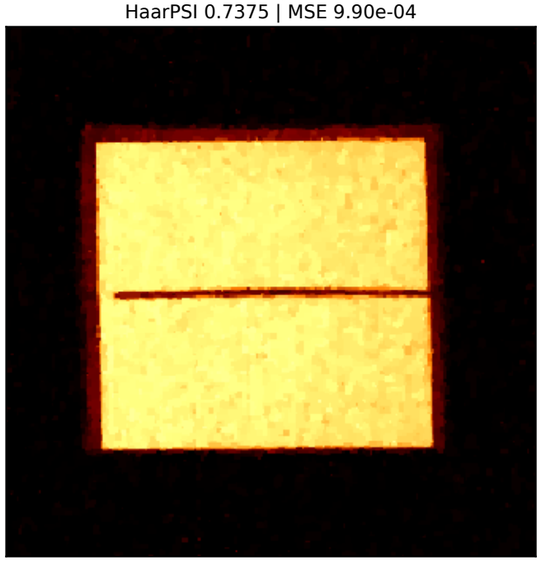
PnP-ADMM tuned-5
PSNR 33.46 dB | SSIM 0.8637
HaarPSI 0.8062 | MSE 4.51e-04



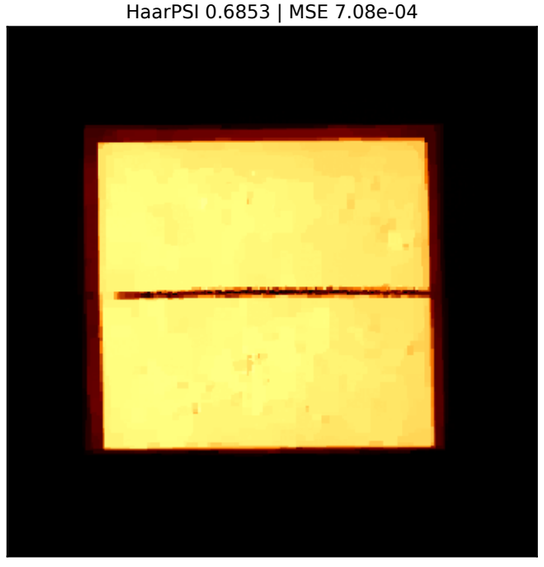
GS-PnP oracle
PSNR 32.86 dB | SSIM 0.8335
HaarPSI 0.7584 | MSE 5.18e-04



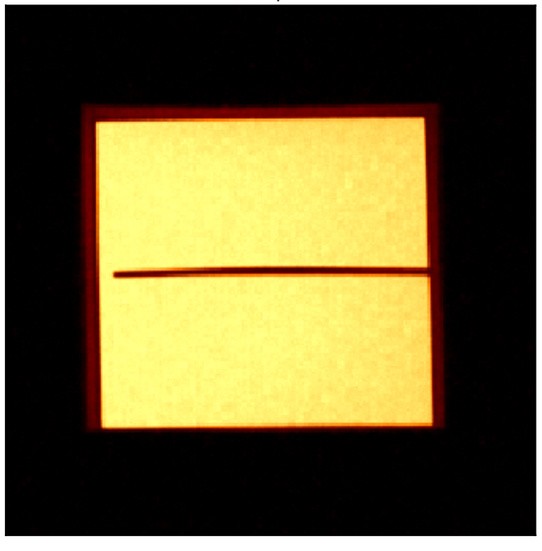
Scalar TV
PSNR 30.04 dB | SSIM 0.6488
HaarPSI 0.7375 | MSE 9.90e-04



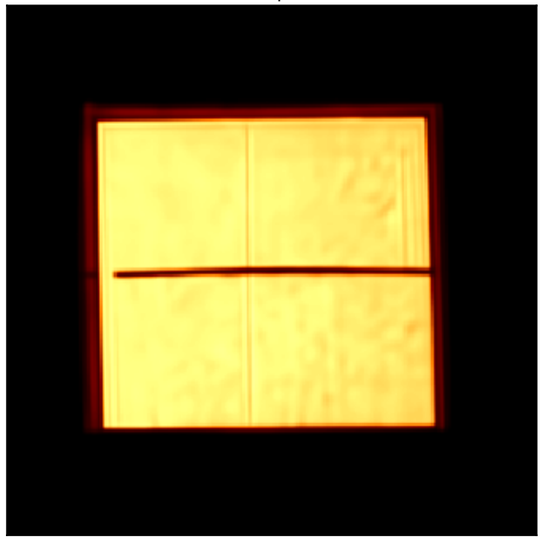
UTV-C60
PSNR 31.50 dB | SSIM 0.8016
HaarPSI 0.6853 | MSE 7.08e-04



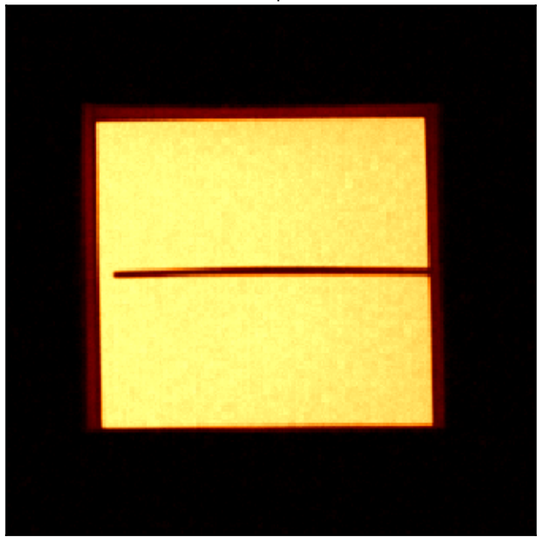
GS-PnP tuned-5
PSNR 30.78 dB | SSIM 0.6826
HaarPSI 0.7150 | MSE 8.35e-04



PnP-ADMM Tune-1 (0008)
PSNR 33.21 dB | SSIM 0.8556
HaarPSI 0.7887 | MSE 4.77e-04

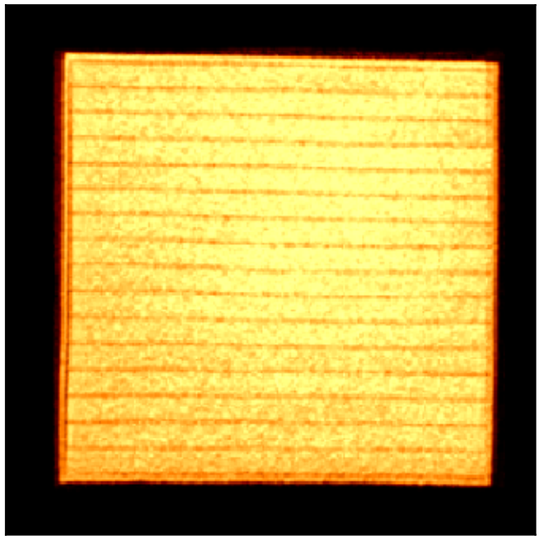


GS-PnP Tune-1 (0008)
PSNR 30.61 dB | SSIM 0.6620
HaarPSI 0.7179 | MSE 8.69e-04

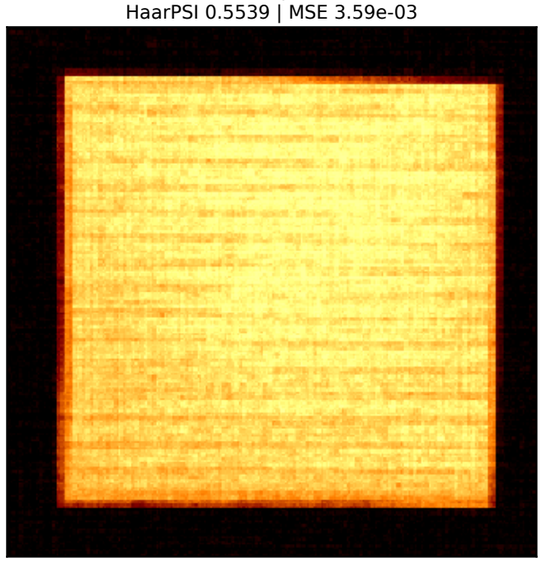


Reference | test image 0702 | 20% Hadamard + NPL-default noise

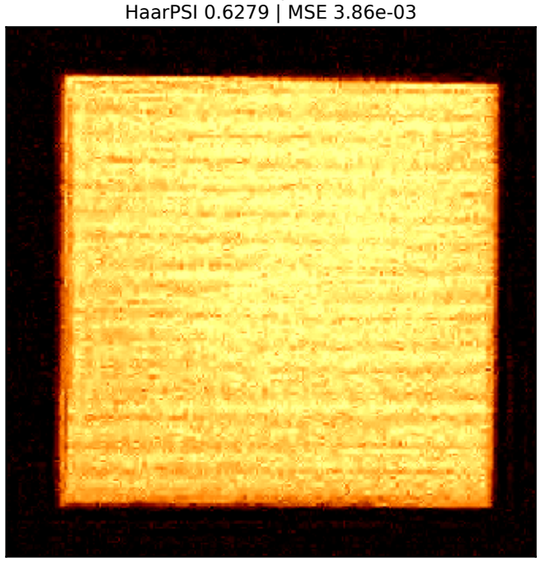
Target
Ground truth



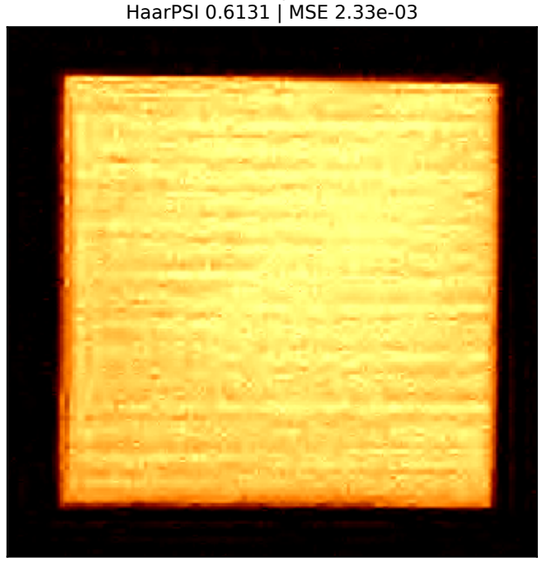
Zero-filled
PSNR 24.45 dB | SSIM 0.5096
HaarPSI 0.5539 | MSE 3.59e-03



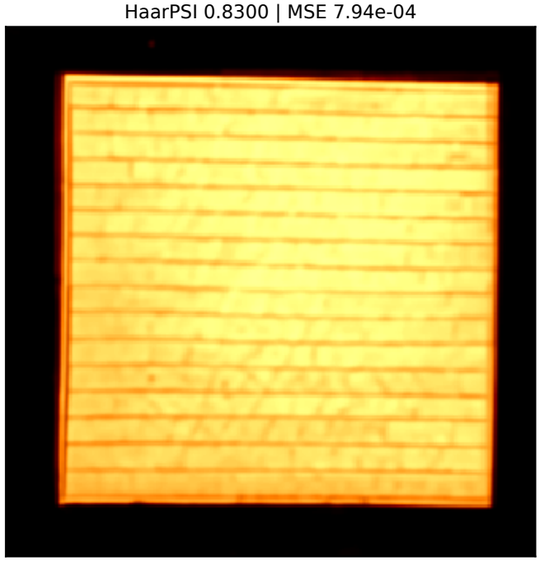
SPGL1-BP
PSNR 24.14 dB | SSIM 0.4450
HaarPSI 0.6279 | MSE 3.86e-03



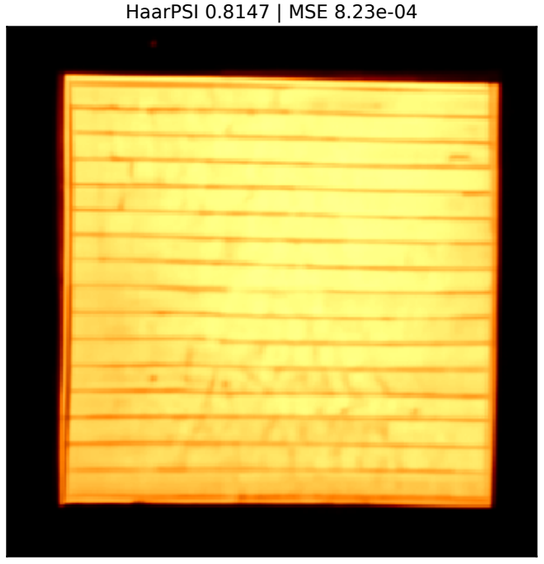
SPGL1-BPDN
PSNR 26.32 dB | SSIM 0.5743
HaarPSI 0.6131 | MSE 2.33e-03



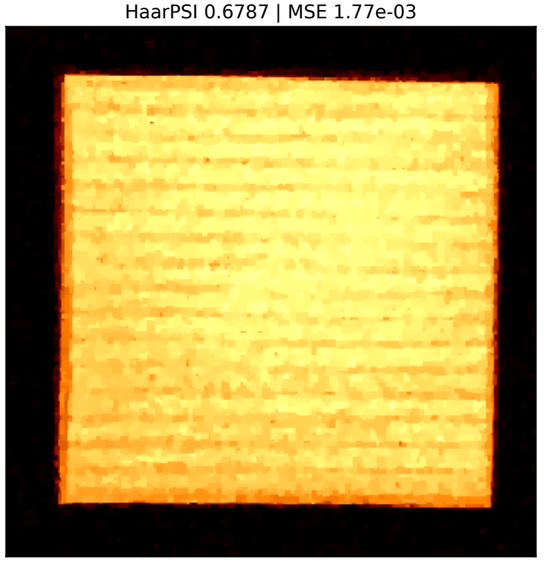
PnP-ADMM tuned-5
PSNR 31.00 dB | SSIM 0.8383
HaarPSI 0.8300 | MSE 7.94e-04



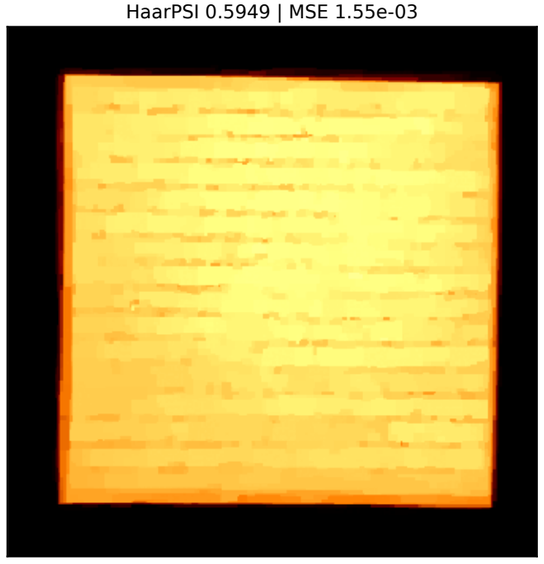
GS-PnP oracle
PSNR 30.85 dB | SSIM 0.8260
HaarPSI 0.8147 | MSE 8.23e-04



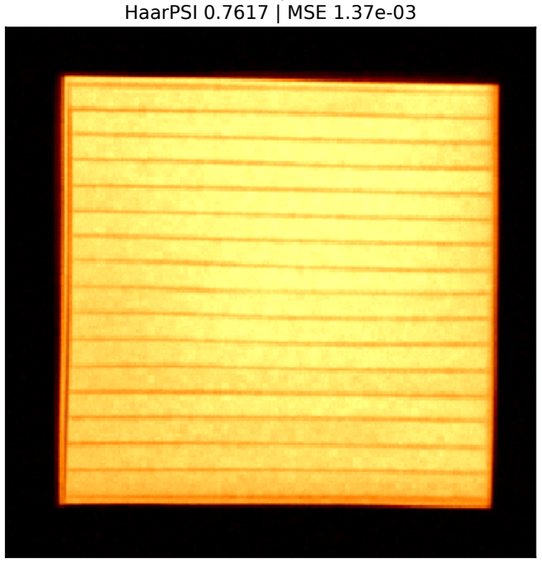
Scalar TV
PSNR 27.51 dB | SSIM 0.6383
HaarPSI 0.6787 | MSE 1.77e-03



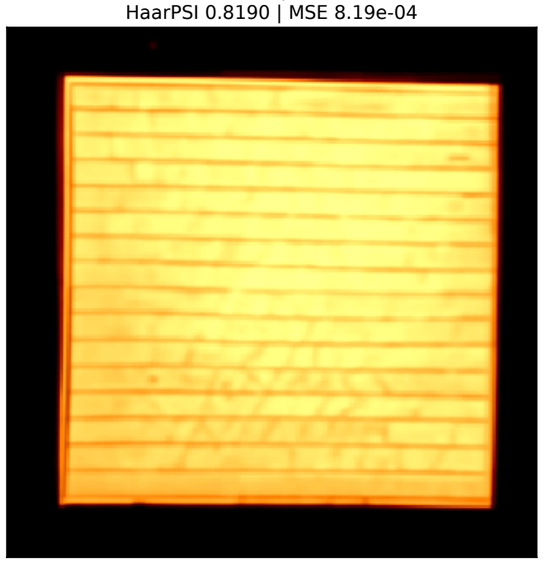
UTV-C60
PSNR 28.09 dB | SSIM 0.6922
HaarPSI 0.5949 | MSE 1.55e-03



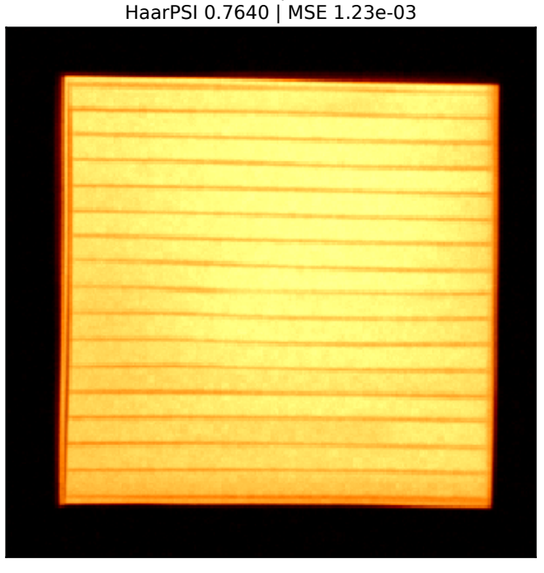
GS-PnP tuned-5
PSNR 28.62 dB | SSIM 0.6954
HaarPSI 0.7617 | MSE 1.37e-03



PnP-ADMM Tune-1 (0008)
PSNR 30.87 dB | SSIM 0.8314
HaarPSI 0.8190 | MSE 8.19e-04

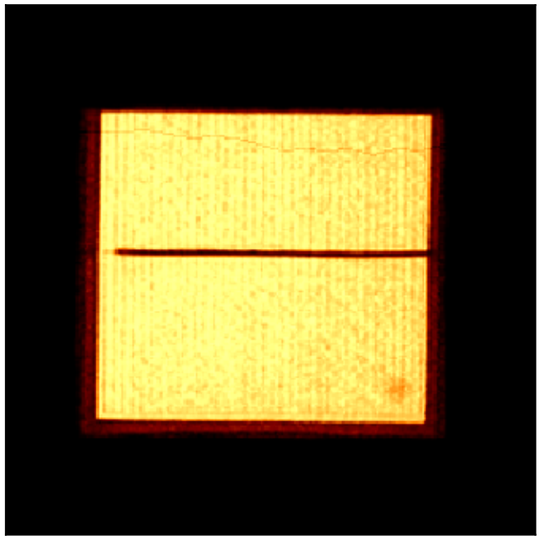


GS-PnP Tune-1 (0008)
PSNR 29.09 dB | SSIM 0.7289
HaarPSI 0.7640 | MSE 1.23e-03

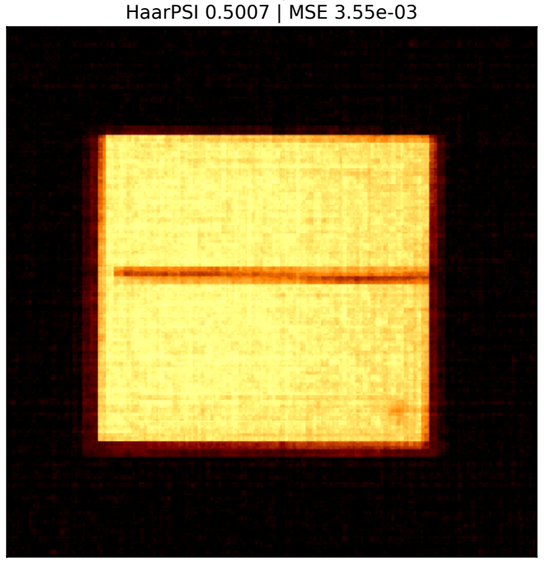


Reference | test image 0793 | 20% Hadamard + NPL-default noise

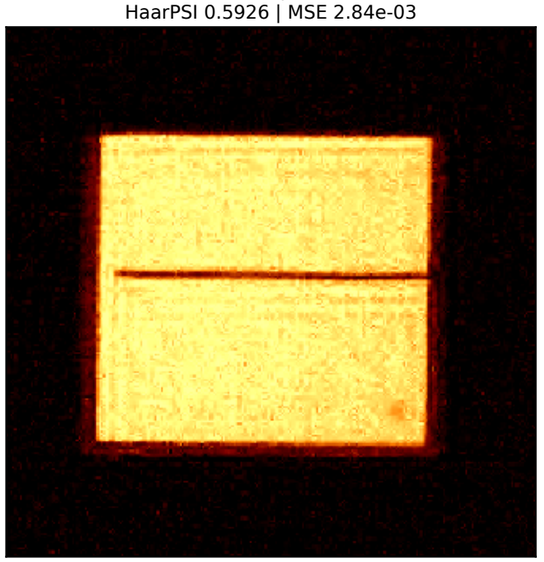
Target
Ground truth



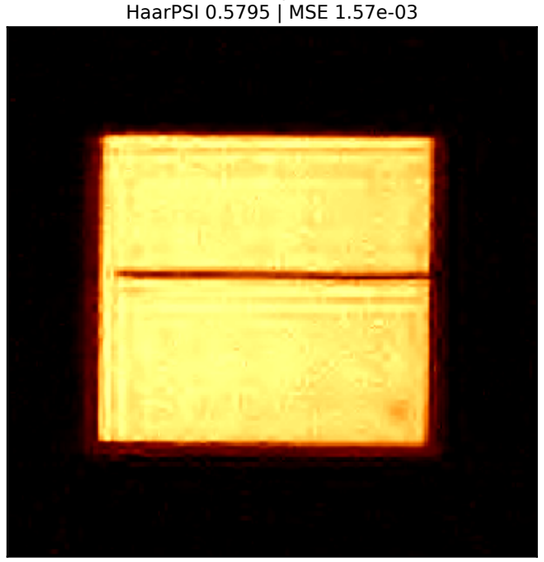
Zero-filled
PSNR 24.50 dB | SSIM 0.4158
HaarPSI 0.5007 | MSE 3.55e-03



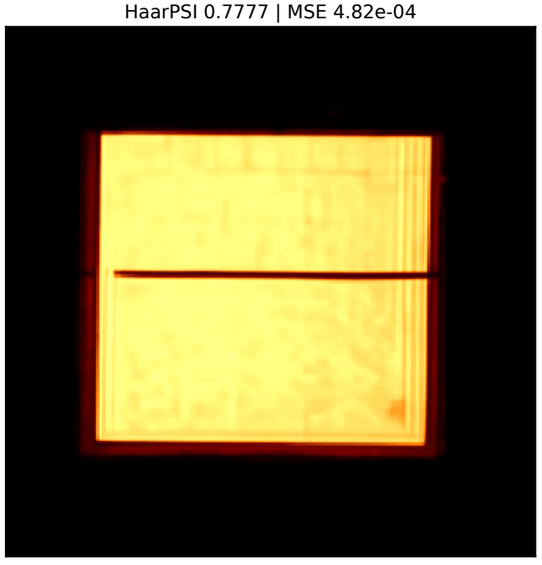
SPGL1-BP
PSNR 25.47 dB | SSIM 0.3849
HaarPSI 0.5926 | MSE 2.84e-03



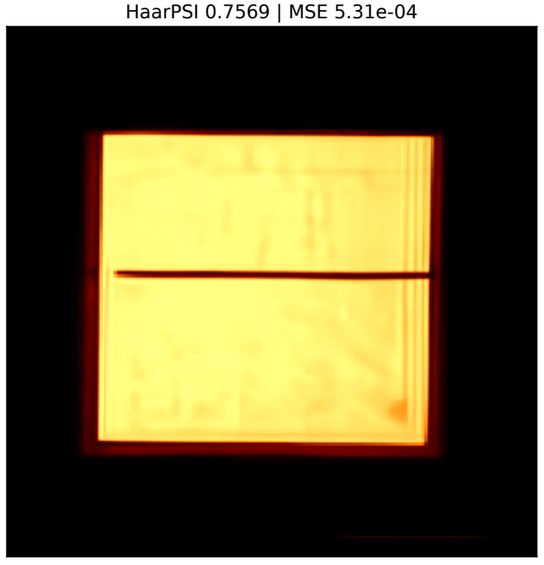
SPGL1-BPDN
PSNR 28.04 dB | SSIM 0.6686
HaarPSI 0.5795 | MSE 1.57e-03



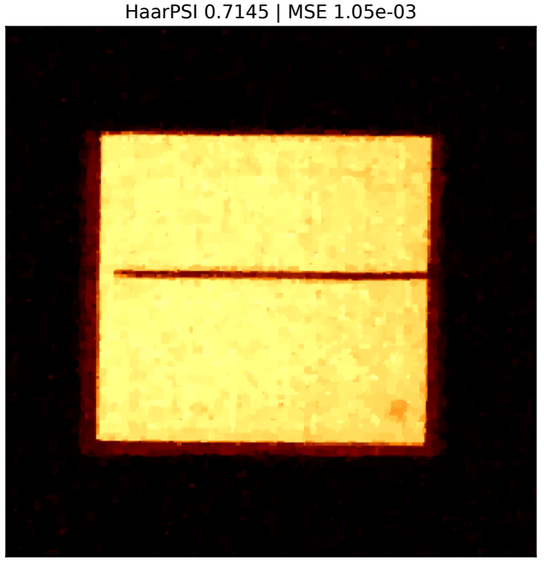
PnP-ADMM tuned-5
PSNR 33.17 dB | SSIM 0.8545
HaarPSI 0.7777 | MSE 4.82e-04



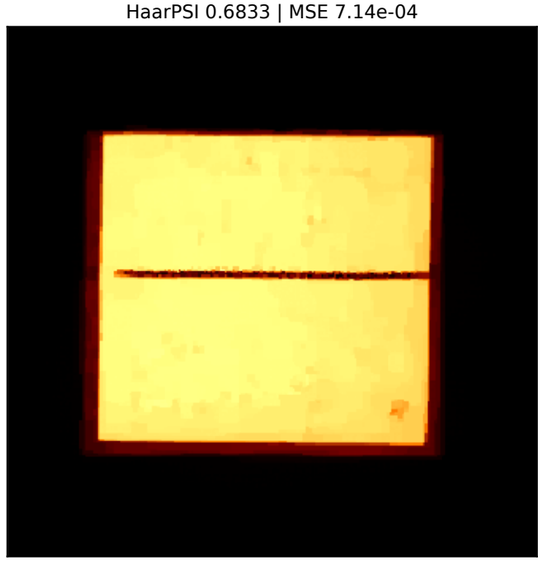
GS-PnP oracle
PSNR 32.75 dB | SSIM 0.8305
HaarPSI 0.7569 | MSE 5.31e-04



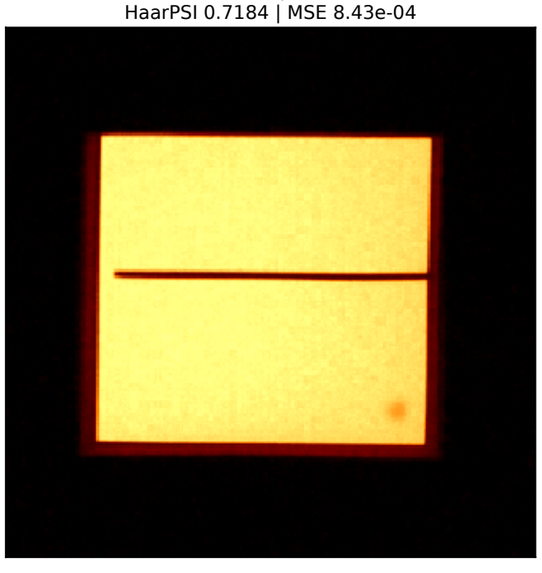
Scalar TV
PSNR 29.78 dB | SSIM 0.6341
HaarPSI 0.7145 | MSE 1.05e-03



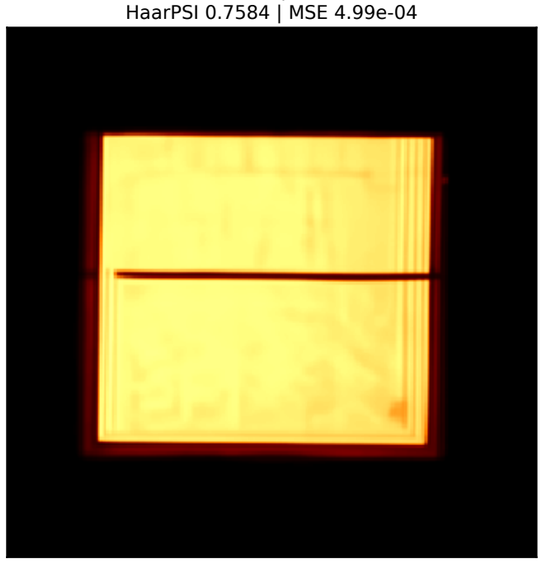
UTV-C60
PSNR 31.46 dB | SSIM 0.7817
HaarPSI 0.6833 | MSE 7.14e-04



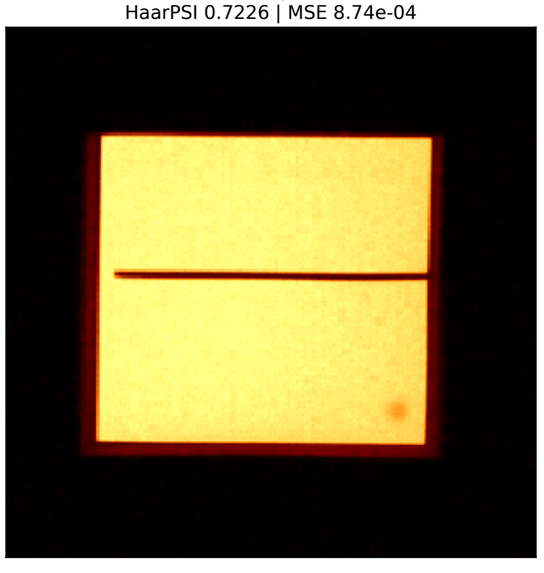
GS-PnP tuned-5
PSNR 30.74 dB | SSIM 0.6793
HaarPSI 0.7184 | MSE 8.43e-04



PnP-ADMM Tune-1 (0008)
PSNR 33.02 dB | SSIM 0.8497
HaarPSI 0.7584 | MSE 4.99e-04

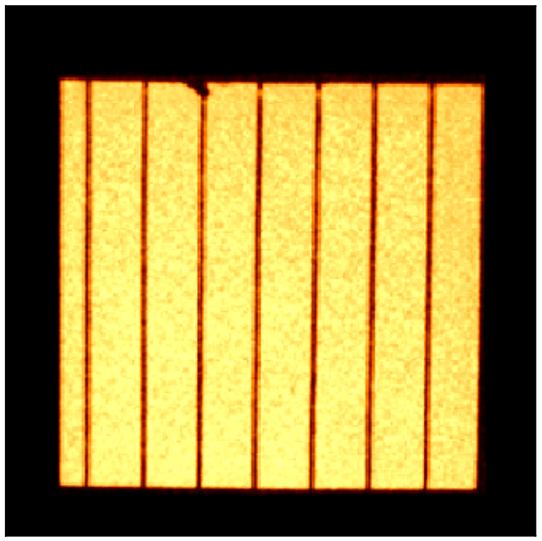


GS-PnP Tune-1 (0008)
PSNR 30.58 dB | SSIM 0.6602
HaarPSI 0.7226 | MSE 8.74e-04

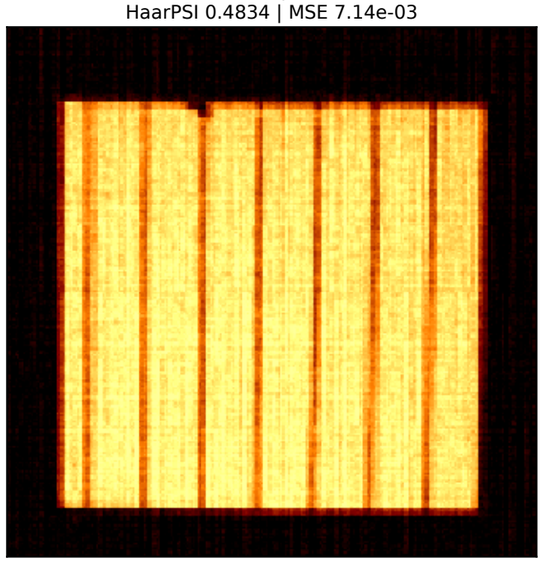


Reference | test image 0946 | 20% Hadamard + NPL-default noise

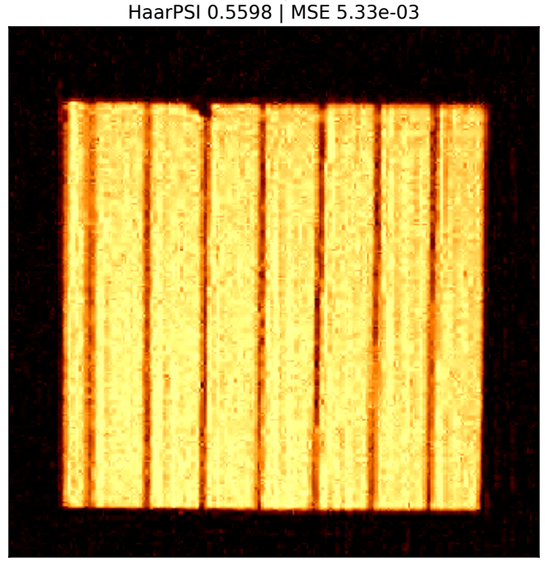
Target
Ground truth



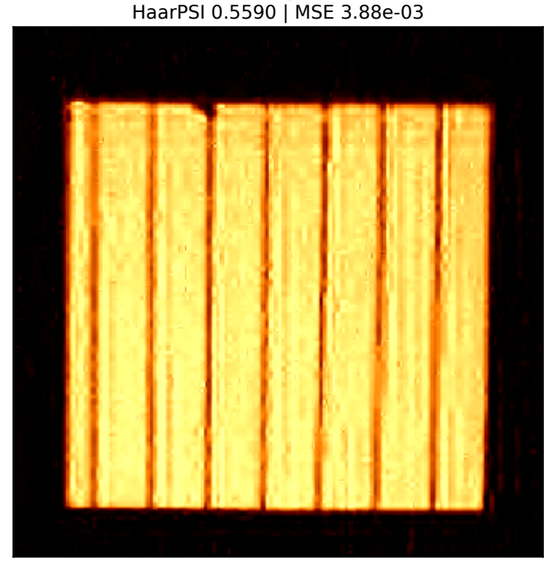
Zero-filled
PSNR 21.46 dB | SSIM 0.4323
HaarPSI 0.4834 | MSE 7.14e-03



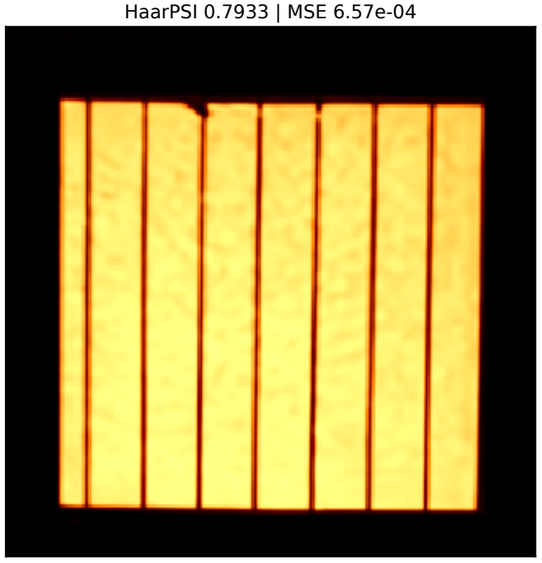
SPGL1-BP
PSNR 22.73 dB | SSIM 0.4447
HaarPSI 0.5598 | MSE 5.33e-03



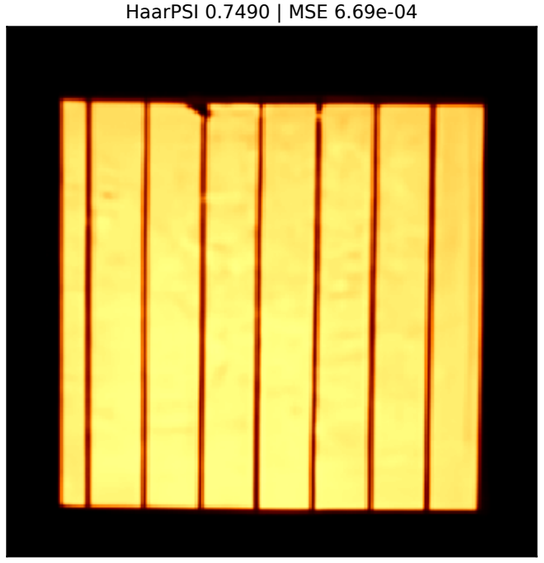
SPGL1-BPDN
PSNR 24.11 dB | SSIM 0.5850
HaarPSI 0.5590 | MSE 3.88e-03



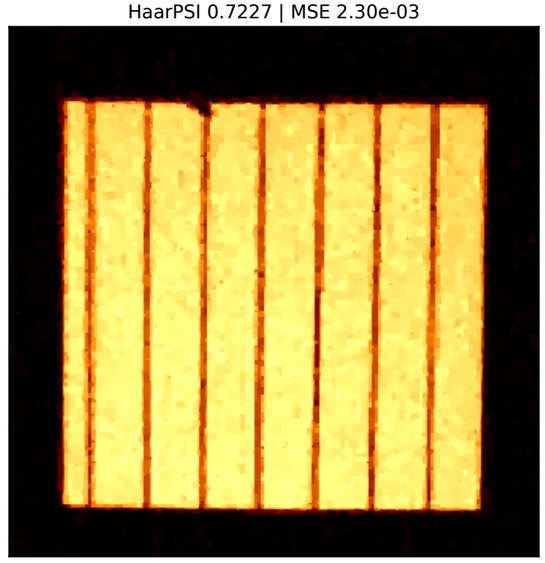
PnP-ADMM tuned-5
PSNR 31.82 dB | SSIM 0.8418
HaarPSI 0.7933 | MSE 6.57e-04



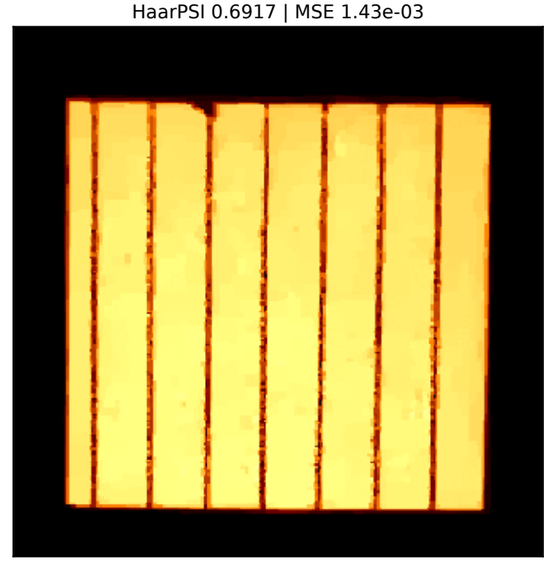
GS-PnP oracle
PSNR 31.75 dB | SSIM 0.8258
HaarPSI 0.7490 | MSE 6.69e-04



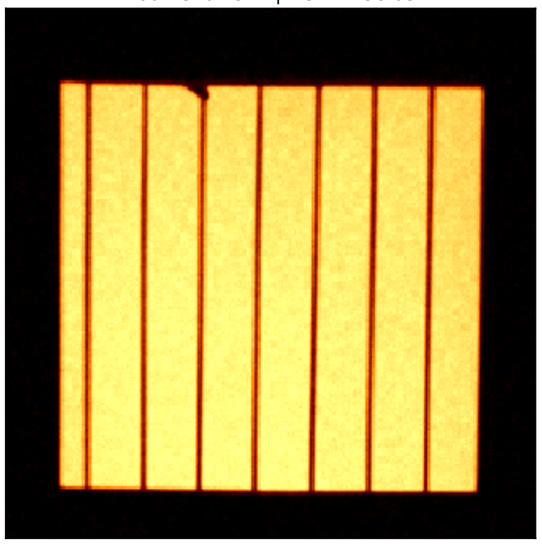
Scalar TV
PSNR 26.38 dB | SSIM 0.6850
HaarPSI 0.7227 | MSE 2.30e-03



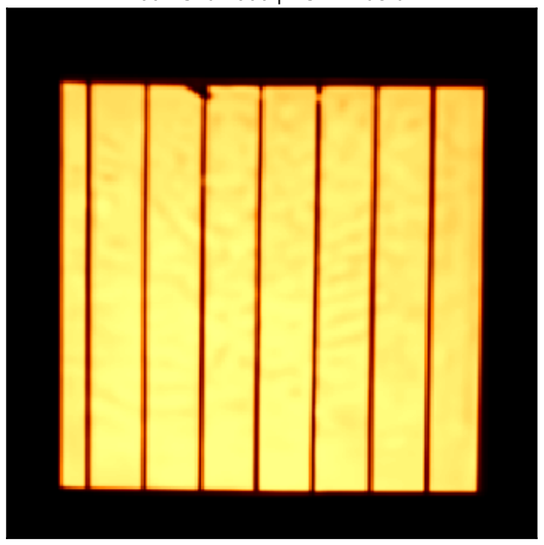
UTV-C60
PSNR 28.45 dB | SSIM 0.7967
HaarPSI 0.6917 | MSE 1.43e-03



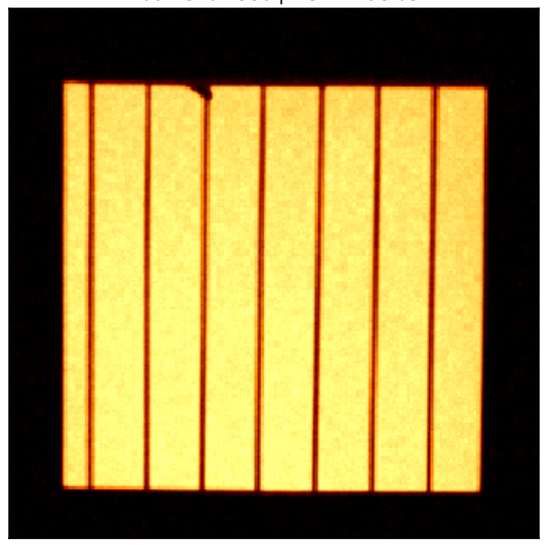
GS-PnP tuned-5
PSNR 29.03 dB | SSIM 0.6405
HaarPSI 0.7571 | MSE 1.25e-03



PnP-ADMM Tune-1 (0008)
PSNR 31.43 dB | SSIM 0.8340
HaarPSI 0.7666 | MSE 7.20e-04

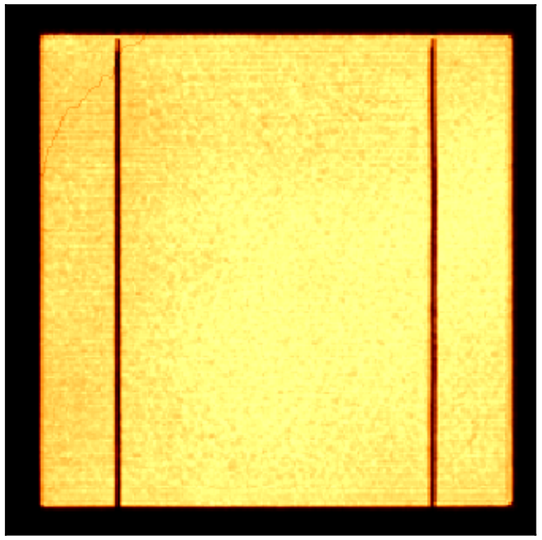


GS-PnP Tune-1 (0008)
PSNR 28.90 dB | SSIM 0.6268
HaarPSI 0.7586 | MSE 1.29e-03

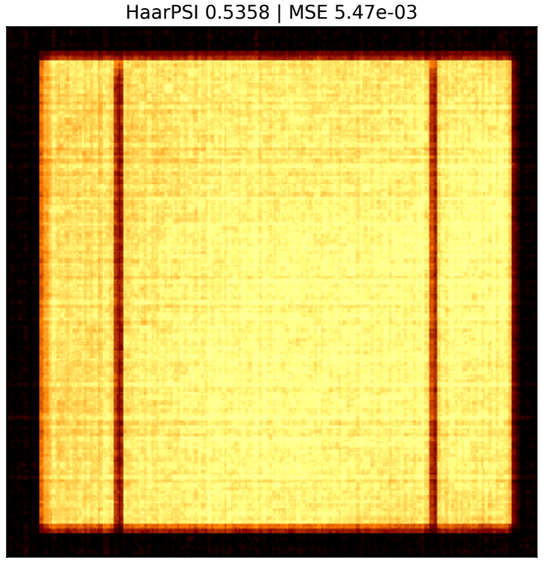


Silicone | test image 0098 | 20% Hadamard + NPL-default noise

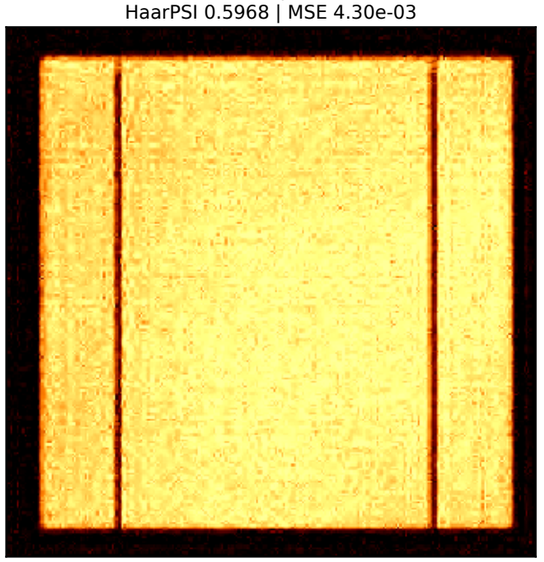
Target
Ground truth



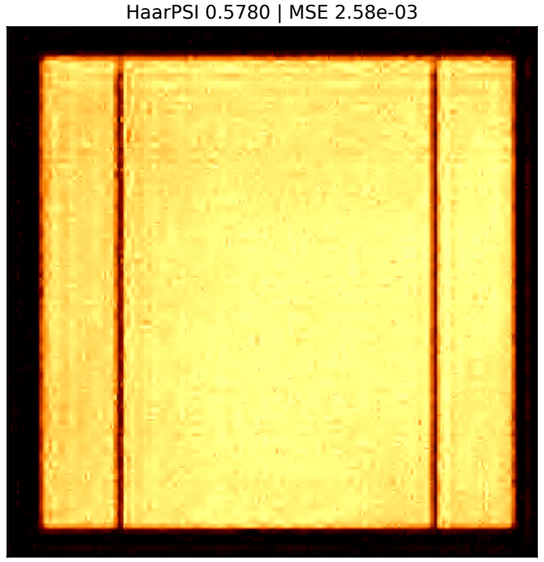
Zero-filled
PSNR 22.62 dB | SSIM 0.4408
HaarPSI 0.5358 | MSE 5.47e-03



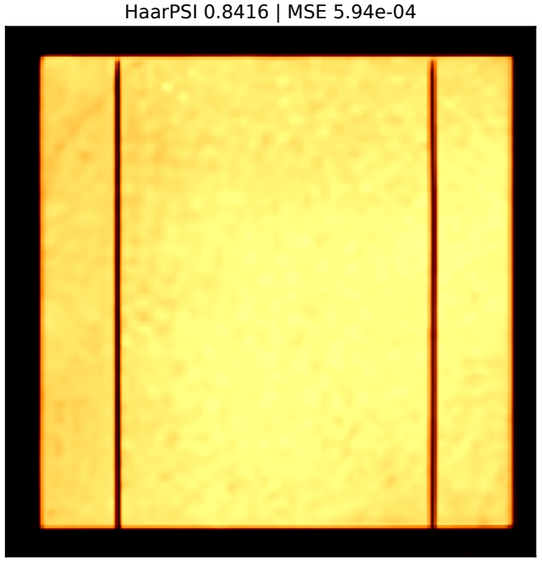
SPGL1-BP
PSNR 23.67 dB | SSIM 0.4082
HaarPSI 0.5968 | MSE 4.30e-03



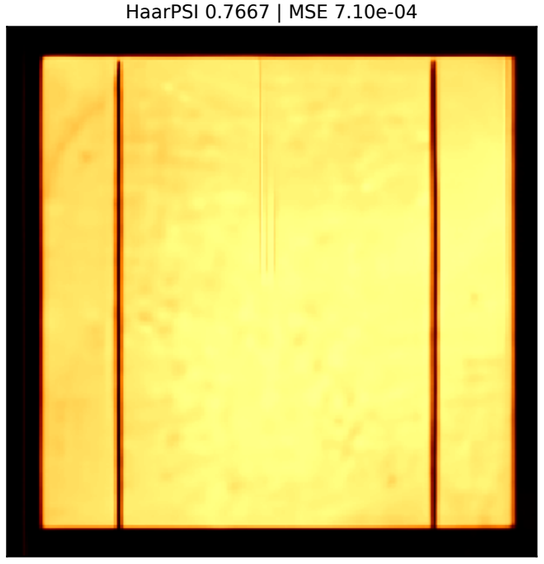
SPGL1-BPDN
PSNR 25.89 dB | SSIM 0.5728
HaarPSI 0.5780 | MSE 2.58e-03



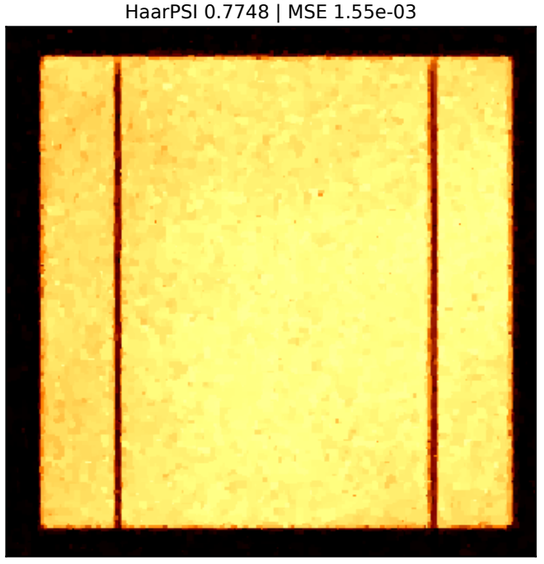
PnP-ADMM tuned-5
PSNR 32.26 dB | SSIM 0.7482
HaarPSI 0.8416 | MSE 5.94e-04



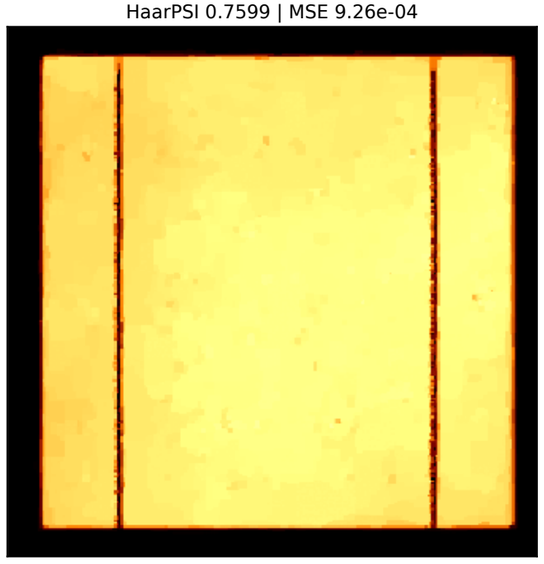
GS-PnP oracle
PSNR 31.48 dB | SSIM 0.7087
HaarPSI 0.7667 | MSE 7.10e-04



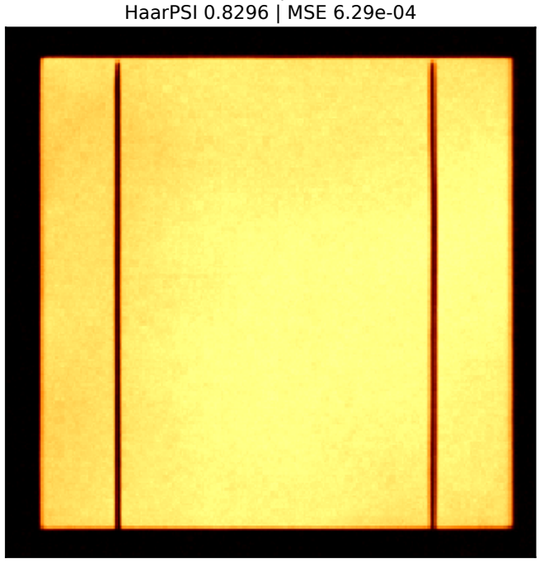
Scalar TV
PSNR 28.09 dB | SSIM 0.6475
HaarPSI 0.7748 | MSE 1.55e-03



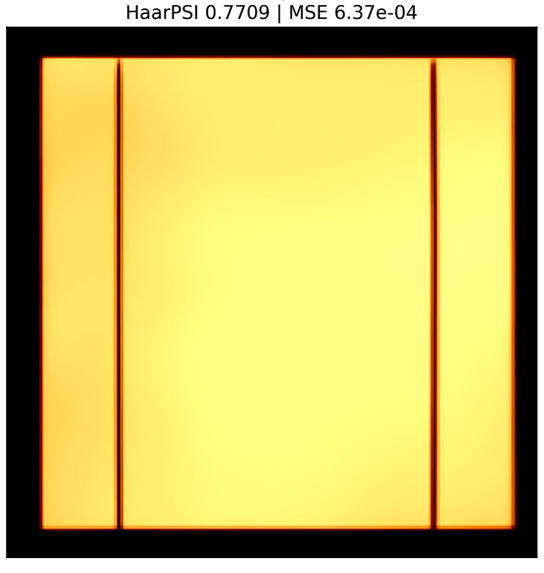
UTV-C60
PSNR 30.33 dB | SSIM 0.7334
HaarPSI 0.7599 | MSE 9.26e-04



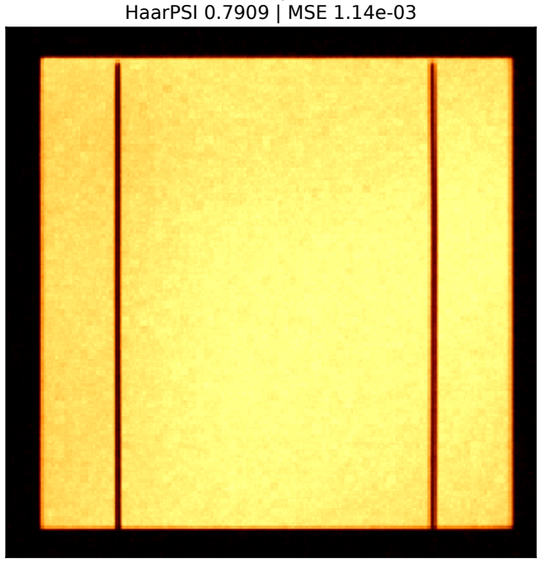
GS-PnP tuned-5
PSNR 32.01 dB | SSIM 0.7224
HaarPSI 0.8296 | MSE 6.29e-04



PnP-ADMM Tune-1 (0008)
PSNR 31.96 dB | SSIM 0.7248
HaarPSI 0.7709 | MSE 6.37e-04

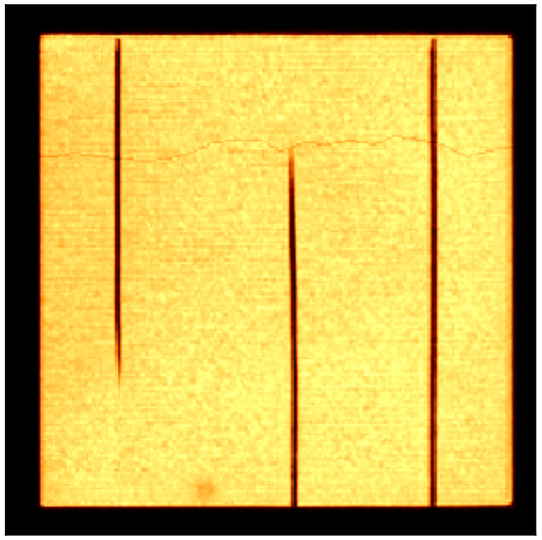


GS-PnP Tune-1 (0008)
PSNR 29.41 dB | SSIM 0.5515
HaarPSI 0.7909 | MSE 1.14e-03

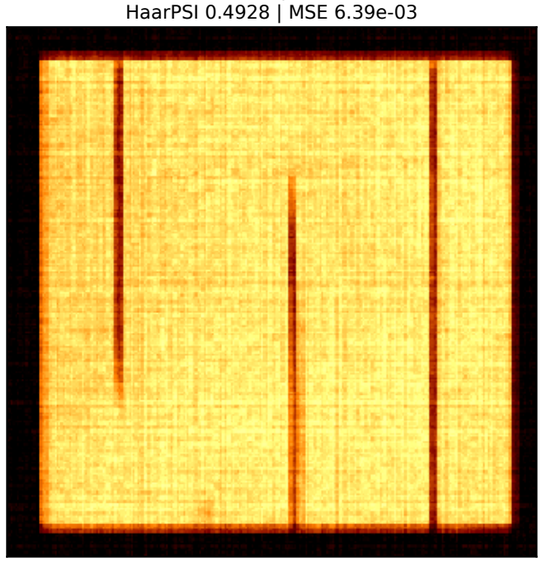


Silicone | test image 0116 | 20% Hadamard + NPL-default noise

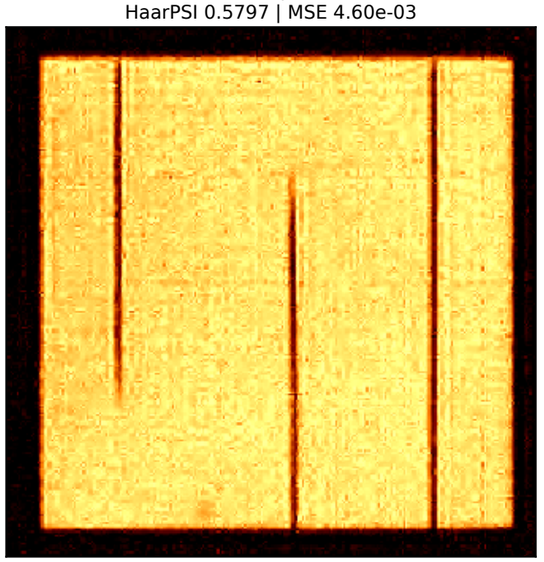
Target
Ground truth



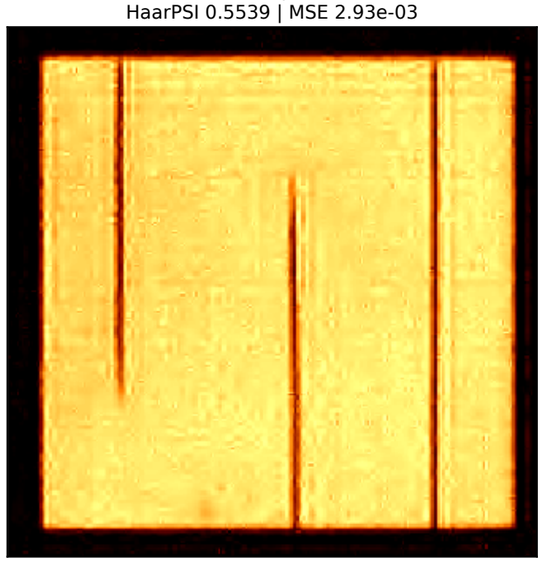
Zero-filled
PSNR 21.95 dB | SSIM 0.4232
HaarPSI 0.4928 | MSE 6.39e-03



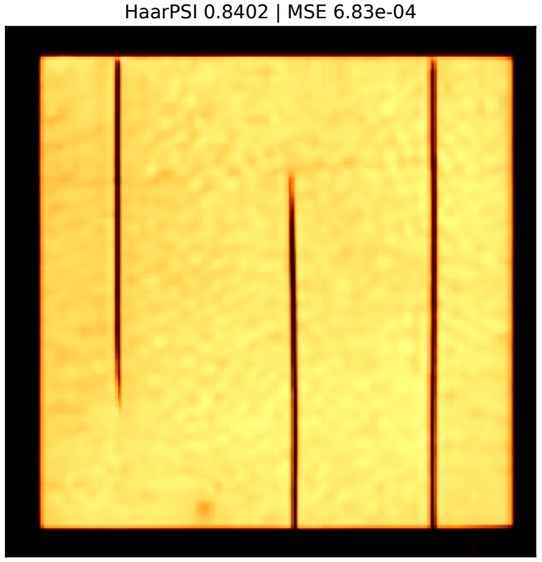
SPGL1-BP
PSNR 23.37 dB | SSIM 0.4157
HaarPSI 0.5797 | MSE 4.60e-03



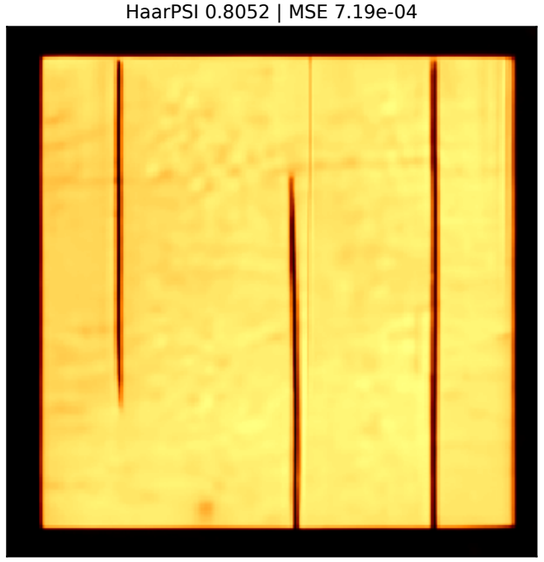
SPGL1-BPDN
PSNR 25.32 dB | SSIM 0.5607
HaarPSI 0.5539 | MSE 2.93e-03



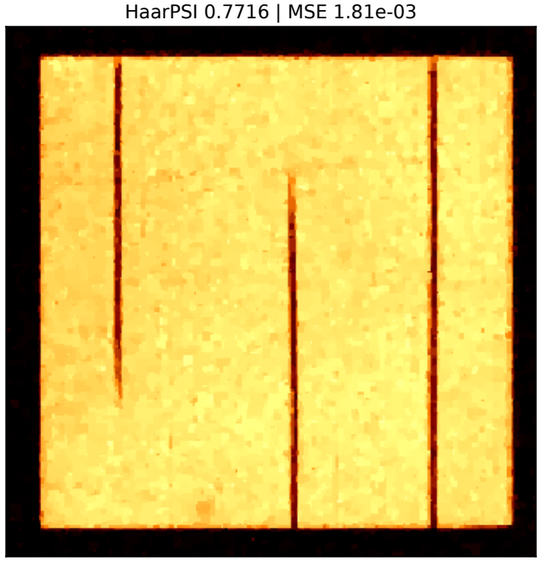
PnP-ADMM tuned-5
PSNR 31.66 dB | SSIM 0.7374
HaarPSI 0.8402 | MSE 6.83e-04



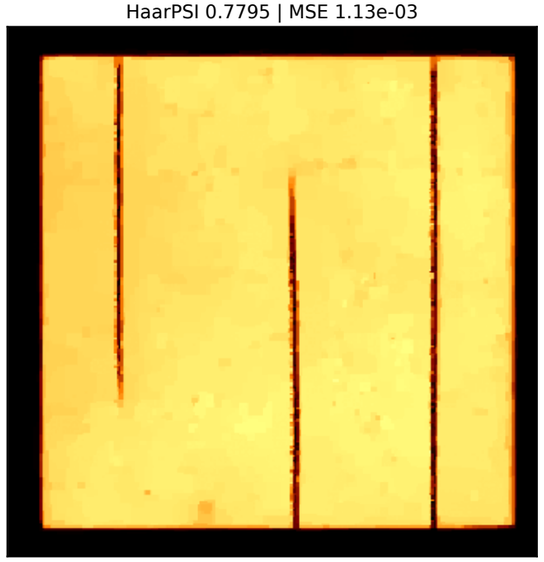
GS-PnP oracle
PSNR 31.43 dB | SSIM 0.7215
HaarPSI 0.8052 | MSE 7.19e-04



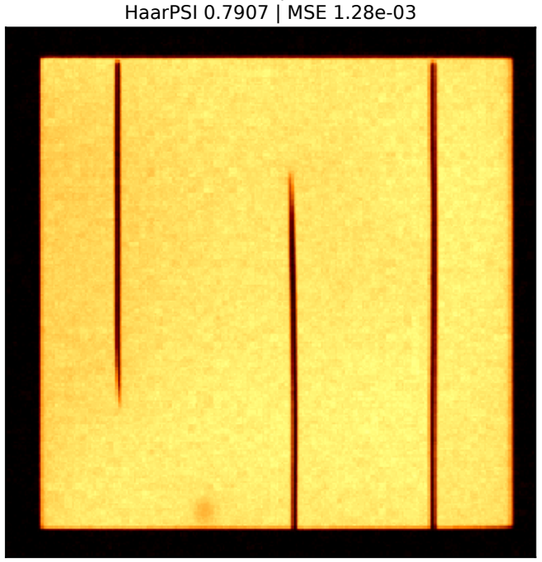
Scalar TV
PSNR 27.42 dB | SSIM 0.6457
HaarPSI 0.7716 | MSE 1.81e-03



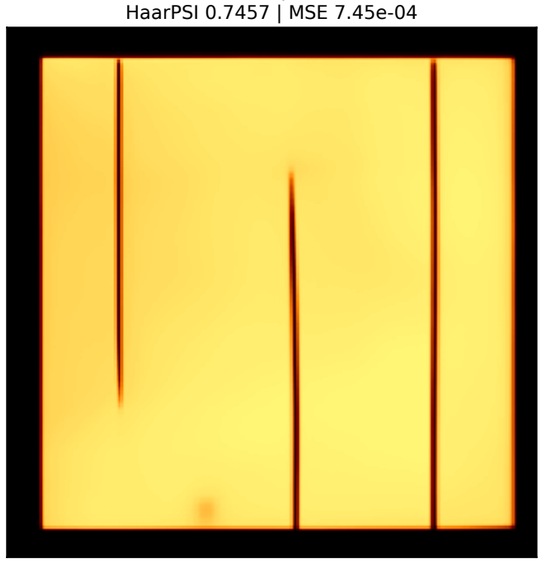
UTV-C60
PSNR 29.48 dB | SSIM 0.7070
HaarPSI 0.7795 | MSE 1.13e-03



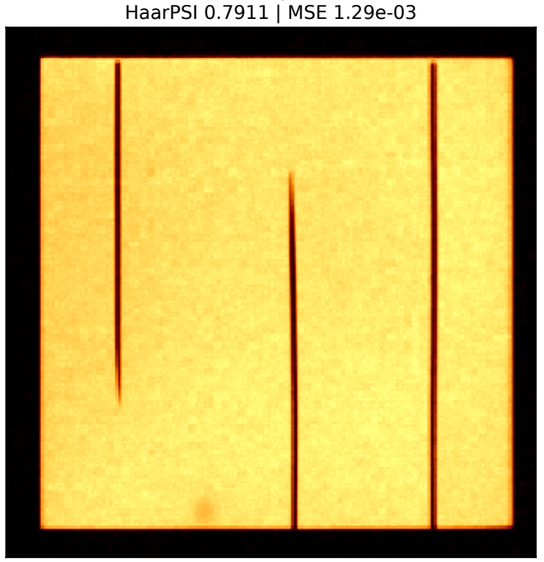
GS-PnP tuned-5
PSNR 28.92 dB | SSIM 0.5245
HaarPSI 0.7907 | MSE 1.28e-03



PnP-ADMM Tune-1 (0008)
PSNR 31.28 dB | SSIM 0.7065
HaarPSI 0.7457 | MSE 7.45e-04

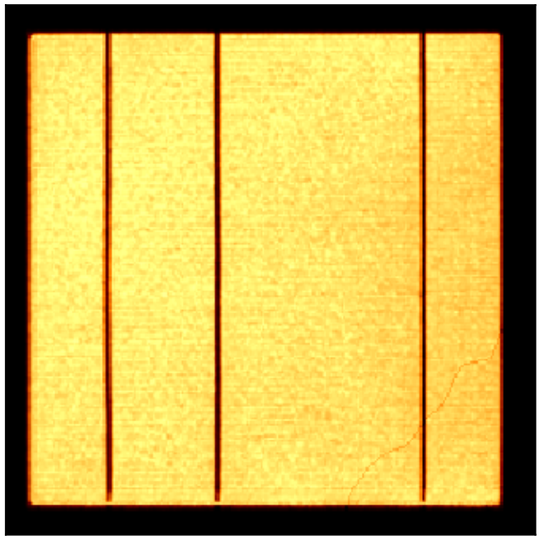


GS-PnP Tune-1 (0008)
PSNR 28.90 dB | SSIM 0.5344
HaarPSI 0.7911 | MSE 1.29e-03

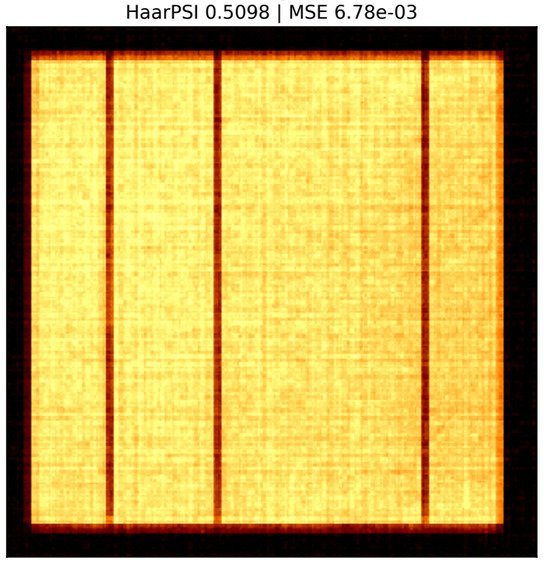


Silicone | test image 0188 | 20% Hadamard + NPL-default noise

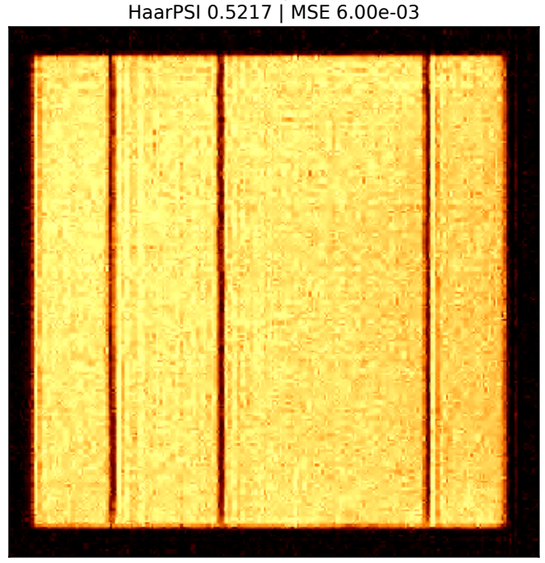
Target
Ground truth



Zero-filled
PSNR 21.69 dB | SSIM 0.4455
HaarPSI 0.5098 | MSE 6.78e-03



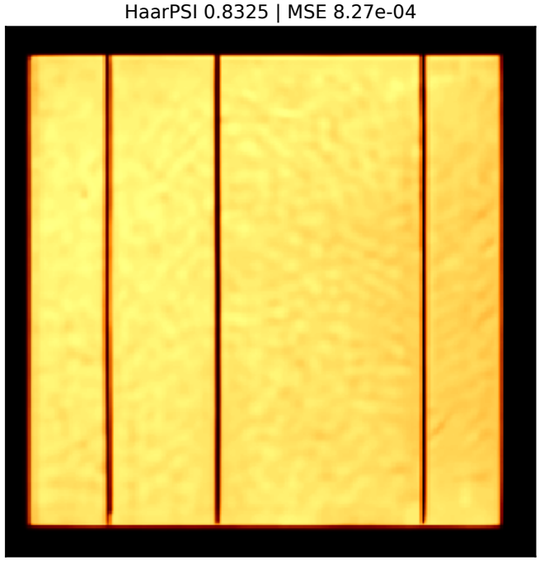
SPGL1-BP
PSNR 22.22 dB | SSIM 0.4144
HaarPSI 0.5217 | MSE 6.00e-03



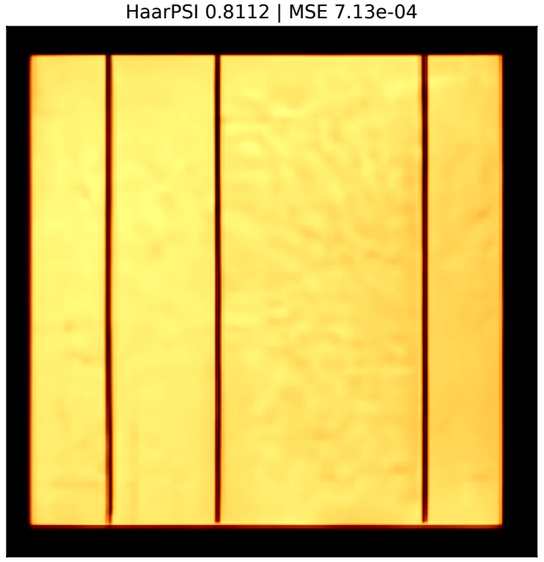
SPGL1-BPDN
PSNR 23.60 dB | SSIM 0.5557
HaarPSI 0.5186 | MSE 4.36e-03



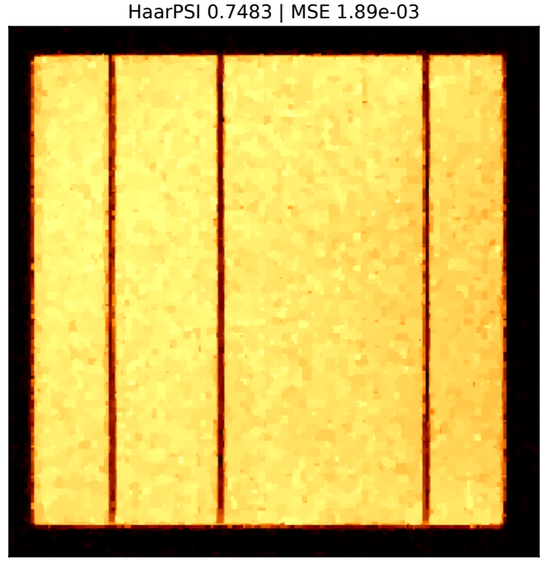
PnP-ADMM tuned-5
PSNR 30.83 dB | SSIM 0.7401
HaarPSI 0.8325 | MSE 8.27e-04



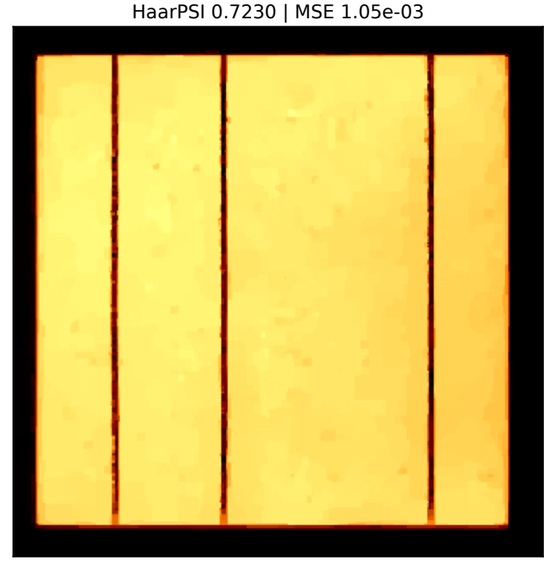
GS-PnP oracle
PSNR 31.47 dB | SSIM 0.7305
HaarPSI 0.8112 | MSE 7.13e-04



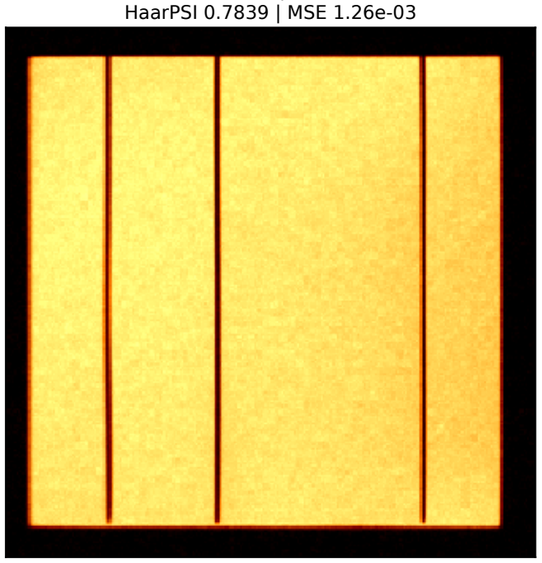
Scalar TV
PSNR 27.24 dB | SSIM 0.6523
HaarPSI 0.7483 | MSE 1.89e-03



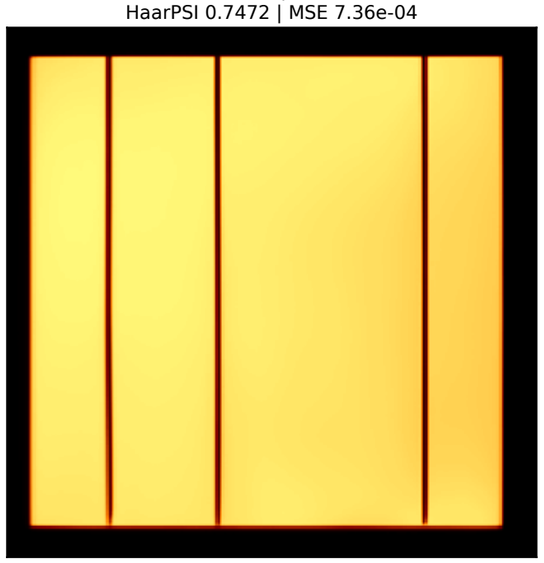
UTV-C60
PSNR 29.77 dB | SSIM 0.7103
HaarPSI 0.7230 | MSE 1.05e-03



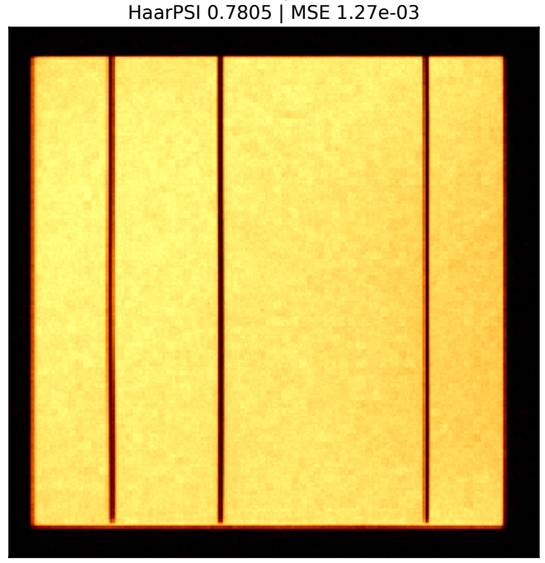
GS-PnP tuned-5
PSNR 29.00 dB | SSIM 0.5386
HaarPSI 0.7839 | MSE 1.26e-03



PnP-ADMM Tune-1 (0008)
PSNR 31.33 dB | SSIM 0.7120
HaarPSI 0.7472 | MSE 7.36e-04

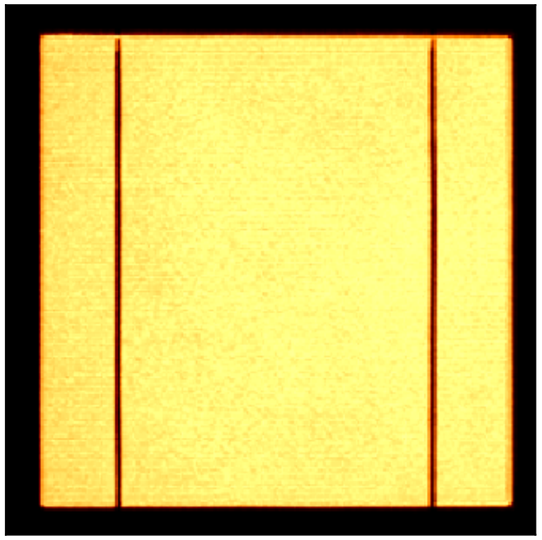


GS-PnP Tune-1 (0008)
PSNR 28.96 dB | SSIM 0.5480
HaarPSI 0.7805 | MSE 1.27e-03

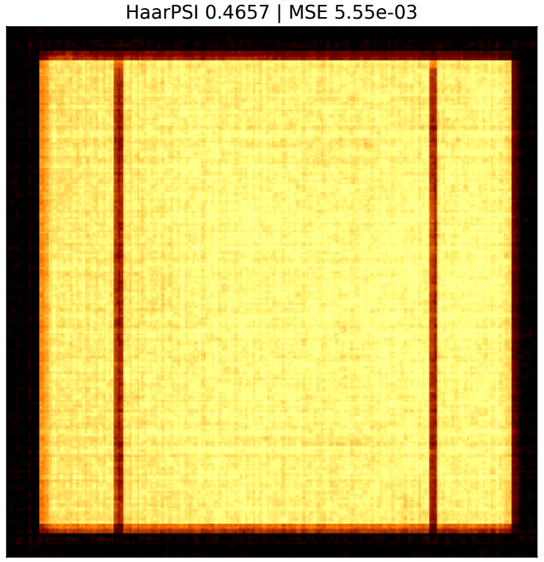


Silicone | test image 0260 | 20% Hadamard + NPL-default noise

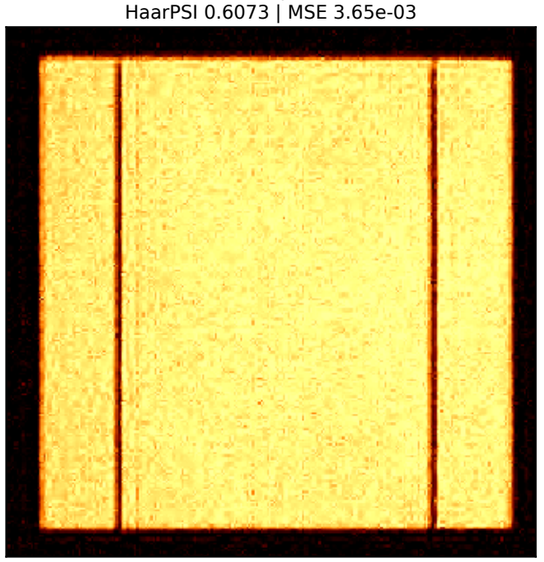
Target
Ground truth



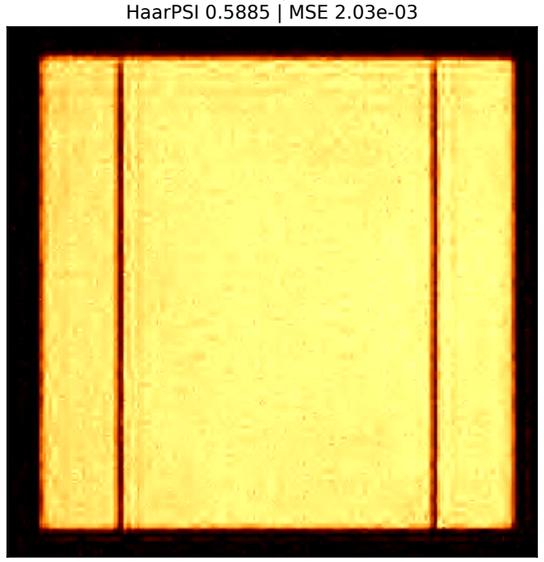
Zero-filled
PSNR 22.55 dB | SSIM 0.4479
HaarPSI 0.4657 | MSE 5.55e-03



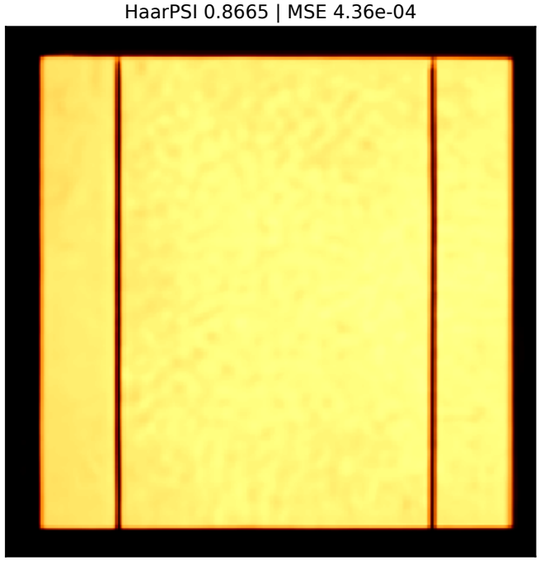
SPGL1-BP
PSNR 24.37 dB | SSIM 0.4230
HaarPSI 0.6073 | MSE 3.65e-03



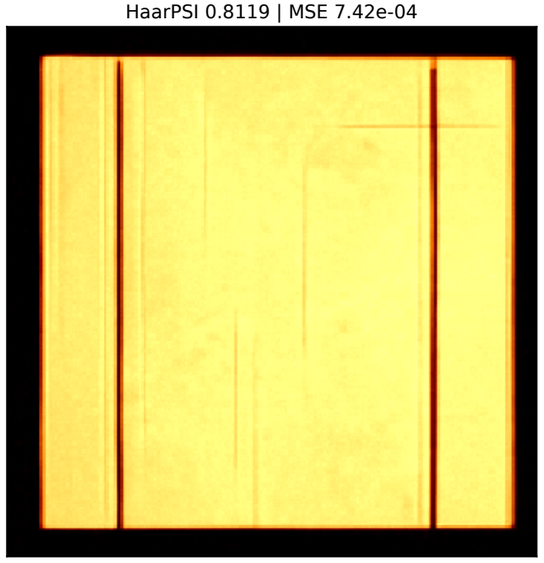
SPGL1-BPDN
PSNR 26.93 dB | SSIM 0.6356
HaarPSI 0.5885 | MSE 2.03e-03



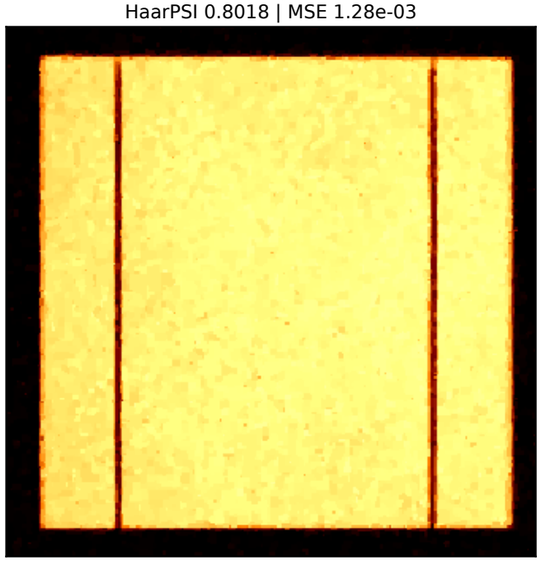
PnP-ADMM tuned-5
PSNR 33.60 dB | SSIM 0.7958
HaarPSI 0.8665 | MSE 4.36e-04



GS-PnP oracle
PSNR 31.30 dB | SSIM 0.7438
HaarPSI 0.8119 | MSE 7.42e-04



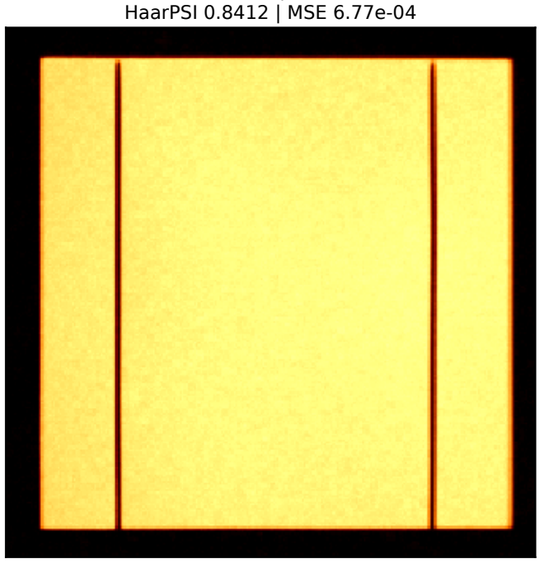
Scalar TV
PSNR 28.92 dB | SSIM 0.6788
HaarPSI 0.8018 | MSE 1.28e-03



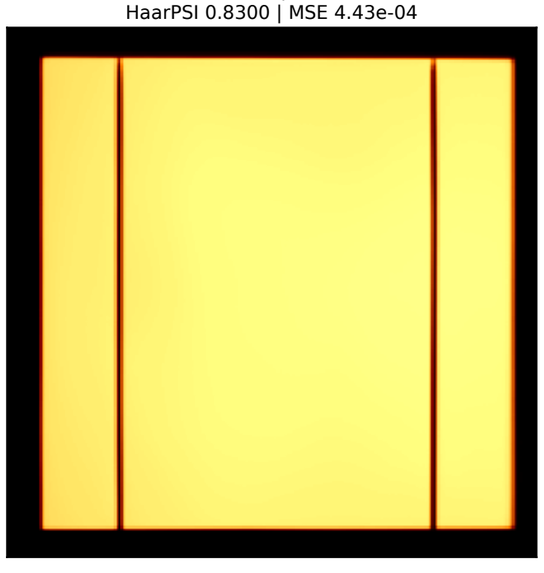
UTV-C60
PSNR 31.13 dB | SSIM 0.7800
HaarPSI 0.8106 | MSE 7.72e-04



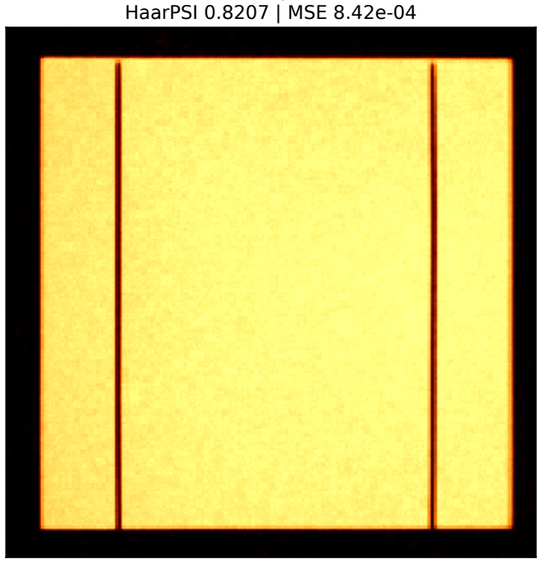
GS-PnP tuned-5
PSNR 31.69 dB | SSIM 0.6709
HaarPSI 0.8412 | MSE 6.77e-04



PnP-ADMM Tune-1 (0008)
PSNR 33.53 dB | SSIM 0.7902
HaarPSI 0.8300 | MSE 4.43e-04

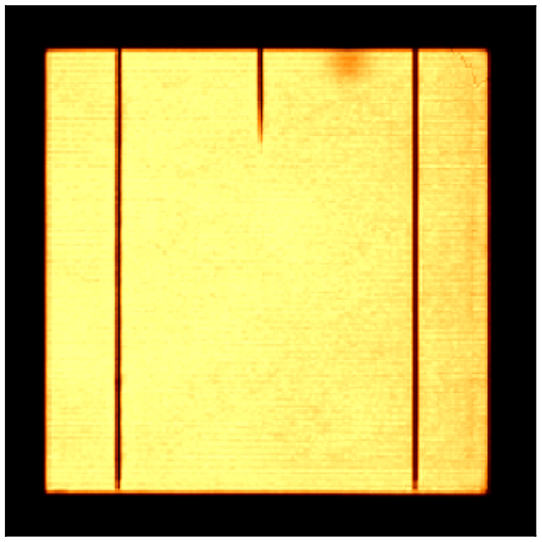


GS-PnP Tune-1 (0008)
PSNR 30.75 dB | SSIM 0.6267
HaarPSI 0.8207 | MSE 8.42e-04

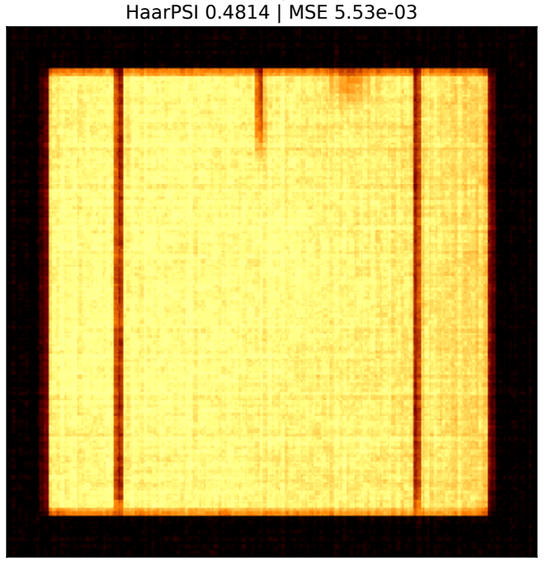


Silicone | test image 0507 | 20% Hadamard + NPL-default noise

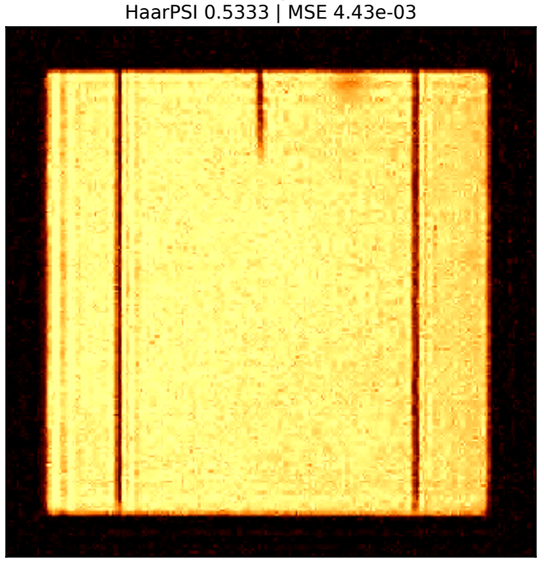
Target
Ground truth



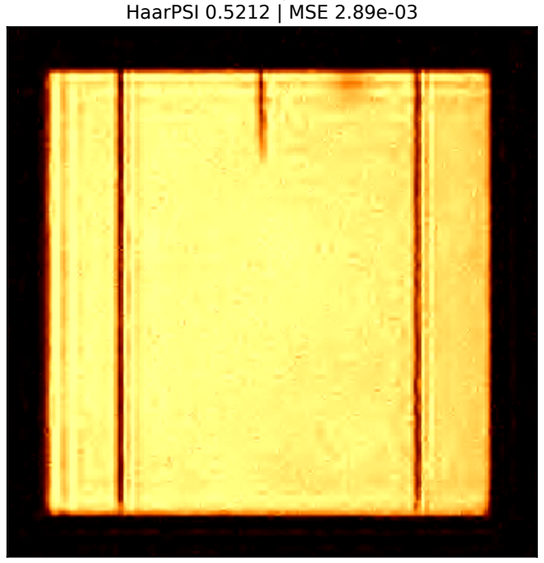
Zero-filled
PSNR 22.57 dB | SSIM 0.4282
HaarPSI 0.4814 | MSE 5.53e-03



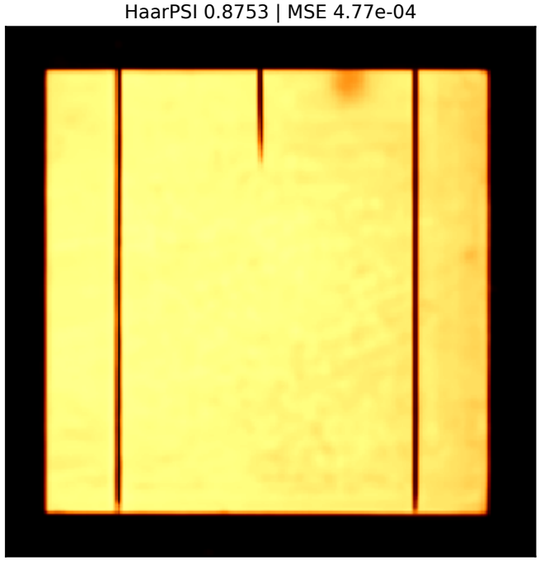
SPGL1-BP
PSNR 23.53 dB | SSIM 0.3908
HaarPSI 0.5333 | MSE 4.43e-03



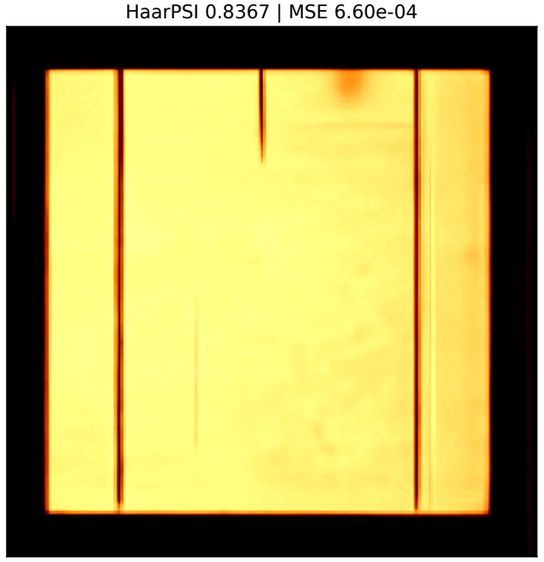
SPGL1-BPDN
PSNR 25.39 dB | SSIM 0.6128
HaarPSI 0.5212 | MSE 2.89e-03



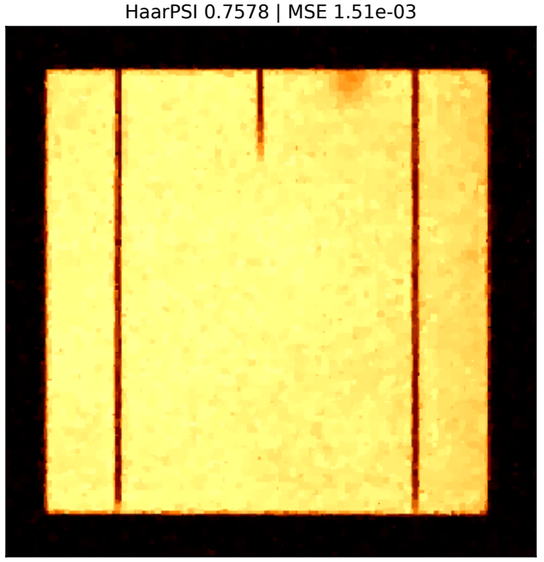
PnP-ADMM tuned-5
PSNR 33.21 dB | SSIM 0.8412
HaarPSI 0.8753 | MSE 4.77e-04



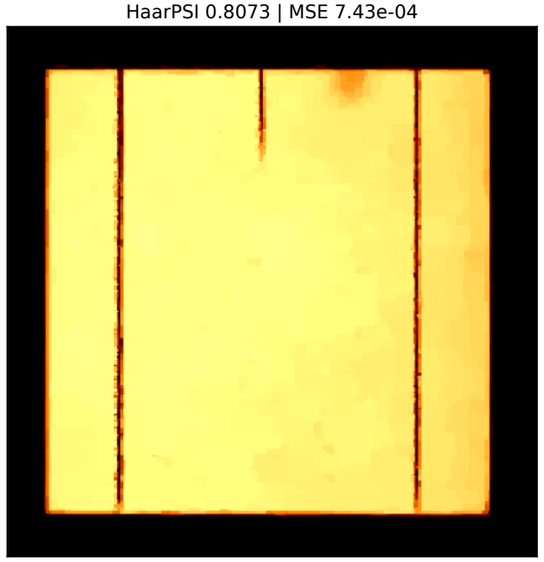
GS-PnP oracle
PSNR 31.81 dB | SSIM 0.8036
HaarPSI 0.8367 | MSE 6.60e-04



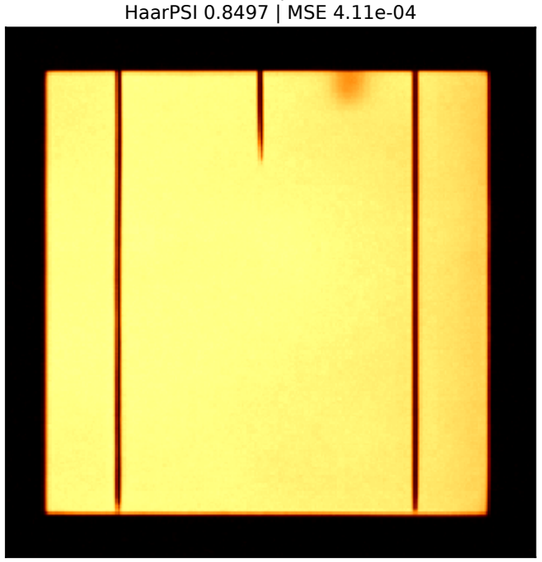
Scalar TV
PSNR 28.22 dB | SSIM 0.6843
HaarPSI 0.7578 | MSE 1.51e-03



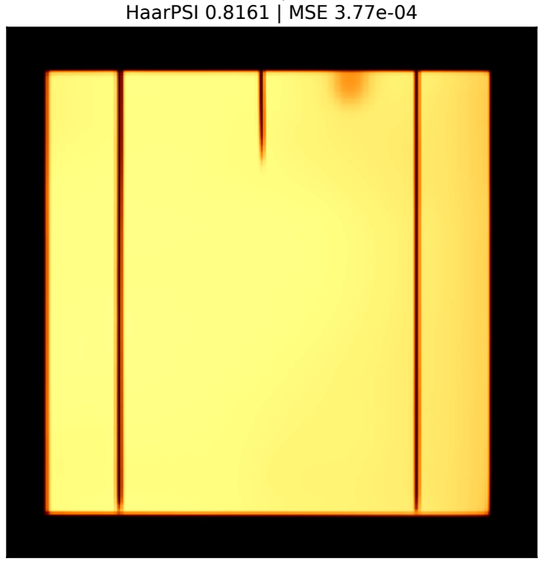
UTV-C60
PSNR 31.29 dB | SSIM 0.8245
HaarPSI 0.8073 | MSE 7.43e-04



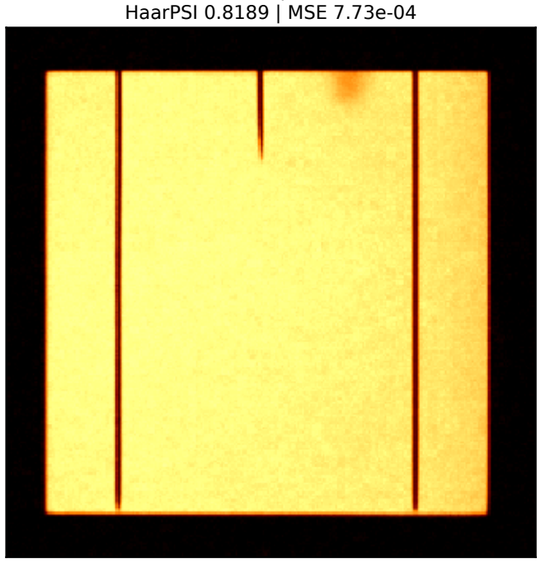
GS-PnP tuned-5
PSNR 33.86 dB | SSIM 0.8223
HaarPSI 0.8497 | MSE 4.11e-04



PnP-ADMM Tune-1 (0008)
PSNR 34.23 dB | SSIM 0.8320
HaarPSI 0.8161 | MSE 3.77e-04

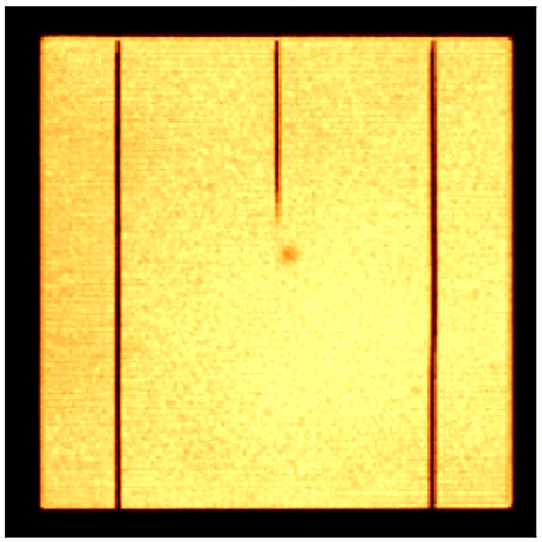


GS-PnP Tune-1 (0008)
PSNR 31.12 dB | SSIM 0.6667
HaarPSI 0.8189 | MSE 7.73e-04

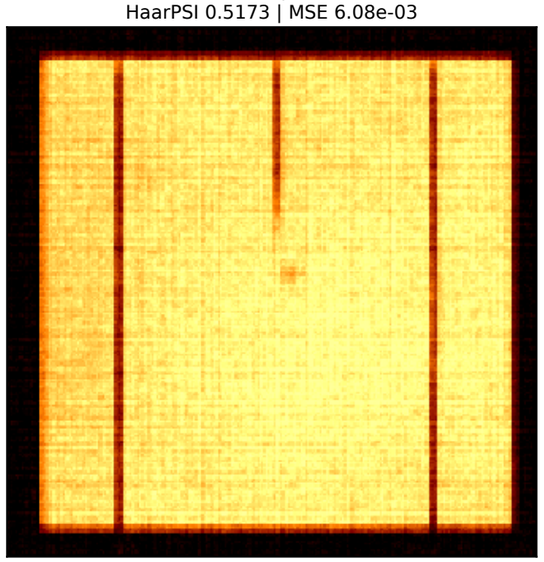


Silicone | test image 0606 | 20% Hadamard + NPL-default noise

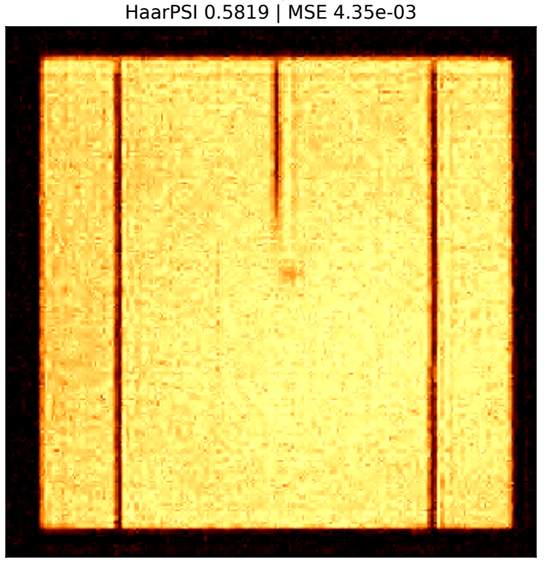
Target
Ground truth



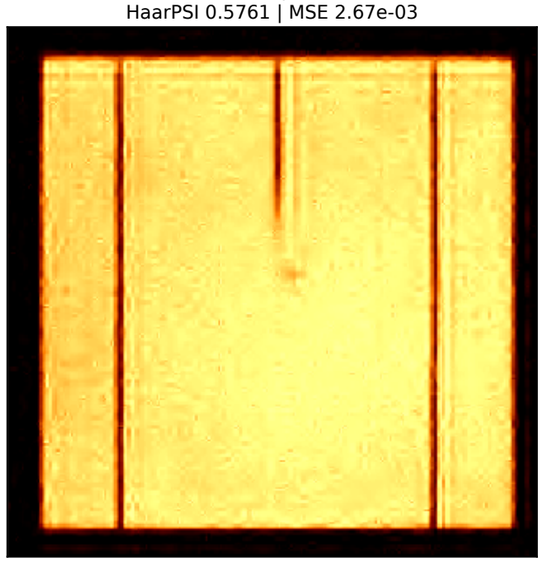
Zero-filled
PSNR 22.16 dB | SSIM 0.4441
HaarPSI 0.5173 | MSE 6.08e-03



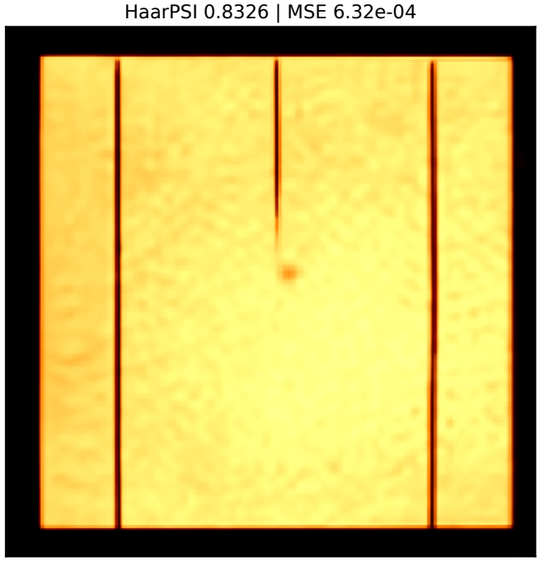
SPGL1-BP
PSNR 23.62 dB | SSIM 0.4212
HaarPSI 0.5819 | MSE 4.35e-03



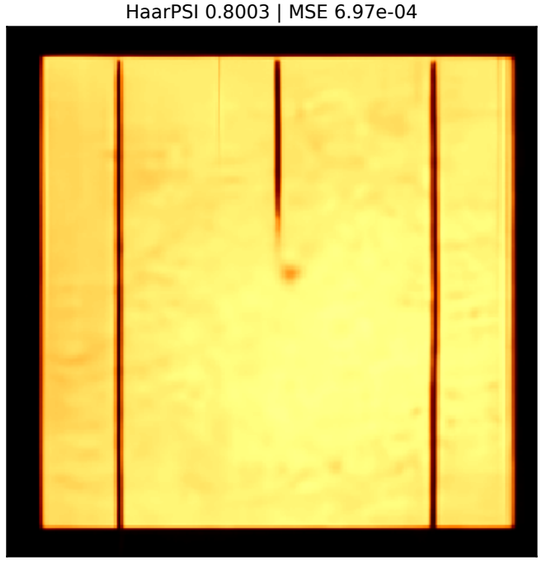
SPGL1-BPDN
PSNR 25.74 dB | SSIM 0.5864
HaarPSI 0.5761 | MSE 2.67e-03



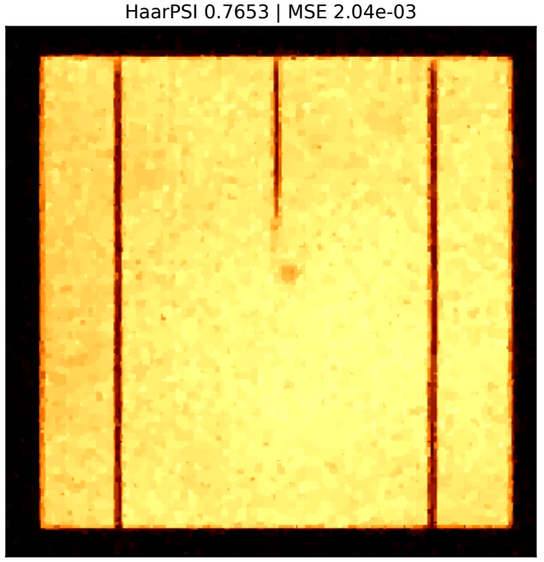
PnP-ADMM tuned-5
PSNR 31.99 dB | SSIM 0.7497
HaarPSI 0.8326 | MSE 6.32e-04



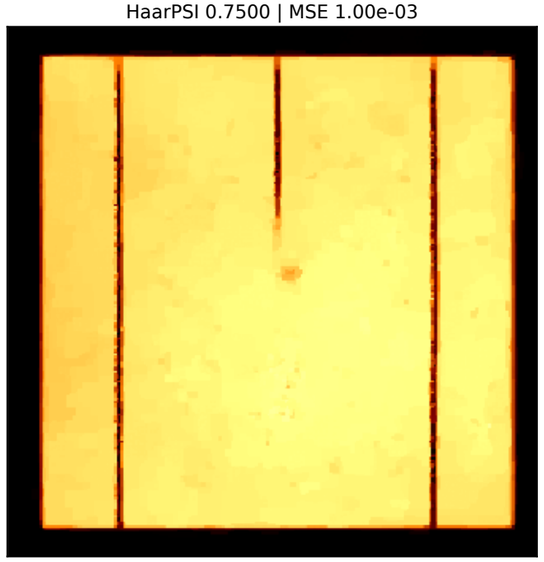
GS-PnP oracle
PSNR 31.57 dB | SSIM 0.7322
HaarPSI 0.8003 | MSE 6.97e-04



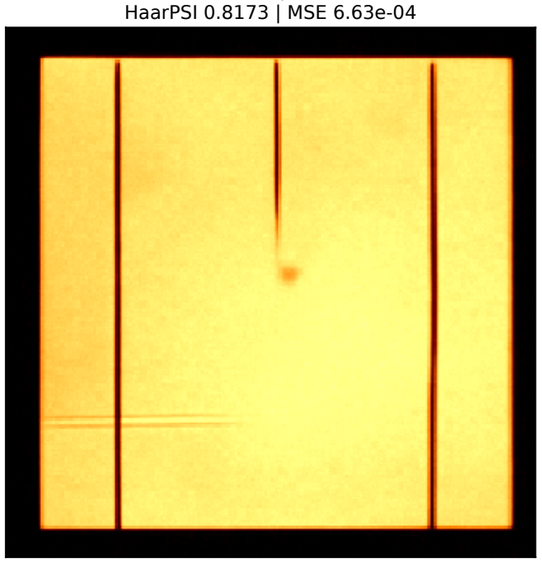
Scalar TV
PSNR 26.91 dB | SSIM 0.6486
HaarPSI 0.7653 | MSE 2.04e-03



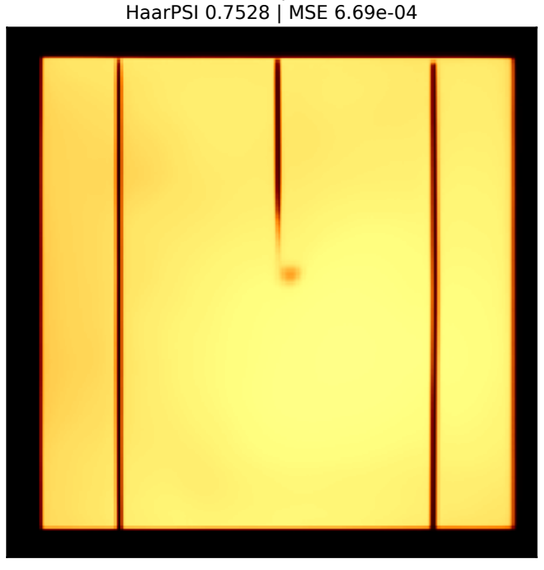
UTV-C60
PSNR 29.99 dB | SSIM 0.7227
HaarPSI 0.7500 | MSE 1.00e-03



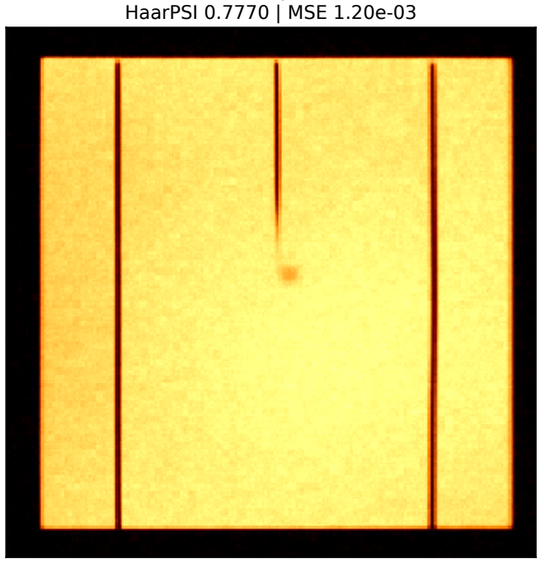
GS-PnP tuned-5
PSNR 31.79 dB | SSIM 0.7280
HaarPSI 0.8173 | MSE 6.63e-04



PnP-ADMM Tune-1 (0008)
PSNR 31.75 dB | SSIM 0.7205
HaarPSI 0.7528 | MSE 6.69e-04

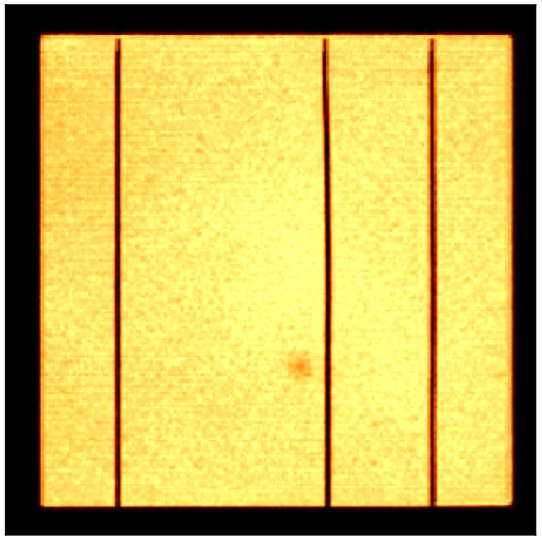


GS-PnP Tune-1 (0008)
PSNR 29.19 dB | SSIM 0.5473
HaarPSI 0.7770 | MSE 1.20e-03

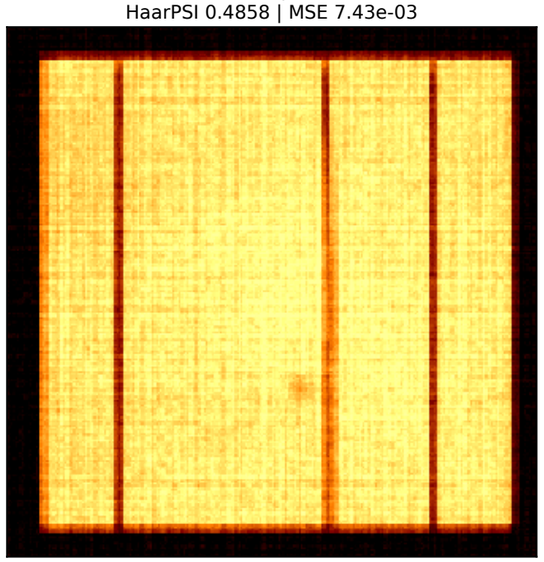


Silicone | test image 0711 | 20% Hadamard + NPL-default noise

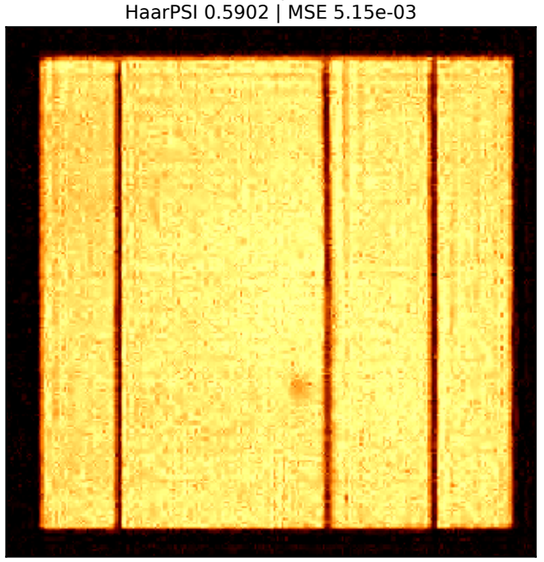
Target
Ground truth



Zero-filled
PSNR 21.29 dB | SSIM 0.4329
HaarPSI 0.4858 | MSE 7.43e-03



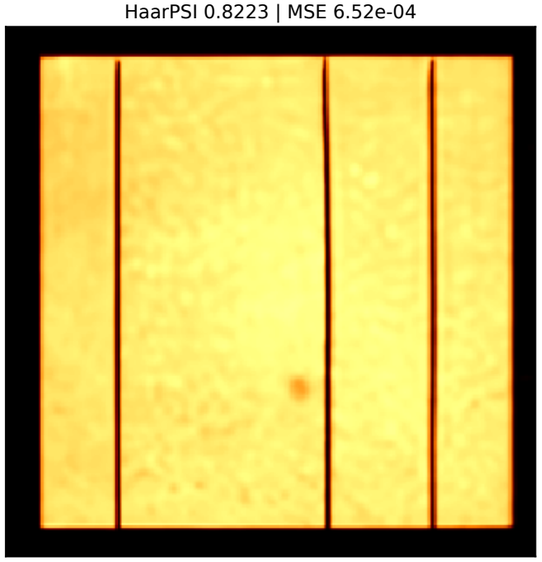
SPGL1-BP
PSNR 22.88 dB | SSIM 0.4228
HaarPSI 0.5902 | MSE 5.15e-03



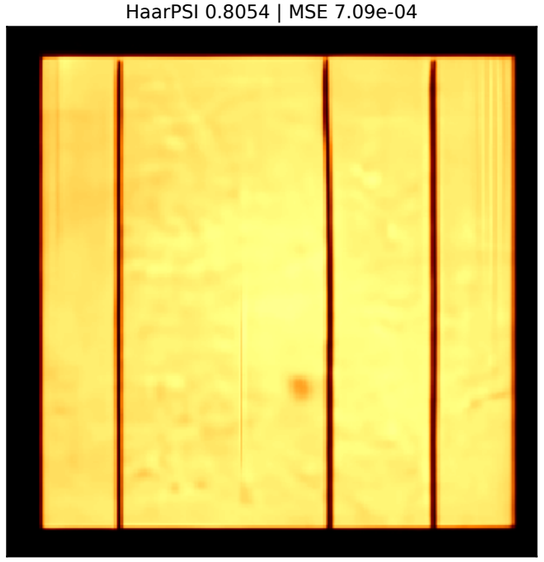
SPGL1-BPDN
PSNR 24.63 dB | SSIM 0.5685
HaarPSI 0.5834 | MSE 3.45e-03



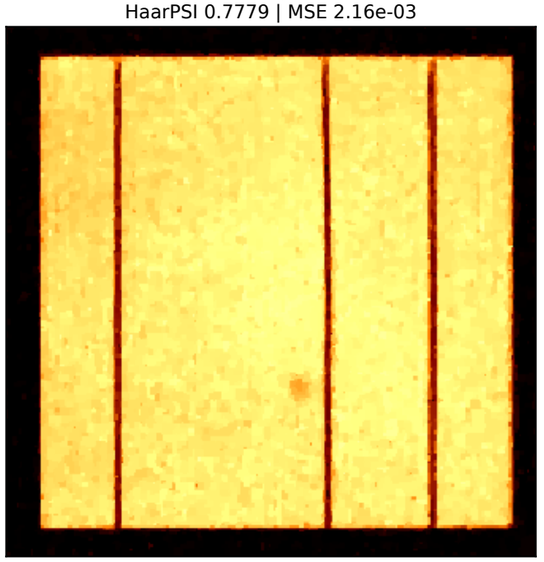
PnP-ADMM tuned-5
PSNR 31.86 dB | SSIM 0.7496
HaarPSI 0.8223 | MSE 6.52e-04



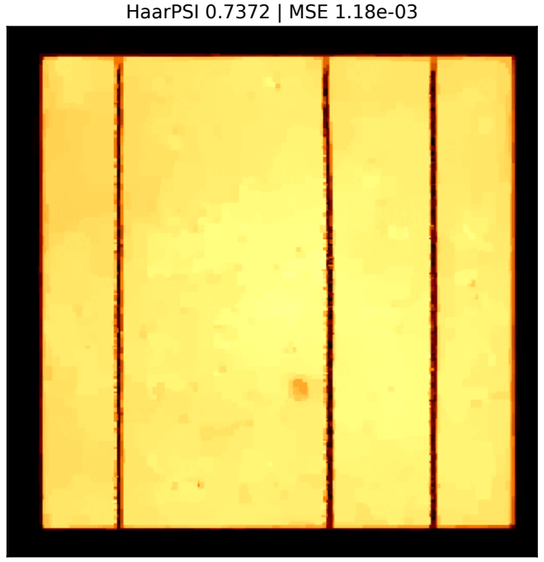
GS-PnP oracle
PSNR 31.49 dB | SSIM 0.7281
HaarPSI 0.8054 | MSE 7.09e-04



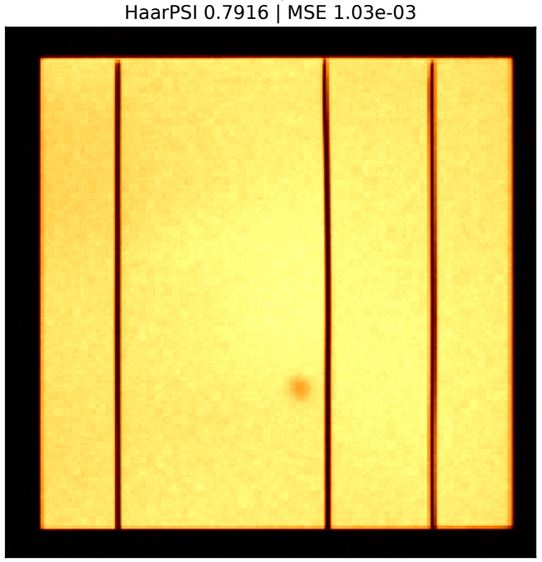
Scalar TV
PSNR 26.66 dB | SSIM 0.6483
HaarPSI 0.7779 | MSE 2.16e-03



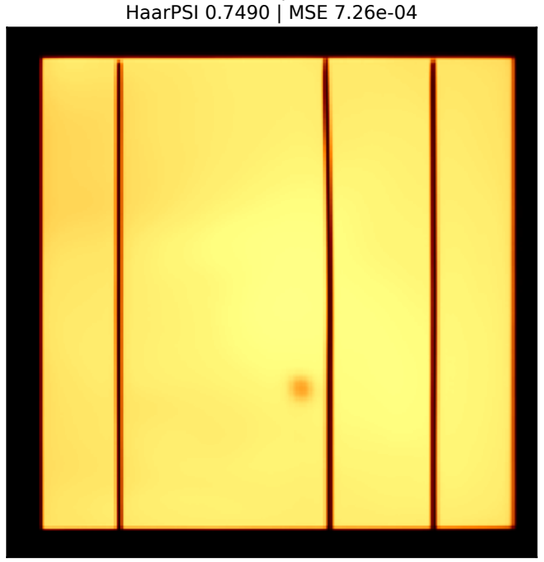
UTV-C60
PSNR 29.27 dB | SSIM 0.7227
HaarPSI 0.7372 | MSE 1.18e-03



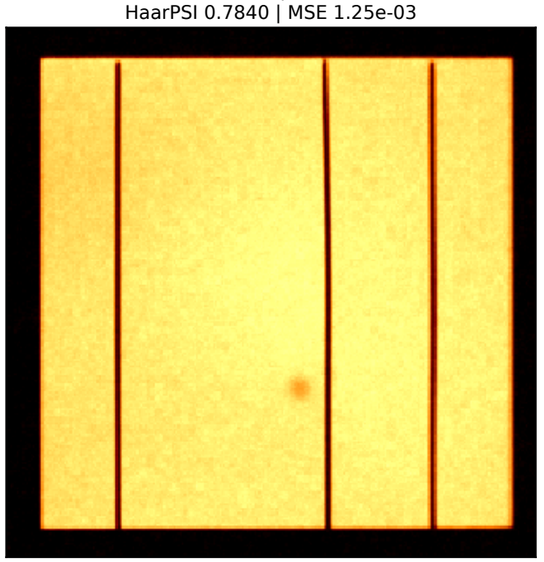
GS-PnP tuned-5
PSNR 29.87 dB | SSIM 0.6002
HaarPSI 0.7916 | MSE 1.03e-03



PnP-ADMM Tune-1 (0008)
PSNR 31.39 dB | SSIM 0.7230
HaarPSI 0.7490 | MSE 7.26e-04

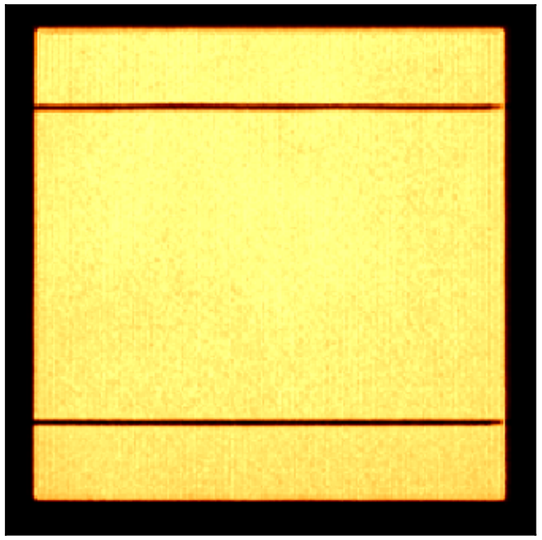


GS-PnP Tune-1 (0008)
PSNR 29.02 dB | SSIM 0.5554
HaarPSI 0.7840 | MSE 1.25e-03

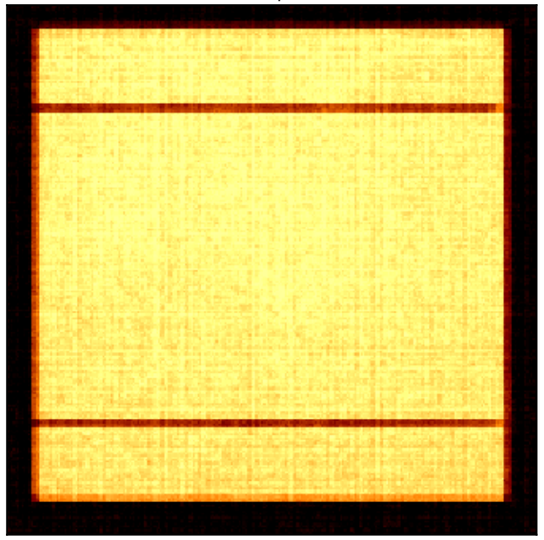


Silicone | test image 0767 | 20% Hadamard + NPL-default noise

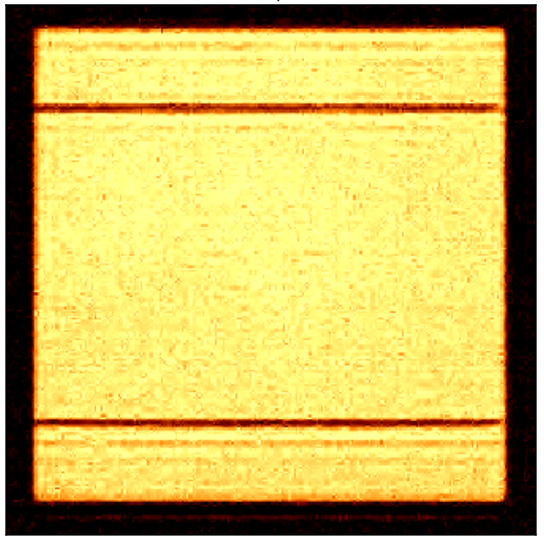
Target
Ground truth



Zero-filled
PSNR 22.40 dB | SSIM 0.4499
HaarPSI 0.4895 | MSE 5.75e-03



SPGL1-BP
PSNR 22.85 dB | SSIM 0.3910
HaarPSI 0.4907 | MSE 5.19e-03



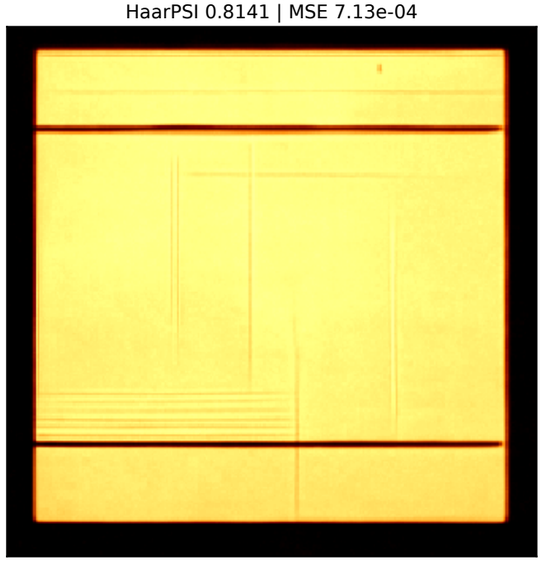
SPGL1-BPDN
PSNR 24.36 dB | SSIM 0.5768
HaarPSI 0.5029 | MSE 3.66e-03



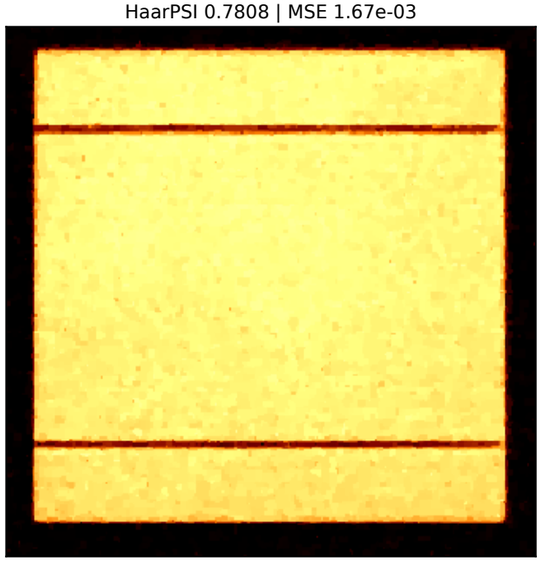
PnP-ADMM tuned-5
PSNR 33.59 dB | SSIM 0.7950
HaarPSI 0.8621 | MSE 4.38e-04



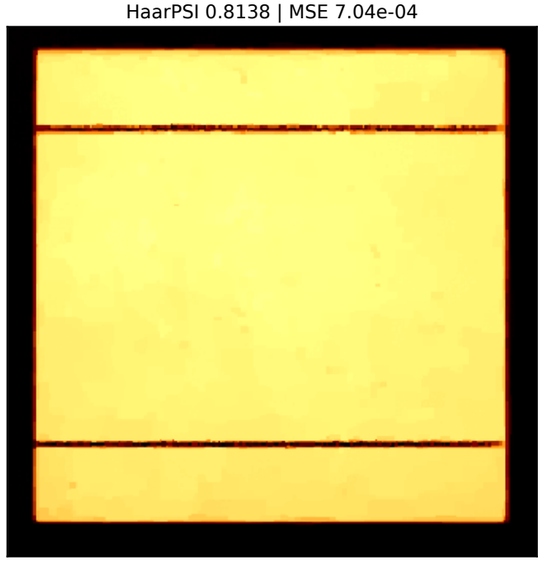
GS-PnP oracle
PSNR 31.47 dB | SSIM 0.7539
HaarPSI 0.8141 | MSE 7.13e-04



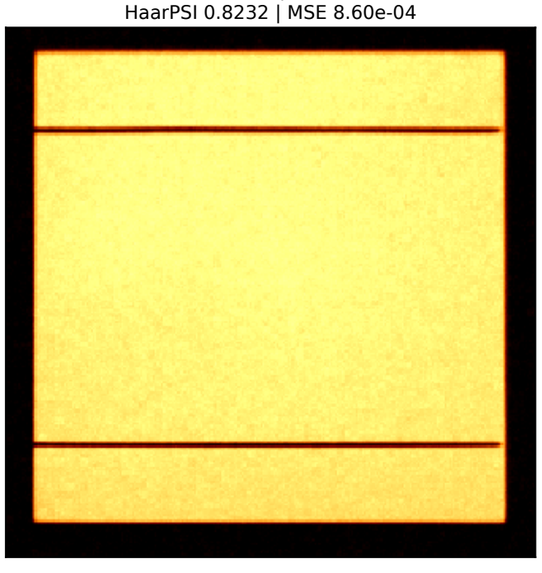
Scalar TV
PSNR 27.78 dB | SSIM 0.6752
HaarPSI 0.7808 | MSE 1.67e-03



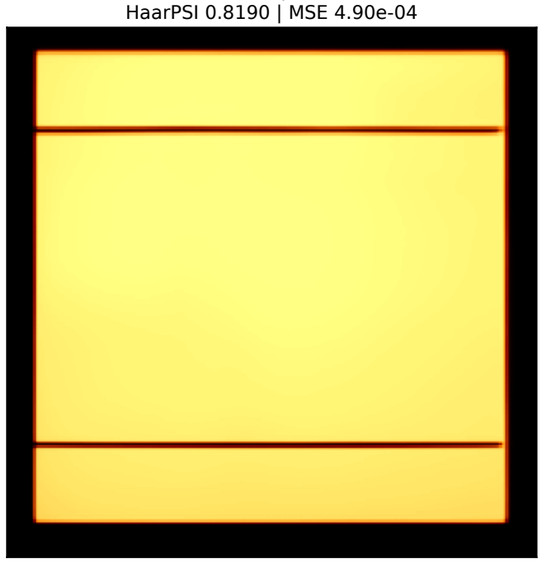
UTV-C60
PSNR 31.53 dB | SSIM 0.7809
HaarPSI 0.8138 | MSE 7.04e-04



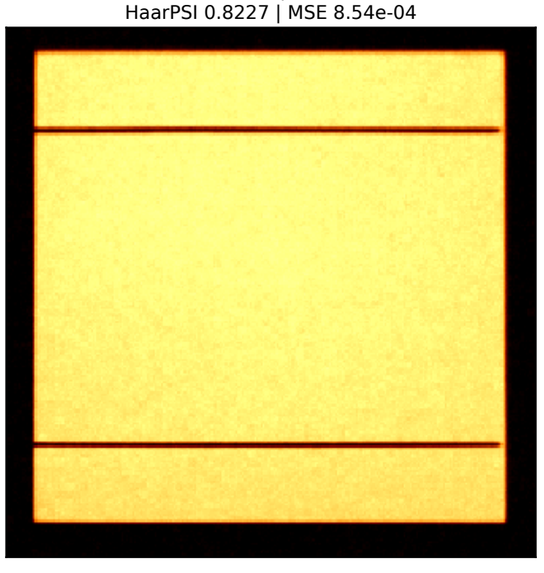
GS-PnP tuned-5
PSNR 30.66 dB | SSIM 0.6221
HaarPSI 0.8232 | MSE 8.60e-04



PnP-ADMM Tune-1 (0008)
PSNR 33.10 dB | SSIM 0.7902
HaarPSI 0.8190 | MSE 4.90e-04

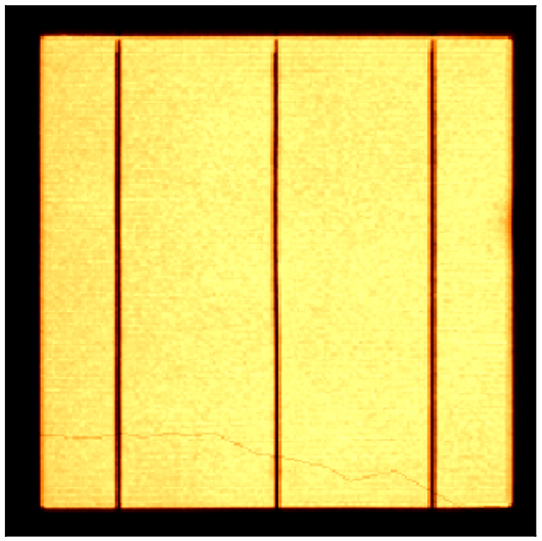


GS-PnP Tune-1 (0008)
PSNR 30.68 dB | SSIM 0.6353
HaarPSI 0.8227 | MSE 8.54e-04

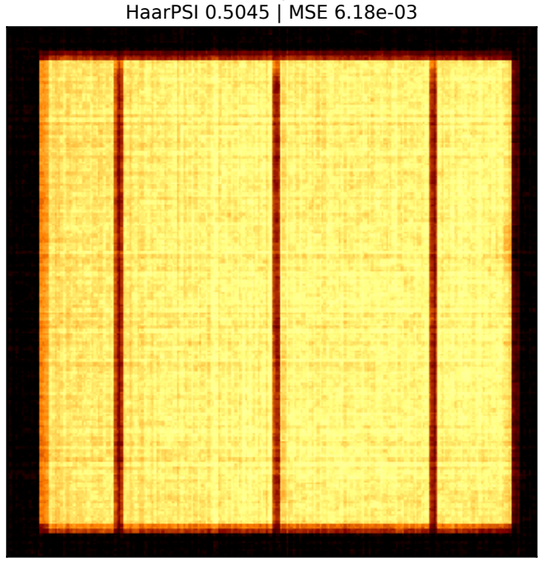


Silicone | test image 0933 | 20% Hadamard + NPL-default noise

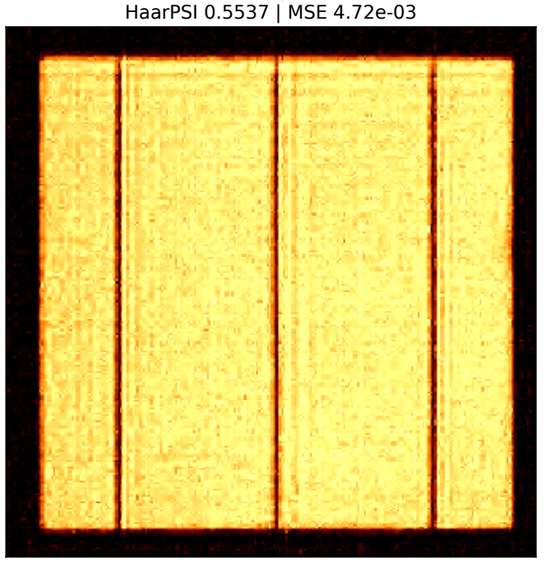
Target
Ground truth



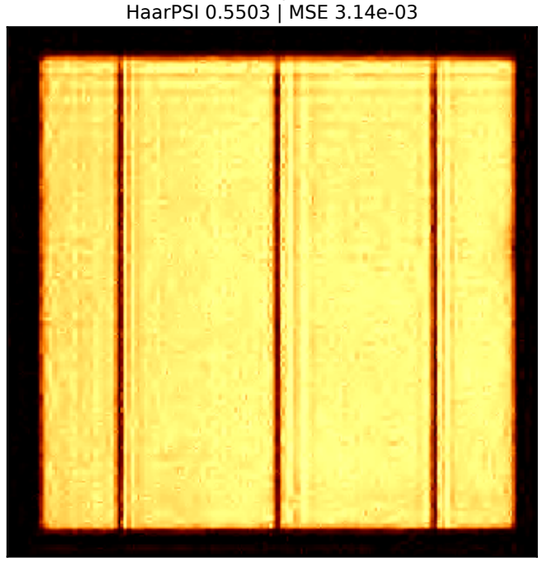
Zero-filled
PSNR 22.09 dB | SSIM 0.4311
HaarPSI 0.5045 | MSE 6.18e-03



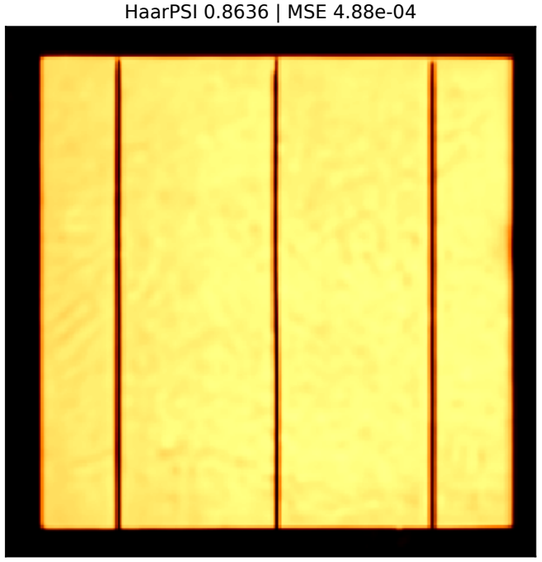
SPGL1-BP
PSNR 23.26 dB | SSIM 0.4103
HaarPSI 0.5537 | MSE 4.72e-03



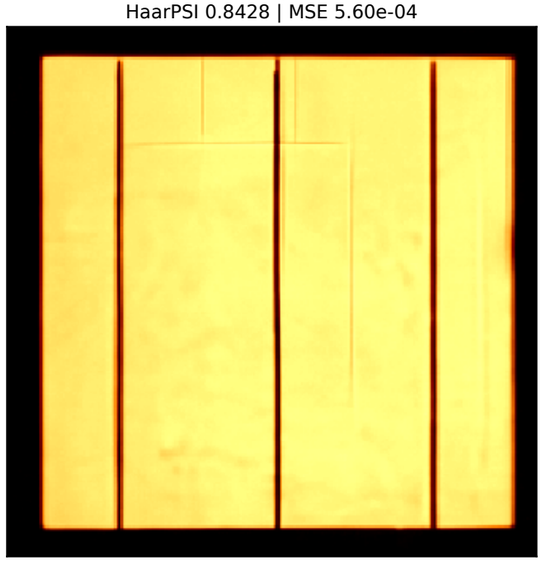
SPGL1-BPDN
PSNR 25.03 dB | SSIM 0.5825
HaarPSI 0.5503 | MSE 3.14e-03



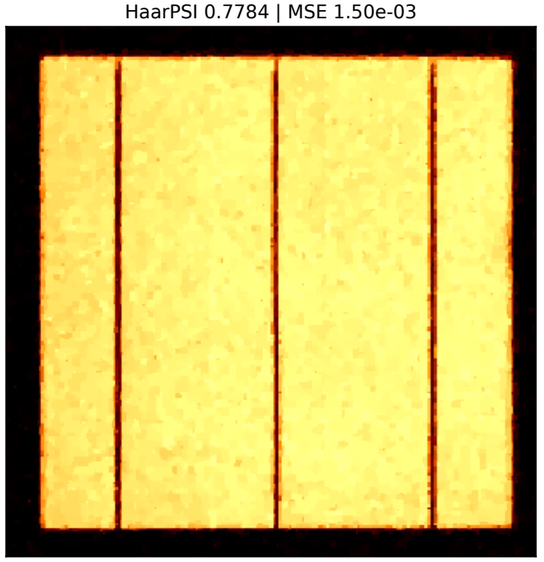
PnP-ADMM tuned-5
PSNR 33.12 dB | SSIM 0.8003
HaarPSI 0.8636 | MSE 4.88e-04



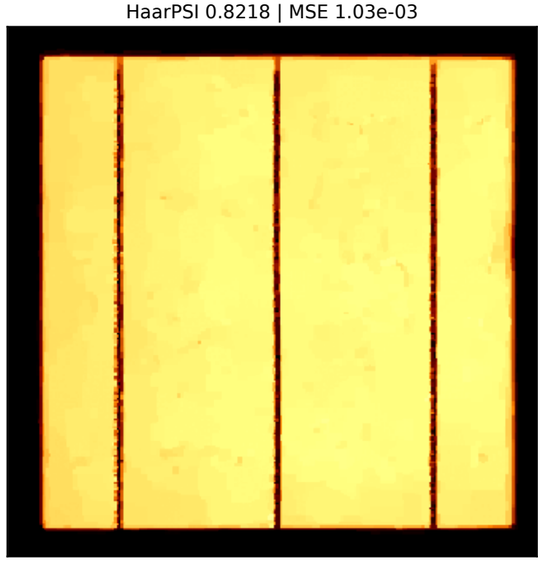
GS-PnP oracle
PSNR 32.52 dB | SSIM 0.7858
HaarPSI 0.8428 | MSE 5.60e-04



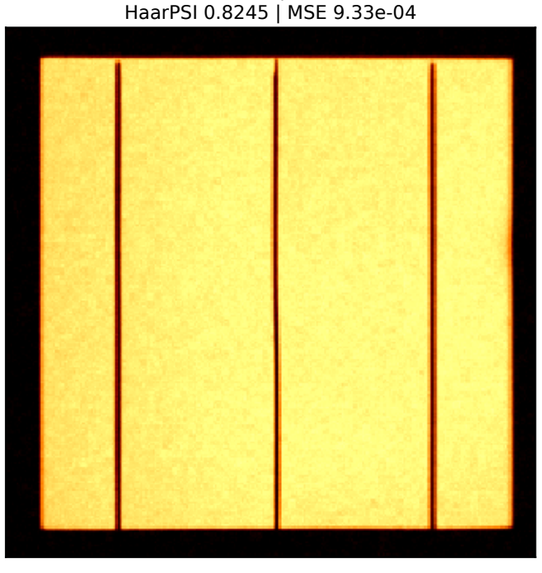
Scalar TV
PSNR 28.24 dB | SSIM 0.6845
HaarPSI 0.7784 | MSE 1.50e-03



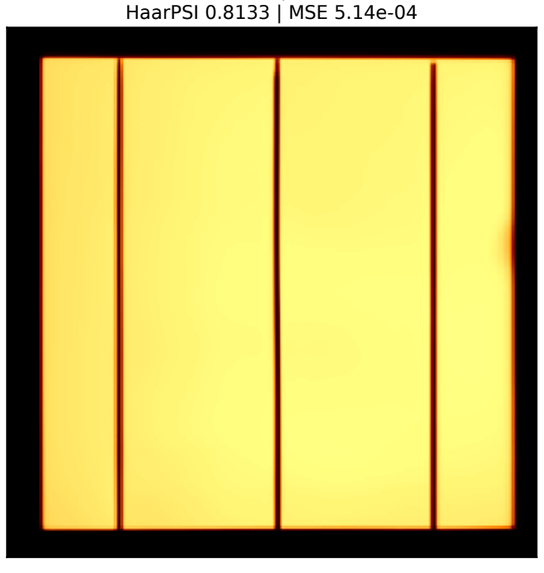
UTV-C60
PSNR 29.87 dB | SSIM 0.7826
HaarPSI 0.8218 | MSE 1.03e-03



GS-PnP tuned-5
PSNR 30.30 dB | SSIM 0.6241
HaarPSI 0.8245 | MSE 9.33e-04



PnP-ADMM Tune-1 (0008)
PSNR 32.89 dB | SSIM 0.7925
HaarPSI 0.8133 | MSE 5.14e-04



GS-PnP Tune-1 (0008)
PSNR 30.19 dB | SSIM 0.6364
HaarPSI 0.8181 | MSE 9.57e-04

